

SMITHIAN FOLD THEORY V3

From One World to Earth

An Exact, Zero-Parameter and Machine-Closed Reconstruction of Earth and Environmental
Sciences from Smithian Fold Theory

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in the same concept DOI lineage. The correction changes no scientific claim,

evidence classification, mathematical value, source identity or machine record.

***Lean verification integration.** The current SFT ordered census passed the*

independent Lean 4 verification layer on 2 August 2026: 2,777 accepted claims,

898,902 candidate records and decisions, 11,108 controls, seventeen branches

and no reported issue. Lean natively proves the two-class operational root and

its unique survivor and constructs proof-bearing acceptance-gate certificates.

The remaining claim content is checked as the complete registered artifact

graph; that distinction is retained throughout this successor.

Abstract

This successor presents the complete current `earth_environment` paper surface: 75 live model-admitted claims, 19,200 generated candidates, 75 unique survivors and 300 passed controls. It integrates 1 claim section(s) not present in the preceding abstract's dated surface and remains complete only to this versioned registered census, with lawful extension open. The independent Lean 4 layer returned PASS for this surface as part of all 2,777 current claims.

The foundation reconstructs Earth-system identity, reservoirs, boundaries, stocks, flows, energy and matter ledgers, geological order and history, interior and geodynamic carriers, rupture and seismic records, volcanism, water and ice, atmosphere, weather, ocean, climate, biogeochemical cycling, environmental transport and quality, observational evidence classes and hazard handoffs. It explicitly distinguishes observation from retrieval, proxy, reconstruction, model and forecast; observed absence from non-detection, censoring and missingness; stock from flow; hazard from risk; and universal upstream law from contingent Earth history.

The quantitative headline is a pre-source exponent prediction for earthquake magnitude-frequency counts. Exact candidate enumeration uniquely selected exponent 1. The first global 2010-2020 mixed-magnitude catalogue was adverse: $N \geq 5/N \geq 6 = 18469/1494$ and its registered exceedance interval $0.076998990840681139-0.084918164588559525$ excludes $1/10$. That result remains adverse and unreclassified. A separately preregistered homogeneous 2020-2026 mww holdout returned $N \geq 6/N \geq 7 = 253/27$, or $27/253$ exceedance; its interval $0.085657503235220589-0.13089188889528297$ contains $1/10$. The branch also derives bounded distinct Earth-ionosphere recurrence modes without importing a dimensional frequency and derives a two-basin tipping structure while rejecting a universal physical half-threshold.

All 356 V1 rows and 407 V2 steps were atomically reviewed. The three Earth-owned prior questions are all reconciled to current receipts. Twenty-one post-seal external source identities contribute ninety-one preregistered features: sixty-seven are present and twenty-four are absent and preserved. One original failed transport remains recorded; an independently preregistered alternate transport is an addendum, not a silent replacement.

Formal admission, implementation validation, observation, measurement, empirical support, confirmation, adverse evidence, unresolved evidence and publication status remain separate classifications. Lean does not reclassify an empirical result, repair an adverse observation or convert a later comparison into an earlier prediction. Lawful criticism, falsification and versioned extension remain open.

Successor headline findings

1. **Independent whole-model result.** Lean 4 returned `PASS` for 2,777/2,777 current claims, 898,902 candidates and decisions, 11,108 controls and all 17 branches, with no issue.
2. **Operational root.** The exact registered two-class grammar is formalised natively, and `presentedOccurrence` is proved to be its unique survivor without an imported or user-declared axiom in the exported theorem's axiom audit.
3. **Typed validation.** The root and generic acceptance implications are native Lean theorems; the remaining scientific claims are checked as complete registered artifacts. Empirical truth is not inferred from proof-assistant acceptance alone.
4. **Fail-closed custody.** A six-file byte-identity mismatch halted the first run. Exact registered bytes were restored, all 385 source bindings were re-audited, and the unchanged verifier then passed.
5. **Current paper surface.** This successor accounts for all 75 current claim(s) in its declared branch ownership boundary; 1 later claim section(s) are integrated in the successor supplement.
6. **Newly integrated results.** `SFT-EARTH-TIPPING-FOLD-LOCK-002` - Normalized Earth tipping Fold lock.

These findings supplement rather than erase the branch-specific headline results retained below. Any adverse, corrected, unresolved, unavailable or chronology-bound result in the preceding scientific record remains governed by its current claim package.

Current status, evidence language and reader map

Status field	Current position
Paper and completion boundary	75 current claims in the declared branch surface. Completion is dated to the current registered census and remains open to lawful versioned extension.
Formal status	The Lean root theorem is kernel checked. Every live model claim has a current admitted receipt and passed the Lean artifact gates. Formal admission does not imply empirical confirmation.
Empirical status	Claim specific. Blind, non-blind, development-observed, holdout, favourable, adverse, null, missing, unavailable, disputed and unresolved distinctions remain exactly as recorded.
Chronology	Claim-specific registration, sealing, custody and observation order controls. Later evidence never becomes an earlier prediction without the registered chronology.
Publication status	Version 1.1.1 is published open access at 10.5281/zenodo.21761661 in the existing Zenodo concept DOI lineage.
Ownership	This paper retains its declared scientific boundary. Lean verification creates no new branch ownership and no application may select a law.
What is not claimed	Permanent closure of science, universal empirical confirmation, conversion of compatibility into confirmation, or superiority to every rival model.

Corpus-wide terminology and evidence key

Term	Reserved meaning in this paper
Theorem	A formally closed proposition at its declared grammar and dependency boundary.
Law	An admitted branch relation with its declared carrier, boundary, dependencies and extension rule.
Claim	The registered unit judged by the engine and bound to one immutable receipt.
Constitution	The rules governing admissible objects, derivations, evidence and publication; not an empirical result.
Derivation	Generation and elimination of the declared candidate space to its surviving structure.
Prediction	A consequence sealed before the matching target is released under the registered custody protocol.
Observation	A source-bound external record; it does not enter candidate generation.
Measurement	An instrument-, method-, condition- and uncertainty-bound observed value.
Reconstruction	A separately implemented regeneration or an explicitly identified inference of a retained state or history.
Exact numerical correspondence	Equality or registered interval relation between exact numerical objects at the declared boundary.
Structural correspondence	A post-derivation relation between forms without a claim of exact numerical prediction.
Boundary correspondence	Agreement restricted to the named interface, limit or ownership boundary.
Compatibility	Non-adverse but non-discriminating evidence.
Support	Relevant evidence that is not unique confirmation.
Confirmation	Used only when the current registered evidence protocol warrants that classification.
Validation	Successful execution of the named formal, computational or empirical test; its kind must be stated.
Adverse result	A registered test result that conflicts with the tested claim at its declared boundary.
Unresolved result	Required evidence remains unavailable, incomplete, disputed or insufficiently classified.
Implementation identity	The hash-bound identity of executable material; implementation success is not empirical confirmation.
Formal, empirical and publication status	Three independent classifications; none substitutes for another.
Foundational closure	Completion of the registered foundation only; not field-wide closure.
Field-wide closure	Completion of the named frozen or registered dated field census.
Current-evidence closure	Closure only to the evidence surface presently registered and preserved.
Extension openness	New questions may enter a later version without rewriting receipts or history.

Proof language is reserved for formal closure; derivation for generated structure; implementation for executable demonstration; prediction for a correctly sealed prospective consequence; observation and measurement for source-bound external records; reconstruction for separately regenerated or inferred states; correspondence for post-derivation relations; compatibility and support for non-unique evidence; confirmation only where the registered protocol authorises it; adverse for conflict; and unresolved where required evidence remains incomplete.

Three reading levels

1. **Conceptual paper.** The abstract, headline findings, branch narrative, major derivations, evidence synthesis, limitations and conclusion provide the readable scientific argument.
2. **Scientific audit layer.** Family and claim sections preserve exact status, dependencies, candidate and survivor counts, controls, chronology, sources, corrections, adverse evidence and reconciliation.
3. **Machine archive.** The repository packages preserve complete candidates, decisions, hashes, receipts, executable traces, source snapshots and certificates. They remain authoritative where prose abbreviates display.

Editorial change control

This successor improves currency, expression, typography and navigation. It does not change scientific meaning, chronology, evidence class, branch ownership, claim status or machine identity. Any conflict between authoritative records must be resolved by explicit supersession or reported to Maria Smith; prose alone cannot manufacture agreement.

Results first: the Earth findings

Headline result	Exact or bounded result	Scientific meaning
Complete foundational closure	74 laws; 18,944 candidates; 74 sole survivors; 296 passing controls	Every foundation obligation has a root trace, unique survivor, independent reconstruction, empirical comparison and immutable receipt.
Earth-system object	bounded joint reservoirs, interfaces, state, scale and time	A convenient variable set cannot silently become the Earth system.
Observation and absence	direct, retrieval, proxy, reconstruction, model and forecast are distinct; absence, non-detection, censoring, missingness and outside-scope are distinct	The symbol 0 may represent declared absence, never ordinary nothing or an unknown value.
Planetary ledger	source + storage change + transfer + export retained at a declared boundary	Conservation cannot be manufactured by deleting or fitting a residual.
Geological history	positive ordered layers plus explicit contacts, interruptions, dates and alternatives	Historical reconstruction preserves gaps instead of converting them into a smooth continuum.
Earthquake size-frequency exponent	unique structural exponent 1; mixed catalogue adverse 18469/1494; homogeneous holdout 253/27	The prediction precedes data; adverse and favourable results coexist without reclassification.
Earth-ionosphere recurrence	bounded carrier forces distinct modes; dimensional frequency remains measured	The model predicts mode structure without fitting a hertz value.
Earth-system tipping	exact two-basin/shared-image witness; no universal physical 1/2 threshold	Fold structure does not overwrite contingent environmental thresholds.
Climate evidence	weather, climate, forcing, feedback, variability and attribution retain timescale and comparison records	Model or forecast output cannot substitute for observation.
Environmental risk	hazard, exposure, vulnerability, capacity and consequence remain separate	Earth science hands social, medical and engineering consequences across explicit boundaries.

Status and scope

PUBLISHED OPEN-ACCESS BRANCH PAPER. The canonical release contains this Markdown paper, its rendered PDF, the evidence/source archive and checksum ledger.

Foundational closure means every question in the frozen seventy-four-obligation Layer One surface has one engine-admitted theorem inside its declared exact grammar, with a complete root path, adverse controls, independent reconstruction and purpose-matched post-seal comparison. It does not claim the complete Layer Two reconstruction of every Earth discipline and it does not permanently lock the branch. Lawful extensions remain open through the same public engine.

Public scientific mission and admission boundary

Ernos Labs is an open-source science movement, verification platform and public tree of knowledge founded by Maria Smith. Its purpose is not to replace one authority with another. It is to make scientific authority narrow, inspectable and revocable: a claim is admitted only through its complete derivation chain, generated alternatives, eliminations, unique survivor, adverse controls, measurement custody where applicable and unchanged-engine receipt. Open criticism is unrestricted and necessary; scientific admission is the separate machine-checked act of satisfying that public standard.

Maria Smith developed Smithian Fold Theory outside formal academic education, institutional research employment and conventional grant funding. That fact is not offered as evidence for a theorem; the derivations and observations carry the entire scientific burden. It is evidence about access. A credential-first system loses more than individual opportunity: it loses unknown questions, methods and discoveries from minds that capital and status never authorise. This work therefore treats Maria Smith's authorship neither as exceptionalism nor as a reason for dismissal, but as an indictment of every scientific contribution lost when financial gatekeeping is presented as rigour.

The institutional argument is empirical, not a claim that every institution or funded researcher acts in bad faith. Published studies document sponsor-linked differences in outcomes and conclusions, commercial influence over research agendas, limits and sensitivities in grant review, underpublication of null results, and inequalities created by paywalls and article-processing charges. Those findings establish that funding, prestige, publication and consensus are selection systems with incentives and failure modes. They cannot substitute for a public proof and evidence chain. Expertise, measurement and adversarial review remain indispensable; institutional permission does not select a fundamental law.

For Earth and environmental science, an expensive sensor, institutional model, consensus map, forecast score or proprietary data product cannot select a natural law. Field observation, instrument output, retrieval, interpolation, proxy reconstruction,

simulation, assimilation and forecast remain separate evidence classes unless a separately derived and tested bridge connects them. Public source and method custody are conditions of admission, not optional documentation.

Maria Smith retains copyright and scientific authorship. Papers and documentation are licensed CC BY 4.0 and code is licensed Apache-2.0, permitting inspection, copying, criticism, reproduction, modification and redistribution with attribution. "Ernos Labs" is a separate, revocable standards-conformance designation. A reuse or fork may use the open work under its licences, but may describe itself as Ernos Labs only while it preserves the public empirical constitution, complete adverse evidence, unchanged admission route, critical review and community standards.

Independent replications, lawful extensions, corrections and attempted invalidations are invited. Credentials cannot rescue a failed gate and lack of credentials cannot prevent a reproducible result from being evaluated. Contact Maria.Smith.Sftoe@gmail.com, submit through <https://discord.gg/ucwGryVxGr>, or inspect the public project at <https://github.com/MettaMazza>.

This position rejects paywalls, credential gates and opaque oracles as substitutes for transparent derivation and inspectable evidence. It does not reject expertise; it requires expertise to remain reproducible and publicly answerable.

Method: one root, one unchanged engine, no target-selected law

The derivational root is There Is No Nothing. The One and Fold, exact arithmetic, information, computation, physics, chemistry, materials and biology enter only through already admitted dependencies. Before any external source identity or outcome was opened, the branch froze seventy-four claims, eight binary preservation axes per claim, 18,944 candidates, all predicted consequence labels and falsification conditions. Each literal 256-form product is complete by construction. One form alone preserves carrier, boundary, relation, record, evidence class, provenance, positive-finite successor closure and absence of an extra rule.

The capability-closed predictor cannot access the filesystem, network, clock, environment, subprocess or target. A distinct custodian releases the source-derived target only after the derivation seal. Every claim is reconstructed by a separate implementation. Four mandatory structural controls reject a false premise, changed source identity, changed candidate artifact and boundary violation. The canonical engine and verification-authority seals were checked before, during and after admission and were not edited.

External evidence constitution

The sources were chosen after the complete derivation seal by question-purpose and authoritative or primary custody. Their ninety-one features were preregistered before capture outcomes. Sixty-seven appeared in the captured surfaces. Twenty-four did not and remain absent. The original Smithsonian transport returned HTTP 403 and remains a failed row; a separately registered WFS addendum succeeded without rewriting it. Earth evidence tests declared consequence boundaries and one exact numerical prediction. It does not select any structural survivor.

V1/V2 atomic reconciliation

The accountability audit reviewed all 763 registered V1/V2 entries. It isolated three Earth-owned questions: Earth-ionosphere recurrence, Earth-system tipping and earthquake magnitude-frequency. All three map to current V3 receipts. Prior answers were never admitted as V3 premises; they ensured that the clean reconstruction did not omit known earlier work.

Reading the exhaustive derivation ledger

Each of the seventy-four sections below states its theorem, receipt dependencies, complete eight-axis grammar, 256-form census, sole survivor, depth-independent base and successor, operational witnesses, scientific meaning, four adverse controls, independent implementation, external evidence scope, complete feature accounting, falsification boundary and immutable evidence identities.

Ledger family 1: Earth-System Identity, Reservoirs, Boundaries and Observation

Earth science begins with a bounded joint carrier, not a map label. Reservoirs, interfaces, state, time, scale and observation class remain explicit. Detected absence, non-detection, censoring, missingness and outside-scope status are different records; the symbol 0 may represent declared absence but may not silently convert an unknown or missing observation into ordinary numerical nothing.

1. Earth-system identity

Claim identity: SFT-EARTH-SYSTEM-001

Question and exact theorem. The Earth system is a declared joint organization of distinguishable reservoirs and interfaces whose state, boundary, scale and observation time remain explicit.

```
nested-earth-reservoir-interface-carrier__earth-state-observation-bounded__bounded
-joint-state-continuation__reservoirs-interfaces-time-and-scale-held__evidence-cla
ss-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-row
s__no-extra-rule
```

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001

- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	nested-earth-reservoir-interface-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	earth-state-observation-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	bounded-joint-state-continuation	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	reservoirs-interfaces-time-and-scale-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: nested-earth-reservoir-interface-carrier__earth-state-observation-bounded__bounded-joint-state-continuation__reservoirs-interfaces-time-and-scale-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- carrier-present: the claim retains a nonempty named carrier; passed true.
- relation-distinct: the carrier and relation are not silently conflated; passed true.

- `evidence-boundary`: the evidence boundary remains distinct from the result carrier; passed `true`.
- `exact-fold-witness`: the family's pre-source exact Fold construction reproduces; passed `true`.

Scientific meaning. The theorem supplies the minimum complete Earth-system object: named reservoirs plus interfaces, state, scale and time. It prevents a convenient variable list from being mistaken for the system it samples.

Earth science begins with a bounded joint carrier, not a map label. Reservoirs, interfaces, state, time, scale and observation class remain explicit. Detected absence, non-detection, censoring, missingness and outside-scope status are different records; the symbol 0 may represent declared absence but may not silently convert an unknown or missing observation into ordinary numerical nothing.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:d964ab5226181573a69b73cdc11e095774d221c3d5b038139a3d125c96509712.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:3af81af13a05b43454668ba8656b9a19b237efc3853b6b4fc27f58b76a10b428.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:f2adb396207e2196e2a67e44e07ebb3fd81ed04a325580d1122fe0098229c1fb.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:b63fad8ec276e91b73ea215f9f42ccfd5722a18c95595606d9ec01c7100770c6.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:6fca232cb0d56e08c5de891abcea56b7b3d60725e76d7d7f9a0ce8705a40e48e; certificate
sha256:29c90626834c04ae31c439570222cea8efb4f94277f25eef391bcab4ca65d14c; external-validation
identity sha256:5abe1026c80dc7b7f01e3f42d5cd7feb3e6f9f5dc11b3cc5eeceee1366e9e7a6.

Post-seal empirical comparison. Directness:

authoritative_observation_or_primary_measurement_boundary_correspondence. Disposition:
purpose_matched_authoritative_source_features_present. Predicted consequence: earth:nested-earth-reservoir-interface-carrier__bounded-joint-state-continuation__reservoirs-interfaces-time-and-scale-held__earth-state-observation-bounded. Source-derived consequence: earth:nested-earth-reservoir-interface-carrier__bounded-joint-state-continuation__reservoirs-interfaces-time-and-scale-held__earth-state-observation-bounded. Exact boundary correspondence: true. External evidence selected the survivor: false. Formal structure relabelled as direct measurement: false. Model, forecast, proxy or retrieval relabelled as observation: false.

Sources and complete feature accounting:

- NASA-EARTHDATA-SEARCH-001 - NASA Earth Science Data and Information System, [Earthdata Search data discovery and access](#); class authoritative_data_catalog; transport captured; features 4/4 present; present: dataset identity, spatial and temporal coverage, instrument or platform provenance, download or service access; absent and preserved: none.
- USGS-LANDSAT-ACCESS-001 - U.S. Geological Survey Landsat Missions, [Landsat Data Access](#); class authoritative_remote_sensing_archive; transport captured; features 0/4 present; present: none; absent and preserved: sensor and mission identity, Earth surface observation, archive access, product and processing distinction.
- NOAA-GHCND-001 - NOAA National Centers for Environmental Information, [Global Historical Climatology Network Daily](#); class authoritative_station_climate_archive; transport captured; features 5/5 present; present: station identity and metadata, daily weather observations, quality flags, record length and variable coverage, archive version; absent and preserved: none.

Comparison record:

- sft-earth-system-001-external-earth-evidence: predicted earth:nested-earth-reservoir-interface-carrier__bounded-joint-state-continuation__reservoirs-interfaces-time-and-scale-held__earth-state-observation-bounded; source-derived earth:nested-earth-reservoir-interface-carrier__bounded-joint-state-continuation__reservoirs-interfaces-time-and-scale-held__earth-state-observation-bounded; exact match True
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if system identity is claimed while a participating reservoir, interface, spatial boundary, timescale or observation class is silently erased.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json;
 source manifest sha256:db820f2d0ff070a2bf441b732a3e2beadd8d0f8be9f18d103b627c01da7ea134;
 derivation seal sha256:9c4c0468ef7b51475940eba855a50d245cd705d97b15aff4da708b9c979bb1c1;
 engine receipt sha256:b2c266ed350d33a40b236d95727e55fcddb38e7537898576bab94041c0391053 at
 receipts/engine/model_admitted/SFT-EARTH-SYSTEM-001-b2c266ed350d33a4.json; empirical
 validation sha256:014d9fcab7520f2364d907f4b07190123e5845de3400e03ebe4e18dc0708ebba;
 measurement receipt
 sha256:faa5494bbbd6c458ca8ea2864177044fbbbea620d55a64cc3cfc4e1c0b729390c; isolation
 sha256:3c25119e7181b35e2bea5fbff89afbfa864a992c9574dbe966b9eedade2c96ca; custody
 sha256:9b0483db5c64176e4659d7ff2cd3cf7a9a2ad0fb7aed7246483a4df4d4a11503.

2. Earth-system reservoir

Claim identity: SFT-EARTH-RESERVOIR-001

Question and exact theorem. An Earth reservoir is a source- and time-bounded carrier with explicit membership, stored quantity, interfaces and observation units.

bounded-stock-membership-carrier__reservoir-definition-bounded__membership-storage
 -and-transfer-relation__members-stock-units-boundary-and-time-held__evidence-class
 -explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows_
 _no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-SYSTEM-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	bounded-stock-membership-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	reservoir-definition-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	membership-storage-and-transfer-relation	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	members-stock-units-boundary-and-time-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: bounded-stock-membership-carrier__reservoir-definition-bounded__membership-storage-and-transfer-relation__members-stock-units-boundary-and-time-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- carrier-present: the claim retains a nonempty named carrier; passed true.
- relation-distinct: the carrier and relation are not silently conflated; passed true.
- evidence-boundary: the evidence boundary remains distinct from the result carrier; passed true.
- exact-fold-witness: the family's pre-source exact Fold construction reproduces; passed true.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar. Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

Earth science begins with a bounded joint carrier, not a map label. Reservoirs, interfaces, state, time, scale and observation class remain explicit. Detected absence, non-detection, censoring, missingness and outside-scope status are different records; the symbol 0 may represent declared absence but may not silently convert an unknown or missing observation into ordinary numerical nothing.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:e83507c600024b2479c4c73555920c5286d18eb02ee26bb3656842b18f1f0633.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:84d83e05113d883b56152d273245ac268870d34fa2e075de9b952f50e1de8400.

- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:6626a3b17207de2ac20917bad0c81c07354a4bc1a4f11731e98fc2fe7ab324b7.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:a930356f7f34094b1140b3e5e07c6285f3826f321c8c11f122efa8e5696aa75c.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:5c1426c97985ee828064013379cb3046e3d4755c779f4f8939c441e5f3fc45b2; certificate
sha256:c51c1c3a2ea5096d998ea28921a81bbbab3efbf624509386f12d6ce24f791045; external-validation
identity sha256:996fc7098d508d16451b779e044f93166eee71939dbad424fc33504949b83b1b.

Post-seal empirical comparison. Directness:

`authoritative_observation_or_primary_measurement_boundary_correspondence`. Disposition:
`purpose_matched_authoritative_source_features_present`. Predicted consequence: `earth:bounded-stock-membership-carrier__membership-storage-and-transfer-relation__members-stock-unit-s-boundary-and-time-held__reservoir-definition-bounded`. Source-derived consequence: `earth:bounded-stock-membership-carrier__membership-storage-and-transfer-relation__members-stock-units-boundary-and-time-held__reservoir-definition-bounded`. Exact boundary correspondence: `true`. External evidence selected the survivor: `false`. Formal structure relabelled as direct measurement: `false`. Model, forecast, proxy or retrieval relabelled as observation: `false`.

Sources and complete feature accounting:

- NASA-EARTHDATA-SEARCH-001 - NASA Earth Science Data and Information System, [Earthdata Search data discovery and access](#); class `authoritative_data_catalog`; transport captured; features 4/4 present; present: dataset identity, spatial and temporal coverage, instrument or platform provenance, download or service access; absent and preserved: none.
- USGS-LANDSAT-ACCESS-001 - U.S. Geological Survey Landsat Missions, [Landsat Data Access](#); class `authoritative_remote_sensing_archive`; transport captured; features 0/4 present; present: none; absent and preserved: sensor and mission identity, Earth surface observation, archive access, product and processing distinction.
- NOAA-GHCND-001 - NOAA National Centers for Environmental Information, [Global Historical Climatology Network Daily](#); class `authoritative_station_climate_archive`; transport captured; features 5/5 present; present: station identity and metadata, daily weather observations, quality flags, record length and variable coverage, archive version; absent and preserved: none.

Comparison record:

- `sft-earth-reservoir-001-external-earth-evidence`: predicted `earth:bounded-stock-membership-carrier__membership-storage-and-transfer-relation__members-stock-units-boundary-and-time-held__reservoir-definition-bounded`; source-derived `earth:bounded-stock-membership-carrier__membership-storage-and-transfer-relation__members-stock-units-boundary-and-time-held__reservoir-definition-bounded`; exact match `True`
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if membership or boundary changes without record, or if an unobserved remainder is treated as absent.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity

- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json;
 source manifest sha256:11aeld653a0616c09cf687315101cfdef000ee068eda2d645a38b7d545c57072;
 derivation seal sha256:9bd096d213eaed4b1f799785e2c4bb62819f1f4795c52160884ddb399f01fa47;
 engine receipt sha256:a54878c96ce4f428686e4ad22ba9dc7e81f23af5205ec5cc4c9b4741b5b24d85 at
 receipts/engine/model_admitted/SFT-EARTH-RESERVOIR-001-a54878c96ce4f428.json; empirical
 validation sha256:5f3199e5c539292915cbe176ce9cc1d2d8507fe0afcc65fbc1060754c936fb2a;
 measurement receipt
 sha256:86f2f02cff16949b91f068d910a2bfd53f7146757c2690d82b3806a3a67c7596; isolation
 sha256:0f79ad45ec7ec2896e155827e33ea8bb93a6c632154f5e5c4ba9f66801990c9d; custody
 sha256:cec7dac8305f85d74b97b9767364fd755242f05f0e1adea57357f608a512b724.

3. Earth-system boundary and interface

Claim identity: SFT-EARTH-BOUNDARY-001

Question and exact theorem. An Earth-system boundary distinguishes adjacent carriers while retaining every registered cross-boundary transfer, orientation and observation limit.

two-reservoir-interface-carrier__interface-resolution-bounded__cross-boundary-transfer-relation__orientation-area-route-and-time-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001

- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-RESERVOIR-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	two-reservoir-interface-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	interface-resolution-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	cross-boundary-transfer-relation	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	orientation-area-route-and-time-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: two-reservoir-interface-carrier__interface-resolution-bounded__cross-boundary-transfer-relation__orientation-area-route-and-time-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retain-all-rows__no-extra-rule.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- carrier-present: the claim retains a nonempty named carrier; passed true.
- relation-distinct: the carrier and relation are not silently conflated; passed true.
- evidence-boundary: the evidence boundary remains distinct from the result carrier; passed true.
- exact-fold-witness: the family's pre-source exact Fold construction reproduces; passed true.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar. Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

Earth science begins with a bounded joint carrier, not a map label. Reservoirs, interfaces, state, time, scale and observation class remain explicit. Detected absence, non-detection, censoring, missingness and outside-scope status are different records; the symbol 0 may represent declared absence but may not silently convert an unknown or missing observation into ordinary numerical nothing.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:12a05c0b4d8b95b57c3f7f28db448854073103cfaeccbb9895c2f637ef27d578.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:1b116c73c0563d763cea5e10382a4bec7dd2e7960877dcc3b665692c52a99661.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:382e2aed9eb89220cd0f0c40572a0f16bb2cec50a14b0a04bed6baad0fe34a32.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:07455afe89ee4b8c71199243422c765233b0eee32e71f254414175a97e25ec8b.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:0ab5d6fb8d2cff354286ffb5b75811d9c381a59d0cb1005b8f01e4ff81b2728d; certificate
sha256:ffc402bf061af4a18eb78b650c344ed779374bde19ffc06317341176d0e8b38d; external-validation
identity sha256:77e64d4d9b8f1a624b35fcc39a64fae08605847f62b93ff34150290c4ecd87b3.

Post-seal empirical comparison. Directness:

authoritative_observation_or_primary_measurement_boundary_correspondence. Disposition:
purpose_matched_authoritative_source_features_present. Predicted consequence: earth:two-reservoir-interface-carrier__cross-boundary-transfer-relation__orientation-area-route-and-

time-held__interface-resolution-bounded. Source-derived consequence: earth:two-reservoir-interface-carrier__cross-boundary-transfer-relation__orientation-area-route-and-time-held__interface-resolution-bounded. Exact boundary correspondence: true. External evidence selected the survivor: false. Formal structure relabelled as direct measurement: false. Model, forecast, proxy or retrieval relabelled as observation: false.

Sources and complete feature accounting:

- NASA-EARTHDATA-SEARCH-001 - NASA Earth Science Data and Information System, [Earthdata Search data discovery and access](#); class authoritative_data_catalog; transport captured; features 4/4 present; present: dataset identity, spatial and temporal coverage, instrument or platform provenance, download or service access; absent and preserved: none.
- USGS-LANDSAT-ACCESS-001 - U.S. Geological Survey Landsat Missions, [Landsat Data Access](#); class authoritative_remote_sensing_archive; transport captured; features 0/4 present; present: none; absent and preserved: sensor and mission identity, Earth surface observation, archive access, product and processing distinction.
- NOAA-GHCND-001 - NOAA National Centers for Environmental Information, [Global Historical Climatology Network Daily](#); class authoritative_station_climate_archive; transport captured; features 5/5 present; present: station identity and metadata, daily weather observations, quality flags, record length and variable coverage, archive version; absent and preserved: none.

Comparison record:

- sft-earth-boundary-001-external-earth-evidence: predicted earth:two-reservoir-interface-carrier__cross-boundary-transfer-relation__orientation-area-route-and-time-held__interface-resolution-bounded; source-derived earth:two-reservoir-interface-carrier__cross-boundary-transfer-relation__orientation-area-route-and-time-held__interface-resolution-bounded; exact match True
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if a transfer is counted on only one side, if orientation is lost, or if the interface is replaced by an ungenerated continuum.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json;
source manifest sha256: 62fba4d883bef295a0a3fa04162b8539b36225e9aeed33602fc6a5fcbfafcbb8;

derivation seal sha256:beb26f4ebd6404ea17b9dala50ba8f51c10b587d74a495c6bcef46f85ac6cb81;
 engine receipt sha256:6e6e0c9566e336744b56a604ccfdab5908911453fd262150562d76b8066e19a6 at
 receipts/engine/model_admitted/SFT-EARTH-BOUNDARY-001-6e6e0c9566e33674.json; empirical
 validation sha256:9ddc30b5864ca24cf81949cdd73a03545a2f4506ab73e519f4cb330c948cebf5;
 measurement receipt
 sha256:542f577585bb63fa29d663657917f4780cf27d491d30ce8f71113231f36c33aa; isolation
 sha256:1720b5f0cb32357447f4f5547e12ea218d46c48e86545869bf57b8f92e5d5363; custody
 sha256:41966b18349e6fe01eb5d736f8d29cee7409f68b143f2c15079544185073e6a6.

4. Earth-system state

Claim identity: SFT-EARTH-STATE-001

Question and exact theorem. An Earth-system state is the complete registered assignment of stocks, relations and observation status at one declared time and scale.

time-indexed-complete-reservoir-carrier__registered-state-vector-bounded__stock-re
 lation-state-assignment__stocks-relations-missingness-and-time-held__evidence-clas
 s-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows
 __no-extra-rule

Routed dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001

- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-BOUNDARY-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	time-indexed-complete-reservoir-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	registered-state-vector-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	stock-relation-state-assignment	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	stocks-relations-missingness-and-time-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: time-indexed-complete-reservoir-carrier__registered-state-vector-bounded__stock-relation-state-assignment__stocks-relations-missingness-and-time-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- carrier-present: the claim retains a nonempty named carrier; passed true.

- `relation-distinct`: the carrier and relation are not silently conflated; passed `true`.
- `evidence-boundary`: the evidence boundary remains distinct from the result carrier; passed `true`.
- `exact-fold-witness`: the family's pre-source exact Fold construction reproduces; passed `true`.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar. Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

Earth science begins with a bounded joint carrier, not a map label. Reservoirs, interfaces, state, time, scale and observation class remain explicit. Detected absence, non-detection, censoring, missingness and outside-scope status are different records; the symbol 0 may represent declared absence but may not silently convert an unknown or missing observation into ordinary numerical nothing.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:7431d9892b043b7a6a283d05d0a6b4f57016fbbd01a700419e5224f1812d41eb.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:44dc005ee32628a9ef3c5704c1c589e18832d2a763eca8f78e1d05d817471302.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:c6b9950af1a706a49bca985201c563f5fa247e31f8180ec4fb1acffab9c9faf2.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:58d4fb648b67d35c3e86df46de885011b232b43956b9bc69de40d6efcc1c8e83.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:e7e688918f428a5092fd32e3fee8ff6557ffea3362b442bdee990e4092e2fbfe; certificate
sha256:a455f4f18047bbceec5b86e030db86cd163cf8e1aab2d521e892efde95551f73; external-validation
identity sha256:b56d891bcc336b34405a57044dc4910403bc3456cf6a3efbac939f19a633698c.

Post-seal empirical comparison. Directness:

`authoritative_observation_or_primary_measurement_boundary_correspondence`. Disposition:
`purpose_matched_authoritative_source_features_present`. Predicted consequence: `earth:time-indexed-complete-reservoir-carrier__stock-relation-state-assignment__stocks-relations-missingness-and-time-held__registered-state-vector-bounded`. Source-derived consequence: `earth:time-indexed-complete-reservoir-carrier__stock-relation-state-assignment__stocks-relations-missingness-and-time-held__registered-state-vector-bounded`. Exact boundary correspondence: `true`. External evidence selected the survivor: `false`. Formal structure relabelled as direct measurement: `false`. Model, forecast, proxy or retrieval relabelled as observation: `false`.

Sources and complete feature accounting:

- NASA-EARTHDATA-SEARCH-001 - NASA Earth Science Data and Information System, [Earthdata Search data discovery and access](#); class `authoritative_data_catalog`; transport captured; features 4/4 present; present: dataset identity, spatial and temporal coverage, instrument or platform provenance, download or service access; absent and preserved: none.
- USGS-LANDSAT-ACCESS-001 - U.S. Geological Survey Landsat Missions, [Landsat Data Access](#); class `authoritative_remote_sensing_archive`; transport captured; features 0/4 present; present: none; absent and preserved: sensor and mission identity, Earth surface observation, archive access, product and processing distinction.
- NOAA-GHCND-001 - NOAA National Centers for Environmental Information, [Global Historical Climatology Network Daily](#); class `authoritative_station_climate_archive`; transport captured; features 5/5 present; present: station identity and metadata, daily weather observations, quality flags, record length and variable coverage, archive

version; absent and preserved: none.

Comparison record:

- sft-earth-state-001-external-earth-evidence: predicted earth:time-indexed-complete-reservoir-carrier__stock-relation-state-assignment__stocks-relations-missingness-and-time-held__registered-state-vector-bounded; source-derived earth:time-indexed-complete-reservoir-carrier__stock-relation-state-assignment__stocks-relations-missingness-and-time-held__registered-state-vector-bounded; exact match True
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if favourable variables alone define the state or if missing and outside-scope entries are collapsed into absence.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json; source manifest sha256:088c417d4ef01db9b2f3ce8ee2998e4d7a6a637d86947a5f0a7e1a7537859084; derivation seal sha256:05e87363c9d2f222a794bd5c2404310f0e58558033c060200d9937237c71eb40; engine receipt sha256:a790fb6e2df910548a230c32b674891cc2fed9a66e07d92608e7f2b04f8f2c3a at receipts/engine/model_admitted/SFT-EARTH-STATE-001-a790fb6e2df91054.json; empirical validation sha256:a7097be3aba1bc52c46d963ce2c320e2908f27447edec6d5fb21667e317f1d0c; measurement receipt sha256:43e968a124e6707b597faf773311042669ec814374e9e682ec0d2f4bdbda17e0; isolation sha256:9911f98e50e7fbbbed053e4063e95d7a14c4511125f5076021199b61f4b933b7; custody sha256:c31ec0ae2be11076756cf0210aa1164ed8b6738fcce204b9faf9cade1d840aed.

5. Earth observation

Claim identity: SFT-EARTH-OBSERVATION-001

Question and exact theorem. An Earth observation is a reconstructible source-to-instrument-to-processing-to-result trace with calibration, uncertainty, exclusions and quality flags retained.

```
source-instrument-processing-result-carrier__method-validity-bounded__source-to-record-trace__location-time-calibration-processing-and-quality-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule
```

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-STATE-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	source-instrument-processing-result-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	method-validity-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	source-to-record-trace	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	location-time-calibration-processing-and-quality-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: `source-instrument-processing-result-carrier__method-validity-bounded__source-to-record-trace__location-time-calibration-processing-and-quality-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule`.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `carrier-present`: the claim retains a nonempty named carrier; passed `true`.
- `relation-distinct`: the carrier and relation are not silently conflated; passed `true`.
- `evidence-boundary`: the evidence boundary remains distinct from the result carrier; passed `true`.
- `exact-fold-witness`: the family's pre-source exact Fold construction reproduces; passed `true`.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar. Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

Earth science begins with a bounded joint carrier, not a map label. Reservoirs, interfaces, state, time, scale and observation class remain explicit. Detected absence, non-detection, censoring, missingness and outside-scope status are different records; the symbol 0 may represent declared absence but may not silently convert an unknown or missing observation into ordinary numerical nothing.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
`sha256:8acdb26b9b03c08898787e05a5745de310878e99b65a887ddaf4e79c3c915878`.

- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:b6e57068ecaab4418294aa43e1134065c039acc6aa11c46417a12d1744e73648.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:808ec7f44ff99be3ff7ec43ac9753850bd43068c779b869f48023935fff18547.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:fe7a0dd832b0b905b6645229e6b89b89166064dd17a9da7322e6ef6adc0c69a6.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:7b41a740b8aedd47d8c46a14062147f4d0aa5c4fa7b306d63d5c0aa7f2d1219a; certificate
sha256:00baabb3ddd782411cf5c9d641a71b57c0ca629d96d89c5359d1cb51fb94ca4c; external-validation
identity sha256:9e39e40e173bb7e5beb3e71f400da4e69ac5f19c7b7765e5ce83f454670e7bee.

Post-seal empirical comparison. Directness:

authoritative_observation_or_primary_measurement_boundary_correspondence. Disposition:
purpose_matched_authoritative_source_features_present. Predicted consequence: earth:source-in
strument-processing-result-carrier__source-to-record-trace__location-time-calibration
-processing-and-quality-held__method-validity-bounded. Source-derived consequence: earth:sourc
e-instrument-processing-result-carrier__source-to-record-trace__location-time-calibra
tion-processing-and-quality-held__method-validity-bounded. Exact boundary correspondence: true.
External evidence selected the survivor: false. Formal structure relabelled as direct measurement: false. Model, forecast,
proxy or retrieval relabelled as observation: false.

Sources and complete feature accounting:

- NASA-EARTHDATA-SEARCH-001 - NASA Earth Science Data and Information System, [Earthdata Search data discovery and access](#); class authoritative_data_catalog; transport captured; features 4/4 present; present: dataset identity, spatial and temporal coverage, instrument or platform provenance, download or service access; absent and preserved: none.
- USGS-LANDSAT-ACCESS-001 - U.S. Geological Survey Landsat Missions, [Landsat Data Access](#); class authoritative_remote_sensing_archive; transport captured; features 0/4 present; present: none; absent and preserved: sensor and mission identity, Earth surface observation, archive access, product and processing distinction.
- NOAA-GHCND-001 - NOAA National Centers for Environmental Information, [Global Historical Climatology Network Daily](#); class authoritative_station_climate_archive; transport captured; features 5/5 present; present: station identity and metadata, daily weather observations, quality flags, record length and variable coverage, archive version; absent and preserved: none.

Comparison record:

- sft-earth-observation-001-external-earth-evidence: predicted earth:source-instrument-processing-result-carrier__source-to-record-trace__location-time-calibration-processing-and-quality-held__method-validity-bounded; source-derived earth:sourc
e-instrument-processing-result-carrier__source-to-record-trace__location-time-calibration-processing-and-quality-held__m
ethod-validity-bounded; exact match True
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if a model output, interpolated field or uncalibrated reading is silently relabelled as direct observation.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity

- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json;
 source manifest sha256:cc8459a719f5fd460a103f97335490b27bb1d128dc9dc8857db1d8934d046d04;
 derivation seal sha256:e670ebd70d7bc7cf80a096563d6a22f9ac700721460f24e53d52339df3116376;
 engine receipt sha256:8b6a8ee6f0aab80b7ea1e9d737248cbc01f20142e1c3de318fbf33b866231b69 at
 receipts/engine/model_admitted/SFT-EARTH-OBSERVATION-001-8b6a8ee6f0aab80b.json; empirical
 validation sha256:6d964795bec3ac7ad88368ceae5ea4fc33a52c304a4c7270697135d61c7645ca;
 measurement receipt
 sha256:1ec0bf5319917e4845251a1d7c742e515fa60affbd76d7ca217ed9c7a022dcf1; isolation
 sha256:9de00da9092a09e06ff953356ae73417c637a4253777fe9e0886f146eabe56bc; custody
 sha256:6a2fe00e3313307a00dc6bd5e0c13949587186e8da5346aca674c333db89e041.

6. Absence, non-detection and missingness

Claim identity: SFT-EARTH-ABSENCE-MISSINGNESS-001

Question and exact theorem. Observed absence, non-detection, censoring, missing data and outside-scope status are distinct Earth evidence classes and may not be merged.

complete-observation-status-carrier__detection-boundary-declared__status-class-distinction__detected-absent-censored-missing-outside-scope-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001

- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-OBSERVATION-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	complete-observation-status-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	detection-boundary-declared	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	status-class-distinction	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	detected-absent-censored-missing-outside-scope-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: complete-observation-status-carrier__detection-boundary-declared__status-class-distinction__detected-absent-censored-missing-outside-scope-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- carrier-present: the claim retains a nonempty named carrier; passed true.
- relation-distinct: the carrier and relation are not silently conflated; passed true.
- evidence-boundary: the evidence boundary remains distinct from the result carrier; passed true.
- exact-fold-witness: the family's pre-source exact Fold construction reproduces; passed true.

Scientific meaning. The symbol 0 is lawful only as an explicit representation of observed absence at a declared detection boundary. Missing, censored, unobserved and outside-scope rows remain distinct and halt any computation that tries to substitute ordinary zero for them.

Earth science begins with a bounded joint carrier, not a map label. Reservoirs, interfaces, state, time, scale and observation class remain explicit. Detected absence, non-detection, censoring, missingness and outside-scope status are different records; the symbol 0 may represent declared absence but may not silently convert an unknown or missing observation into ordinary numerical nothing.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:f8f9157355df037e6d50e197e173b373b143315d8acdbc31010884b4817412c9.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:ceaa11b61ffbb5cd504133cf5a5c5c63cb6c51227a4fcb0f5b71204d54a9150e.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:5b93ba0f12ca7c3606ea88a24ddc900bb3bd26948ad8d9d59a7ce2f55ecc3875.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:1e3c517d5ebca4803869b817c1a3b2747816ce0b2c1a261d505372dabb109252.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:2ff5db30f5387e2db7fbcce0731f0763384157e135d306715018dd417835381c; certificate
sha256:bb40769201f0f471568ff0a5488aeacd191d80a4241c6fa47a21d44e31d85af2; external-validation
identity sha256:b19c298a7452ae4e420537379036eb91a4993938dc761310f14a5d15d3cac435.

Post-seal empirical comparison. Directness:

authoritative_observation_or_primary_measurement_boundary_correspondence. Disposition:

purpose_matched_authoritative_source_features_present. Predicted consequence: earth:complete-observation-status-carrier__status-class-distinction__detected-absent-censored-missing-outside-scope-held__detection-boundary-declared. Source-derived consequence: earth:complete-observation-status-carrier__status-class-distinction__detected-absent-censored-missing-outside-scope-held__detection-boundary-declared. Exact boundary correspondence: true. External evidence selected the survivor: false. Formal structure relabelled as direct measurement: false. Model, forecast, proxy or retrieval relabelled as observation: false.

Sources and complete feature accounting:

- NASA-EARTHDATA-SEARCH-001 - NASA Earth Science Data and Information System, [Earthdata Search data discovery and access](#); class authoritative_data_catalog; transport captured; features 4/4 present; present: dataset identity, spatial and temporal coverage, instrument or platform provenance, download or service access; absent and preserved: none.
- USGS-LANDSAT-ACCESS-001 - U.S. Geological Survey Landsat Missions, [Landsat Data Access](#); class authoritative_remote_sensing_archive; transport captured; features 0/4 present; present: none; absent and preserved: sensor and mission identity, Earth surface observation, archive access, product and processing distinction.
- NOAA-GHCND-001 - NOAA National Centers for Environmental Information, [Global Historical Climatology Network Daily](#); class authoritative_station_climate_archive; transport captured; features 5/5 present; present: station identity and metadata, daily weather observations, quality flags, record length and variable coverage, archive version; absent and preserved: none.

Comparison record:

- sft-earth-absence-missingness-001-external-earth-evidence: predicted earth:complete-observation-status-carrier__status-class-distinction__detected-absent-censored-missing-outside-scope-held__detection-boundary-declared; source-derived earth:complete-observation-status-carrier__status-class-distinction__detected-absent-censored-missing-outside-scope-held__detection-boundary-declared; exact match True
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if any unavailable or unresolved observation is converted into ordinary numerical zero or favourable absence.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

```
experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json;
source manifest sha256:95dd2802a82742acd74cf341f61af5322f2968a2a893277ff5a3314f638a47e8;
derivation seal sha256:e3a90106ba9d4e5bf87ce1376b8ebfc9532209f794951768edd29c777dfe5525;
engine receipt sha256:2e5cf46bef731f8ef72f0b0bc6cb746ff7024b67d45c808e41bda4bc36f1efa4 at re
ceipts/engine/model_admitted/SFT-EARTH-ABSENCE-MISSINGNESS-001-2e5cf46bef731f8e.json;
empirical validation sha256:9ba42654d8bf933007d421abaae92ab0327780d571f2f22d4decc5e43b50af99;
measurement receipt
sha256:561355d50ea28cfd31fe1995ead408b6d6a98206914be8e9991726b2abc06046; isolation
sha256:f515c89d48177fc3f0b886f0cce1f8d76e9452a43c4a4ef5d78576716564db7e; custody
sha256:7a7d528918a9e749ce87ef0ba5ae6a5ef99b2c655e308d1ca52eed552b148a24.
```

Ledger family 2: Planetary Energy, Matter and Information Budgets

A planetary budget is a complete bounded ledger. Stock, flow, source, sink, storage, transformation and export are separate coordinates. Closure can be tested only after units, time intervals, interfaces, missing terms and residuals are retained; a fitted residual cannot manufacture conservation.

7. Earth-system energy budget

Claim identity: SFT-EARTH-ENERGY-BUDGET-001

Question and exact theorem. An Earth energy budget retains every registered source, storage change, transfer and export across a declared boundary before testing closure.

```
bounded-energy-stock-transfer-carrier__budget-resolution-bounded__source-storage-t
ransfer-loss-balance__all-terms-units-boundaries-and-times-held__evidence-class-ex
plicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no
-extra-rule
```

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001

- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-ABSENCE-MISSINGNESS-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	bounded-energy-stock-transfer-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	budget-resolution-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	source-storage-transfer-loss-balance	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	all-terms-units-boundaries-and-times-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflicted	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: bounded-energy-stock-transfer-carrier__budget-resolution-bounded__source-storage-transfer-loss-balance__all-terms-units-boundaries

-and-times-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite
-successor-retains-all-rows__no-extra-rule.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- carrier-present: the claim retains a nonempty named carrier; passed true.
- relation-distinct: the carrier and relation are not silently conflated; passed true.
- evidence-boundary: the evidence boundary remains distinct from the result carrier; passed true.
- exact-fold-witness: the family's pre-source exact Fold construction reproduces; passed true.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar.

Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

A planetary budget is a complete bounded ledger. Stock, flow, source, sink, storage, transformation and export are separate coordinates. Closure can be tested only after units, time intervals, interfaces, missing terms and residuals are retained; a fitted residual cannot manufacture conservation.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:f696fd9c7a541b7d2cb34094c22db27664fccb26e37bfbff1a4a175b69dd4aa8.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:9b070e42516bb2f71601839664c76233fef6b5d2b6429d8302905794285aa078.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:26cd56e4fc4ceaeafe6a8cceef07655f236477b207a3f229da8fa04c04b632750.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:82fd5e5069cb211878f063f4944d76312c97c3dd8a49d6ab8ac504c51e8b555d.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:cedcf2a5ade23828b56559c796a17357bed5a8f11a882bbcb577d9d7cf34fb7; certificate
sha256:ebc3345e4a684d8c48bb7f39d5839de4d7006cbfcdaf5b90ffc324dc7cddc51a; external-validation
identity sha256:1736957221fc724b66e398e648074ff5dfe92152b4db69a04a5ea7bb07f2620c.

Post-seal empirical comparison. Directness:

authoritative_observation_or_primary_measurement_boundary_correspondence. Disposition:
purpose_matched_authoritative_source_features_present. Predicted consequence: earth:bounded-e
nergy-stock-transfer-carrier__source-storage-transfer-loss-balance__all-terms-units-b
oundaries-and-times-held__budget-resolution-bounded. Source-derived consequence: earth:bounded
-energy-stock-transfer-carrier__source-storage-transfer-loss-balance__all-terms-units
-boundaries-and-times-held__budget-resolution-bounded. Exact boundary correspondence: true.

External evidence selected the survivor: false. Formal structure relabelled as direct measurement: false. Model, forecast, proxy or retrieval relabelled as observation: false.

Sources and complete feature accounting:

- NASA-CERES-EBAF-001 - NASA Langley Research Center CERES, **CERES Energy Balanced and Filled data products**; class `authoritative_radiative_budget_data_product`; transport captured; features 5/5 present; present: top-of-atmosphere and surface boundary, shortwave and longwave flux distinction, cloud and clear-sky distinction, spatial and temporal resolution, data-quality documentation; absent and preserved: none.
- NOAA-GML-CO2-001 - NOAA Global Monitoring Laboratory, **Carbon-cycle greenhouse-gas measurements**; class `authoritative_atmospheric_composition_archive`; transport captured; features 3/5 present; present: gas and measurement programme identity, site and altitude, calibration and data-use policy; absent and preserved: time and frequency, source and sink interpretation separated from measurement.
- NOAA-WOD-001 - NOAA National Centers for Environmental Information, **World Ocean Database**; class `authoritative_ocean_profile_archive`; transport captured; features 5/5 present; present: cast cruise and accession identity, depth location and time, temperature salinity nutrients and plankton, quality flags, original-source provenance; absent and preserved: none.
- USGS-NWIS-001 - U.S. Geological Survey Water Resources, **National Water Information System**; class `authoritative_hydrological_observation_network`; transport captured; features 2/4 present; present: site identity, surface water and groundwater observations; absent and preserved: time series, parameter units and methods.

Comparison record:

- `sft-earth-energy-budget-001-external-earth-evidence`: predicted earth:bounded-energy-stock-transfer-carrier__source-storage-transfer-loss-balance__all-terms-units-boundaries-and-times-held__budget-resolution-bounded; source-derived earth:bounded-energy-stock-transfer-carrier__source-storage-transfer-loss-balance__all-terms-units-boundaries-and-times-held__budget-resolution-bounded; exact match True
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if closure is asserted after omitting a flux, changing units, mixing timescales or fitting a residual away.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json;
 source manifest sha256:b43a689ebf60a0048da5e79df80b34441b509108604f2ce689fcdd6ed456c94f;
 derivation seal sha256:aa24722fbb709378a13c4aa77406f749d300f1d1fba9f54ccc803bd530bb07a8;
 engine receipt sha256:de5ea4b0a621a9a47f9f0f272702584353ffa09185cdb951a7b3e55b90963b36 at

receipts/engine/model_admitted/SFT-EARTH-ENERGY-BUDGET-001-de5ea4b0a621a9a4.json;
 empirical validation sha256:131155bcc2ff17cd5972d93038b839b92c2f55e6dc2ebe62cfd9333f7ab8029;
 measurement receipt
 sha256:0e0dcdf40768c395d2b8e45e7a67909e48d9a582ab19941d8eb8174c1ac278e4; isolation
 sha256:6df7f9a1981e3c7d86ba272cbca1db457e76beffc4eea88314bfd2a31d14fb38; custody
 sha256:ab63a6df6486e631c6e288a8f234889ac27a3535ae140d0df9c158f7f914fcc4.

8. Earth-system matter budget

Claim identity: SFT-EARTH-MATTER-BUDGET-001

Question and exact theorem. An Earth matter budget follows distinguishable material carriers through storage, transformation, transfer and export without permitting disappearance.

species-resolved-stock-transfer-carrier__matter-domain-bounded__source-storage-transformation-export-balance__species-phase-location-time-and-uncertainty-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001

- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-ABSENCE-MISSINGNESS-001
- SFT-EARTH-ENERGY-BUDGET-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	species-resolved-stock-transfer-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	matter-domain-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	source-storage-transformation-export-balance	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	species-phase-location-time-and-uncertainty-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: species-resolved-stock-transfer-carrier__matter-domain-bounded__source-storage-transformation-export-balance__species-phase-location-time-and-uncertainty-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- carrier-present: the claim retains a nonempty named carrier; passed true.

- `relation-distinct`: the carrier and relation are not silently conflated; passed `true`.
- `evidence-boundary`: the evidence boundary remains distinct from the result carrier; passed `true`.
- `exact-fold-witness`: the family's pre-source exact Fold construction reproduces; passed `true`.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar. Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

A planetary budget is a complete bounded ledger. Stock, flow, source, sink, storage, transformation and export are separate coordinates. Closure can be tested only after units, time intervals, interfaces, missing terms and residuals are retained; a fitted residual cannot manufacture conservation.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:55373db81cbf333fd9a5710d82f8c1c02110ab7c3bd2f80e29c8714c9b55b4a4.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:3f711318dc8a71fa3a00007f05cba24a7c02c9d561a98e371951a61ddf63171f.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:1c3cedaa6ee394b10f29ae49f5b49f6b05c065ab5cf77c7186d4b608ba8c54ca.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:9900af0d5514ced7d4c376e744f58833a3f45ab3cdb428d5aea04aafe40b1b1b.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:44aa7656c846b20a74ce1af464d27e0e808a78dc4ef30a4850ebfcbd5339d38c; certificate
sha256:8c0dc11aca3519d0068610a4af2d565ef04b7ecb181d1f7a10e558c685b233e9; external-validation
identity sha256:f4dbee5d82863e859eb8de4f5da2c1295112e014631c2892dad5f1b63c084bb6.

Post-seal empirical comparison. Directness:

`authoritative_observation_or_primary_measurement_boundary_correspondence`. Disposition:
`purpose_matched_authoritative_source_features_present`. Predicted consequence: `earth:species-resolved-stock-transfer-carrier__source-storage-transformation-export-balance__species-phase-location-time-and-uncertainty-held__matter-domain-bounded`. Source-derived consequence:
`earth:species-resolved-stock-transfer-carrier__source-storage-transformation-export-balance__species-phase-location-time-and-uncertainty-held__matter-domain-bounded`. Exact
boundary correspondence: `true`. External evidence selected the survivor: `false`. Formal structure relabelled as direct
measurement: `false`. Model, forecast, proxy or retrieval relabelled as observation: `false`.

Sources and complete feature accounting:

- NASA-CERES-EBAF-001 - NASA Langley Research Center CERES, **CERES Energy Balanced and Filled data products**; class `authoritative_radiative_budget_data_product`; transport captured; features 5/5 present; present: top-of-atmosphere and surface boundary, shortwave and longwave flux distinction, cloud and clear-sky distinction, spatial and temporal resolution, data-quality documentation; absent and preserved: none.
- NOAA-GML-CO2-001 - NOAA Global Monitoring Laboratory, **Carbon-cycle greenhouse-gas measurements**; class `authoritative_atmospheric_composition_archive`; transport captured; features 3/5 present; present: gas and measurement programme identity, site and altitude, calibration and data-use policy; absent and preserved: time and frequency, source and sink interpretation separated from measurement.
- NOAA-WOD-001 - NOAA National Centers for Environmental Information, **World Ocean Database**; class `authoritative_ocean_profile_archive`; transport captured; features 5/5 present; present: cast cruise and accession identity, depth location and time, temperature salinity nutrients and plankton, quality flags, original-source

provenance; absent and preserved: none.

- USGS-NWIS-001 - U.S. Geological Survey Water Resources, **National Water Information System**; class authoritative_hydrological_observation_network; transport captured; features 2/4 present; present: site identity, surface water and groundwater observations; absent and preserved: time series, parameter units and methods.

Comparison record:

- sft-earth-matter-budget-001-external-earth-evidence: predicted earth:species-resolved-stock-transfer-carrier__source-storage-e-transformation-export-balance__species-phase-location-time-and-uncertainty-held__matter-domain-bounded; source-derived earth:species-resolved-stock-transfer-carrier__source-storage-transformation-export-balance__species-phase-location-time-and-uncertainty-held__matter-domain-bounded; exact match True
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if chemical identity, phase, location, boundary transfer or unresolved stock is erased.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json;
 source manifest sha256:653743ef1c19efd5cde7d4bbd2f2ccba87d585a9fd0dc50914f9a71f2d062ae1;
 derivation seal sha256:1bdaa74e0ef95a0eb9ce1e03d81a4e9782411b944d73e8a98da7b22a26c4ca23;
 engine receipt sha256:34942b19640ee03f34cdae7d7ea01bb258668febf291a4f03487786ee80fe34e at
 receipts/engine/model_admitted/SFT-EARTH-MATTER-BUDGET-001-34942b19640ee03f.json;
 empirical validation sha256:e74fb5170bf25a9448ab1c1d7d0ae08e0a9b253abb9909a10359b26ecf90ac29;
 measurement receipt
 sha256:9b753bc99aab35011585a4fdffc71b3f013b60339722fe5af14673b1bd20ad87; isolation
 sha256:134131b3adea0f2a249460b34f1bef6406603d0daa14c20d987b86d2be10fda5; custody
 sha256:c5464812cd7510be287e21cd4d7f1cad0446a833cecef14b001f6d9cfa9e446a.

9. Earth-system information budget

Claim identity: SFT-EARTH-INFORMATION-BUDGET-001

Question and exact theorem. An Earth information budget records which environmental distinctions are retained, merged, reconstructed or lost by observation and transformation.

distinction-observation-transformation-carrier__observation-process-bounded__retention-loss-and-reconstruction-relation__source-resolution-processing-and-lost-labels-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-ABSENCE-MISSINGNESS-001
- SFT-EARTH-MATTER-BUDGET-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	distinction-observation-transformation-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	observation-process-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	retention-loss-and-reconstruction-relation	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	source-resolution-processing-and-lost-labels-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: `distinction-observation-transformation-carrier_observation-process-bounded_retention-loss-and-reconstruction-relation_source-resolution-processing-and-lost-labels-held_evidence-class-explicit_root-bound-forward-forcing_positive-finite-successor-retains-all-rows_no-extra-rule`.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `carrier-present`: the claim retains a nonempty named carrier; passed `true`.
- `relation-distinct`: the carrier and relation are not silently conflated; passed `true`.
- `evidence-boundary`: the evidence boundary remains distinct from the result carrier; passed `true`.
- `exact-fold-witness`: the family's pre-source exact Fold construction reproduces; passed `true`.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar. Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

A planetary budget is a complete bounded ledger. Stock, flow, source, sink, storage, transformation and export are separate coordinates. Closure can be tested only after units, time intervals, interfaces, missing terms and residuals are retained; a fitted residual cannot manufacture conservation.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:f81f193ce97ee5215a31263f9c2515817aca15503b6f3704f4b4b0993961c2fa.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:80efe889e53ffd06560418f8f673bdad0386bd696ab79b4eb0309d776f7db13d.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:c41b1a3bccfa2bc16d8a884746460cb1f2aa1427936ecb382f4633583914ce62.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:ca3abef12bf16713ae9d21a2d6061321fd28da2ff2ffc41ac57b25f9df017d59.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:e52a9be98e081cabc66acc7612d887202ce78743923f238e0d4c738ea8386ca9; certificate
sha256:bb4edded30586b23b6c4abb3edad26fa79fe98c824f379c8cf90341c23c97ed3; external-validation
identity sha256:ca3efa62b7ff35b94d23ee6aed050c159c95e2942c00a7ba1662b979e7bf6ca6.

Post-seal empirical comparison. Directness:

authoritative_observation_or_primary_measurement_boundary_correspondence. Disposition:
purpose_matched_authoritative_source_features_present. Predicted consequence: earth:distinction-
on-observation-transformation-carrier__retention-loss-and-reconstruction-relation__so
urce-resolution-processing-and-lost-labels-held__observation-process-bounded.
Source-derived consequence: earth:distinction-observation-transformation-carrier__retention-l
oss-and-reconstruction-relation__source-resolution-processing-and-lost-labels-held__o
bservation-process-bounded. Exact boundary correspondence: true. External evidence selected the survivor:
false. Formal structure relabelled as direct measurement: false. Model, forecast, proxy or retrieval relabelled as
observation: false.

Sources and complete feature accounting:

- NASA-CERES-EBAF-001 - NASA Langley Research Center CERES, **CERES Energy Balanced and Filled data products**; class authoritative_radiative_budget_data_product; transport captured; features 5/5 present; present: top-of-atmosphere and surface boundary, shortwave and longwave flux distinction, cloud and clear-sky distinction, spatial and temporal resolution, data-quality documentation; absent and preserved: none.
- NOAA-GML-CO2-001 - NOAA Global Monitoring Laboratory, **Carbon-cycle greenhouse-gas measurements**; class authoritative_atmospheric_composition_archive; transport captured; features 3/5 present; present: gas and measurement programme identity, site and altitude, calibration and data-use policy; absent and preserved: time and frequency, source and sink interpretation separated from measurement.
- NOAA-WOD-001 - NOAA National Centers for Environmental Information, **World Ocean Database**; class authoritative_ocean_profile_archive; transport captured; features 5/5 present; present: cast cruise and accession identity, depth location and time, temperature salinity nutrients and plankton, quality flags, original-source provenance; absent and preserved: none.
- USGS-NWIS-001 - U.S. Geological Survey Water Resources, **National Water Information System**; class authoritative_hydrological_observation_network; transport captured; features 2/4 present; present: site identity, surface water and groundwater observations; absent and preserved: time series, parameter units and methods.

Comparison record:

- sft-earth-information-budget-001-external-earth-evidence: predicted earth:distinction-observation-transformation-carrier__r
etention-loss-and-reconstruction-relation__source-resolution-processing-and-lost-labels-held__observation-process-bounde
d; source-derived earth:distinction-observation-transformation-carrier__retention-loss-and-reconstruction-relation__source
-resolution-processing-and-lost-labels-held__observation-process-bounded; exact match True
- source features: 67 present; 24 absent preserved; 91 registered

- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if increased resolution is claimed without a source distinction or if merged histories are treated as uniquely recoverable without a record.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json;
 source manifest sha256:36404c063c4b20b800f374f3ca2d25420e02c33ac8f57a0823a36dfb1850a967;
 derivation seal sha256:2274ce011a5862852226987b9ee038f617a4fd117bf3a85d1bfa7ec8951952ce;
 engine receipt sha256:14aa00f28ff6c71073c77ad1aca80c55d42b311ba49c189c0d0d2cc8e5c0a564 at re
 cepts/engine/model_admitted/SFT-EARTH-INFORMATION-BUDGET-001-14aa00f28ff6c710.json;
 empirical validation sha256:2bb82f22cb7189c03e3f918d8e2d0c7e547c62abddcf1925a68f19adb31d0be2;
 measurement receipt
 sha256:235f260f3b547ba05478006cc48664f666ffaaffc43345bb01761c8dbc46ecb7e; isolation
 sha256:e4c3a5d6bd6264a8401754f77a9a7b3a532432ade93a4062bd4d646430e363cf; custody
 sha256:8405526a7e5728e7f148e9fadf4d76712eb4353ae072cbc89877a1088e6db5ca.

10. Stock and flow distinction

Claim identity: SFT-EARTH-STOCK-FLOW-001

Question and exact theorem. A stock is retained occupancy within a boundary; a flow is a counted transfer across an interface during a declared interval, and neither may substitute for the other.

```
reservoir-stock-interface-flow-carrier__quantity-and-interval-bounded__time-integr
ated-transfer-relation__stock-basis-flow-interval-and-boundary-held__evidence-clas
s-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows
__no-extra-rule
```

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001

- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-ABSENCE-MISSINGNESS-001
- SFT-EARTH-INFORMATION-BUDGET-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	reservoir-stock-interface-flow-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
boundary	boundary-or-interface-erased	quantity-and-interval-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	time-integrated-transfer-relation	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	stock-basis-flow-interval-and-boundary-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: `reservoir-stock-interface-flow-carrier__quantity-and-interval-bounded__time-integrated-transfer-relation__stock-basis-flow-interval-and-boundary-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule`.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `carrier-present`: the claim retains a nonempty named carrier; passed `true`.
- `relation-distinct`: the carrier and relation are not silently conflated; passed `true`.
- `evidence-boundary`: the evidence boundary remains distinct from the result carrier; passed `true`.
- `exact-fold-witness`: the family's pre-source exact Fold construction reproduces; passed `true`.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar.

Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

A planetary budget is a complete bounded ledger. Stock, flow, source, sink, storage, transformation and export are separate coordinates. Closure can be tested only after units, time intervals, interfaces, missing terms and residuals are retained; a fitted residual cannot manufacture conservation.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
`sha256:536b9af9313cfbe4ccc86fbc098bac454807615c0c4bd74955b3d78829351f35`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
`sha256:a0627b129684fff633cc9a95a2e22d9e4710e644d7ba483d47ac93ec1369c6e0`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
`sha256:7654c5225e5b14cb30ce45647dfd21b4eb489704e3c1a1798216901b6d2a97d2`.

- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:df3f252f86e16aabd497834751b1d8978f20c26ef8116db8e1055c5fffee3dd1.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:c3b8694cf183c82e96acca99fdd15792f50cd3b9c5217f7ff144e1be4403f711; certificate
sha256:19ea0030f13a713a7883b1b14b27690097422e8aff15d8e39901de13d9712b47; external-validation
identity sha256:e2d373e9686e237742a8999419b3750ec80e4e590ed155d91ef0190fe62bb65e.

Post-seal empirical comparison. Directness:

authoritative_observation_or_primary_measurement_boundary_correspondence. Disposition:
purpose_matched_authoritative_source_features_present. Predicted consequence: earth:reservoir-
stock-interface-flow-carrier__time-integrated-transfer-relation__stock-basis-flow-in
terval-and-boundary-held__quantity-and-interval-bounded. Source-derived consequence: earth:res
ervoir-stock-interface-flow-carrier__time-integrated-transfer-relation__stock-basis-f
low-interval-and-boundary-held__quantity-and-interval-bounded. Exact boundary correspondence:
true. External evidence selected the survivor: false. Formal structure relabelled as direct measurement: false. Model,
forecast, proxy or retrieval relabelled as observation: false.

Sources and complete feature accounting:

- NASA-CERES-EBAF-001 - NASA Langley Research Center CERES, **CERES Energy Balanced and Filled data products**; class authoritative_radiative_budget_data_product; transport captured; features 5/5 present; present: top-of-atmosphere and surface boundary, shortwave and longwave flux distinction, cloud and clear-sky distinction, spatial and temporal resolution, data-quality documentation; absent and preserved: none.
- NOAA-GML-CO2-001 - NOAA Global Monitoring Laboratory, **Carbon-cycle greenhouse-gas measurements**; class authoritative_atmospheric_composition_archive; transport captured; features 3/5 present; present: gas and measurement programme identity, site and altitude, calibration and data-use policy; absent and preserved: time and frequency, source and sink interpretation separated from measurement.
- NOAA-WOD-001 - NOAA National Centers for Environmental Information, **World Ocean Database**; class authoritative_ocean_profile_archive; transport captured; features 5/5 present; present: cast cruise and accession identity, depth location and time, temperature salinity nutrients and plankton, quality flags, original-source provenance; absent and preserved: none.
- USGS-NWIS-001 - U.S. Geological Survey Water Resources, **National Water Information System**; class authoritative_hydrological_observation_network; transport captured; features 2/4 present; present: site identity, surface water and groundwater observations; absent and preserved: time series, parameter units and methods.

Comparison record:

- sft-earth-stock-flow-001-external-earth-evidence: predicted earth:reservoir-stock-interface-flow-carrier__time-integrated-tr
ansfer-relation__stock-basis-flow-interval-and-boundary-held__quantity-and-interval-bounded; source-derived earth:reserv
oir-stock-interface-flow-carrier__time-integrated-transfer-relation__stock-basis-flow-interval-and-boundary-held__quantit
y-and-interval-bounded; exact match True
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if an instantaneous rate is added to a stock without an interval or if cumulative transfer is presented as simultaneous occupancy.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value

- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json;
 source manifest sha256:d7ec9470e0314d39806043bbd1047ce561a2b4abe17521505428da6819f3b165;
 derivation seal sha256:dd65606801d58951028802deb8750698ac3b731c447221c4356af610c26ed3f8;
 engine receipt sha256:2a21e28b8288bdd73f95a06bc6950e7376cc707c5c6df04b99641f0d5b57484f at
 receipts/engine/model_admitted/SFT-EARTH-STOCK-FLOW-001-2a21e28b8288bdd7.json; empirical
 validation sha256:f8b7754dc2b86b7bac3a169b9ba5a4b22d2fc439028d5ffcdcfb3de387ee3d8f9;
 measurement receipt
 sha256:06f2f54ea3824f9d3d58974af30dalb9e9a94f83c97cfffac9acc8972a17d5bc5; isolation
 sha256:426b54a7b7240e7560617003fac3953852803024d7ff5c52a778624723bdf6f6; custody
 sha256:a0ad7f5e93625b5175d59c9e95dee552b4f0847f89cedc74d899a0f5c801065a.

11. Source, sink and storage roles

Claim identity: SFT-EARTH-SOURCE-SINK-STORAGE-001

Question and exact theorem. Source, sink and storage are boundary- and interval-relative roles assigned from the complete transfer record rather than permanent labels attached to an object.

process-reservoir-role-carrier__system-boundary-bounded__boundary-relative-role-as
 signment__direction-duration-capacity-and-reversal-held__evidence-class-explicit__
 root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-r
 ule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001

- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-ABSENCE-MISSINGNESS-001
- SFT-EARTH-STOCK-FLOW-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	process-reservoir-role-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	system-boundary-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	boundary-relative-role-assignment	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	direction-duration-capacity-and-reversal-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: process-reservoir-role-carrier__system-boundary-bounded__boundary-relative-role-assignment__direction-duration-capacity-and-reversal-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- carrier-present: the claim retains a nonempty named carrier; passed true.
- relation-distinct: the carrier and relation are not silently conflated; passed true.
- evidence-boundary: the evidence boundary remains distinct from the result carrier; passed true.
- exact-fold-witness: the family's pre-source exact Fold construction reproduces; passed true.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar.

Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

A planetary budget is a complete bounded ledger. Stock, flow, source, sink, storage, transformation and export are separate coordinates. Closure can be tested only after units, time intervals, interfaces, missing terms and residuals are retained; a fitted residual cannot manufacture conservation.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:86c13d3ed16aab4758b8cdd0bedaddab93ca72edeb7df369a104550ca85ec475.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:d28fc2e5fda6db875584d22974c63f1076eaf5385d5e72844d4339f8a305afbc.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:e43405efe3be471b1d6def4ccc19b73a0cfd56e139d3e1b6ca3728addfe77968.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:f97a923e65aac4d2cc63928f6642ebfced9fc990f0e42f6678b16ed06f65f14e.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:68f3e858226a68918ae98e0ce3c5a19e754a56eddc8b67480fcf2d6d98da97a9; certificate
sha256:22c0a8e31b9537ef6ba7d53538c2d3836c86a15de659e975acf99d44ecb33d79; external-validation
identity sha256:86899ea66465909210031905e598e58f9c675f6005ecf39a5a30a41a0234bd9d.

Post-seal empirical comparison. Directness:

authoritative_observation_or_primary_measurement_boundary_correspondence. Disposition:
purpose_matched_authoritative_source_features_present. Predicted consequence: earth:process-r

eservoir-role-carrier__boundary-relative-role-assignment__direction-duration-capacity-and-reversal-held__system-boundary-bounded. Source-derived consequence: earth:process-reservoir-role-carrier__boundary-relative-role-assignment__direction-duration-capacity-and-reversal-held__system-boundary-bounded. Exact boundary correspondence: true. External evidence selected the survivor: false. Formal structure relabelled as direct measurement: false. Model, forecast, proxy or retrieval relabelled as observation: false.

Sources and complete feature accounting:

- NASA-CERES-EBAF-001 - NASA Langley Research Center CERES, **CERES Energy Balanced and Filled data products**; class authoritative_radiative_budget_data_product; transport captured; features 5/5 present; present: top-of-atmosphere and surface boundary, shortwave and longwave flux distinction, cloud and clear-sky distinction, spatial and temporal resolution, data-quality documentation; absent and preserved: none.
- NOAA-GML-CO2-001 - NOAA Global Monitoring Laboratory, **Carbon-cycle greenhouse-gas measurements**; class authoritative_atmospheric_composition_archive; transport captured; features 3/5 present; present: gas and measurement programme identity, site and altitude, calibration and data-use policy; absent and preserved: time and frequency, source and sink interpretation separated from measurement.
- NOAA-WOD-001 - NOAA National Centers for Environmental Information, **World Ocean Database**; class authoritative_ocean_profile_archive; transport captured; features 5/5 present; present: cast cruise and accession identity, depth location and time, temperature salinity nutrients and plankton, quality flags, original-source provenance; absent and preserved: none.
- USGS-NWIS-001 - U.S. Geological Survey Water Resources, **National Water Information System**; class authoritative_hydrological_observation_network; transport captured; features 2/4 present; present: site identity, surface water and groundwater observations; absent and preserved: time series, parameter units and methods.

Comparison record:

- sft-earth-source-sink-storage-001-external-earth-evidence: predicted earth:process-reservoir-role-carrier__boundary-relative-role-assignment__direction-duration-capacity-and-reversal-held__system-boundary-bounded; source-derived earth:process-reservoir-role-carrier__boundary-relative-role-assignment__direction-duration-capacity-and-reversal-held__system-boundary-bounded; exact match True
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if a role is assigned without direction and interval or retained after the measured transfer reverses.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law

- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json;
 source manifest sha256:a08573ae07a0d7820cba2de043ceea24b06d3a251a68c34453356c6bcf474e9d;
 derivation seal sha256:fe802e3014bd9df18400440f144567aa45208e6a843216063fb2da87a33c4235;
 engine receipt sha256:00938fce626f4c74780542f114907934621c2b42350f40a5a4372c43e9e4dc9b at re
 ceipts/engine/model_admitted/SFT-EARTH-SOURCE-SINK-STORAGE-001-00938fce626f4c74.json;
 empirical validation sha256:cf86e875e1933f8cd7f90328eca3a235b2439286e6b80958b9fb183d22f44c64;
 measurement receipt
 sha256:0e0e7fc19534d74a67fbb172b8b3726fc74f2061a734d008bbdb64394079f27b; isolation
 sha256:3cd1723df1ab0aca9bbca061cf03a203aeb72beaa39f3930fac71de9324c86a7; custody
 sha256:08da58e5a3d3ff1966e3a1f972f2e31c02a91fe386231ce86d9c48f5bfd6f3a3.

12. Budget closure and residual

Claim identity: SFT-EARTH-BUDGET-CLOSURE-001

Question and exact theorem. Budget closure is exact agreement of independently reconstructed registered routes; any unexplained residual remains evidence rather than being distributed by fit.

complete-registered-ledger-carrier__declared-ledger-bounded__independent-route-bal
 ance-comparison__measurement-uncertainty-missing-terms-and-residual-held__evidence
 -class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all
 -rows__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001

- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-ABSENCE-MISSINGNESS-001
- SFT-EARTH-SOURCE-SINK-STORAGE-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	complete-registered-ledger-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	declared-ledger-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	independent-route-balance-comparison	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	measurement-uncertainty-missing-terms-and-residual-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: complete-registered-ledger-carrier__declared-ledger-bounded__independent-route-balance-comparison__measurement-uncertainty-missing-terms-and-residual-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `carrier-present`: the claim retains a nonempty named carrier; passed `true`.
- `relation-distinct`: the carrier and relation are not silently conflated; passed `true`.
- `evidence-boundary`: the evidence boundary remains distinct from the result carrier; passed `true`.
- `exact-fold-witness`: the family's pre-source exact Fold construction reproduces; passed `true`.

Scientific meaning. Budget closure is a comparison of independently reconstructible routes. It cannot be achieved by deleting an unresolved term or fitting the residual to the desired answer.

A planetary budget is a complete bounded ledger. Stock, flow, source, sink, storage, transformation and export are separate coordinates. Closure can be tested only after units, time intervals, interfaces, missing terms and residuals are retained; a fitted residual cannot manufacture conservation.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:012c9da6c5a5026a0f052a213b9aafad390dd9691580e509e43277d4ea7c98bf.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:18c00af091506e2d25cfb1cbdb13490d1b7caa09d9b22a066c60857affaafbe.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:c13ba6665a54a1dec0532c123237eb523af2074bb7d49148094c07cd71746274.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:4a56e073154221698e56b770ac11eddb6bec3e6c410b063102aca8396b2197e4.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:0736413062b9484174432d16cad2f9e5dcdeb5b3e53e565700cf8aac69c9b4b; certificate
sha256:63dd25055b99e59afa2b6bd65f4c9511434d4fc9df35eccfff59f0f211a74d4a; external-validation
identity sha256:f771121b1966baca8a2d71a1e21d147da963cfe62fce28da657805df86d6609f.

Post-seal empirical comparison. Directness:

authoritative_observation_or_primary_measurement_boundary_correspondence. Disposition:
purpose_matched_authoritative_source_features_present. Predicted consequence: earth:complete-
registered-ledger-carrier__independent-route-balance-comparison__measurement-uncertai-
nty-missing-terms-and-residual-held__declared-ledger-bounded. Source-derived consequence: eart
h:complete-registered-ledger-carrier__independent-route-balance-comparison__measureme
nt-uncertainty-missing-terms-and-residual-held__declared-ledger-bounded. Exact boundary
correspondence: true. External evidence selected the survivor: false. Formal structure relabelled as direct measurement:
false. Model, forecast, proxy or retrieval relabelled as observation: false.

Sources and complete feature accounting:

- NASA-CERES-EBAF-001 - NASA Langley Research Center CERES, CERES Energy Balanced and Filled data products; class authoritative_radiative_budget_data_product; transport captured; features 5/5 present; present: top-of-atmosphere and surface boundary, shortwave and longwave flux distinction, cloud and clear-sky distinction, spatial and temporal resolution, data-quality documentation; absent and preserved: none.
- NOAA-GML-CO2-001 - NOAA Global Monitoring Laboratory, Carbon-cycle greenhouse-gas measurements; class authoritative_atmospheric_composition_archive; transport captured; features 3/5 present; present:

gas and measurement programme identity, site and altitude, calibration and data-use policy; absent and preserved: time and frequency, source and sink interpretation separated from measurement.

- NOAA-WOD-001 - NOAA National Centers for Environmental Information, [World Ocean Database](#); class `authoritative_ocean_profile_archive`; transport captured; features 5/5 present; present: cast cruise and accession identity, depth location and time, temperature salinity nutrients and plankton, quality flags, original-source provenance; absent and preserved: none.
- USGS-NWIS-001 - U.S. Geological Survey Water Resources, [National Water Information System](#); class `authoritative_hydrological_observation_network`; transport captured; features 2/4 present; present: site identity, surface water and groundwater observations; absent and preserved: time series, parameter units and methods.

Comparison record:

- `sft-earth-budget-closure-001-external-earth-evidence`: predicted earth:complete-registered-ledger-carrier__independent-route-balance-comparison__measurement-uncertainty-missing-terms-and-residual-held__declared-ledger-bounded; source-derived earth:complete-registered-ledger-carrier__independent-route-balance-comparison__measurement-uncertainty-missing-terms-and-residual-held__declared-ledger-bounded; exact match True
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if a residual is deleted, reallocated by tuning or called closure outside the declared uncertainty and source boundary.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

```
experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json;
source manifest sha256:f397ff3b65957b486e75b3f92dc35f76b45bcf85443f7c31b5de7757da1116bb;
derivation seal sha256:9b7c377fab672915ee86d0f8da494b129008612bcc5d7438598822d43e512f87;
engine receipt sha256:61658ad6107b07e20188fdd35f26c03f8f06bd248b11619f2bde057c081ad167 at
receipts/engine/model_admitted/SFT-EARTH-BUDGET-CLOSURE-001-61658ad6107b07e2.json;
empirical validation sha256:4059f06ef6c44981a5df23b71857c487c9ecf3e658a52e4656122a28446f5eea;
measurement receipt
sha256:d601e3ea6db153772b21d788306dbac6b766461daa2d6725292e629330c17941; isolation
sha256:ff5f122420984b3ff3006fab0fc881aa940cc64eab95161f00528df22eef4dd6; custody
sha256:3eddf0e1905708d040f8b8477c1fd53e789cd52d26573eb1d51910c83d3ce4cc.
```

Ledger family 3: Geological Material, Layer, Provenance and History

Geological identity requires specimen, composition, phase, texture, provenance and method. Layers are positive ordered carriers with explicit contacts and interruptions. Earth history is a dated multi-proxy reconstruction whose gaps and alternatives remain evidence, not empty numerical entries.

13. Geological material identity

Claim identity: SFT-EARTH-GEOLOGICAL-MATERIAL-001

Question and exact theorem. A geological material is a source-bound specimen class retaining chemical composition, material phases, texture, structure, provenance and observation scale.

```
specimen-composition-phase-texture-carrier__specimen-method-bounded__material-class-equivalence__provenance-mineralogy-chemistry-structure-and-scale-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule
```

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001

- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-BUDGET-CLOSURE-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	specimen-composition-phase-texture-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	specimen-method-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	material-class-equivalence	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	provenance-mineralogy-chemistry-structure-and-scale-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: specimen-composition-phase-texture-carrier__specimen-method-bounded__material-class-equivalence__provenance-mineralogy-chemistry-structure-and-scale-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- carrier-present: the claim retains a nonempty named carrier; passed true.
- relation-distinct: the carrier and relation are not silently conflated; passed true.

- `evidence-boundary`: the evidence boundary remains distinct from the result carrier; passed `true`.
- `exact-fold-witness`: the family's pre-source exact Fold construction reproduces; passed `true`.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar. Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

Geological identity requires specimen, composition, phase, texture, provenance and method. Layers are positive ordered carriers with explicit contacts and interruptions. Earth history is a dated multi-proxy reconstruction whose gaps and alternatives remain evidence, not empty numerical entries.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:fdcc193069b636b5566c946fec6c0d7a458b267360640528f94ec1c4a26dbad7.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:60b2290ac99160dd6ab737d8887203582830549684ea9d0b553981e5e2bda600.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:cb676506c4d518c23dd271806dd60ca6655b13e55a4a045422969ae9552a0325.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:944f510ffcd5d1012a7e31856e4b8a9472229f6451633b1d22b602b35d334d04.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:ee73c35ea2221aa1b58b268a50b76a0dd2c57f439202dcce572522b5fc3dd576; certificate
sha256:92de62e9df8fc3560558cd5e00610e012ec0722ef6dd5a56e6c2b489b52edb2e; external-validation
identity sha256:d368b1c10f508743df7a54b47658707b45d64d49f9d527740289ff215f7fcda9.

Post-seal empirical comparison. Directness:

authoritative_observation_or_primary_measurement_boundary_correspondence. Disposition:
purpose_matched_authoritative_source_features_present. Predicted consequence: earth:specimen-
composition-phase-texture-carrier__material-class-equivalence__provenance-mineralogy-
chemistry-structure-and-scale-held__specimen-method-bounded. Source-derived consequence: earth
:specimen-composition-phase-texture-carrier__material-class-equivalence__provenance-m
ineralogy-chemistry-structure-and-scale-held__specimen-method-bounded. Exact boundary
correspondence: true. External evidence selected the survivor: false. Formal structure relabelled as direct measurement:
false. Model, forecast, proxy or retrieval relabelled as observation: false.

Sources and complete feature accounting:

- USGS-NGMDB-001 - U.S. Geological Survey National Cooperative Geologic Mapping Program, [National Geologic Map Database](#); class authoritative_geoscience_archive; transport captured; features 0/4 present; present: none; absent and preserved: spatially referenced geology, map and report provenance, stratigraphic information, geochemistry geophysics palaeontology and geochronology coverage.
- USGS-SOIL-GEOCHEM-001 - U.S. Geological Survey, [Soil geochemistry and mineralogy data portal](#); class authoritative_soil_observation_archive; transport captured; features 3/4 present; present: soil specimen location, geochemical composition, mineralogical composition; absent and preserved: spatial coverage.
- NOAA-PALEO-001 - NOAA National Centers for Environmental Information World Data Service for Paleoclimatology, [Paleoclimatology data archive](#); class authoritative_proxy_archive; transport captured; features 3/4 present; present: archive and study identity, proxy and reconstructed-variable distinction, location and investigator provenance; absent and preserved: dating and temporal coverage.

- SMITHSONIAN-GVP-VOTW-001 - Smithsonian Institution Global Volcanism Program, [Volcanoes of the World web services](#); class authoritative_volcano_eruption_database; transport captured_via_addendum_prior_failure_preserved; features 3/4 present; present: volcano identity, eruption identity and time, database version and citation; absent and preserved: location and morphology.

Comparison record:

- sft-earth-geological-material-001-external-earth-evidence: predicted earth:specimen-composition-phase-texture-carrier__material-class-equivalence__provenance-mineralogy-chemistry-structure-and-scale-held__specimen-method-bounded; source-derived earth:specimen-composition-phase-texture-carrier__material-class-equivalence__provenance-mineralogy-chemistry-structure-and-scale-held__specimen-method-bounded; exact match True
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if a hand specimen name erases heterogeneous phases, alteration, provenance or measurement resolution.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json;
 source manifest sha256:e06740579e7637a8512e74268ad67e2d9e3a9ef44608290a35fa5c52ee7b9d33;
 derivation seal sha256:1011bd842dbad8341ec95137ec214f080952645ca127d7b85cb78ccdc6c9acf5;
 engine receipt sha256:701d119a2a29bfa4979b7fe0cdf7dec6b83e1d3063f69bb73e344d777b266ae9 at receipts/engine/model_admitted/SFT-EARTH-GEOLOGICAL-MATERIAL-001-701d119a2a29bfa4.json;
 empirical validation sha256:7bacdbf0b2c1c0f193ecefaf00e705b6f65097f3a92d10ad09f113e917268f879;
 measurement receipt
 sha256:6df9da576890164ef8bfe57934a350b4c95966b7abb72e090dd2f0099bd46559; isolation
 sha256:af991b4101ba5c1c5fe83aaeee470df2c0d0d4ae6ca88957e12735fc49e8088a; custody
 sha256:bbd94d99ee828763b0c22bf5ce0e48aaa30e8702c9b75e309a1744238f786eac.

14. Geological layer

Claim identity: SFT-EARTH-GEOLOGICAL-LAYER-001

Question and exact theorem. A geological layer is a mapped material body whose extent, contacts, continuity, internal variation and uncertainty are retained.

bounded-spatial-material-carrier__mapping-resolution-bounded__adjacency-continuity
 -and-contact-relation__extent-thickness-contacts-facies-and-uncertainty-held__evid
 ence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains
 -all-rows__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-BUDGET-CLOSURE-001
- SFT-EARTH-GEOLOGICAL-MATERIAL-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	bounded-spatial-material-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	mapping-resolution-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	adjacency-continuity-and-contact-relation	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	extent-thickness-contacts-facies-and-uncertainty-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: bounded-spatial-material-carrier__mapping-resolution-bounded__adjacency-continuity-and-contact-relation__extent-thickness-contacts-facies-and-uncertainty-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- carrier-present: the claim retains a nonempty named carrier; passed true.
- relation-distinct: the carrier and relation are not silently conflated; passed true.
- evidence-boundary: the evidence boundary remains distinct from the result carrier; passed true.
- exact-fold-witness: the family's pre-source exact Fold construction reproduces; passed true.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar. Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

Geological identity requires specimen, composition, phase, texture, provenance and method. Layers are positive ordered carriers with explicit contacts and interruptions. Earth history is a dated multi-proxy reconstruction whose gaps and alternatives remain evidence, not empty numerical entries.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:520ae26615e59e57fe46bfe5d625c935109ba4829c95a05af63a7daf74bc45cd.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:7ca0852d61d914a29133cb9b0225de19e80e76f2b29b8ef0ebdd9a059adfe386.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:4a41d83537a2857cc7b24883a59386634cab7406ee466d78fc73a39e146be539.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:135e9ed5766c2a205ad7d769ac55ddbec0dd4dee44393f3f7d8aa2ae2721aa9e.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:878b3b7b896d6333a097d302b25259382a35949294ba2a9b2995e4b6e417433e; certificate

sha256:08f84b40a253fce9edd5cdf9461968f390b87e270e57f617065d7f763903847; external-validation identity sha256:f4ecf57cf2cf0b2888d955ac1f9bb20905df852e135b839c7ace18ee2850e2fb.

Post-seal empirical comparison. Directness:

authoritative_observation_or_primary_measurement_boundary_correspondence. Disposition: purpose_matched_authoritative_source_features_present. Predicted consequence: earth:bounded-spatial-material-carrier_adjacency-continuity-and-contact-relation_extent-thickness-contacts-facies-and-uncertainty-held_mapping-resolution-bounded. Source-derived consequence: earth:bounded-spatial-material-carrier_adjacency-continuity-and-contact-relation_extent-thickness-contacts-facies-and-uncertainty-held_mapping-resolution-bounded. Exact boundary correspondence: true. External evidence selected the survivor: false. Formal structure relabelled as direct measurement: false. Model, forecast, proxy or retrieval relabelled as observation: false.

Sources and complete feature accounting:

- USGS-NGMDB-001 - U.S. Geological Survey National Cooperative Geologic Mapping Program, [National Geologic Map Database](#); class authoritative_geoscience_archive; transport captured; features 0/4 present; present: none; absent and preserved: spatially referenced geology, map and report provenance, stratigraphic information, geochemistry geophysics palaeontology and geochronology coverage.
- USGS-SOIL-GEOCHEM-001 - U.S. Geological Survey, [Soil geochemistry and mineralogy data portal](#); class authoritative_soil_observation_archive; transport captured; features 3/4 present; present: soil specimen location, geochemical composition, mineralogical composition; absent and preserved: spatial coverage.
- NOAA-PALEO-001 - NOAA National Centers for Environmental Information World Data Service for Paleoclimatology, [Paleoclimatology data archive](#); class authoritative_proxy_archive; transport captured; features 3/4 present; present: archive and study identity, proxy and reconstructed-variable distinction, location and investigator provenance; absent and preserved: dating and temporal coverage.
- SMITHSONIAN-GVP-VOTW-001 - Smithsonian Institution Global Volcanism Program, [Volcanoes of the World web services](#); class authoritative_volcano_eruption_database; transport captured_via_addendum_prior_failure_preserved; features 3/4 present; present: volcano identity, eruption identity and time, database version and citation; absent and preserved: location and morphology.

Comparison record:

- sft-earth-geological-layer-001-external-earth-evidence: predicted earth:bounded-spatial-material-carrier_adjacency-continuity-and-contact-relation_extent-thickness-contacts-facies-and-uncertainty-held_mapping-resolution-bounded; source-derived earth:bounded-spatial-material-carrier_adjacency-continuity-and-contact-relation_extent-thickness-contacts-facies-and-uncertainty-held_mapping-resolution-bounded; exact match True
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved

- a deliberately changed external consequence was rejected

Falsification boundary: Reject if continuity is inferred across an unobserved gap without recording the correlation rule and alternatives.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json;
 source manifest sha256:fa4b5bc908f84a92fad5677d694756d6cbc81ad816154c99cacb6f64b510c08f;
 derivation seal sha256:da0daef3d056097dd736aaae332f5cdc4dabf9221ce9889d9f3229bc52b48e22;
 engine receipt sha256:116ab73eab4c1f79ef6119f2ca671f234e4a678551a5b77e39cfa8aee9aae177 at
 receipts/engine/model_admitted/SFT-EARTH-GEOLOGICAL-LAYER-001-116ab73eab4c1f79.json;
 empirical validation sha256:fdee6e00d871cd0e652515d86511ef07e1d1493003e5eb89ed0680751ed77a22;
 measurement receipt
 sha256:a5de59057f6468193e72812a21dd448a14c105a372e327ac52b22bdfc6648517; isolation
 sha256:755fe471aff327b1382c84e85bb330c128003baa928ec56031038b83f3f2c4a8; custody
 sha256:0f7ecc7147da0d6d40413eb85debfcdb19065afadd5d3897b15eafeb4cc8a61.

15. Stratigraphic order

Claim identity: SFT-EARTH-STRATIGRAPHIC-ORDER-001

Question and exact theorem. Stratigraphic order is a partial order over observed layers and events that retains unconformities, faults, intrusions and unresolved correlations.

ordered-layer-event-carrier__section-and-correlation-bounded__before-after-and-correlation-relation__contacts-unconformities-intrusions-faults-and-dates-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001

- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-BUDGET-CLOSURE-001
- SFT-EARTH-GEOLOGICAL-LAYER-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	ordered-layer-event-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	section-and-correlation-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
relation	imported-fitted-or-missing-relation	before-after-and-correlation-relation	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	contacts-unconformities-intrusions-faults-and-dates-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: `ordered-layer-event-carrier__section-and-correlation-bounded__before-after-and-correlation-relation__contacts-unconformities-intrusions-faults-and-dates-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule`.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `carrier-present`: the claim retains a nonempty named carrier; passed `true`.
- `relation-distinct`: the carrier and relation are not silently conflated; passed `true`.
- `evidence-boundary`: the evidence boundary remains distinct from the result carrier; passed `true`.
- `exact-fold-witness`: the family's pre-source exact Fold construction reproduces; passed `true`.

Scientific meaning. Positive ordered layers force a partial history while contacts, unconformities, intrusions and faults preserve the evidence that interrupts simple succession.

Geological identity requires specimen, composition, phase, texture, provenance and method. Layers are positive ordered carriers with explicit contacts and interruptions. Earth history is a dated multi-proxy reconstruction whose gaps and alternatives remain evidence, not empty numerical entries.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:0e64120bb4ed52382f37662c0bd86d9e5c56aed1bc4722064e5679c648a7e3dc.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:d64743f2dd80c5fdf47906617474544411aa233163c0558a7e6a1c10f65126a0.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:fd277463493d25a89f6bf062456baa61c6024a8526776c3b66fa35d8fd69ddfd.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:3f3bd65e8cf148295e1823fc51acb36e106a03eeeff3e23716aa16494faf1789.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
 sha256:6d682044436b55adef77d1db0c73bf424ecc83348bf652f51aa21ac4eedc5d34; certificate
 sha256:dfebbbc99c9621f4762c8f25007ef761c60f91a025842b826b0d90eecb3054ea; external-validation
 identity sha256:77ec106fd237292322f55aa853651ea036e92467e1d0123217bad1a5a8d43f2.

Post-seal empirical comparison. Directness:

authoritative_observation_or_primary_measurement_boundary_correspondence. Disposition:
 purpose_matched_authoritative_source_features_present. Predicted consequence: earth:ordered-l
 ayer-event-carrier__before-after-and-correlation-relation__contacts-unconformities-in
 trusions-faults-and-dates-held__section-and-correlation-bounded. Source-derived consequence: e
 arth:ordered-layer-event-carrier__before-after-and-correlation-relation__contacts-unc
 onformities-intrusions-faults-and-dates-held__section-and-correlation-bounded. Exact
 boundary correspondence: true. External evidence selected the survivor: false. Formal structure relabelled as direct
 measurement: false. Model, forecast, proxy or retrieval relabelled as observation: false.

Sources and complete feature accounting:

- USGS-NGMDB-001 - U.S. Geological Survey National Cooperative Geologic Mapping Program, [National Geologic Map Database](#); class authoritative_geoscience_archive; transport captured; features 0/4 present; present: none; absent and preserved: spatially referenced geology, map and report provenance, stratigraphic information, geochemistry geophysics palaeontology and geochronology coverage.
- USGS-SOIL-GEOCHEM-001 - U.S. Geological Survey, [Soil geochemistry and mineralogy data portal](#); class authoritative_soil_observation_archive; transport captured; features 3/4 present; present: soil specimen location, geochemical composition, mineralogical composition; absent and preserved: spatial coverage.
- NOAA-PALEO-001 - NOAA National Centers for Environmental Information World Data Service for Paleoclimatology, [Paleoclimatology data archive](#); class authoritative_proxy_archive; transport captured; features 3/4 present; present: archive and study identity, proxy and reconstructed-variable distinction, location and investigator provenance; absent and preserved: dating and temporal coverage.
- SMITHSONIAN-GVP-VOTW-001 - Smithsonian Institution Global Volcanism Program, [Volcanoes of the World web services](#); class authoritative_volcano_eruption_database; transport captured_via_addendum_prior_failure_preserved; features 3/4 present; present: volcano identity, eruption identity and time, database version and citation; absent and preserved: location and morphology.

Comparison record:

- sft-earth-stratigraphic-order-001-external-earth-evidence: predicted earth:ordered-layer-event-carrier__before-after-and-correlation-relation__contacts-unconformities-intrusions-faults-and-dates-held__section-and-correlation-bounded; source-derived earth:ordered-layer-event-carrier__before-after-and-correlation-relation__contacts-unconformities-intrusion-s-faults-and-dates-held__section-and-correlation-bounded; exact match True
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if one local sequence is silently promoted to a global total order or if a contradictory contact is discarded.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter

- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json;
 source manifest sha256:e53f687fd2e11c8b2aa7e3b413e69cd3e3a29aaf406bf55d488f744e70b28b4c;
 derivation seal sha256:9fca03d16e6c3ac4f08c94e1f13c79f0711f081b9939b1895a455ff0338a931f;
 engine receipt sha256:41f826d3bb75313530365272560df0e48149f66522a01f8300bcf154afa3752e at re
 ceipts/engine/model_admitted/SFT-EARTH-STRATIGRAPHIC-ORDER-001-41f826d3bb753135.json;
 empirical validation sha256:56c28aa0d9530484b0402e04a5fadab3793b91ff105642050fb13ce2884b6ffd;
 measurement receipt
 sha256:ec67abe84729f45ce39e02ca6143ada1f4813da04a6cb3151190581cad11b920; isolation
 sha256:edf8db025b29db27201a4dbd688e079ba7264b0bee70f3c06bbd3f99d047f735; custody
 sha256:44f592893f635ace5b410c30c55830034323dad7774396bc3b32b09966305285.

16. Geological transformation cycle

Claim identity: SFT-EARTH-GEOLOGICAL-CYCLE-001

Question and exact theorem. A geological cycle is a reconstructible network of material-state transitions whose carriers, paths, transfers, losses and recurrence boundary remain explicit.

material-state-transition-network__process-network-bounded__weathering-transport-d
 eposition-transformation-relation__inputs-products-paths-losses-and-times-held__ev
 idence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retai
 ns-all-rows__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001

- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-BUDGET-CLOSURE-001
- SFT-EARTH-STRATIGRAPHIC-ORDER-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	material-state-transition-network	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	process-network-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	weathering-transport-deposition-transformation-relation	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	inputs-products-paths-losses-and-times-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: material-state-transition-network__process-network-bounded__weathering-transport-deposition-transformation-relation__inputs-products-paths-losses-and-times-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- carrier-present: the claim retains a nonempty named carrier; passed true.
- relation-distinct: the carrier and relation are not silently conflated; passed true.
- evidence-boundary: the evidence boundary remains distinct from the result carrier; passed true.
- exact-fold-witness: the family's pre-source exact Fold construction reproduces; passed true.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar.

Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

Geological identity requires specimen, composition, phase, texture, provenance and method. Layers are positive ordered carriers with explicit contacts and interruptions. Earth history is a dated multi-proxy reconstruction whose gaps and alternatives remain evidence, not empty numerical entries.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:e4d4a239a511d6cb4f4b4a0545b2357f2bf4aa9c86b356e68448447262057882.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:6ec882583262d07549e38c49b0cc6ad10b08083d6f1b0b26ed817f2ad04b10a6.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:61ebac1e7d57d5e4b6f3b9c1ad693835dc73882d0d8b4d97fda5191b71f241c3.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:0316b3a9a92fbedece81451833de635ece90aa8c017828957f7dd306d7e9d34f.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:8d26c813671715ef799708225816ee8ba194ecf020f92a0941ba758badc4ee0e; certificate
sha256:878fde2c3b2a3ab7963c8a6cec89bf2536afc583358807d5845d2c3d8c83f306; external-validation
identity sha256:4a9735752efa7adb1afaec608aafba7e2a5b061438fd79ad36585b118d743945.

Post-seal empirical comparison. Directness:

authoritative_observation_or_primary_measurement_boundary_correspondence. Disposition:
purpose_matched_authoritative_source_features_present. Predicted consequence: earth:material-
state-transition-network__weathering-transport-deposition-transformation-relation__in
puts-products-paths-losses-and-times-held__process-network-bounded. Source-derived

consequence: earth:material-state-transition-network__weathering-transport-deposition-transformation-relation__inputs-products-paths-losses-and-times-held__process-network-bounded. Exact boundary correspondence: true. External evidence selected the survivor: false. Formal structure relabelled as direct measurement: false. Model, forecast, proxy or retrieval relabelled as observation: false.

Sources and complete feature accounting:

- USGS-NGMDB-001 - U.S. Geological Survey National Cooperative Geologic Mapping Program, [National Geologic Map Database](#); class authoritative_geoscience_archive; transport captured; features 0/4 present; present: none; absent and preserved: spatially referenced geology, map and report provenance, stratigraphic information, geochemistry geophysics palaeontology and geochronology coverage.
- USGS-SOIL-GEOCHEM-001 - U.S. Geological Survey, [Soil geochemistry and mineralogy data portal](#); class authoritative_soil_observation_archive; transport captured; features 3/4 present; present: soil specimen location, geochemical composition, mineralogical composition; absent and preserved: spatial coverage.
- NOAA-PALEO-001 - NOAA National Centers for Environmental Information World Data Service for Paleoclimatology, [Paleoclimatology data archive](#); class authoritative_proxy_archive; transport captured; features 3/4 present; present: archive and study identity, proxy and reconstructed-variable distinction, location and investigator provenance; absent and preserved: dating and temporal coverage.
- SMITHSONIAN-GVP-VOTW-001 - Smithsonian Institution Global Volcanism Program, [Volcanoes of the World web services](#); class authoritative_volcano_eruption_database; transport captured_via_addendum_prior_failure_preserved; features 3/4 present; present: volcano identity, eruption identity and time, database version and citation; absent and preserved: location and morphology.

Comparison record:

- sft-earth-geological-cycle-001-external-earth-evidence: predicted earth:material-state-transition-network__weathering-transport-deposition-transformation-relation__inputs-products-paths-losses-and-times-held__process-network-bounded; source-derived earth:material-state-transition-network__weathering-transport-deposition-transformation-relation__inputs-products-paths-losses-and-times-held__process-network-bounded; exact match True
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if a named cycle implies that every carrier returns or if intermediate transformations and exports are omitted.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

```
experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json;
source manifest sha256:0f9004843766ac7bb0f89cd3ee04bee509914df328d128ac532234c88e89522c;
derivation seal sha256:faf28e264910b43205dd04b43148d642abd1f135e105146cb26ab3b816dd13e5;
engine receipt sha256:82a3a688e27dc2d90df87dba81ef18f300524956d4c02c3de5bf267fab8ff3a1 at
receipts/engine/model_admitted/SFT-EARTH-GEOLOGICAL-CYCLE-001-82a3a688e27dc2d9.json;
empirical validation sha256:14e5cae111f11e40feaa7f3e8b1b724fd783c81e729602df11ca21f0e5c484f3;
measurement receipt
sha256:96774802bea84a38cfcde1defab5b8b49765d39e5ec6cbb524d2169b190ba9c4; isolation
sha256:b3876d0bf5ba399b8e38a04887a42d48c18e5cdf29e2678da71227d3eb71a539; custody
sha256:4b28adea78b89d5c9f9dc0e3bffe3ede79d3938a91826d93672cf7009b17e576.
```

17. Geological provenance

Claim identity: SFT-EARTH-GEOLOGICAL-PROVENANCE-001

Question and exact theorem. Geological provenance is a source-to-path-to-deposit reconstruction supported by retained composition, age, alteration and competing-source evidence.

```
source-path-deposit-carrier__method-and-basin-bounded__source-to-observed-material
-reconstruction__composition-age-route-alteration-and-alternatives-held__evidence-
class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-
rows__no-extra-rule
```

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001

- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-BUDGET-CLOSURE-001
- SFT-EARTH-GEOLOGICAL-CYCLE-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	source-path-deposit-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	method-and-basin-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	source-to-observed-material-reconstruction	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	composition-age-route-alteration-and-alternatives-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: source-path-deposit-carrier__method-and-basin-bounded__source-to-observed-material-reconstruction__composition-age-route-alteration-and-alternatives-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `carrier-present`: the claim retains a nonempty named carrier; passed `true`.
- `relation-distinct`: the carrier and relation are not silently conflated; passed `true`.
- `evidence-boundary`: the evidence boundary remains distinct from the result carrier; passed `true`.
- `exact-fold-witness`: the family's pre-source exact Fold construction reproduces; passed `true`.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar. Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

Geological identity requires specimen, composition, phase, texture, provenance and method. Layers are positive ordered carriers with explicit contacts and interruptions. Earth history is a dated multi-proxy reconstruction whose gaps and alternatives remain evidence, not empty numerical entries.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:ca2871a8bf4b75538c8b6457aa4372850a4c864c7b4ba0b3160e8131ef3d31d8.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:5d3a83b9b56043964642fb1ea4eaf8e00c9b5047e8fe35b55155f21f8c2ff114.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:b6ce2b88256130da71431725de7e72eb7987c97cd174d29a97052f1890165e85.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:be35a22ba05a6e724c1ea9daa54064c32788a1e42a6a2c27359302e4e3b3942b.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:f3ebf178c6b82a59dcb9b7161ffbc228f727fd79be6774fa99647c2a1549c49d; certificate
sha256:7bde77e8dd6bf99ccc51d99a8aac9e49383079fdc0f18344888c540ef86f2c7f; external-validation
identity sha256:5c5c7508bfe5a3f61fae33723c5b02859c05842828d6827fe4dd7b3b47607c78.

Post-seal empirical comparison. Directness:

`authoritative_observation_or_primary_measurement_boundary_correspondence`. Disposition:
`purpose_matched_authoritative_source_features_present`. Predicted consequence: `earth:source-path-deposit-carrier__source-to-observed-material-reconstruction__composition-age-route-alteration-and-alternatives-held__method-and-basin-bounded`. Source-derived consequence: `earth:source-path-deposit-carrier__source-to-observed-material-reconstruction__composition-age-route-alteration-and-alternatives-held__method-and-basin-bounded`. Exact boundary correspondence: `true`. External evidence selected the survivor: `false`. Formal structure relabelled as direct measurement: `false`. Model, forecast, proxy or retrieval relabelled as observation: `false`.

Sources and complete feature accounting:

- USGS-NGMDB-001 - U.S. Geological Survey National Cooperative Geologic Mapping Program, [National Geologic Map Database](#); class `authoritative_geoscience_archive`; transport captured; features 0/4 present; present: none; absent and preserved: spatially referenced geology, map and report provenance, stratigraphic information, geochemistry geophysics palaeontology and geochronology coverage.
- USGS-SOIL-GEOCHEM-001 - U.S. Geological Survey, [Soil geochemistry and mineralogy data portal](#); class `authoritative_soil_observation_archive`; transport captured; features 3/4 present; present: soil specimen location, geochemical composition, mineralogical composition; absent and preserved: spatial coverage.

- NOAA-PALEO-001 - NOAA National Centers for Environmental Information World Data Service for Paleoclimatology, [Paleoclimatology data archive](#); class authoritative_proxy_archive; transport captured; features 3/4 present; present: archive and study identity, proxy and reconstructed-variable distinction, location and investigator provenance; absent and preserved: dating and temporal coverage.
- SMITHSONIAN-GVP-VOTW-001 - Smithsonian Institution Global Volcanism Program, [Volcanoes of the World web services](#); class authoritative_volcano_eruption_database; transport captured_via_addendum_prior_failure_preserved; features 3/4 present; present: volcano identity, eruption identity and time, database version and citation; absent and preserved: location and morphology.

Comparison record:

- sft-earth-geological-provenance-001-external-earth-evidence: predicted earth:source-path-deposit-carrier__source-to-observed-material-reconstruction__composition-age-route-alteration-and-alternatives-held__method-and-basin-bounded; source-derived earth:source-path-deposit-carrier__source-to-observed-material-reconstruction__composition-age-route-alteration-and-alternatives-held__method-and-basin-bounded; exact match True
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if one matching property uniquely identifies a source while generated alternatives remain viable.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json; source manifest sha256:1e5c7eca4f3f57e69aafce41dc4fa2dbe100ace2d828b2123734bc02861ef833; derivation seal sha256:436fa128a478d1b504f96974ce09f013bc2faa84c0d47b0f06a1549f19a222ca; engine receipt sha256:6326865fe7807b2645a6a402c7b87f32ccf7cccaa8cb9446f72865a2cccae46f at receipts/engine/model_admitted/SFT-EARTH-GEOLOGICAL-PROVENANCE-001-6326865fe7807b26.json; empirical validation sha256:ad8a411fd344cad252ae5f4f51eb4304196ef7f3739b942512ca93aa3420a03f; measurement receipt sha256:05888a10f4e8627abbb0fd050d76e08ab42e42d8372a51dfca74268ac62a4df7; isolation sha256:b9913c4a7e1a4cfa4eaaccee0d5b3e332c51b641b7bb333ae52c528b0bf5d84f; custody sha256:17d4a20c2d4c38561e29ea88c5b41d2d554c51112d7f43984acdc46b1e9fe00a.

18. Earth historical record

Claim identity: SFT-EARTH-EARTH-HISTORY-001

Question and exact theorem. An Earth-history claim is a source-dated reconstruction over preserved records with temporal uncertainty, gaps, resolution and alternative histories retained.

```
time-ordered-multi-proxy-carrier__dated-record-bounded__event-state-reconstruction
__dating-provenance-gaps-resolution-and-alternatives-held__evidence-class-explicit
__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra
-rule
```

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-BUDGET-CLOSURE-001
- SFT-EARTH-GEOLOGICAL-PROVENANCE-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	time-ordered-multi-proxy-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	dated-record-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	event-state-reconstruction	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	dating-provenance-gaps-resolution-and-alternatives-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: time-ordered-multi-proxy-carrier__dated-record-bounded__event-state-reconstruction__dating-provenance-gaps-resolution-and-alternatives-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- carrier-present: the claim retains a nonempty named carrier; passed true.
- relation-distinct: the carrier and relation are not silently conflated; passed true.
- evidence-boundary: the evidence boundary remains distinct from the result carrier; passed true.
- exact-fold-witness: the family's pre-source exact Fold construction reproduces; passed true.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar. Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

Geological identity requires specimen, composition, phase, texture, provenance and method. Layers are positive ordered carriers with explicit contacts and interruptions. Earth history is a dated multi-proxy reconstruction whose gaps and alternatives remain evidence, not empty numerical entries.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:534de55a1877a02d01eb6aa5d8eba704ba2032185d417ed474dadcd433701ce25.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:b3f1d3adef5c2818f7bc7efc73a804191298f8db1a145461e4e303414b22139c.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:093b36324e8dbe611ff74bf0642f226b5d7fb5f867f9ee6991a768d9e699072b.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:60f2bdaf5915da137f16cf9d93a550c027ae55465e2a667a151b9dab7602a2aa.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:0d9f22a2f728328014b3e2c4458981ff7907702db89521d52a025f0edc5ea67b; certificate
sha256:0cd2f1a6ceeeaca48a7a75d809042174ccb7a593d46916cc97ad629d3a9c8d84; external-validation
identity sha256:4baa81eb282829d4cb90b329b16cf43faa1651f2f79f9c87847254f20bc8a472.

Post-seal empirical comparison. Directness:

authoritative_observation_or_primary_measurement_boundary_correspondence. Disposition:
purpose_matched_authoritative_source_features_present. Predicted consequence: earth:time-ordered-multi-proxy-carrier__event-state-reconstruction__dating-provenance-gaps-resolution-and-alternatives-held__dated-record-bounded. Source-derived consequence: earth:time-ordered-multi-proxy-carrier__event-state-reconstruction__dating-provenance-gaps-resolution-and-alternatives-held__dated-record-bounded. Exact boundary correspondence: true. External evidence selected the survivor: false. Formal structure relabelled as direct measurement: false. Model, forecast, proxy or retrieval relabelled as observation: false.

Sources and complete feature accounting:

- USGS-NGMDB-001 - U.S. Geological Survey National Cooperative Geologic Mapping Program, [National Geologic Map Database](#); class authoritative_geoscience_archive; transport captured; features 0/4 present; present: none; absent and preserved: spatially referenced geology, map and report provenance, stratigraphic information, geochemistry geophysics palaeontology and geochronology coverage.
- USGS-SOIL-GEOCHEM-001 - U.S. Geological Survey, [Soil geochemistry and mineralogy data portal](#); class authoritative_soil_observation_archive; transport captured; features 3/4 present; present: soil specimen location, geochemical composition, mineralogical composition; absent and preserved: spatial coverage.
- NOAA-PALEO-001 - NOAA National Centers for Environmental Information World Data Service for Paleoclimatology, [Paleoclimatology data archive](#); class authoritative_proxy_archive; transport captured; features 3/4 present; present: archive and study identity, proxy and reconstructed-variable distinction, location and investigator provenance; absent and preserved: dating and temporal coverage.
- SMITHSONIAN-GVP-VOTW-001 - Smithsonian Institution Global Volcanism Program, [Volcanoes of the World web services](#); class authoritative_volcano_eruption_database; transport captured_via_addendum_prior_failure_preserved; features 3/4 present; present: volcano identity, eruption identity and time, database version and citation; absent and preserved: location and morphology.

Comparison record:

- sft-earth-earth-history-001-external-earth-evidence: predicted earth:time-ordered-multi-proxy-carrier__event-state-reconstruction__dating-provenance-gaps-resolution-and-alternatives-held__dated-record-bounded; source-derived earth:time-order

ed-multi-proxy-carrier__event-state-reconstruction__dating-provenance-gaps-resolution-and-alternatives-held__dated-record-bounded; exact match True

- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if absence of record becomes proof of absence, if dating uncertainty is erased, or if a universal law is inferred from one contingent history alone.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json;
 source manifest sha256:d2f0ae3f15517340dad872cf2de8c3bbb6ce98962bfe23b79f3f24b9daca2e86;
 derivation seal sha256:073119c5386cc748cf7af210bdfd7f420f1aaf4d988781a5418af5d99d033525;
 engine receipt sha256:5bc50a62a3e5eb2354c01a146acd248a2b55437433c916ea0dd80e1da029a887 at
 receipts/engine/model_admitted/SFT-EARTH-EARTH-HISTORY-001-5bc50a62a3e5eb23.json;
 empirical validation sha256:6b9ffce6fee3dc51b5a60151f65ea2a22b7228ee5cda99c215e4d12b0040ff8e;
 measurement receipt
 sha256:14ac1627e7b1940ec0c2657adeb25c922763d76ba556c04fd0eb6ee2d29700ed; isolation
 sha256:31df2985caa2a7e09ef9ce2ce47ecd4134e8d7ac423db0bf36e43bb9d452c08b; custody
 sha256:62211e47fe89d4ff426b07b43b8dea422ab905f90c0593b08c2e70409b77a277.

Ledger family 4: Planetary Interior and Geodynamics

Interior compartments and geodynamic relations are inference-bounded. Depth, composition, state, contrast, source, transfer and uncertainty remain reconstructible. Universal mechanics and thermodynamics enter only through prior admitted receipts; Earth-specific observations do not select those laws.

19. Earth interior compartments

Claim identity: SFT-EARTH-INTERIOR-COMPARTMENT-001

Question and exact theorem. An inferred Earth-interior compartment requires a reproducible property contrast across a declared boundary with its observation and inversion limits retained.

nested-interior-region-carrier__geophysical-inference-bounded__boundary-and-property-contrast-relation__depth-composition-state-observation-and-uncertainty-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retai

ns-all-rows__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-EARTH-HISTORY-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	nested-interior-region-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	geophysical-inference-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	boundary-and-property-contrast-relation	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	depth-composition-state-observation-and-uncertainty-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: nested-interior-region-carrier__geophysical-inference-bounded__boundary-and-property-contrast-relation__depth-composition-state-observation-and-uncertainty-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- carrier-present: the claim retains a nonempty named carrier; passed true.
- relation-distinct: the carrier and relation are not silently conflated; passed true.
- evidence-boundary: the evidence boundary remains distinct from the result carrier; passed true.
- exact-fold-witness: the family's pre-source exact Fold construction reproduces; passed true.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar.

Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

Interior compartments and geodynamic relations are inference-bounded. Depth, composition, state, contrast, source, transfer and uncertainty remain reconstructible. Universal mechanics and thermodynamics enter only through prior admitted receipts; Earth-specific observations do not select those laws.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:1b15e33de99c1ebd21aa8e845507eeb5a82dee1fdc84e0f6eafaa53154bd9db2.

- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:651443552c35a4e0f44c057bd117f78f523dcc9da19a7080bdb3f6bebdd6a85.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:74b3624135aac4ceff634681be3881b62b2a8b7183527e504ced80a9e56f7c15.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:aa050101e3680dcfd994f6e7e976628c0ab205e8cfc7cb3a95abc2bda1d0e281.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:59c99bcd9fca3c870928f1b410b598391ee7903a5558aff2ebfcb4ba2314b173; certificate
sha256:87f81b8a93f7993b0b14611f14aa1bb2079f22bf14766112f47f0fbb59f6c988; external-validation
identity sha256:9ffb885585e25aa9a545ad6f908d09431006b2ee7970c0050c52b21270c212a3.

Post-seal empirical comparison. Directness:

authoritative_observation_or_primary_measurement_boundary_correspondence. Disposition:
purpose_matched_authoritative_source_features_present. Predicted consequence: earth:nested-in
terior-region-carrier__boundary-and-property-contrast-relation__depth-composition-sta
te-observation-and-uncertainty-held__geophysical-inference-bounded. Source-derived
consequence: earth:nested-interior-region-carrier__boundary-and-property-contrast-relatio
n__depth-composition-state-observation-and-uncertainty-held__geophysical-inference-bo
unded. Exact boundary correspondence: true. External evidence selected the survivor: false. Formal structure relabelled as
direct measurement: false. Model, forecast, proxy or retrieval relabelled as observation: false.

Sources and complete feature accounting:

- EARTHSCOPE-SAGE-SERVICES-001 - NSF SAGE Facility operated by EarthScope Consortium, **Seismological and geophysical web services**; class authoritative_geophysical_data_service; transport captured; features 3/4 present; present: station and channel identity, waveform and event access, instrument response; absent and preserved: observed and derived product distinction.
- USGS-GEOMAG-OBSERVATORIES-001 - U.S. Geological Survey Geomagnetism Program, **USGS geomagnetic observatory operations**; class authoritative_geomagnetic_observation_network; transport captured; features 0/4 present; present: none; absent and preserved: observatory identity, time-bounded geomagnetic observation, station status, read-only access.
- USGS-NGMDB-001 - U.S. Geological Survey National Cooperative Geologic Mapping Program, **National Geologic Map Database**; class authoritative_geoscience_archive; transport captured; features 0/4 present; present: none; absent and preserved: spatially referenced geology, map and report provenance, stratigraphic information, geochemistry geophysics palaeontology and geochronology coverage.

Comparison record:

- sft-earth-interior-compartment-001-external-earth-evidence: predicted earth:nested-interior-region-carrier__boundary-and-property-contrast-relation__depth-composition-state-observation-and-uncertainty-held__geophysical-inference-bounded; source-derived earth:nested-interior-region-carrier__boundary-and-property-contrast-relation__depth-composition-state-observation-and-uncertainty-held__geophysical-inference-bounded; exact match True
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if an inversion model is presented as direct access or if incompatible observations and nonunique interiors are erased.

Explicit exclusions.

- semantic numerical zero

- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json;
 source manifest sha256:4fef92ef408ece82fc2f0367ec8e8559796625b306fba72b9eb11c80c74e0f41;
 derivation seal sha256:6df55fc37fc6ad21130f6fad6ca1548cf518c2e0327c2d4f074211b4f6203fe6;
 engine receipt sha256:054996179e5ec7a8745973f9c68fa170b60e1565eefd2e2f65fbe2dab1da4509 at re
 cepts/engine/model_admitted/SFT-EARTH-INTERIOR-COMPARTMENT-001-054996179e5ec7a8.json
 ; empirical validation
 sha256:95da05f51e4279633347b483715517a374db082229904b4828f27bcfc5f777c7; measurement
 receipt sha256:63416fa182d0f4343ece93207064b2fec961962c587d19574ea8a1711a1c4865; isolation
 sha256:d210527fe181503f074873535cf5bcc1fc66c3dc0a8fae07eb5f4a512b001dbf; custody
 sha256:3789edb6632f38d787d75845a59aabe3e89175f81276c2f6369d8767be9e40e1.

20. Earth internal heat sources

Claim identity: SFT-EARTH-INTERNAL-HEAT-001

Question and exact theorem. Earth internal heat is a complete source, storage and transport ledger distinguishing inherited, transformation and internally generated contributions.

```
source-reservoir-thermal-carrier__thermal-ledger-bounded__production-storage-trans
fer-relation__source-identity-location-rate-history-and-uncertainty-held__evidence
-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all
-rows__no-extra-rule
```

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001

- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-EARTH-HISTORY-001
- SFT-EARTH-INTERIOR-COMPARTMENT-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	source-reservoir-thermal-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	thermal-ledger-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	production-storage-transfer-relation	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	source-identity-location-ratio-history-and-uncertainty-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: `source-reservoir-thermal-carrier__thermal-ledger-bounded__production-storage-transfer-relation__source-identity-location-rate-history-and-uncertainty-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule`.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `carrier-present`: the claim retains a nonempty named carrier; passed `true`.
- `relation-distinct`: the carrier and relation are not silently conflated; passed `true`.
- `evidence-boundary`: the evidence boundary remains distinct from the result carrier; passed `true`.
- `exact-fold-witness`: the family's pre-source exact Fold construction reproduces; passed `true`.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar.

Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

Interior compartments and geodynamic relations are inference-bounded. Depth, composition, state, contrast, source, transfer and uncertainty remain reconstructible. Universal mechanics and thermodynamics enter only through prior admitted receipts; Earth-specific observations do not select those laws.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:439a74e5fa7d93ecad5b55f516d1af936301e96fe4febdd07b46a24e6b80f690.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:elfcf8927814042156b2d89fe00e2513f83076cb4c9aa00f28302a7800ff03e8.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:cfa4b0dc9585e31ad2b519f4868bf342f8bc1f55d8ca3fd9292536cdb2db0cd5.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:c1f63eeef0ff34e6bf2b1418fad0cbf5e9e83a29ccf4739277b41e3fbf2e8fa9.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:ee5ca1771822395ad1a30de988ce7e00cc4025e24a26a49d0a31f0f3701fefe8; certificate
sha256:2d5bf0816a2a3ca15b56a0f690ebbfb3fc2fa15ae2e64f8a2dc1fbfc8cdd1e06; external-validation
identity sha256:5121cab7190754501e2c66d57e785de7ff75fab8a35c3a8d1995b61b205e0327.

Post-seal empirical comparison. Directness:

authoritative_observation_or_primary_measurement_boundary_correspondence. Disposition: purpose_matched_authoritative_source_features_present. Predicted consequence: earth:source-reservoir-thermal-carrier__production-storage-transfer-relation__source-identity-location-rate-history-and-uncertainty-held__thermal-ledger-bounded. Source-derived consequence: earth:source-reservoir-thermal-carrier__production-storage-transfer-relation__source-identity-location-rate-history-and-uncertainty-held__thermal-ledger-bounded. Exact boundary correspondence: true. External evidence selected the survivor: false. Formal structure relabelled as direct measurement: false. Model, forecast, proxy or retrieval relabelled as observation: false.

Sources and complete feature accounting:

- EARTHSCOPE-SAGE-SERVICES-001 - NSF SAGE Facility operated by EarthScope Consortium, **Seismological and geophysical web services**; class authoritative_geophysical_data_service; transport captured; features 3/4 present; present: station and channel identity, waveform and event access, instrument response; absent and preserved: observed and derived product distinction.
- USGS-GEOMAG-OBSERVATORIES-001 - U.S. Geological Survey Geomagnetism Program, **USGS geomagnetic observatory operations**; class authoritative_geomagnetic_observation_network; transport captured; features 0/4 present; present: none; absent and preserved: observatory identity, time-bounded geomagnetic observation, station status, read-only access.
- USGS-NGMDB-001 - U.S. Geological Survey National Cooperative Geologic Mapping Program, **National Geologic Map Database**; class authoritative_geoscience_archive; transport captured; features 0/4 present; present: none; absent and preserved: spatially referenced geology, map and report provenance, stratigraphic information, geochemistry geophysics palaeontology and geochronology coverage.

Comparison record:

- sft-earth-internal-heat-001-external-earth-evidence: predicted earth:source-reservoir-thermal-carrier__production-storage-transfer-relation__source-identity-location-rate-history-and-uncertainty-held__thermal-ledger-bounded; source-derived earth:source-reservoir-thermal-carrier__production-storage-transfer-relation__source-identity-location-rate-history-and-uncertainty-held__thermal-ledger-bounded; exact match True
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if present surface flux is assigned to a unique interior source without a registered transport history.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law

- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json;
 source manifest sha256:9134cc47c777f9ba55216ee9c14c009102dd08b5e13c95e7712088c40d438f33;
 derivation seal sha256:985cee9fa2bd082d9f065048b7892079cdfae9afaed4e062bb5697b2ceb09d3d;
 engine receipt sha256:3357bcff272a96cb45d8f3d98e7694e6258ade8b1571f6d215bafc9c5a3082a6 at
 receipts/engine/model_admitted/SFT-EARTH-INTERNAL-HEAT-001-3357bcff272a96cb.json;
 empirical validation sha256:31e55b99045ea62b3f160f5c17f7bbb03fa3b76967cef8d8e9f40ed888cb1a90;
 measurement receipt
 sha256:99b36d0ef971774816b71dd7e51540804746886d53d189f1a242dbb2fa90f56c; isolation
 sha256:a0f0f98a419edc9313486c692152fd56b0d0312c546033e9b33006753f17b322; custody
 sha256:9fdfdf81fdb9dd13da53d422ba54cccc425dfbcb81d9af38a081b6135a9c9cfa.

21. Interior heat transport

Claim identity: SFT-EARTH-INTERIOR-HEAT-TRANSPORT-001

Question and exact theorem. Interior heat transport composes admitted material conduction with counted carrier motion and phase transitions under explicit geometry and boundaries.

temperature-material-motion-carrier__interior-scale-bounded__conduction-advection-
 and-transition-relation__geometry-properties-boundaries-time-and-flux-held__eviden
 ce-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-a
 ll-rows__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001

- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-EARTH-HISTORY-001
- SFT-EARTH-INTERNAL-HEAT-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	temperature-material-motion-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	interior-scale-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	conduction-advection-and-transition-relation	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	geometry-properties-boundaries-time-and-flux-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: temperature-material-motion-carrier__interior-scale-bounded__conduction-advection-and-transition-relation__geometry-properties-boundaries-time-and-flux-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `carrier-present`: the claim retains a nonempty named carrier; passed `true`.
- `relation-distinct`: the carrier and relation are not silently conflated; passed `true`.
- `evidence-boundary`: the evidence boundary remains distinct from the result carrier; passed `true`.
- `exact-fold-witness`: the family's pre-source exact Fold construction reproduces; passed `true`.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar.

Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

Interior compartments and geodynamic relations are inference-bounded. Depth, composition, state, contrast, source, transfer and uncertainty remain reconstructible. Universal mechanics and thermodynamics enter only through prior admitted receipts; Earth-specific observations do not select those laws.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:95d722eb9fd3aae4de204b33e3125d3770f6c23ccf5161725ff74845ac3771e9.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:4d0d62a0815fda3fa0f69aa043249ebff3b3bb11f9123bbdca4b940d186ef4d4.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:b49bbafd25d8a2e1eb3392b333d64556df4d08672b07e01c6231622a93acb399.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:5064167961e791f90e77cf214f847bf7b60472f2fc0194af30576186be08f024.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:8d4ac0c74a82ebe1e256b3bea83021dfdfd5601a018940a5a29bd229eb5da42f; certificate

sha256:729ecdff3d08e3f5a3ccbd1421644cb1fe00836ca1f6c29349ac5d3bede1d97; external-validation identity sha256:a686f87e3ab0b2e0d5afc0a4dde5c497722f27dbe970702065ddc129a4ef6dee.

Post-seal empirical comparison. Directness:

`authoritative_observation_or_primary_measurement_boundary_correspondence`. Disposition: `purpose_matched_authoritative_source_features_present`. Predicted consequence: `earth:temperature-material-motion-carrier__conduction-advection-and-transition-relation__geometry-properties-boundaries-time-and-flux-held__interior-scale-bounded`. Source-derived consequence: `earth:temperature-material-motion-carrier__conduction-advection-and-transition-relation__geometry-properties-boundaries-time-and-flux-held__interior-scale-bounded`. Exact boundary correspondence: `true`. External evidence selected the survivor: `false`. Formal structure relabelled as direct measurement: `false`. Model, forecast, proxy or retrieval relabelled as observation: `false`.

Sources and complete feature accounting:

- EARTHSCOPE-SAGE-SERVICES-001 - NSF SAGE Facility operated by EarthScope Consortium, **Seismological and geophysical web services**; class `authoritative_geophysical_data_service`; transport captured; features 3/4 present; present: station and channel identity, waveform and event access, instrument response; absent and preserved: observed and derived product distinction.

- USGS-GEOMAG-OBSERVATORIES-001 - U.S. Geological Survey Geomagnetism Program, **USGS geomagnetic observatory operations**; class authoritative_geomagnetic_observation_network; transport captured; features 0/4 present; present: none; absent and preserved: observatory identity, time-bounded geomagnetic observation, station status, read-only access.
- USGS-NGMDB-001 - U.S. Geological Survey National Cooperative Geologic Mapping Program, **National Geologic Map Database**; class authoritative_geoscience_archive; transport captured; features 0/4 present; present: none; absent and preserved: spatially referenced geology, map and report provenance, stratigraphic information, geochemistry geophysics palaeontology and geochronology coverage.

Comparison record:

- sft-earth-interior-heat-transport-001-external-earth-evidence: predicted earth:temperature-material-motion-carrier__conduction-advection-and-transition-relation__geometry-properties-boundaries-time-and-flux-held__interior-scale-bounded; source-derived earth:temperature-material-motion-carrier__conduction-advection-and-transition-relation__geometry-properties-boundaries-time-and-flux-held__interior-scale-bounded; exact match True
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if a fitted effective parameter replaces the missing carrier path or if distinct transport modes are silently merged.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

```
experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json;
source manifest sha256:de6abd8a0d4453aacaffec63449a94b068be4c03ba3f47802bb55e9e835c8c9c;
derivation seal sha256:6c53580d4a0a32a79a8427131dd1894e5114af493c32d3c74489171e78a0cef0;
engine receipt sha256:5f0c7741ccc3031a6da52d220ad80cd144380214ba556a5de084c06e47ffa4fe at receipts/engine/model_admitted/SFT-EARTH-INTERIOR-HEAT-TRANSPORT-001-5f0c7741ccc3031a.json; empirical validation
sha256:eb7bd631397dbf17d7e55dedd9d5e939e25c27e99cfccc305daae2bfd24c13cf; measurement
receipt sha256:f0bfe5f0bbf88c7ac945deaf00b56600863d119ad9a2a59d5303b71ad9852201; isolation
sha256:0243f76df78a078f7768f3185c25e91a68637754b123899876c5567878ac8ccb; custody
sha256:946b319143fe5d2c84943698be72a95e53208007af805f6bbe7cfea7cab6216c.
```

22. Crust-mantle boundary

Claim identity: SFT-EARTH-CRUST-MANTLE-BOUNDARY-001

Question and exact theorem. The crust-mantle boundary is an observation-bounded interface identified through retained composition, structure and propagation contrasts, not a globally uniform surface by premise.

```
solid-earth-interface-carrier__regional-observation-bounded__composition-structure
-and-wave-contrast__depth-lateral-variation-method-and-uncertainty-held__evidence-
class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-
rows__no-extra-rule
```

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001

- SFT-EARTH-EARTH-HISTORY-001
- SFT-EARTH-INTERIOR-HEAT-TRANSPORT-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	solid-earth-interface-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	regional-observation-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	composition-structure-and-wave-contrast	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	depth-lateral-variation-method-and-uncertainty-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: solid-earth-interface-carrier__regional-observation-bounded__composition-structure-and-wave-contrast__depth-lateral-variation-method-and-uncertainty-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- carrier-present: the claim retains a nonempty named carrier; passed true.
- relation-distinct: the carrier and relation are not silently conflated; passed true.
- evidence-boundary: the evidence boundary remains distinct from the result carrier; passed true.
- exact-fold-witness: the family's pre-source exact Fold construction reproduces; passed true.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar.

Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

Interior compartments and geodynamic relations are inference-bounded. Depth, composition, state, contrast, source, transfer and uncertainty remain reconstructible. Universal mechanics and thermodynamics enter only through prior admitted receipts; Earth-specific observations do not select those laws.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:ee03efff41e6484892ef08cacacae2f4b7e9cdeaad1e5f9e4b6732f346dfb1.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:651488e56c2e1c0319ef42110cac7c1fdab7853f379070c7b495dcff5d12a3a0.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:b262dbcd5c2a99fbf4b7b9bd673d417ef39df268c617dffcb88afc2d1bbf64db.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:a7beea4929f9319ef8095157a7de288c43ac5036c219403b536106cdf75efa57.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:2978f7a9e11b155b8fba838493dd8348417077ac1f0395b308fa823891fe009f; certificate
sha256:4080c898752ff40631742675d41c00c54367a5e3216454bbd7b9cec1d41dbb0f; external-validation
identity sha256:29be5e2993f036ea2e8568a0a0b1f47e8383ba937412341945a0f07dafda8c79.

Post-seal empirical comparison. Directness:

authoritative_observation_or_primary_measurement_boundary_correspondence. Disposition:
purpose_matched_authoritative_source_features_present. Predicted consequence: earth:solid-earth-interface-carrier__composition-structure-and-wave-contrast__depth-lateral-variation-method-and-uncertainty-held__regional-observation-bounded. Source-derived consequence: earth:solid-earth-interface-carrier__composition-structure-and-wave-contrast__depth-lateral-variation-method-and-uncertainty-held__regional-observation-bounded. Exact boundary correspondence: true. External evidence selected the survivor: false. Formal structure relabelled as direct measurement: false. Model, forecast, proxy or retrieval relabelled as observation: false.

Sources and complete feature accounting:

- EARTHSOPE-SAGE-SERVICES-001 - NSF SAGE Facility operated by EarthScope Consortium, **Seismological and geophysical web services**; class authoritative_geophysical_data_service; transport captured; features 3/4 present; present: station and channel identity, waveform and event access, instrument response; absent and preserved: observed and derived product distinction.
- USGS-GEOMAG-OBSERVATORIES-001 - U.S. Geological Survey Geomagnetism Program, **USGS geomagnetic observatory operations**; class authoritative_geomagnetic_observation_network; transport captured; features 0/4 present; present: none; absent and preserved: observatory identity, time-bounded geomagnetic observation, station status, read-only access.
- USGS-NGMDB-001 - U.S. Geological Survey National Cooperative Geologic Mapping Program, **National Geologic Map Database**; class authoritative_geoscience_archive; transport captured; features 0/4 present; present: none; absent and preserved: spatially referenced geology, map and report provenance, stratigraphic information, geochemistry geophysics palaeontology and geochronology coverage.

Comparison record:

- sft-earth-crust-mantle-boundary-001-external-earth-evidence: predicted earth:solid-earth-interface-carrier__composition-structure-and-wave-contrast__depth-lateral-variation-method-and-uncertainty-held__regional-observation-bounded;

source-derived earth:solid-earth-interface-carrier__composition-structure-and-wave-contrast__depth-lateral-variation-method-and-uncertainty-held__regional-observation-bounded; exact match True

- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if local depth is generalised globally without evidence or if method-dependent alternatives are discarded.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json;
 source manifest sha256:9bf3958cdf31249eda582e171f929046485f144296fea7b1c5ba3cc2c380b0c8;
 derivation seal sha256:4e08247a95ad537ed9647226c7305bf57573891700b3ce3ad0b4fc951dc358da;
 engine receipt sha256:9d2618cb35c99fe7687503ae36f2493e9caaf3d5f89db4335d9ce42e72ab0236 at receipts/engine/model_admitted/SFT-EARTH-CRUST-MANTLE-BOUNDARY-001-9d2618cb35c99fe7.json;
 empirical validation
 sha256:4c5f4937433c5c07f1fffc75d32465c1192b64943f598cf260b449d5ea7f06aa; measurement
 receipt sha256:7930eef8b70d27a3ee40142eac3e9ddb76f8c64331de7e1daefc2be73bda51e; isolation
 sha256:fe7bd277f03530bcf50bec73d5d054bcca5bae6f09f91d5249da5be4bd553cbb; custody
 sha256:eb617dad7f8a7a3fb1b3072caaa8abb81aded49d1d05ba19168e4ce0c099a593.

23. Tectonic plate boundary

Claim identity: SFT-EARTH-PLATE-BOUNDARY-001

Question and exact theorem. A tectonic plate boundary is a zone of measured relative motion and deformation whose geometry, rate, distributed strain and observation interval are retained.

moving-region-interface-carrier__geodetic-geologic-window-bounded__relative-motion-and-deformation-relation__orientation-rate-strain-events-and-uncertainty-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001

- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-EARTH-HISTORY-001
- SFT-EARTH-CRUST-MANTLE-BOUNDARY-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	moving-region-interface-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	geodetic-geologic-window-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	relative-motion-and-deformation-relation	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	orientation-rate-strain-events-and-uncertainty-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: `moving-region-interface-carrier__geodetic-geologic-window-bounded__relative-motion-and-deformation-relation__orientation-rate-strain-events-and-uncertainty-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule`.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `carrier-present`: the claim retains a nonempty named carrier; passed `true`.
- `relation-distinct`: the carrier and relation are not silently conflated; passed `true`.
- `evidence-boundary`: the evidence boundary remains distinct from the result carrier; passed `true`.
- `exact-fold-witness`: the family's pre-source exact Fold construction reproduces; passed `true`.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar.

Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

Interior compartments and geodynamic relations are inference-bounded. Depth, composition, state, contrast, source, transfer and uncertainty remain reconstructible. Universal mechanics and thermodynamics enter only through prior admitted receipts; Earth-specific observations do not select those laws.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
`sha256:1a8984bcb4b77677498514b778973b69775d6f63379c4d64ef889810ff443887`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
`sha256:9998e135fd3065dad0b10753736641dde6498616c55066a5366d1878f6544c38`.

- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:eec53e96c1e5919dd4f4678aeea3a899bd4102367601c594d859785cbd5d18f1.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:058d6e0270e48e8aa775762f85d85d3328f6ffa74d349ead7c7f0259ee103635.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:307f8f5c30412e63cb9e4b0612bc1015d5c1984ecd46a42a7b4a499fcc2fc8de; certificate
sha256:2efa6af3aaf26ba9d1ca5eed4802c45fdd94048592d7f7b42d84ad17cdc8d013; external-validation
identity sha256:46d35b712e98ed61d4f5f3edb994a3264ae9dc2d049748af8f414c94c427727e.

Post-seal empirical comparison. Directness:

authoritative_observation_or_primary_measurement_boundary_correspondence. Disposition:
purpose_matched_authoritative_source_features_present. Predicted consequence: earth:moving-region-interface-carrier__relative-motion-and-deformation-relation__orientation-rate-strain-events-and-uncertainty-held__geodetic-geologic-window-bounded. Source-derived consequence: earth:moving-region-interface-carrier__relative-motion-and-deformation-relation__orientation-rate-strain-events-and-uncertainty-held__geodetic-geologic-window-bounded. Exact boundary correspondence: true. External evidence selected the survivor: false. Formal structure relabelled as direct measurement: false. Model, forecast, proxy or retrieval relabelled as observation: false.

Sources and complete feature accounting:

- EARTHSCOPE-SAGE-SERVICES-001 - NSF SAGE Facility operated by EarthScope Consortium, **Seismological and geophysical web services**; class authoritative_geophysical_data_service; transport captured; features 3/4 present; present: station and channel identity, waveform and event access, instrument response; absent and preserved: observed and derived product distinction.
- USGS-GEOMAG-OBSERVATORIES-001 - U.S. Geological Survey Geomagnetism Program, **USGS geomagnetic observatory operations**; class authoritative_geomagnetic_observation_network; transport captured; features 0/4 present; present: none; absent and preserved: observatory identity, time-bounded geomagnetic observation, station status, read-only access.
- USGS-NGMDB-001 - U.S. Geological Survey National Cooperative Geologic Mapping Program, **National Geologic Map Database**; class authoritative_geoscience_archive; transport captured; features 0/4 present; present: none; absent and preserved: spatially referenced geology, map and report provenance, stratigraphic information, geochemistry geophysics palaeontology and geochronology coverage.

Comparison record:

- sft-earth-plate-boundary-001-external-earth-evidence: predicted earth:moving-region-interface-carrier__relative-motion-and-deformation-relation__orientation-rate-strain-events-and-uncertainty-held__geodetic-geologic-window-bounded; source-derived earth:moving-region-interface-carrier__relative-motion-and-deformation-relation__orientation-rate-strain-events-and-uncertainty-held__geodetic-geologic-window-bounded; exact match True
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if a sharp line replaces an observed deformation zone or if motion is inferred without a time-indexed reference frame.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value

- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json;
 source manifest sha256:4efcade11c3e642519a73e2ba7253d8e27b3cda7dd476c66adb12121962ffba5;
 derivation seal sha256:debf76431f0525e990e067e738b266e48dc0324855aa862706e3fa4e907ae7ef;
 engine receipt sha256:d9dcd8021473f468bc7cf0569f708fa1f18b10c402308a0868b76509d0b6dfcb at
 receipts/engine/model_admitted/SFT-EARTH-PLATE-BOUNDARY-001-d9dcd8021473f468.json;
 empirical validation sha256:38e3be5ada9be847a44b248146c0f7fbe64cc8df06714de497d59655e40ee658;
 measurement receipt
 sha256:8c857ddc472ecc70d04b1cc8be7bdbc91d4d12d095a2a1c894a26a67df134723; isolation
 sha256:e93b5f2748a68e7196b590b29ff5451079da0c4171a62943eea2a068b9e764eb; custody
 sha256:f14b81b85c3f2306b2ec1078623a78bba32455cbb5d7dd264e3228b20301e71f.

24. Tectonic motion and deformation

Claim identity: SFT-EARTH-TECTONIC-MOTION-001

Question and exact theorem. Tectonic motion is a reference-frame-bound history of relative displacement and deformation reconstructed from complete geodetic and geological records.

time-indexed-solid-earth-position-carrier__spatial-temporal-window-bounded__relati
 ve-displacement-strain-transition__reference-frame-path-rate-events-and-errors-hel
 d__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-
 retains-all-rows__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001

- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-EARTH-HISTORY-001
- SFT-EARTH-PLATE-BOUNDARY-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	time-indexed-solid-earth-position-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	spatial-temporal-window-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	relative-displacement-strain-transition	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	reference-frame-path-rate-events-and-errors-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: time-indexed-solid-earth-position-carrier__spatial-temporal-window-bounded__relative-displacement-strain-transition__reference-frame-path-rate-events-and-errors-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- carrier-present: the claim retains a nonempty named carrier; passed true.
- relation-distinct: the carrier and relation are not silently conflated; passed true.
- evidence-boundary: the evidence boundary remains distinct from the result carrier; passed true.
- exact-fold-witness: the family's pre-source exact Fold construction reproduces; passed true.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar.

Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

Interior compartments and geodynamic relations are inference-bounded. Depth, composition, state, contrast, source, transfer and uncertainty remain reconstructible. Universal mechanics and thermodynamics enter only through prior admitted receipts; Earth-specific observations do not select those laws.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:d6dd33b4e48c2ca5a77829a2f4e033878be3a776d721c12fde1fbb154a893678.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:9419d029f014d7d26acbdcb77d14a06c410492b1363c0f2d460d1e4e70ac941.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:714b4de6061b9b52171ea434621ae5d4406980e50b2439237a9cad9d2e0a21c5.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:92ee8e114f7d27c96c4f806448461a326785f062625881fac272977954c8378d.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:5fc394288eb7adf5c189dc7e99a4ecd8a23b9b4544420150cbb21a53d28e7843; certificate
sha256:a80511fa74884c897f3c054b326fff89d3a4fbfb90321bf1e838b3ca2c767b83; external-validation
identity sha256:451e64d136ae8a13bfff795055efe05770351373b944a45cc95754d9e15a9b870.

Post-seal empirical comparison. Directness:

authoritative_observation_or_primary_measurement_boundary_correspondence. Disposition:
purpose_matched_authoritative_source_features_present. Predicted consequence: earth:time-inde

xed-solid-earth-position-carrier__relative-displacement-strain-transition__reference-frame-path-rate-events-and-errors-held__spatial-temporal-window-bounded. Source-derived consequence: earth:time-indexed-solid-earth-position-carrier__relative-displacement-strain-transition__reference-frame-path-rate-events-and-errors-held__spatial-temporal-window-bounded. Exact boundary correspondence: true. External evidence selected the survivor: false. Formal structure relabelled as direct measurement: false. Model, forecast, proxy or retrieval relabelled as observation: false.

Sources and complete feature accounting:

- EARTHSCOPE-SAGE-SERVICES-001 - NSF SAGE Facility operated by EarthScope Consortium, **Seismological and geophysical web services**; class authoritative_geophysical_data_service; transport captured; features 3/4 present; present: station and channel identity, waveform and event access, instrument response; absent and preserved: observed and derived product distinction.
- USGS-GEOMAG-OBSERVATORIES-001 - U.S. Geological Survey Geomagnetism Program, **USGS geomagnetic observatory operations**; class authoritative_geomagnetic_observation_network; transport captured; features 0/4 present; present: none; absent and preserved: observatory identity, time-bounded geomagnetic observation, station status, read-only access.
- USGS-NGMDB-001 - U.S. Geological Survey National Cooperative Geologic Mapping Program, **National Geologic Map Database**; class authoritative_geoscience_archive; transport captured; features 0/4 present; present: none; absent and preserved: spatially referenced geology, map and report provenance, stratigraphic information, geochemistry geophysics palaeontology and geochronology coverage.

Comparison record:

- sft-earth-tectonic-motion-001-external-earth-evidence: predicted earth:time-indexed-solid-earth-position-carrier__relative-displacement-strain-transition__reference-frame-path-rate-events-and-errors-held__spatial-temporal-window-bounded; source-derived earth:time-indexed-solid-earth-position-carrier__relative-displacement-strain-transition__reference-frame-path-rate-events-and-errors-held__spatial-temporal-window-bounded; exact match True
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if reference-frame change creates the claimed motion or if episodic and unresolved intervals are smoothed away.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

```
experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json;
source manifest sha256:db916a5d709c8f4683061432a09520db6a12f41626a5274870fe7635fff09ec;
derivation seal sha256:55a7b8bb744437fed196b83e1b45b309ce702952f7d1783f51088f65cceff39e;
engine receipt sha256:dc57e68ba400a59ff492830f106e034b6bb73aae7d79a4045179583c7ad596c0 at
receipts/engine/model_admitted/SFT-EARTH-TECTONIC-MOTION-001-dc57e68ba400a59f.json;
empirical validation sha256:07815e00a4107d6cfee4740dfae27e848bafb189b56777c9ef3b8254f1740c0a;
measurement receipt
sha256:b5c07da9b782ea6370f712950086041f445194f7247130ab82d62190313ef926; isolation
sha256:fd116c51ca074f1ec02e9718e67c7bc3f6f594313cfda41db9a41442960ed415; custody
sha256:da2082820d6bfccdbff13b8f8793a05b5eea0a5269dfbcc87f2ff4d4d3eb063d.
```

Ledger family 5: Stress, Rupture, Seismicity and Volcanism

Stress, strain, rupture, waves, catalogue construction and volcanism remain distinct. The exact size-count grammar enumerates exponents before data. A heterogeneous catalogue result that opposes the unit-exponent prediction remains adverse; a later homogeneous holdout may add evidence but cannot rewrite it.

25. Geophysical stress and strain

Claim identity: SFT-EARTH-STRESS-STRAIN-001

Question and exact theorem. Geophysical stress and strain retain load orientation, deformation, material state, path, rate and scale rather than reducing both to an unsigned scalar.

```
material-load-deformation-carrier__specimen-and-scale-bounded__load-to-deformation
-relation__orientation-path-rate-history-and-material-state-held__evidence-class-e
xplicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__n
o-extra-rule
```

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001

- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-TECTONIC-MOTION-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	material-load-deformation-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	specimen-and-scale-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	load-to-deformation-relation	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	orientation-path-rate-history-and-material-state-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: material-load-deformation-carrier__specimen-and-scale-bounded__load-to-deformation-relation__orientation-path-rate-history-and-material-state-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- carrier-present: the claim retains a nonempty named carrier; passed true.
- relation-distinct: the carrier and relation are not silently conflated; passed true.
- evidence-boundary: the evidence boundary remains distinct from the result carrier; passed true.
- exact-fold-witness: the family's pre-source exact Fold construction reproduces; passed true.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar.

Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

Stress, strain, rupture, waves, catalogue construction and volcanism remain distinct. The exact size-count grammar enumerates exponents before data. A heterogeneous catalogue result that opposes the unit-exponent prediction remains adverse; a later homogeneous holdout may add evidence but cannot rewrite it.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:e36c5cf4544cf52d1bd66e317108798c45e7e85c3e567761eaa13e422ee1d9cc.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:b02cd7e39a794bc3780f653929cfda4bbcc79574a99066235ce9cfc25bc5e06e.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:48142346d780ec3a5451762fbb576825bdee469ad2b83adc83f5c8a0344a7d5c.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:33a57fe3d198f30a972bf93d5f916aed78f8fde6a6a2219c295ab910682f417f.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:d87e76ff18dee6409498f2b60e1f32467d1f97258c0ffc44f1474c1d3554cd7e; certificate
sha256:0c3fcb3227a5fcee1fb210bdc7834d7d0dce0ee93b62e13df913a8c9e2b7828; external-validation
identity sha256:7082404237c6e4c47a233ea4e0d0857fbed669048041113bf1eb737celaeafb9.

Post-seal empirical comparison. Directness:

authoritative_observation_or_primary_measurement_boundary_correspondence. Disposition:
purpose_matched_authoritative_source_features_present. Predicted consequence: earth:material-
load-deformation-carrier__load-to-deformation-relation__orientation-path-rate-history
-and-material-state-held__specimen-and-scale-bounded. Source-derived consequence: earth:materi-
al-load-deformation-carrier__load-to-deformation-relation__orientation-path-rate-hist-
ory-and-material-state-held__specimen-and-scale-bounded. Exact boundary correspondence: true.
External evidence selected the survivor: false. Formal structure relabelled as direct measurement: false. Model, forecast,
proxy or retrieval relabelled as observation: false.

Sources and complete feature accounting:

- USGS-EARTHQUAKE-FDSN-001 - U.S. Geological Survey Earthquake Hazards Program, [FDSN Earthquake Catalog web service](#); class authoritative_event_catalog_api; transport captured; features 5/5 present; present:
event identity, origin time and location, magnitude and magnitude type, catalog query boundary, service version; absent and

preserved: none.

- EARTHSCOPE-SAGE-SERVICES-001 - NSF SAGE Facility operated by EarthScope Consortium, **Seismological and geophysical web services**; class authoritative_geophysical_data_service; transport captured; features 3/4 present; present: station and channel identity, waveform and event access, instrument response; absent and preserved: observed and derived product distinction.
- SMITHSONIAN-GVP-VOTW-001 - Smithsonian Institution Global Volcanism Program, **Volcanoes of the World web services**; class authoritative_volcano_eruption_database; transport captured_via_addendum_prior_failure_preserved; features 3/4 present; present: volcano identity, eruption identity and time, database version and citation; absent and preserved: location and morphology.

Comparison record:

- sft-earth-stress-strain-001-external-earth-evidence: predicted earth:material-load-deformation-carrier__load-to-deformation-relation__orientation-path-rate-history-and-material-state-held__specimen-and-scale-bounded; source-derived earth:material-load-deformation-carrier__load-to-deformation-relation__orientation-path-rate-history-and-material-state-held__specimen-and-scale-bounded; exact match True
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if orientation or loading history is lost or if a laboratory relation is applied across scale without an admitted handoff.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json;
 source manifest sha256:393f7c8f0c6a7f89181be20a1e2d2cc6aa627a20dfef3b1022b598d23683a5cd;
 derivation seal sha256:b47ecd81aef9bd62f3eed28b31c54406d33f6a581f6d70868690b34db6257c3d;
 engine receipt sha256:6286cab2f1d868af4443f1b473770b1312c9f2580d77c0472be2666f316113ef at
 receipts/engine/model_admitted/SFT-EARTH-STRESS-STRAIN-001-6286cab2f1d868af.json;
 empirical validation sha256:19d195405b80a66bfe4951904335cb32226a14075e617128b3b8ac67eb7e5dd1;
 measurement receipt
 sha256:2eb92a0f6a37ead87c5d03580f1fe9c3c23535736557787d90361ef76e22542d; isolation
 sha256:bbb8db987d2c5737a1c5d1476e4155c597bc30f38965dcc3b6fba73c6067a593; custody
 sha256:514ad6c4706ea29446b2c5684e51577c0a04465a2b1b7928357844f9971beca3.

26. Earth rupture process

Claim identity: SFT-EARTH-RUPTURE-001

Question and exact theorem. Rupture is a source-bound transition from stored deformation to propagating and arrested slip with fault geometry, loading history and unresolved segments retained.

```
fault-state-transition-carrier__event-and-network-bounded__stored-deformation-to-slip-transition__geometry-loading-nucleation-propagation-and-arrest-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule
```

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001

- SFT-EARTH-TECTONIC-MOTION-001
- SFT-EARTH-STRESS-STRAIN-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	fault-state-transition-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	event-and-network-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	stored-deformation-to-slip-transition	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	geometry-loading-nucleation-propagation-and-arrest-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: fault-state-transition-carrier__event-and-network-bounded__stored-deformation-to-slip-transition__geometry-loading-nucleation-propagation-and-arrest-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- carrier-present: the claim retains a nonempty named carrier; passed true.
- relation-distinct: the carrier and relation are not silently conflated; passed true.
- evidence-boundary: the evidence boundary remains distinct from the result carrier; passed true.
- exact-fold-witness: the family's pre-source exact Fold construction reproduces; passed true.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar. Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

Stress, strain, rupture, waves, catalogue construction and volcanism remain distinct. The exact size-count grammar enumerates exponents before data. A heterogeneous catalogue result that opposes the unit-exponent prediction remains adverse; a later homogeneous holdout may add evidence but cannot rewrite it.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:5edac00e830614a52469cc9ad5511cbc53db88835461493bf54861e9a2e822b4.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:7a944706a42bd261383b64e07f3e436f5c4585c07d23c558f915ba5f57aa2349.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:7a94ae9bfc6ad0f4ed4a10d17ad35cf515206c6f8d35b57d65685c6702eae1aa.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:c6f0ef64d25cb8a78a25f5e8916d88e2144c487d6a6ce634edf34fc8a294eb00.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:0d6a49f0d7d80462b8014787cac165a3941eac74a176f4238243dbd9f01034bf; certificate
sha256:27c08f257f0b3831ff59a0588b1238de94dc17e03ab9be7aaf74122462d6a593; external-validation
identity sha256:6e4f439477252a034ba9ed0b37deb3af06a5de3321b8a9c8586f3b7897d340cd.

Post-seal empirical comparison. Directness:

authoritative_observation_or_primary_measurement_boundary_correspondence. Disposition:
purpose_matched_authoritative_source_features_present. Predicted consequence: earth:fault-state-transition-carrier__stored-deformation-to-slip-transition__geometry-loading-nucleation-propagation-and-arrest-held__event-and-network-bounded. Source-derived consequence: earth:fault-state-transition-carrier__stored-deformation-to-slip-transition__geometry-loading-nucleation-propagation-and-arrest-held__event-and-network-bounded. Exact boundary correspondence: true. External evidence selected the survivor: false. Formal structure relabelled as direct measurement: false. Model, forecast, proxy or retrieval relabelled as observation: false.

Sources and complete feature accounting:

- USGS-EARTHQUAKE-FDSN-001 - U.S. Geological Survey Earthquake Hazards Program, [FDSN Earthquake Catalog web service](#); class authoritative_event_catalog_api; transport captured; features 5/5 present; present: event identity, origin time and location, magnitude and magnitude type, catalog query boundary, service version; absent and preserved: none.
- EARTHSCOPE-SAGE-SERVICES-001 - NSF SAGE Facility operated by EarthScope Consortium, [Seismological and geophysical web services](#); class authoritative_geophysical_data_service; transport captured; features 3/4 present; present: station and channel identity, waveform and event access, instrument response; absent and preserved: observed and derived product distinction.
- SMITHSONIAN-GVP-VOTW-001 - Smithsonian Institution Global Volcanism Program, [Volcanoes of the World web services](#); class authoritative_volcano_eruption_database; transport captured_via_addendum_prior_failure_preserved; features 3/4 present; present: volcano identity, eruption identity and time, database version and citation; absent and preserved: location and morphology.

Comparison record:

- sft-earth-rupture-001-external-earth-evidence: predicted earth:fault-state-transition-carrier__stored-deformation-to-slip-transition__geometry-loading-nucleation-propagation-and-arrest-held__event-and-network-bounded; source-derived earth:fault-state-transition-carrier__stored-deformation-to-slip-transition__geometry-loading-nucleation-propagation-and-arrest-held__event-and-network-bounded.

t-state-transition-carrier__stored-deformation-to-slip-transition__geometry-loading-nucleation-propagation-and-arrest-held__event-and-network-bounded; exact match True

- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if catalogue time alone is treated as the rupture mechanism or if unobserved nucleation is asserted as measured fact.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json;
 source manifest sha256:5c896cf8fedc7a1987b03937a190b25722910bba0a275f06858c3f57cd0be6fd;
 derivation seal sha256:641c6737a2bea5ee01842f49ce83fd5e00581328b175c0cbeb60e97d4fb8c367;
 engine receipt sha256:cbbfa3cca4c6fd0471e81a76c31b8cd236534de3b39132a35d755233fd42c8ea at
 receipts/engine/model_admitted/SFT-EARTH-RUPTURE-001-cbbfa3cca4c6fd04.json; empirical
 validation sha256:83795f0e704b53c9d76e4bf8f1fe353ac8a0e83ec9a93b3d5cdaa4b61f608d2f;
 measurement receipt
 sha256:cf5b8c2bdea06002fc9a55427a46ed0ccd5261d76e9ce41a6f9f681b2289a0c9; isolation
 sha256:173ae6c2823661a82221668c7beda6d2b0690cdb3ee098bb970c07c71ce6f8b3; custody
 sha256:f8fb204134e8fbfbc86121074014889e55e7dddc3676c225a8656d860eaf018.

27. Seismic propagation record

Claim identity: SFT-EARTH-SEISMIC-WAVE-001

Question and exact theorem. A seismic record is a source-medium-receiver propagation trace with phase identity, path, timing, calibration and alternative source or medium explanations retained.

source-medium-receiver-carrier__network-resolution-bounded__source-to-receiver-propagation-relation__phase-path-time-instrument-and-uncertainty-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001

- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-TECTONIC-MOTION-001
- SFT-EARTH-RUPTURE-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	source-medium-receiver-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	network-resolution-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	source-to-receiver-propagation-relation	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	phase-path-time-instrument-and-uncertainty-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: `source-medium-receiver-carrier__network-resolution-bounded__source-to-receiver-propagation-relation__phase-path-time-instrument-and-uncertainty-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule`.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `carrier-present`: the claim retains a nonempty named carrier; passed `true`.
- `relation-distinct`: the carrier and relation are not silently conflated; passed `true`.
- `evidence-boundary`: the evidence boundary remains distinct from the result carrier; passed `true`.
- `exact-fold-witness`: the family's pre-source exact Fold construction reproduces; passed `true`.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar.

Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

Stress, strain, rupture, waves, catalogue construction and volcanism remain distinct. The exact size-count grammar enumerates exponents before data. A heterogeneous catalogue result that opposes the unit-exponent prediction remains adverse; a later homogeneous holdout may add evidence but cannot rewrite it.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
`sha256:4afc0e3408dcb0e537188ffad8cb9b3ce6e5c3578767778684ff5163ba30daca`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
`sha256:ccb6820ea27b1e280aef9355cbc804e9ebbb42d9a0802e8db751fa5eb6c093e`.

- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:ee055327fe2815c778084db4a50dca2a5a229cacff17ca427e1898cc8eaf7145.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:c1bcfad2f329a11bd302c5891465d9389f6da49bee4a73bb56202cd60b2bb6f4.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:af7253bc2b4e9927788506e5c410c8c37b21833d27263d04c0a9baec3d1392ef; certificate
sha256:2e87f31618819fddda09fe67429fd6f06f3cba4e3912f7147bd2b33a8b588194; external-validation
identity sha256:620d80358ebdefef9e8f768602aa7c16bb7435dd7d8ca0df2b000f271c02e44a.

Post-seal empirical comparison. Directness:

authoritative_observation_or_primary_measurement_boundary_correspondence. Disposition:
purpose_matched_authoritative_source_features_present. Predicted consequence: earth:source-me
dium-receiver-carrier__source-to-receiver-propagation-relation__phase-path-time-instr
ument-and-uncertainty-held__network-resolution-bounded. Source-derived consequence: earth:sour
ce-medium-receiver-carrier__source-to-receiver-propagation-relation__phase-path-time-
instrument-and-uncertainty-held__network-resolution-bounded. Exact boundary correspondence:
true. External evidence selected the survivor: false. Formal structure relabelled as direct measurement: false. Model,
forecast, proxy or retrieval relabelled as observation: false.

Sources and complete feature accounting:

- USGS-EARTHQUAKE-FDSN-001 - U.S. Geological Survey Earthquake Hazards Program, [FDSN Earthquake Catalog web service](#); class authoritative_event_catalog_api; transport captured; features 5/5 present; present: event identity, origin time and location, magnitude and magnitude type, catalog query boundary, service version; absent and preserved: none.
- EARTHSCOPE-SAGE-SERVICES-001 - NSF SAGE Facility operated by EarthScope Consortium, [Seismological and geophysical web services](#); class authoritative_geophysical_data_service; transport captured; features 3/4 present; present: station and channel identity, waveform and event access, instrument response; absent and preserved: observed and derived product distinction.
- SMITHSONIAN-GVP-VOTW-001 - Smithsonian Institution Global Volcanism Program, [Volcanoes of the World web services](#); class authoritative_volcano_eruption_database; transport captured_via_addendum_prior_failure_preserved; features 3/4 present; present: volcano identity, eruption identity and time, database version and citation; absent and preserved: location and morphology.

Comparison record:

- sft-earth-seismic-wave-001-external-earth-evidence: predicted earth:source-medium-receiver-carrier__source-to-receiver-propagation-relation__phase-path-time-instrument-and-uncertainty-held__network-resolution-bounded; source-derived earth:source-medium-receiver-carrier__source-to-receiver-propagation-relation__phase-path-time-instrument-and-uncertainty-held__network-resolution-bounded; exact match True
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if a processed arrival is treated as a direct interior image or if missing stations and phase ambiguity are erased.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value

- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json;
 source manifest sha256:63e247907c2b04021d38ede071e7bc36051afd192e4c34293990b4b0c6b4e8b8;
 derivation seal sha256:207f20e9e14860504e0dd958802e16c994b8e9b04c778e2f222a91d4f498f610;
 engine receipt sha256:8e147c5eec7c1286f7b4d84e4a4ea05850e943b83707fa28dc461dee999b9952 at
 receipts/engine/model_admitted/SFT-EARTH-SEISMIC-WAVE-001-8e147c5eec7c1286.json;
 empirical validation sha256:fd3be4b72cabdf63d22c8b8f4f7a8ebc0ae84aee2869eed7743904d0b70a49bb;
 measurement receipt
 sha256:fca1f1bebbdb053a340f380835429bb2966069723f0f82249dfa2e028d66523c; isolation
 sha256:37334f93f2e601a11614aef13386cc63de6386b3904db52255c92b64ba8f8a47; custody
 sha256:3c3fd34c951bf32bfd4434d0c75579b5f41771674b0f0eea47cbd2b1d2014328.

28. Earthquake catalogue

Claim identity: SFT-EARTH-EARTHQUAKE-CATALOG-001

Question and exact theorem. An earthquake catalog is a method- and interval-bounded event set retaining detection threshold, location, magnitude scale, duplicates, uncertainty and completeness changes.

event-detection-location-magnitude-carrier__catalog-completeness-bounded__catalog-membership-and-equivalence-relation__network-threshold-method-duplicates-errors-and-time-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001

- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-TECTONIC-MOTION-001
- SFT-EARTH-SEISMIC-WAVE-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	event-detection-location-magnitude-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	catalog-completeness-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	catalog-membership-and-equivalence-relation	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	network-threshold-method-duplicates-errors-and-time-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: event-detection-location-magnitude-carrier__catalog-completeness-bounded__catalog-membership-and-equivalence-relation__network-threshold-method-duplicates-errors-and-time-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- carrier-present: the claim retains a nonempty named carrier; passed true.
- relation-distinct: the carrier and relation are not silently conflated; passed true.
- evidence-boundary: the evidence boundary remains distinct from the result carrier; passed true.
- exact-fold-witness: the family's pre-source exact Fold construction reproduces; passed true.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar.

Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

Stress, strain, rupture, waves, catalogue construction and volcanism remain distinct. The exact size-count grammar enumerates exponents before data. A heterogeneous catalogue result that opposes the unit-exponent prediction remains adverse; a later homogeneous holdout may add evidence but cannot rewrite it.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:8bdd4af93a9a1c5d10ec4794632471dd2a3b18c748fe440d0d97f9a0c6b2eba9.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:6b453f7fa41aa6f3406886e958f0dd82432f44c3e97d4c70303390adc7923691.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:elaec6c3afe82a679bb4e61cbf1a666a4c698cccb818291d8bbd497b61162ed7.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:3385762506e720b27a36e09b5c1454be133f668ca8ea253495385d10a7606075.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:c34c219576386142403c2c209d131b96fc2874d1523a675607cb37745c5e7107; certificate
sha256:cdabcbcd5ae8d5cd792c0815e3daf771b35a06ee5f6736b29f5e5d447aa85422; external-validation
identity sha256:39ce4bde5110e328ccc0709828e5e25287d41b433baaa0a9ad5de0eaf761b5b3.

Post-seal empirical comparison. Directness:

authoritative_observation_or_primary_measurement_boundary_correspondence. Disposition:
purpose_matched_authoritative_source_features_present. Predicted consequence: earth:event-det

ection-location-magnitude-carrier__catalog-membership-and-equivalence-relation__network-threshold-method-duplicates-errors-and-time-held__catalog-completeness-bounded. Source-derived consequence: earth:event-detection-location-magnitude-carrier__catalog-membership-and-equivalence-relation__network-threshold-method-duplicates-errors-and-time-held__catalog-completeness-bounded. Exact boundary correspondence: true. External evidence selected the survivor: false. Formal structure relabelled as direct measurement: false. Model, forecast, proxy or retrieval relabelled as observation: false.

Sources and complete feature accounting:

- USGS-EARTHQUAKE-FDSN-001 - U.S. Geological Survey Earthquake Hazards Program, [FDSN Earthquake Catalog web service](#); class authoritative_event_catalog_api; transport captured; features 5/5 present; present: event identity, origin time and location, magnitude and magnitude type, catalog query boundary, service version; absent and preserved: none.
- EARTHSCOPE-SAGE-SERVICES-001 - NSF SAGE Facility operated by EarthScope Consortium, [Seismological and geophysical web services](#); class authoritative_geophysical_data_service; transport captured; features 3/4 present; present: station and channel identity, waveform and event access, instrument response; absent and preserved: observed and derived product distinction.
- SMITHSONIAN-GVP-VOTW-001 - Smithsonian Institution Global Volcanism Program, [Volcanoes of the World web services](#); class authoritative_volcano_eruption_database; transport captured_via_addendum_prior_failure_preserved; features 3/4 present; present: volcano identity, eruption identity and time, database version and citation; absent and preserved: location and morphology.

Comparison record:

- sft-earth-earthquake-catalogue-001-external-earth-evidence: predicted earth:event-detection-location-magnitude-carrier__catalog-membership-and-equivalence-relation__network-threshold-method-duplicates-errors-and-time-held__catalog-completeness-bounded; source-derived earth:event-detection-location-magnitude-carrier__catalog-membership-and-equivalence-relation__network-threshold-method-duplicates-errors-and-time-held__catalog-completeness-bounded; exact match True
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if catalogues with different thresholds or magnitude scales are pooled without a registered translation.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json;
 source manifest sha256:1651bc7575dcddcf59abccdeb0dd322638e5b3df260bb4fdabfeeb4da0630559;
 derivation seal sha256:b5b377aed318bd8e799e1ebc96428d60489e47eeca4e6eb77e9bbe66fb3d9e0a;
 engine receipt sha256:0b92793e1b4f37ec12872ebc82ddbba312eae97e40f4fc73d51ccfe23f23f152 at re
 ceipts/engine/model_admitted/SFT-EARTH-EARTHQUAKE-CATALOG-001-0b92793e1b4f37ec.json;
 empirical validation sha256:f4d30ed1288d521191a21199bad47d92907b46462d6ca683e45e0bb89812ab7a;
 measurement receipt
 sha256:71054b7fd99f175eaf24709a74f2206a99744fd59922423c5bfc227b99e9e2aa; isolation
 sha256:d7058af6bcc8d2023f6d4e62b8acd1ed4a340e64079d08079ce380ca6b2f6395; custody
 sha256:ca28b744f2f95cbd3a56411328a6a98015deb56463f0ff71b45401a754c07398.

29. Earthquake magnitude-frequency law

Claim identity: SFT-EARTH-QUAKE-MAGNITUDE-FREQUENCY-001

Question and exact theorem. An earthquake magnitude-frequency claim is an exact size-count relation over a complete declared catalog, with its exponent derived before blind comparison and all departures retained.

complete-catalog-size-count-carrier__catalog-completeness-bounded__scale-to-freque
 ncy-recurrence__magnitude-scale-threshold-region-time-and-residuals-held__evidence
 -class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all
 -rows__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001

- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-TECTONIC-MOTION-001
- SFT-EARTH-EARTHQUAKE-CATALOG-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	complete-catalog-size-count-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	catalog-completeness-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	scale-to-frequency-recurrence	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	magnitude-scale-threshold-region-time-and-residuals-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflicted	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: complete-catalog-size-count-carrier__catalog-completeness-bounded__scale-to-frequency-recurrence__magnitude-scale-threshold-region-time-and-residuals-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- carrier-present: the claim retains a nonempty named carrier; passed true.
- relation-distinct: the carrier and relation are not silently conflated; passed true.
- evidence-boundary: the evidence boundary remains distinct from the result carrier; passed true.
- exact-fold-witness: the family's pre-source exact Fold construction reproduces; passed true.

Scientific meaning. The generated size-count candidates are exponents 1, 2 and 3. Exact enumeration uniquely selects exponent 1 before opening the catalogue. The first global mixed-type test was adverse: $N \geq 5/N \geq 6 = 18469/1494$ and the registered interval 0.076998990840681139-0.084918164588559525 excludes 1/10. Because magnitude-type and event-status heterogeneity were preserved, a separately preregistered 2020-2026 mww holdout tested $N \geq 6/N \geq 7 = 253/27$, equivalently exceedance 27/253; its interval 0.085657503235220589-0.13089188889528297 contains 1/10. The first result remains adverse and unreclassified.

Stress, strain, rupture, waves, catalogue construction and volcanism remain distinct. The exact size-count grammar enumerates exponents before data. A heterogeneous catalogue result that opposes the unit-exponent prediction remains adverse; a later homogeneous holdout may add evidence but cannot rewrite it.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:a5e7990e67ffe0aadd2bc5e6aaacble89fc1b96fc67171fbc73262ec31c836cb.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:8d3e40c689a0fc0ba3c499b92eedbb87eb099deba0e82faf48f2567e9f8926b9.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:b775bc1cdb50968e8d058ced3766a8a42f08deb2bbff31b0790411f9942a8ea4.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:ffe08510768c261e8573334e1e51436291dfbf9c5d725a8fc700d62278366150.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:48cd7330943decf947c5fe3df229f0b4a217e4ec6444c514edab529b9d842104; certificate
sha256:522b6e5f95286602bbe5c57cc3e507041f97fd3f138d7f583e8a5c52c38e0b3a; external-validation
identity sha256:c6b3e159a80675f8b70b810e672b8e32e8a0ac9d1612359dcd7f156a2348ff45.

Post-seal empirical comparison. Directness:

authoritative_observation_or_primary_measurement_boundary_correspondence. Disposition:
mixed_catalog_adverse_preserved_homogeneous_holdout_compatible. Predicted consequence: earth:
complete-catalog-size-count-carrier__scale-to-frequency-recurrence__magnitude-scale-threshold-region-time-and-residuals-held__catalog-completeness-bounded__exact-unit-exponent-presealed. Source-derived consequence: earth:complete-catalog-size-count-carrier__scale-to-frequency-recurrence__magnitude-scale-threshold-region-time-and-residuals-held__catalog-completeness-bounded__exact-unit-exponent-presealed. Exact boundary correspondence: true. External evidence selected the survivor: false. Formal structure relabelled as direct measurement: false. Model, forecast, proxy or retrieval relabelled as observation: false.

Complete quantitative comparison. {"first_adverse_result_reclassified": false,
"first_mixed_catalog_result": {"adverse_conditions": {"catalog_truncated": false,
"event_status_heterogeneity_present": true,
"heterogeneity_rows_preserved_not_filtered": true,
"magnitude_type_heterogeneity_present": true, "missing_or_nonfinite_magnitudes":

```
false}, "interval": {"lower": "0.076998990840681139", "upper":
"0.084918164588559525"}, "observed_count_ratio_exact": {"denominator": 1494,
"numerator": 18469}, "passed": false, "result_hash":
"sha256:4657a82ad8b4263fa5394ad83a490fc7ea23bcb3a91afca04fb20cea7b7902c7"},
"independent_homogeneous_holdout": {"interval": {"lower": "0.085657503235220589",
"upper": "0.13089188889528297"}, "observed_count_ratio_exact": {"denominator": 27,
"numerator": 253}, "passed": true, "registered_magnitude_label": "mww",
"result_hash":
"sha256:2f69b1d7e56a2a1a6615b83bba99802186e7356f97cd953fd0f530191e146646"}}
```

Sources and complete feature accounting:

- USGS-EARTHQUAKE-FDSN-001 - U.S. Geological Survey Earthquake Hazards Program, [FDSN Earthquake Catalog web service](#); class authoritative_event_catalog_api; transport captured; features 5/5 present; present: event identity, origin time and location, magnitude and magnitude type, catalog query boundary, service version; absent and preserved: none.

Comparison record:

- sft-earth-quake-magnitude-frequency-001-external-earth-evidence: predicted earth:complete-catalogue-size-count-carrier__scale-to-frequency-recurrence__magnitude-scale-threshold-region-time-and-residuals-held__catalog-completeness-bounded__exact-unit-exponent-presealed; source-derived earth:complete-catalogue-size-count-carrier__scale-to-frequency-recurrence__magnitude-scale-threshold-region-time-and-residuals-held__catalog-completeness-bounded__exact-unit-exponent-presealed; exact match True
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected
- earthquake unit-exponent test: mixed catalogue adverse retained; separately preregistered homogeneous mww holdout compatible

Falsification boundary: Reject if the catalogue selects the exponent, if incomplete small events are treated as absent, or if one fitted line erases scale-dependent departures.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

```
experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json;
source manifest sha256:17c81640b6f369940e46af4f8cc1366bc4bc48b873186e4ded871ae2b8fa8bfa;
```

derivation seal sha256:c81b35ab12973adbf829fcc5bc630ff4878db9ed7403186d96495b2274e3e319;
 engine receipt sha256:e17ee1c4fb75debd7ac64a4cba308926415ef124df4f45908bf51f83084b60d6 at re
 cepts/engine/model_admitted/SFT-EARTH-QUAKE-MAGNITUDE-FREQUENCY-001-e17ee1c4fb75debd
 .json; empirical validation
 sha256:54c9721201830512a2a3a6d9c2117602b9bf9ef21e83e0a605e6ea4f00fad2c9; measurement
 receipt sha256:9cd2a75a62bc03b4719eba7d11001b4c5fd7710a96024f219930128327628199; isolation
 sha256:d1099e340cee23d535fc24031e3e396d5dec438ec2c6fc05b846b19d6e46885; custody
 sha256:54b56c157abe6a97f9aa0df05a2687bb10f14adfe35a35bdf88b551870eb41ac.

30. Volcanic process and eruption record

Claim identity: SFT-EARTH-VOLCANIC-PROCESS-001

Question and exact theorem. A volcanic process is a complete material and state-transition record from generation through storage, transport and release, including failed and non-eruptive paths.

magma-pathway-eruption-carrier__volcanic-system-bounded__generation-storage-transp
 ort-release-transition__composition-pressure-geometry-time-products-and-failures-h
 eld__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successo
 r-retains-all-rows__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001

- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-TECTONIC-MOTION-001
- SFT-EARTH-QUAKE-MAGNITUDE-FREQUENCY-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	magma-pathway-eruption-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	volcanic-system-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	generation-storage-transport-release-transition	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	composition-pressure-geometry-time-products-and-failures-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: magma-pathway-eruption-carrier__volcanic-system-bounded__generation-storage-transport-release-transition__composition-pressure-geometry-time-products-and-failures-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `carrier-present`: the claim retains a nonempty named carrier; passed `true`.
- `relation-distinct`: the carrier and relation are not silently conflated; passed `true`.
- `evidence-boundary`: the evidence boundary remains distinct from the result carrier; passed `true`.
- `exact-fold-witness`: the family's pre-source exact Fold construction reproduces; passed `true`.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar. Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

Stress, strain, rupture, waves, catalogue construction and volcanism remain distinct. The exact size-count grammar enumerates exponents before data. A heterogeneous catalogue result that opposes the unit-exponent prediction remains adverse; a later homogeneous holdout may add evidence but cannot rewrite it.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:3aa8eba5ca771a725c8cda3722b3a29f21188c9a3f5029f753a16461418ca364.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:4c968bbdf719f7fc6cd3d95151cef1fa0281e9d4a7834b2dea2e9656eab69693.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:b1b97c69e89e696eaf7f889c1684a8bab94723f1566df2625a388be9cad1af20.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:85f88b10cf0ae1cee660dfe34ebdbfcd18970744a5f2655f9cc6c575b13293ee.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:2d304d57bfaf5a2412c61379b239f5656ac67a65b3e12d3ea1ff92fc25386e3d; certificate
sha256:47c536dc07e995c55578a7bfe5e63a187b79dc792d156277bb8a6dc370e3d757; external-validation
identity sha256:be911fbff653a5b8b6e3f88ba3e1aa5dedf298b2d94d884d7820535122e2764.

Post-seal empirical comparison. Directness:

authoritative_observation_or_primary_measurement_boundary_correspondence. Disposition:
purpose_matched_authoritative_source_features_present. Predicted consequence: earth:magma-pathway-eruption-carrier__generation-storage-transport-release-transition__composition-pressure-geometry-time-products-and-failures-held__volcanic-system-bounded. Source-derived consequence: earth:magma-pathway-eruption-carrier__generation-storage-transport-release-transition__composition-pressure-geometry-time-products-and-failures-held__volcanic-system-bounded. Exact boundary correspondence: true. External evidence selected the survivor: false. Formal structure relabelled as direct measurement: false. Model, forecast, proxy or retrieval relabelled as observation: false.

Sources and complete feature accounting:

- USGS-EARTHQUAKE-FDSN-001 - U.S. Geological Survey Earthquake Hazards Program, [FDSN Earthquake Catalog web service](#); class authoritative_event_catalog_api; transport captured; features 5/5 present; present: event identity, origin time and location, magnitude and magnitude type, catalog query boundary, service version; absent and preserved: none.
- EARTHSCOPE-SAGE-SERVICES-001 - NSF SAGE Facility operated by EarthScope Consortium, [Seismological and geophysical web services](#); class authoritative_geophysical_data_service; transport captured; features 3/4 present; present: station and channel identity, waveform and event access, instrument response; absent and preserved: observed and derived product distinction.
- SMITHSONIAN-GVP-VOTW-001 - Smithsonian Institution Global Volcanism Program, [Volcanoes of the World web services](#); class authoritative_volcano_eruption_database; transport

captured_via_addendum_prior_failure_preserved; features 3/4 present; present: volcano identity, eruption identity and time, database version and citation; absent and preserved: location and morphology.

Comparison record:

- sft-earth-volcanic-process-001-external-earth-evidence: predicted earth:magma-pathway-eruption-carrier__generation-storage-transport-release-transition__composition-pressure-geometry-time-products-and-failures-held__volcanic-system-bounded; source-derived earth:magma-pathway-eruption-carrier__generation-storage-transport-release-transition__composition-pressure-geometry-time-products-and-failures-held__volcanic-system-bounded; exact match True
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if surface products uniquely determine an unobserved pathway without alternative reconstructions.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json;
 source manifest sha256:eb2607b3fc8361f222686b9c8f1346de029d72ba8065f6e3c21e764d652b46f5;
 derivation seal sha256:a8731489a073db54ee6009d93e890dc6a1f271917c0cdb0772384e6244b7a0a7;
 engine receipt sha256:5ae1862b28dafa2c35858c64775482d17fe6b572a82c855a31e85dee9e104c3e at
 receipts/engine/model_admitted/SFT-EARTH-VOLCANIC-PROCESS-001-5ae1862b28dafa2c.json;
 empirical validation sha256:9c73d94a1e1eb205ff48c625ffa0f82e689128e5b8d52546455a0099cc5cb25a;
 measurement receipt
 sha256:1f529c471ec891088ce0bae8610f9707bf82958898c99ec46d854b76617aa1be; isolation
 sha256:b48c959b71b5b3a2e9ebc20c9ea469deb62323abc24dc404ef1c689ab528eb4d; custody
 sha256:9d32bf85dbeef30d1cc3f4351c0249c05b31dd0237b952eff53bb1ef812116b0.

Ledger family 6: Hydrosphere and Cryosphere

Water reservoirs, phase handoffs, surface routes, groundwater routes and ice carriers form one coupled ledger without losing phase, location, interval or uncertainty. An unobserved store is not zero and a flow is not a stock.

31. Water reservoir identity

Claim identity: SFT-EARTH-WATER-RESERVOIR-001

Question and exact theorem. A water reservoir retains amount, phase, composition, location, boundary, residence interval and exchanges as distinct registered properties.

phase-location-bounded-water-carrier__hydrological-domain-bounded__storage-members
hip-and-exchange-relation__amount-phase-composition-boundary-and-time-held__eviden
ce-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-a
ll-rows__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-VOLCANIC-PROCESS-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	phase-location-bounded-water-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	hydrological-domain-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	storage-membership-and-exchange-relation	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	amount-phase-composition-boundary-and-time-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: phase-location-bounded-water-carrier__hydrological-domain-bounded__storage-membership-and-exchange-relation__amount-phase-composition-boundary-and-time-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- carrier-present: the claim retains a nonempty named carrier; passed true.
- relation-distinct: the carrier and relation are not silently conflated; passed true.
- evidence-boundary: the evidence boundary remains distinct from the result carrier; passed true.
- exact-fold-witness: the family's pre-source exact Fold construction reproduces; passed true.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar. Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

Water reservoirs, phase handoffs, surface routes, groundwater routes and ice carriers form one coupled ledger without losing phase, location, interval or uncertainty. An unobserved store is not zero and a flow is not a stock.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:85cf6d4351bdeded9d3033c53fbc0143aeba4ef58c8912fbcd4f8923b51fae1d.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:12aa7db2721b1efc9344f20d609616dfa7dcd7eb2ccc0799e4cf40f99a5c0473.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:cd3ead4987efbdb6f7c1b06f8342b0893f515b618e4d2941fa8e309c17a3cf3d.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:f0ba852a3048750d56ab96a3ff94c49c5bba08e90f68b2d8705dd6acf2318ca6.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:79da813df6823792cf4b4f1de795f0744a025ada0bd6eff10c71d6de21069be4; certificate
sha256:8634bb2ac2d976deab00627555009909c3a962ed535957948e20bdda5f5c0030; external-validation
identity sha256:1f6fcd5ea95c065dcaf61bc043ac4fb65aea096499355c2db3eac907bc9f21b4.

Post-seal empirical comparison. Directness:

`authoritative_observation_or_primary_measurement_boundary_correspondence`. Disposition:
`purpose_matched_authoritative_source_features_present`. Predicted consequence: `earth:phase-location-bounded-water-carrier__storage-membership-and-exchange-relation__amount-phase-composition-boundary-and-time-held__hydrological-domain-bounded`. Source-derived consequence: `earth:phase-location-bounded-water-carrier__storage-membership-and-exchange-relation__amount-phase-composition-boundary-and-time-held__hydrological-domain-bounded`. Exact
boundary correspondence: `true`. External evidence selected the survivor: `false`. Formal structure relabelled as direct
measurement: `false`. Model, forecast, proxy or retrieval relabelled as observation: `false`.

Sources and complete feature accounting:

- USGS-NWIS-001 - U.S. Geological Survey Water Resources, [National Water Information System](#); class `authoritative_hydrological_observation_network`; transport captured; features 2/4 present; present: site identity, surface water and groundwater observations; absent and preserved: time series, parameter units and methods.
- WQP-WEB-SERVICES-001 - U.S. Geological Survey, U.S. Environmental Protection Agency and National Water Quality Monitoring Council, [Water Quality Portal web services](#); class `authoritative_water_quality_federation`; transport captured; features 4/4 present; present: station and organization identity, sample and result identity, characteristic and unit, quality and metadata fields; absent and preserved: none.
- NSIDC-SEA-ICE-INDEX-001 - National Snow and Ice Data Center, [Sea Ice Index data and image archive](#); class `authoritative_cryosphere_data_product`; transport captured; features 4/4 present; present: hemisphere and date identity, extent and area distinction, daily and monthly data, missing-data and product-version documentation; absent and preserved: none.

Comparison record:

- `sft-earth-water-reservoir-001-external-earth-evidence`: predicted `earth:phase-location-bounded-water-carrier__storage-membership-and-exchange-relation__amount-phase-composition-boundary-and-time-held__hydrological-domain-bounded`; source-derived `earth:phase-location-bounded-water-carrier__storage-membership-and-exchange-relation__amount-phase-composition-boundary-and-time-held__hydrological-domain-bounded`; exact match `True`
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if water outside the observation boundary is treated as absent or if phase and chemical composition are erased.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json;
 source manifest sha256:bdb5315716222440616829fd18157048eb43cba8f8b6ddd8121752c8015397e4;
 derivation seal sha256:652599eed09a65728978068b08cbb6019199a106de578e51e954329b50bb5051;
 engine receipt sha256:f2eee3e63ecba04faed36bcf828c203aead5d1b3a4d49ed749ff3000afc86af7 at
 receipts/engine/model_admitted/SFT-EARTH-WATER-RESERVOIR-001-f2eee3e63ecba04f.json;
 empirical validation sha256:d89a6930e88390c70b3111d32d27cc8a391f44237840bba5c93dd1cea5566d21;
 measurement receipt
 sha256:0862ce5d1c1d023f22b31deaacf308ec41ed81bb9543dfc71131a726d1bfff0b4; isolation
 sha256:631fb48c1f6250c6b67b955874c5e2fdc789282afa5b10ea04bd48c70e28bbe9; custody
 sha256:0b5adb740337943fb875937c51e1dad39d226631a5f8772c902beb77ec6599ab.

32. Environmental water phase handoff

Claim identity: SFT-EARTH-WATER-PHASE-HANDOFF-001

Question and exact theorem. An environmental water phase handoff cites admitted physical and chemical phase laws while retaining source, destination, energy, solutes, boundary and time.

water-state-interface-carrier__interface-condition-bounded__phase-transition-and-t
 ransfer-relation__source-destination-energy-solutes-and-time-held__evidence-class-
 explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__
 no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001

- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-VOLCANIC-PROCESS-001
- SFT-EARTH-WATER-RESERVOIR-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	water-state-interface-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	interface-condition-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
relation	imported-fitted-or-missing-relation	phase-transition-and-transfer-relation	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	source-destination-energy-solutes-and-time-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: `water-state-interface-carrier__interface-condition-bounded__phase-transition-and-transfer-relation__source-destination-energy-solutes-and-time-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule`.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `carrier-present`: the claim retains a nonempty named carrier; passed `true`.
- `relation-distinct`: the carrier and relation are not silently conflated; passed `true`.
- `evidence-boundary`: the evidence boundary remains distinct from the result carrier; passed `true`.
- `exact-fold-witness`: the family's pre-source exact Fold construction reproduces; passed `true`.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar.

Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

Water reservoirs, phase handoffs, surface routes, groundwater routes and ice carriers form one coupled ledger without losing phase, location, interval or uncertainty. An unobserved store is not zero and a flow is not a stock.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:822d1a20d43b018fa3b86180d08493a23b1069d0f327c4d272828dd23d69881d.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:1257895db1bbce983f2e46b3bb7e78aecfb09b5df78bcd51495f89d29b60e54b.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:21141592e39d2bf4b189caaddbdb0950b0ad2b0596e986d39c85d7b7545129a8.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:2ad6a64584f0911accdf73c7390676bad2faa833408a92cd04df70d73cd75701.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation sha256:0fad5f0c1312fa136e20a269bc7f1bc65cba502b6a6fed0db17b1a5e9a6b337a; certificate sha256:c34806cf7375ec85604a9539510f7c9a9e96838ded63b09d91be18b5b2b86cd5; external-validation identity sha256:9f2d0aaf978c72f1911d96aa88df8b3879e7bec826056415ee9ac70bc816522e.

Post-seal empirical comparison. Directness:

authoritative_observation_or_primary_measurement_boundary_correspondence. Disposition: purpose_matched_authoritative_source_features_present. Predicted consequence: earth:water-state-interface-carrier__phase-transition-and-transfer-relation__source-destination-energy-solutes-and-time-held__interface-condition-bounded. Source-derived consequence: earth:water-state-interface-carrier__phase-transition-and-transfer-relation__source-destination-energy-solutes-and-time-held__interface-condition-bounded. Exact boundary correspondence: true. External evidence selected the survivor: false. Formal structure relabelled as direct measurement: false. Model, forecast, proxy or retrieval relabelled as observation: false.

Sources and complete feature accounting:

- USGS-NWIS-001 - U.S. Geological Survey Water Resources, [National Water Information System](#); class authoritative_hydrological_observation_network; transport captured; features 2/4 present; present: site identity, surface water and groundwater observations; absent and preserved: time series, parameter units and methods.
- WQP-WEB-SERVICES-001 - U.S. Geological Survey, U.S. Environmental Protection Agency and National Water Quality Monitoring Council, [Water Quality Portal web services](#); class authoritative_water_quality_federation; transport captured; features 4/4 present; present: station and organization identity, sample and result identity, characteristic and unit, quality and metadata fields; absent and preserved: none.
- NSIDC-SEA-ICE-INDEX-001 - National Snow and Ice Data Center, [Sea Ice Index data and image archive](#); class authoritative_cryosphere_data_product; transport captured; features 4/4 present; present: hemisphere and date identity, extent and area distinction, daily and monthly data, missing-data and product-version documentation; absent and preserved: none.

Comparison record:

- sft-earth-water-phase-handoff-001-external-earth-evidence: predicted earth:water-state-interface-carrier__phase-transition-and-transfer-relation__source-destination-energy-solutes-and-time-held__interface-condition-bounded; source-derived earth:water-state-interface-carrier__phase-transition-and-transfer-relation__source-destination-energy-solutes-and-time-held__interface-condition-bounded; exact match True
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if universal phase behaviour is renowned by Earth Science or if environmental carriers and impurities are omitted.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise

- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

```
experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json;
source manifest sha256:c0c1916a378bec3e9f5fa4ccf703dae1037277f6d079f48a2963553474d1608b;
derivation seal sha256:9c49e0172385555b84d4e7c413e281a5d9bd982e30f0fa5a359170b9fd1a7cb7;
engine receipt sha256:7f037ea8c6d5a29616899ca27bf8d35c7eaa003b016ce1b146ccb3bd42855f1 at re
ceipts/engine/model_admitted/SFT-EARTH-WATER-PHASE-HANDOFF-001-7f037ea8c6d5a296.json;
empirical validation sha256:8ff2274066f26309159eb966a67479dba2f72ac65d0141cb806db43a27dd2ca2;
measurement receipt
sha256:cbec10270225f6f0e71cad83c9fc8bc5c0ae0294f21878872a74ffcaf6e55f7b; isolation
sha256:a9b6caeef02645422eb9e8e3caa5d0783c721c29415d63230eb6b7dcc34f6a4c; custody
sha256:63bb1f435872e3f83611cabe8b68034c3e2ecd3ed242f0557240011f55cbab13.
```

33. Hydrologic cycle

Claim identity: SFT-EARTH-HYDROLOGIC-CYCLE-001

Question and exact theorem. The hydrologic cycle is a bounded reservoir-transfer network whose stocks, phase changes, atmospheric transport, precipitation, runoff, infiltration and exports remain counted.

```
water-reservoir-transfer-network__basin-and-time-bounded__evaporation-transport-pr
ecipitation-runoff-infiltration-relation__stocks-flows-phases-boundaries-and-losse
s-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-succe
ssor-retains-all-rows__no-extra-rule
```

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001

- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-VOLCANIC-PROCESS-001
- SFT-EARTH-WATER-PHASE-HANDOFF-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	water-reservoir-transfer-net work	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	basin-and-time-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	evaporation-transport-precipitation-runoff-infiltration-relation	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	stocks-flows-phases-boundaries-and-losses-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: water-reservoir-transfer-network__basin-and-time-bounded__evaporation-transport-precipitation-runoff-infiltration-relation__stocks-flows-phases-boundaries-and-losses-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- carrier-present: the claim retains a nonempty named carrier; passed true.
- relation-distinct: the carrier and relation are not silently conflated; passed true.
- evidence-boundary: the evidence boundary remains distinct from the result carrier; passed true.
- exact-fold-witness: the family's pre-source exact Fold construction reproduces; passed true.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar.

Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

Water reservoirs, phase handoffs, surface routes, groundwater routes and ice carriers form one coupled ledger without losing phase, location, interval or uncertainty. An unobserved store is not zero and a flow is not a stock.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:d531aedd691067e2951f4abb601744459501a3024b65fab9f9257b1ccd1175fe.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:7980a471b1894d5024070f40e59f9f8e8140f200dee139db571fb20cfcdda2ee.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:eea6df5380613a6a3ec5ef6cac73623025e66a167a3807ed8ac94876746e48f5.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:cd7eab7f43f393578663ccaac5c90d2b98b6fedd15918a2481092100a609ac4b.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:6525a4332af87b9dca03ef7a1e6c9cdc4f633632485c5b6f4999f8601676e21f; certificate

sha256:e10640e629d80a1210d55fbb6efb70ccb0fe1fc259aef0be95ba409e270964f; external-validation identity sha256:7efcbca5ea5e595447be9bd6f962be58e9dc09596571a979cc42e38e9470a4c1.

Post-seal empirical comparison. Directness:

authoritative_observation_or_primary_measurement_boundary_correspondence. Disposition: purpose_matched_authoritative_source_features_present. Predicted consequence: earth:water-reservoir-transfer-network__evaporation-transport-precipitation-runoff-infiltration-relation__stocks-flows-phases-boundaries-and-losses-held__basin-and-time-bounded.

Source-derived consequence: earth:water-reservoir-transfer-network__evaporation-transport-precipitation-runoff-infiltration-relation__stocks-flows-phases-boundaries-and-losses-held__basin-and-time-bounded. Exact boundary correspondence: true. External evidence selected the survivor: false. Formal structure relabelled as direct measurement: false. Model, forecast, proxy or retrieval relabelled as observation: false.

Sources and complete feature accounting:

- USGS-NWIS-001 - U.S. Geological Survey Water Resources, [National Water Information System](#); class `authoritative_hydrological_observation_network`; transport captured; features 2/4 present; present: site identity, surface water and groundwater observations; absent and preserved: time series, parameter units and methods.
- WQP-WEB-SERVICES-001 - U.S. Geological Survey, U.S. Environmental Protection Agency and National Water Quality Monitoring Council, [Water Quality Portal web services](#); class `authoritative_water_quality_federation`; transport captured; features 4/4 present; present: station and organization identity, sample and result identity, characteristic and unit, quality and metadata fields; absent and preserved: none.
- NSIDC-SEA-ICE-INDEX-001 - National Snow and Ice Data Center, [Sea Ice Index data and image archive](#); class `authoritative_cryosphere_data_product`; transport captured; features 4/4 present; present: hemisphere and date identity, extent and area distinction, daily and monthly data, missing-data and product-version documentation; absent and preserved: none.

Comparison record:

- `sft-earth-hydrologic-cycle-001-external-earth-evidence`: `predicted earth:water-reservoir-transfer-network__evaporation-transport-precipitation-runoff-infiltration-relation__stocks-flows-phases-boundaries-and-losses-held__basin-and-time-bounded`; `source-derived earth:water-reservoir-transfer-network__evaporation-transport-precipitation-runoff-infiltration-relation__stocks-flows-phases-boundaries-and-losses-held__basin-and-time-bounded`; exact match True
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if the name cycle implies closure while storage change, groundwater exchange or boundary export remains unregistered.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

```
experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json;
source manifest sha256:218a3a603dce80c747a56290ff1d8f358c9c8a9725905f730d7de8464766629a;
derivation seal sha256:fd405a71ca5e4a9de46c4f3cf98414e75d4d59c6996ae55f6eb181c6bf6f6e1;
engine receipt sha256:c984bb56ec2018dae267b635ac5e35e860451a99d5f3526b5ac1bca70842fe49 at
receipts/engine/model_admitted/SFT-EARTH-HYDROLOGIC-CYCLE-001-c984bb56ec2018da.json;
empirical validation sha256:6db4c322a8640789a22f918e4f5f336bd69cf9cd685f2f22da3c9f91d2e393c8;
```

measurement receipt

sha256:73d1d1c36ef02bec26fbee326fec2395b507d2d7ab71089194c5ad4f44f9ef92; isolation
 sha256:8b60e318c492d6fa802bc1062afe63e59a03adc9953bbfa386d9250411705149; custody
 sha256:484d93439e162d350639ead7b9ec45b56d7f4339fce728750790552a11fc44ae.

34. Groundwater system

Claim identity: SFT-EARTH-GROUNDWATER-001

Question and exact theorem. A groundwater system is a material-bounded storage and transfer organization with recharge, paths, composition, residence and discharge observations retained.

subsurface-water-material-carrier__aquifer-observation-bounded__storage-gradient-p
 ath-and-exchange-relation__porosity-composition-boundaries-recharge-discharge-and-
 time-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-su
 ccessor-retains-all-rows__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001

- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-VOLCANIC-PROCESS-001
- SFT-EARTH-HYDROLOGIC-CYCLE-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	subsurface-water-material-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	aquifer-observation-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	storage-gradient-path-and-exchange-relation	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	porosity-composition-boundaries-recharge-discharge-and-time-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: subsurface-water-material-carrier__aquifer-observation-bounded__storage-gradient-path-and-exchange-relation__porosity-composition-boundaries-recharge-discharge-and-time-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- carrier-present: the claim retains a nonempty named carrier; passed true.
- relation-distinct: the carrier and relation are not silently conflated; passed true.

- evidence-boundary: the evidence boundary remains distinct from the result carrier; passed true.
- exact-fold-witness: the family's pre-source exact Fold construction reproduces; passed true.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar. Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

Water reservoirs, phase handoffs, surface routes, groundwater routes and ice carriers form one coupled ledger without losing phase, location, interval or uncertainty. An unobserved store is not zero and a flow is not a stock.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:ce512b6f42afad4aafe214bbb34266a1f4d49f03728f30facf3f7ac9744b53b9.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:06f429caf7b80601678e5a68619a44b9e0fe48b05b11ac957da3f6db7267e1b1.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:62cb71b37f4b970f35d2f1d9cf0519e17ce9bdf3fa56c50a87a6c959e99dc95b.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:ae4d54af9724ae8838d3843d5dd8090bee71343a16494f0b900f05fa2b15d773.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:884207ddfa3a7c0867075c3af11311bebf2daa98a4774f55153ce8a023f104a1; certificate
sha256:8ffc7d6c3c3dc7bb03bac95844d577aad3c9e8b905f670b7fa9881d9ca8afe7f; external-validation
identity sha256:0e89c19f7510888b8386a6fe83f87d3e49c32747d7305d27389f275e1227cdabb.

Post-seal empirical comparison. Directness:

authoritative_observation_or_primary_measurement_boundary_correspondence. Disposition:
purpose_matched_authoritative_source_features_present. Predicted consequence: earth:subsurface-water-material-carrier__storage-gradient-path-and-exchange-relation__porosity-composition-boundaries-recharge-discharge-and-time-held__aquifer-observation-bounded.
Source-derived consequence: earth:subsurface-water-material-carrier__storage-gradient-path-and-exchange-relation__porosity-composition-boundaries-recharge-discharge-and-time-held__aquifer-observation-bounded. Exact boundary correspondence: true. External evidence selected the survivor: false. Formal structure relabelled as direct measurement: false. Model, forecast, proxy or retrieval relabelled as observation: false.

Sources and complete feature accounting:

- USGS-NWIS-001 - U.S. Geological Survey Water Resources, [National Water Information System](#); class authoritative_hydrological_observation_network; transport captured; features 2/4 present; present: site identity, surface water and groundwater observations; absent and preserved: time series, parameter units and methods.
- WQP-WEB-SERVICES-001 - U.S. Geological Survey, U.S. Environmental Protection Agency and National Water Quality Monitoring Council, [Water Quality Portal web services](#); class authoritative_water_quality_federation; transport captured; features 4/4 present; present: station and organization identity, sample and result identity, characteristic and unit, quality and metadata fields; absent and preserved: none.
- NSIDC-SEA-ICE-INDEX-001 - National Snow and Ice Data Center, [Sea Ice Index data and image archive](#); class authoritative_cryosphere_data_product; transport captured; features 4/4 present; present: hemisphere and date identity, extent and area distinction, daily and monthly data, missing-data and product-version documentation; absent and preserved: none.

Comparison record:

- sft-earth-groundwater-001-external-earth-evidence: predicted earth:subsurface-water-material-carrier__storage-gradient-path-and-exchange-relation__porosity-composition-boundaries-recharge-discharge-and-time-held__aquifer-observation-bounded; source-derived earth:subsurface-water-material-carrier__storage-gradient-path-and-exchange-relation__porosity-composition-boundaries-recharge-discharge-and-time-held__aquifer-observation-bounded; exact match True
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if hydraulic head or a model surface is substituted for direct water identity, path and source evidence.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json; source manifest sha256:548a9f78f5b902ec287d7241bed7a9a3ee5cdc4a71c440135b0141bd0248fd84; derivation seal sha256:523f4268bc95d105869a6a30b34866a84b61a72dae6db42fbfb8361dedf4217d; engine receipt sha256:061ec0ea77701328ac3b8326a9d75aa69b3eb66fd308bd487cf5870c94687139 at receipts/engine/model_admitted/SFT-EARTH-GROUNDWATER-001-061ec0ea77701328.json; empirical validation sha256:6645792ab97a78b7de66f47325b442626f200844d7e7adc7192e6b1b88ed20ff; measurement receipt sha256:7c93392a09f7ed75cdbb5ad200cf114ff2538202806c9e0a334d6846ef33a003; isolation sha256:e5774843af71b65f6db23e78438dd892a528f8471a442df6cf97cb65b10b3e09; custody sha256:86683c5f94572d09e9e0559b2142b9bb96ff05d1f89092ae881fd889d3a0d377.

35. Rivers, lakes and surface-water networks

Claim identity: SFT-EARTH-SURFACE-WATER-001

Question and exact theorem. Surface-water organization is a directed storage and transfer network over rivers, lakes, wetlands and interfaces with complete catchment and observation boundaries.

surface-water-node-edge-carrier__catchment-window-bounded__storage-routing-and-exchange-relation__network-geometry-flow-quality-boundaries-and-time-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-VOLCANIC-PROCESS-001
- SFT-EARTH-GROUNDWATER-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	surface-water-node-edge-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	catchment-window-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	storage-routing-and-exchange-relation	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	network-geometry-flow-quality-boundaries-and-time-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: surface-water-node-edge-carrier__catchment-window-bounded__storage-routing-and-exchange-relation__network-geometry-flow-quality-boundaries-and-time-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- carrier-present: the claim retains a nonempty named carrier; passed true.
- relation-distinct: the carrier and relation are not silently conflated; passed true.
- evidence-boundary: the evidence boundary remains distinct from the result carrier; passed true.
- exact-fold-witness: the family's pre-source exact Fold construction reproduces; passed true.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar.

Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

Water reservoirs, phase handoffs, surface routes, groundwater routes and ice carriers form one coupled ledger without losing phase, location, interval or uncertainty. An unobserved store is not zero and a flow is not a stock.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 94a7ad14a3a943432e5baba16e26b5be9eb090442e0a0c32fc835c555e91e09f.

- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:2618ebef41473db2fb81dc5cf64b91247f45ece0770e572ce0293d7eb0f4c929.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:6c9507199095fc3fe9de6c5108f20b5e93eceb520182d80ec932df525cd28c42.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:4c32bfff893d314a29e60039013a691c0ab3521d7f06cc0a33fdec0ff8cf5e523.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:6219989cdf3380524e052103a1ac003bfe792cd71413466c50748cf07deedac4; certificate
sha256:7c709948d76619b9dfb99f0e13b5318abca512ba29bd891dd88dd7a5514cdd2e; external-validation
identity sha256:12b33df7efe7f59b80aadf5440e190158685f07e235406fc6880a6a27048585a.

Post-seal empirical comparison. Directness:

authoritative_observation_or_primary_measurement_boundary_correspondence. Disposition:
purpose_matched_authoritative_source_features_present. Predicted consequence: earth:surface-water-node-edge-carrier__storage-routing-and-exchange-relation__network-geometry-flow-quality-boundaries-and-time-held__catchment-window-bounded. Source-derived consequence: earth:surface-water-node-edge-carrier__storage-routing-and-exchange-relation__network-geometry-flow-quality-boundaries-and-time-held__catchment-window-bounded. Exact boundary correspondence: true. External evidence selected the survivor: false. Formal structure relabelled as direct measurement: false. Model, forecast, proxy or retrieval relabelled as observation: false.

Sources and complete feature accounting:

- USGS-NWIS-001 - U.S. Geological Survey Water Resources, [National Water Information System](#); class authoritative_hydrological_observation_network; transport captured; features 2/4 present; present: site identity, surface water and groundwater observations; absent and preserved: time series, parameter units and methods.
- WQP-WEB-SERVICES-001 - U.S. Geological Survey, U.S. Environmental Protection Agency and National Water Quality Monitoring Council, [Water Quality Portal web services](#); class authoritative_water_quality_federation; transport captured; features 4/4 present; present: station and organization identity, sample and result identity, characteristic and unit, quality and metadata fields; absent and preserved: none.
- NSIDC-SEA-ICE-INDEX-001 - National Snow and Ice Data Center, [Sea Ice Index data and image archive](#); class authoritative_cryosphere_data_product; transport captured; features 4/4 present; present: hemisphere and date identity, extent and area distinction, daily and monthly data, missing-data and product-version documentation; absent and preserved: none.

Comparison record:

- sft-earth-surface-water-001-external-earth-evidence: predicted earth:surface-water-node-edge-carrier__storage-routing-and-exchange-relation__network-geometry-flow-quality-boundaries-and-time-held__catchment-window-bounded; source-derived earth:surface-water-node-edge-carrier__storage-routing-and-exchange-relation__network-geometry-flow-quality-boundaries-and-time-held__catchment-window-bounded; exact match True
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if ungauged reaches, diversions, floodplain storage or groundwater exchange are silently omitted.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity

- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json;
 source manifest sha256:19a8ed8e9508ed57968ddfc76d5954ca481b8f19207556a4541dfa27eb283f23;
 derivation seal sha256:5bc6a5c37967f68775b5560eb0870b8277b762f0df9941b23a7539398887c3ae;
 engine receipt sha256:532157a251356bed60f616f20dc1d9548c0fe3c71416a5f00ce41bc02c999a79 at
 receipts/engine/model_admitted/SFT-EARTH-SURFACE-WATER-001-532157a251356bed.json;
 empirical validation sha256:ea549f46422026ad568168681189e6a4c93546587d4af4e449ad706ee0538d28;
 measurement receipt
 sha256:8d20250bf10f136aa7381384b8c24adfdb0ce9ca75a9a78a536cce454739783f; isolation
 sha256:0316cb10a7e942d9e91863da1fd565ffa8cccf869a8c77e29660c88e16d095d0; custody
 sha256:c7ba7532cc771258284b87d3c373ca83b9662542a6e20cec22ab0c611cbebf4.

36. Cryosphere state and transition

Claim identity: SFT-EARTH-CRYOSPHERE-001

Question and exact theorem. A cryosphere state retains snow, ice and frozen-ground identities, mass, extent, thickness, motion, age, phase transitions and measurement uncertainty.

snow-ice-frozen-ground-carrier__method-region-time-bounded__accumulation-transport
 -phase-and-loss-relation__extent-thickness-mass-motion-age-and-uncertainty-held__e
 vidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-reta
 ins-all-rows__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001

- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-VOLCANIC-PROCESS-001
- SFT-EARTH-SURFACE-WATER-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	snow-ice-frozen-ground-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	method-region-time-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	accumulation-transport-phase-and-loss-relation	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	extent-thickness-mass-motion-age-and-uncertainty-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: snow-ice-frozen-ground-carrier__method-region-time-bounded__accumulation-transport-phase-and-loss-relation__extent-thickness-mass-motion-age-and-uncertainty-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- carrier-present: the claim retains a nonempty named carrier; passed true.
- relation-distinct: the carrier and relation are not silently conflated; passed true.
- evidence-boundary: the evidence boundary remains distinct from the result carrier; passed true.
- exact-fold-witness: the family's pre-source exact Fold construction reproduces; passed true.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar.

Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

Water reservoirs, phase handoffs, surface routes, groundwater routes and ice carriers form one coupled ledger without losing phase, location, interval or uncertainty. An unobserved store is not zero and a flow is not a stock.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:24a7c89a654f6a006694434a3eb3a723dc966141d766363f2d1fecf877b44c40.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:434106ccb10fa3fe565c8e30cac129035f700a58731f99199c771cd07f0588bd.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:bd034df7d06804d16cdad86b86f9c38b52d4eb6bd46072a4f4b5859a67acb789.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:75dc3b7711a859a3acd198e13d2f16577bef924dd2b74699dae0c70a82941beb.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:92b105e68a756ab48770cc972b99c63878b466533e534c06d552b09c2f44e684; certificate
sha256:417308116e7c13d9c6291b0bf58d7fc48b04235cdb489bfa1ca9b565a207c8de; external-validation
identity sha256:e46f3ff67c5f90e0671984b12e3a6392e2467973974073d3b83131add9b0dodd.

Post-seal empirical comparison. Directness:

authoritative_observation_or_primary_measurement_boundary_correspondence. Disposition: purpose_matched_authoritative_source_features_present. Predicted consequence: earth:snow-ice-frozen-ground-carrier__accumulation-transport-phase-and-loss-relation__extent-thickness-mass-motion-age-and-uncertainty-held__method-region-time-bounded. Source-derived consequence: earth:snow-ice-frozen-ground-carrier__accumulation-transport-phase-and-loss-relation__extent-thickness-mass-motion-age-and-uncertainty-held__method-region-time-bounded. Exact boundary correspondence: true. External evidence selected the survivor: false. Formal structure relabelled as direct measurement: false. Model, forecast, proxy or retrieval relabelled as observation: false.

Sources and complete feature accounting:

- USGS-NWIS-001 - U.S. Geological Survey Water Resources, [National Water Information System](#); class authoritative_hydrological_observation_network; transport captured; features 2/4 present; present: site identity, surface water and groundwater observations; absent and preserved: time series, parameter units and methods.
- WQP-WEB-SERVICES-001 - U.S. Geological Survey, U.S. Environmental Protection Agency and National Water Quality Monitoring Council, [Water Quality Portal web services](#); class authoritative_water_quality_federation; transport captured; features 4/4 present; present: station and organization identity, sample and result identity, characteristic and unit, quality and metadata fields; absent and preserved: none.
- NSIDC-SEA-ICE-INDEX-001 - National Snow and Ice Data Center, [Sea Ice Index data and image archive](#); class authoritative_cryosphere_data_product; transport captured; features 4/4 present; present: hemisphere and date identity, extent and area distinction, daily and monthly data, missing-data and product-version documentation; absent and preserved: none.

Comparison record:

- sft-earth-cryosphere-001-external-earth-evidence: predicted earth:snow-ice-frozen-ground-carrier__accumulation-transport-phase-and-loss-relation__extent-thickness-mass-motion-age-and-uncertainty-held__method-region-time-bounded; source-derived earth:snow-ice-frozen-ground-carrier__accumulation-transport-phase-and-loss-relation__extent-thickness-mass-motion-age-and-uncertainty-held__method-region-time-bounded; exact match True
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if area alone is substituted for mass, if unavailable thickness becomes absence, or if distinct ice reservoirs are pooled without record.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law

- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json;
 source manifest sha256:58ee9fc967436e2c57e9a4aa7ae49c3c0b34d1d1d5f8795720d3d46165dc7470;
 derivation seal sha256:3962ee470282be4343e8457346e98ca7b8f14749472e4da8f5b96499eb7bf42d;
 engine receipt sha256:185ff3a0c30cbf254e05c303479804c40909ffbd79d0b8f2128ca1f2c6a426b at
 receipts/engine/model_admitted/SFT-EARTH-CRYOSPHERE-001-185ff3a0c30cbf25.json; empirical
 validation sha256:ddcf09313b0aeed82fb72846ac390fe180fcf045e441e9da2ddb7b04bfaf7186;
 measurement receipt
 sha256:7e5984ec9c05c1f3db3da13a8b874d544fdd468540731d6a3e619ba6e9c36ac0; isolation
 sha256:dac2b171eaf4f0e624c19a93ec76efb7ea298b5eb0831ef843556c6e3356a738; custody
 sha256:5120291b5fdb44a9ee5ad0fe64de38a793edde89c5b58a682afef91dba54e2c6.

Ledger family 7: Atmosphere, Weather and Earth-Ionosphere Recurrence

Atmospheric composition, radiative transfer, moisture, circulation and weather are source- and timescale-bounded. A bounded Earth-ionosphere cavity forces distinct recurrent modes structurally; dimensional frequencies remain separately measured records rather than imported proof parameters.

37. Atmospheric system identity

Claim identity: SFT-EARTH-ATMOSPHERE-001

Question and exact theorem. The atmosphere is a layered, moving material and energy carrier whose composition, state, boundaries and exchanges with other Earth reservoirs remain explicit.

bounded-gas-aerosol-energy-carrier__atmospheric-observation-bounded__layered-composition-motion-and-exchange-relation__composition-state-height-boundaries-and-time-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-success-or-retains-all-rows__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001

- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-CRYOSPHERE-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	bounded-gas-aerosol-energy-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	atmospheric-observation-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	layered-composition-motion-and-exchange-relation	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	composition-state-height-boundaries-and-time-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: bounded-gas-aerosol-energy-carrier__atmospheric-observation-bounded__layered-composition-motion-and-exchange-relation__composition-state-height-boundaries-and-time-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- carrier-present: the claim retains a nonempty named carrier; passed true.
- relation-distinct: the carrier and relation are not silently conflated; passed true.
- evidence-boundary: the evidence boundary remains distinct from the result carrier; passed true.
- exact-fold-witness: the family's pre-source exact Fold construction reproduces; passed true.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar.

Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

Atmospheric composition, radiative transfer, moisture, circulation and weather are source- and timescale-bounded. A bounded Earth-ionosphere cavity forces distinct recurrent modes structurally; dimensional frequencies remain separately measured records rather than imported proof parameters.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:9bc1a0aab8a0ba2166a8b17db6b1cce105431cba9b1a15e81fc5222dcc192fd4.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:8fed5c8fd4afc48222e2ca3e692d9b3e2095be3c7b2ad773b78ab95b63d847c2.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:2d71e10d862ba3808956b8455a8e3a7209970f3effae13903113797971f2e790.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:827675631c444725533ae176760c05774d141dd4edddbe57405062a456877293.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:c07bbe159b0f981038ed55023b1684b31d2d53b0a4a9432c2a889d8f650b978f; certificate

sha256:4b7f7a707d45360e8cd200a7a61aa3d9978893d15f3de6cae664fe49a7cae98e; external-validation
identity sha256:040126207715100a7b119ed6270bfde06533ec6d3f77ec1885caf623a28f5177.

Post-seal empirical comparison. Directness:

authoritative_observation_or_primary_measurement_boundary_correspondence. Disposition:
purpose_matched_authoritative_source_features_present. Predicted consequence: earth:bounded-gas-aerosol-energy-carrier__layered-composition-motion-and-exchange-relation__composition-state-height-boundaries-and-time-held__atmospheric-observation-bounded. Source-derived consequence: earth:bounded-gas-aerosol-energy-carrier__layered-composition-motion-and-exchange-relation__composition-state-height-boundaries-and-time-held__atmospheric-observation-bounded. Exact boundary correspondence: true. External evidence selected the survivor: false. Formal structure relabelled as direct measurement: false. Model, forecast, proxy or retrieval relabelled as observation: false.

Sources and complete feature accounting:

- NOAA-GHCND-001 - NOAA National Centers for Environmental Information, **Global Historical Climatology Network Daily**; class `authoritative_station_climate_archive`; transport captured; features 5/5 present; present: station identity and metadata, daily weather observations, quality flags, record length and variable coverage, archive version; absent and preserved: none.
- NASA-CERES-EBAF-001 - NASA Langley Research Center CERES, **CERES Energy Balanced and Filled data products**; class `authoritative_radiative_budget_data_product`; transport captured; features 5/5 present; present: top-of-atmosphere and surface boundary, shortwave and longwave flux distinction, cloud and clear-sky distinction, spatial and temporal resolution, data-quality documentation; absent and preserved: none.
- NOAA-GML-CO2-001 - NOAA Global Monitoring Laboratory, **Carbon-cycle greenhouse-gas measurements**; class `authoritative_atmospheric_composition_archive`; transport captured; features 3/5 present; present: gas and measurement programme identity, site and altitude, calibration and data-use policy; absent and preserved: time and frequency, source and sink interpretation separated from measurement.
- EPA-AIRDATA-001 - U.S. Environmental Protection Agency, **AirData pre-generated monitoring files**; class `authoritative_air_quality_archive`; transport captured; features 4/5 present; present: site and monitor identity, pollutant parameter and units, hourly daily and annual distinction, blank and missing records; absent and preserved: method and quality metadata.

Comparison record:

- `sft-earth-atmosphere-001-external-earth-evidence: predicted earth:bounded-gas-aerosol-energy-carrier__layered-composition-motion-and-exchange-relation__composition-state-height-boundaries-and-time-held__atmospheric-observation-bounded`; source-derived `earth:bounded-gas-aerosol-energy-carrier__layered-composition-motion-and-exchange-relation__composition-state-height-boundaries-and-time-held__atmospheric-observation-bounded`; exact match True
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if a global mean erases vertical, regional or temporal structure required by the claim.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

`experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json`;
 source manifest `sha256:b014781612a26c22ba73b35f26997a3fa7f52799b2dc3efe10e5c47659bfa725`;
 derivation seal `sha256:091ecb00e6fdc2b8458de25810adee0f79ee15d0b3bd25f87c7e723525873f68`;

engine receipt sha256:328297e6b5903b149bc20f75b35537c5eeac3a851795aa86d273eec580d3dd6f at
 receipts/engine/model_admitted/SFT-EARTH-ATMOSPHERE-001-328297e6b5903b14.json; empirical
 validation sha256:a8a2e520967c922c2344964f712667f6013c14f9a79630dacfa18c950d764e93;
 measurement receipt
 sha256:ef13d3484b636d42bee2326022939e8f8e3b8b8eeead5f846dd0666f79787e96; isolation
 sha256:54bea25abd8f70c4ccbb2c9ffa680d339b5d1e28e9dd44ebaa1fafad39b3623a; custody
 sha256:8eaad102bb64bc65adb768c797f690adaeac164e7df1d9f49c4ee7628558d454.

38. Atmospheric composition

Claim identity: SFT-EARTH-ATMOSPHERIC-COMPOSITION-001

Question and exact theorem. Atmospheric composition is a source- and method-bound mixture record retaining species, amount basis, altitude, location, time, transformation and uncertainty.

species-aerosol-mixture-carrier__sampling-network-bounded__amount-fraction-and-tra
 nsformation-relation__species-units-altitude-location-time-and-method-held__eviden
 ce-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-a
 ll-rows__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001

- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-CRYOSPHERE-001
- SFT-EARTH-ATMOSPHERE-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	species-aerosol-mixture-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	sampling-network-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	amount-fraction-and-transformation-relation	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	species-units-altitude-location-time-and-method-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: species-aerosol-mixture-carrier__sampling-network-bounded__amount-fraction-and-transformation-relation__species-units-altitude-location-time-and-method-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `carrier-present`: the claim retains a nonempty named carrier; passed `true`.
- `relation-distinct`: the carrier and relation are not silently conflated; passed `true`.
- `evidence-boundary`: the evidence boundary remains distinct from the result carrier; passed `true`.
- `exact-fold-witness`: the family's pre-source exact Fold construction reproduces; passed `true`.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar. Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

Atmospheric composition, radiative transfer, moisture, circulation and weather are source- and timescale-bounded. A bounded Earth-ionosphere cavity forces distinct recurrent modes structurally; dimensional frequencies remain separately measured records rather than imported proof parameters.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:1068311e6e254480c0934885ae091ce0a3673b057817b63e3037f0a3f9c24f23.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:00c7b712c256c6192f24da980e510c7fa89507349f4e62cabb7739d287d94dd0.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:6893831f59c9056ff022f7880dad99b8b938474b229dc7747e08ff186c584b.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:9ceba2c785c62a69cd4043ba58aa768faad2e3bfcaf976ccce549830d068fd03.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:e9cab25379eb34dd01f808a160248a779e0a5043a8706f0b723370b171e4fc24; certificate
sha256:27891afff9134025d526261f489913db7140440d63360eafecb087934a268cac; external-validation
identity sha256:5f98df70d3e67ff58b1f91e715e2b8ec918e03a5c9b27377932a227e35aecdc.

Post-seal empirical comparison. Directness:

`authoritative_observation_or_primary_measurement_boundary_correspondence`. Disposition:
`purpose_matched_authoritative_source_features_present`. Predicted consequence: `earth:species-aerosol-mixture-carrier__amount-fraction-and-transformation-relation__species-units-altitude-location-time-and-method-held__sampling-network-bounded`. Source-derived consequence: `earth:species-aerosol-mixture-carrier__amount-fraction-and-transformation-relation__species-units-altitude-location-time-and-method-held__sampling-network-bounded`. Exact boundary correspondence: `true`. External evidence selected the survivor: `false`. Formal structure relabelled as direct measurement: `false`. Model, forecast, proxy or retrieval relabelled as observation: `false`.

Sources and complete feature accounting:

- NOAA-GHCND-001 - NOAA National Centers for Environmental Information, [Global Historical Climatology Network Daily](#); class `authoritative_station_climate_archive`; transport captured; features 5/5 present; present: station identity and metadata, daily weather observations, quality flags, record length and variable coverage, archive version; absent and preserved: none.
- NASA-CERES-EBAF-001 - NASA Langley Research Center CERES, [CERES Energy Balanced and Filled data products](#); class `authoritative_radiative_budget_data_product`; transport captured; features 5/5 present; present: top-of-atmosphere and surface boundary, shortwave and longwave flux distinction, cloud and clear-sky distinction, spatial and temporal resolution, data-quality documentation; absent and preserved: none.
- NOAA-GML-CO2-001 - NOAA Global Monitoring Laboratory, [Carbon-cycle greenhouse-gas measurements](#); class `authoritative_atmospheric_composition_archive`; transport captured; features 3/5 present; present:

gas and measurement programme identity, site and altitude, calibration and data-use policy; absent and preserved: time and frequency, source and sink interpretation separated from measurement.

- EPA-AIRDATA-001 - U.S. Environmental Protection Agency, **AirData pre-generated monitoring files**; class `authoritative_air_quality_archive`; transport captured; features 4/5 present; present: site and monitor identity, pollutant parameter and units, hourly daily and annual distinction, blank and missing records; absent and preserved: method and quality metadata.

Comparison record:

- `sft-earth-atmospheric-composition-001-external-earth-evidence: predicted earth:species-aerosol-mixture-carrier__amount-fraction-and-transformation-relation__species-units-altitude-location-time-and-method-held__sampling-network-bounded; source-derived earth:species-aerosol-mixture-carrier__amount-fraction-and-transformation-relation__species-units-altitude-location-time-and-method-held__sampling-network-bounded; exact match True`
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if dry, wet, local and global composition bases are mixed or if below-detection values become ordinary zero.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

`experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json`;
 source manifest sha256:b4c0d7747423c2a7419ac4ce35911c3220b488014d09c9c8033b9b12a05142ea;
 derivation seal sha256:5774652a8fa1e0090a4abc69a2059492f5b9f3472d8f2e0d98f205a358f67efc;
 engine receipt sha256:0bc028829580795fd5d7152cf74d8d55cf930eb481a1ceef65c064a407a4062a at receipts/engine/model_admitted/SFT-EARTH-ATMOSPHERIC-COMPOSITION-001-0bc028829580795f.json; empirical validation
 sha256:e060cbdf77685a1cfc935292242e5aa4e73f9f88e93e0bff8e7d937b1c904e1; measurement receipt sha256:ea16596a69992b7c71e7e4743be0e2b44c7a24c14fb2eec24cd92b449b20ee0c; isolation sha256:63015cde821a8e9b7c0879d78d653d144f40fcc5530a0b16552c8efalecc8eb0; custody sha256:4b11f56907ac7439de5c6b7ea2586c795995b31219aa5bf5c369cb6dd03b33f7.

39. Atmospheric radiative budget

Claim identity: SFT-EARTH-ATMOSPHERIC-RADIATIVE-BUDGET-001

Question and exact theorem. An atmospheric radiative budget is a directional and spectral transfer ledger through atmosphere and surface with clouds, aerosols, storage and unresolved terms retained.

```
spectral-source-layer-boundary-carrier__spectral-spatial-window-bounded__absorptio
n-emission-scattering-transfer-relation__spectrum-direction-cloud-aerosol-surface-
and-time-held__evidence-class-explicit__root-bound-forward-forcing__positive-finit
e-successor-retains-all-rows__no-extra-rule
```

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-CRYOSPHERE-001
- SFT-EARTH-ATMOSPHERIC-COMPOSITION-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	spectral-source-layer-boundary-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	spectral-spatial-window-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	absorption-emission-scattering-transfer-relation	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	spectrum-direction-cloud-aerosol-surface-and-time-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: spectral-source-layer-boundary-carrier__spectral-spatial-window-bounded__absorption-emission-scattering-transfer-relation__spectrum-direction-cloud-aerosol-surface-and-time-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- carrier-present: the claim retains a nonempty named carrier; passed true.
- relation-distinct: the carrier and relation are not silently conflated; passed true.
- evidence-boundary: the evidence boundary remains distinct from the result carrier; passed true.
- exact-fold-witness: the family's pre-source exact Fold construction reproduces; passed true.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar. Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

Atmospheric composition, radiative transfer, moisture, circulation and weather are source- and timescale-bounded. A bounded Earth-ionosphere cavity forces distinct recurrent modes structurally; dimensional frequencies remain separately measured records rather than imported proof parameters.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:e02e321b2bfc90f59582bc5bfc0712c4047e7af0a9bb17f7bdac79f6f512fc35.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:2c37afc380385be74518d40b6f67073a0ef0d58f0590cfb29ca383249d5a85dd.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:7fc572ef67f43e4b621b1a430b041dd1a7b21211a9e191c0babdc7b46981973d.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:07c39645dd6826a7e44fae719288414c13c044d0e0e597fa0157606e14b21beb.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:265038cd3cd487f4e75fa4390076a29bedfcef228c98a3153dd02078adbd09a0; certificate
sha256:e5d68d821394718c48f969a637028f6a5f1d1f5202a69d36e1770b0915538825; external-validation
identity sha256:d7f5922754a7b4d607e1b8156a1f87c94c6c5e267ae0f3f1d0e2130a83bb208c.

Post-seal empirical comparison. Directness:

authoritative_observation_or_primary_measurement_boundary_correspondence. Disposition:
purpose_matched_authoritative_source_features_present. Predicted consequence: earth:spectral-
source-layer-boundary-carrier__absorption-emission-scattering-transfer-relation__spec-
trum-direction-cloud-aerosol-surface-and-time-held__spectral-spatial-window-bounded.
Source-derived consequence: earth:spectral-source-layer-boundary-carrier__absorption-emission-
scattering-transfer-relation__spectrum-direction-cloud-aerosol-surface-and-time-held__
spectral-spatial-window-bounded. Exact boundary correspondence: true. External evidence selected the
survivor: false. Formal structure relabelled as direct measurement: false. Model, forecast, proxy or retrieval relabelled as
observation: false.

Sources and complete feature accounting:

- NOAA-GHCND-001 - NOAA National Centers for Environmental Information, **Global Historical Climatology Network Daily**; class authoritative_station_climate_archive; transport captured; features 5/5 present; present: station identity and metadata, daily weather observations, quality flags, record length and variable coverage, archive version; absent and preserved: none.
- NASA-CERES-EBAF-001 - NASA Langley Research Center CERES, **CERES Energy Balanced and Filled data products**; class authoritative_radiative_budget_data_product; transport captured; features 5/5 present; present: top-of-atmosphere and surface boundary, shortwave and longwave flux distinction, cloud and clear-sky distinction, spatial and temporal resolution, data-quality documentation; absent and preserved: none.
- NOAA-GML-CO2-001 - NOAA Global Monitoring Laboratory, **Carbon-cycle greenhouse-gas measurements**; class authoritative_atmospheric_composition_archive; transport captured; features 3/5 present; present: gas and measurement programme identity, site and altitude, calibration and data-use policy; absent and preserved: time and frequency, source and sink interpretation separated from measurement.
- EPA-AIRDATA-001 - U.S. Environmental Protection Agency, **AirData pre-generated monitoring files**; class authoritative_air_quality_archive; transport captured; features 4/5 present; present: site and monitor identity, pollutant parameter and units, hourly daily and annual distinction, blank and missing records; absent and preserved: method and quality metadata.

Comparison record:

- sft-earth-atmospheric-radiative-budget-001-external-earth-evidence: predicted earth:spectral-source-layer-boundary-carrier__absorption-emission-scattering-transfer-relation__spectrum-direction-cloud-aerosol-surface-and-time-held__spectral-spatial-window-bounded; source-derived earth:spectral-source-layer-boundary-carrier__absorption-emission-scattering-transfer-relation__spectrum-direction-cloud-aerosol-surface-and-time-held__spectral-spatial-window-bounded; exact match True
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if a bulk fitted parameter substitutes for the registered carriers or if incoming and outgoing quantities use incompatible boundaries.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json; source manifest sha256:cc74964ead136fc7dac5678054233ee12aa5bdc1c97cd65506decd09585ff15; derivation seal sha256:7f75166e721c9c9888169f4622265ddf5b8e88ad77e8a42da33ea214ee460a5d; engine receipt sha256:6d88b381038fc08752beff62eafb1942936de895f479cd772025b710ed2c884b at receipts/engine/model_admitted/SFT-EARTH-ATMOSPHERIC-RADIATIVE-BUDGET-001-6d88b381038fc087.json; empirical validation sha256:2794c3aff6867e270b0a569ed18f17ed077b07128622698acdba7ceefa76944b; measurement receipt sha256:d12f604bf36e4531247e910f3aee491e90b41039950fcbb88e4611ab9addc11b; isolation sha256:43e91e1b534e2c2446f49406964adcd83d941ac8855518f9c0351774761dcfd4; custody sha256:06383ba8de95d62677d33264e07141e08268ad65d6df34c9d9a1c9c6b794faf7.

40. Atmospheric moisture and cloud organisation

Claim identity: SFT-EARTH-MOISTURE-CLOUD-001

Question and exact theorem. Atmospheric moisture and clouds are phase-resolved water organizations coupled to air motion, particles and radiation under explicit observation limits.

water-vapour-droplet-ice-air-carrier__cloud-observation-bounded__transport-phase-nucleation-and-precipitation-relation__phase-size-location-time-aerosol-and-uncertainty-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-CRYOSPHERE-001
- SFT-EARTH-ATMOSPHERIC-RADIATIVE-BUDGET-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	water-vapour-droplet-ice-air-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	cloud-observation-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	transport-phase-nucleation-and-precipitation-relation	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	phase-size-location-time-aerosol-and-uncertainty-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: `water-vapour-droplet-ice-air-carrier__cloud-observation-bounded__transport-phase-nucleation-and-precipitation-relation__phase-size-location-time-aerosol-and-uncertainty-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule`.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `carrier-present`: the claim retains a nonempty named carrier; passed `true`.
- `relation-distinct`: the carrier and relation are not silently conflated; passed `true`.
- `evidence-boundary`: the evidence boundary remains distinct from the result carrier; passed `true`.
- `exact-fold-witness`: the family's pre-source exact Fold construction reproduces; passed `true`.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar. Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

Atmospheric composition, radiative transfer, moisture, circulation and weather are source- and timescale-bounded. A bounded Earth-ionosphere cavity forces distinct recurrent modes structurally; dimensional frequencies remain separately measured records rather than imported proof parameters.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 919a7d43310cd19414046d6c767b49ace7ac6a6cc92efbad037230a23e5e5fe5.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt

sha256:71908c63beb1c64f2453a18d1aa89077c661a82942e6f7b108f065eb02720110.

- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:bd23f0f06e38daeec4f6508bf39839d2870c013a061343b715289029cbf7f278.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:f52d9e1b5dcf6ddf84ab7272c7285cd3c1d4cc1e79b76355fc16e2e97100ca19.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:519be6f9ca3ffc53100ebd891c913b932547f2d360a638dc063358067aac12d4; certificate

sha256:2c750661fbc8dc4c0e9bd511a39cb2a4ed754e1d64d9c6cb98445b6e036429ff; external-validation identity sha256:f0b5377c5f8da355e42a5a5a21f4a3cb916e851fbcaffb23cd08533acc2fa07f.

Post-seal empirical comparison. Directness:

authoritative_observation_or_primary_measurement_boundary_correspondence. Disposition: purpose_matched_authoritative_source_features_present. Predicted consequence: earth:water-vapour-droplet-ice-air-carrier__transport-phase-nucleation-and-precipitation-relation__phase-size-location-time-aerosol-and-uncertainty-held__cloud-observation-bounded. Source-derived consequence: earth:water-vapour-droplet-ice-air-carrier__transport-phase-nucleation-and-precipitation-relation__phase-size-location-time-aerosol-and-uncertainty-held__cloud-observation-bounded. Exact boundary correspondence: true. External evidence selected the survivor: false. Formal structure relabelled as direct measurement: false. Model, forecast, proxy or retrieval relabelled as observation: false.

Sources and complete feature accounting:

- NOAA-GHCND-001 - NOAA National Centers for Environmental Information, [Global Historical Climatology Network Daily](#); class authoritative_station_climate_archive; transport captured; features 5/5 present; present: station identity and metadata, daily weather observations, quality flags, record length and variable coverage, archive version; absent and preserved: none.
- NASA-CERES-EBAF-001 - NASA Langley Research Center CERES, [CERES Energy Balanced and Filled data products](#); class authoritative_radiative_budget_data_product; transport captured; features 5/5 present; present: top-of-atmosphere and surface boundary, shortwave and longwave flux distinction, cloud and clear-sky distinction, spatial and temporal resolution, data-quality documentation; absent and preserved: none.
- NOAA-GML-CO2-001 - NOAA Global Monitoring Laboratory, [Carbon-cycle greenhouse-gas measurements](#); class authoritative_atmospheric_composition_archive; transport captured; features 3/5 present; present: gas and measurement programme identity, site and altitude, calibration and data-use policy; absent and preserved: time and frequency, source and sink interpretation separated from measurement.
- EPA-AIRDATA-001 - U.S. Environmental Protection Agency, [AirData pre-generated monitoring files](#); class authoritative_air_quality_archive; transport captured; features 4/5 present; present: site and monitor identity, pollutant parameter and units, hourly daily and annual distinction, blank and missing records; absent and preserved: method and quality metadata.

Comparison record:

- sft-earth-moisture-cloud-001-external-earth-evidence: predicted earth:water-vapour-droplet-ice-air-carrier__transport-phase-nucleation-and-precipitation-relation__phase-size-location-time-aerosol-and-uncertainty-held__cloud-observation-bounded; source-derived earth:water-vapour-droplet-ice-air-carrier__transport-phase-nucleation-and-precipitation-relation__phase-size-location-time-aerosol-and-uncertainty-held__cloud-observation-bounded; exact match True
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if cloud occurrence alone fixes its microphysics or if unresolved sub-grid structure is treated as absent.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json;
 source manifest sha256:af91eb2ad3f337f99918911be2122fa731836b2ed6c5a68faba3b33c25a6bf14;
 derivation seal sha256:6e54e86cda91bfe3311f6b734f430fe664cd2c0e8694b14fdde3e5ba2beb9e2e;
 engine receipt sha256:7b957974667ff8c2dadcb9cb82a88a1c993ed86ad3f73da5f4c9ff9bf7fd1cc5 at
 receipts/engine/model_admitted/SFT-EARTH-MOISTURE-CLOUD-001-7b957974667ff8c2.json;
 empirical validation sha256:918457e376289528ec347ddd045f73ea9cd27adb2022c1483822c030b7858a4a;
 measurement receipt
 sha256:8eacc5304804e18119a750a997a5894d6c538bd325e1dcf2d8fa464a335e33ec; isolation
 sha256:0d30b0717661091977a65a5035c2879f6fecdd5ff9f8fd4bb03113f512d7fc6d0; custody
 sha256:99540695dc3f99d66e59079d38d0ad0c588304b6efc1339d7361bd77d265fa01.

41. Atmospheric circulation

Claim identity: SFT-EARTH-ATMOSPHERIC-CIRCULATION-001

Question and exact theorem. Atmospheric circulation is a time-indexed network of air-mass motion and exchange retaining orientation, transported carriers, boundaries and unresolved scales.

time-indexed-air-mass-motion-carrier__spatial-temporal-window-bounded__transport-r
 ecurrance-and-exchange-network__position-orientation-moisture-energy-boundaries-an
 d-time-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-
 successor-retains-all-rows__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001

- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-CRYOSPHERE-001
- SFT-EARTH-MOISTURE-CLOUD-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	time-indexed-air-mass-motion-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	spatial-temporal-window-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
relation	imported-fitted-or-missing-relation	transport-recurrence-and-exchange-network	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	position-orientation-moisture-energy-boundaries-and-time-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: `time-indexed-air-mass-motion-carrier__spatial-temporal-window-bounded__transport-recurrence-and-exchange-network__position-orientation-moisture-energy-boundaries-and-time-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule`.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `carrier-present`: the claim retains a nonempty named carrier; passed `true`.
- `relation-distinct`: the carrier and relation are not silently conflated; passed `true`.
- `evidence-boundary`: the evidence boundary remains distinct from the result carrier; passed `true`.
- `exact-fold-witness`: the family's pre-source exact Fold construction reproduces; passed `true`.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar.

Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

Atmospheric composition, radiative transfer, moisture, circulation and weather are source- and timescale-bounded. A bounded Earth-ionosphere cavity forces distinct recurrent modes structurally; dimensional frequencies remain separately measured records rather than imported proof parameters.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:2bc181eb990d9695aaa77b3bd7cf003a55c77c9723fdcc4cc74febfb33f07f4e1.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:df85e5ff9c9a1ef6e2ca830d13dab02132ff593e596cfe8fd8495f57d0e47f43.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:4135203f1b1893d9ca00f6e8a7d3d14d1a9a2d9b5f1491f48d7a63509daa8474.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt

sha256:5bdb85458ee8e7b431a7239413b78c99fb007170f232e40a91bfff3bd836d0f10.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:0acf35592c0c9e751000f9104210e7411a90d3eb68c2360c316773c0b7cce321; certificate

sha256:2b35a598687d9e5e8213693149005afa2ae7f12f3fd7d3ac98818d6346692543; external-validation identity sha256:01b1d7d4af5ab6a6505d245b7310e74322b13fb54dd49ce084ae209c493557e1.

Post-seal empirical comparison. Directness:

authoritative_observation_or_primary_measurement_boundary_correspondence. Disposition: purpose_matched_authoritative_source_features_present. Predicted consequence: earth:time-indexed-air-mass-motion-carrier__transport-recurrence-and-exchange-network__position-orientation-moisture-energy-boundaries-and-time-held__spatial-temporal-window-bounded. Source-derived consequence: earth:time-indexed-air-mass-motion-carrier__transport-recurrence-and-exchange-network__position-orientation-moisture-energy-boundaries-and-time-held__spatial-temporal-window-bounded. Exact boundary correspondence: true. External evidence selected the survivor: false. Formal structure relabelled as direct measurement: false. Model, forecast, proxy or retrieval relabelled as observation: false.

Sources and complete feature accounting:

- NOAA-GHCND-001 - NOAA National Centers for Environmental Information, **Global Historical Climatology Network Daily**; class authoritative_station_climate_archive; transport captured; features 5/5 present; present: station identity and metadata, daily weather observations, quality flags, record length and variable coverage, archive version; absent and preserved: none.
- NASA-CERES-EBAF-001 - NASA Langley Research Center CERES, **CERES Energy Balanced and Filled data products**; class authoritative_radiative_budget_data_product; transport captured; features 5/5 present; present: top-of-atmosphere and surface boundary, shortwave and longwave flux distinction, cloud and clear-sky distinction, spatial and temporal resolution, data-quality documentation; absent and preserved: none.
- NOAA-GML-CO2-001 - NOAA Global Monitoring Laboratory, **Carbon-cycle greenhouse-gas measurements**; class authoritative_atmospheric_composition_archive; transport captured; features 3/5 present; present: gas and measurement programme identity, site and altitude, calibration and data-use policy; absent and preserved: time and frequency, source and sink interpretation separated from measurement.
- EPA-AIRDATA-001 - U.S. Environmental Protection Agency, **AirData pre-generated monitoring files**; class authoritative_air_quality_archive; transport captured; features 4/5 present; present: site and monitor identity, pollutant parameter and units, hourly daily and annual distinction, blank and missing records; absent and preserved: method and quality metadata.

Comparison record:

- sft-earth-atmospheric-circulation-001-external-earth-evidence: predicted earth:time-indexed-air-mass-motion-carrier__transport-recurrence-and-exchange-network__position-orientation-moisture-energy-boundaries-and-time-held__spatial-temporal-window-bounded; source-derived earth:time-indexed-air-mass-motion-carrier__transport-recurrence-and-exchange-network__position-orientation-moisture-energy-boundaries-and-time-held__spatial-temporal-window-bounded; exact match True
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if one averaged velocity field is treated as the complete carrier history or if opposite orientations are encoded as negative proof magnitudes.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value

- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json;
 source manifest sha256:a60ffeb6365480922a4aef352ec1b1f8ff5b5b3626c6b2f70b58f3bfa8d407dc;
 derivation seal sha256:d84372a5ea5d98e901c2c03258a208fc66555b31db1aff7c9ea4a97022975873;
 engine receipt sha256:0cdc723b6c39a6e6ee1414ca050b4482461094dc3fb69e90468584da713f27b5 at re
 ceipts/engine/model_admitted/SFT-EARTH-ATMOSPHERIC-CIRCULATION-001-0cdc723b6c39a6e6.j
 son; empirical validation
 sha256:d98d391a1cf9393a15922edd72968cdc71058a10a1b37aaa07da10d4ddb146fd; measurement
 receipt sha256:e49a02293724b48b49e79cfac9da2dd2d483e6f4fecdd787c3f0c9612a4f565c; isolation
 sha256:cad0f531c4b4755a5ca7bbadfcdaa94d92128d18db5c97bbbf50d04b105c5471; custody
 sha256:0d54b568e2baf6dbbecb68f0240c30a186c3e913f3c9a9600922c05d7b2e3091.

42. Weather state and forecast boundary

Claim identity: SFT-EARTH-WEATHER-001

Question and exact theorem. Weather is the observed atmospheric state and its finite-time transition record; a forecast is a sealed conditional consequence, not an observation or natural law.

finite-time-atmospheric-state-carrier__forecast-horizon-bounded__state-transition-
 and-observation-relation__initial-state-forcing-resolution-ensemble-and-outcomes-h
 eld__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successo
 r-retains-all-rows__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001

- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-CRYOSPHERE-001
- SFT-EARTH-ATMOSPHERIC-CIRCULATION-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	finite-time-atmospheric-state-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	forecast-horizon-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	state-transition-and-observation-relation	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	initial-state-forcing-resolution-ensemble-and-outcomes-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: `finite-time-atmospheric-state-carrier__forecast-horizon-bounded__state-transition-and-observation-relation__initial-state-forcing-resolution-ensemble-and-outcomes-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule`.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `carrier-present`: the claim retains a nonempty named carrier; passed `true`.
- `relation-distinct`: the carrier and relation are not silently conflated; passed `true`.
- `evidence-boundary`: the evidence boundary remains distinct from the result carrier; passed `true`.
- `exact-fold-witness`: the family's pre-source exact Fold construction reproduces; passed `true`.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar.

Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

Atmospheric composition, radiative transfer, moisture, circulation and weather are source- and timescale-bounded. A bounded Earth-ionosphere cavity forces distinct recurrent modes structurally; dimensional frequencies remain separately measured records rather than imported proof parameters.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:b9d0a70f92ecf8494e16271d5109a9e2d2b9227c95d176b3bf316fbfbf6a4f40.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:208ad202fa5b8cce073a8ef4048f2299694733500c595cc9b5213d2f03364f45.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:be39b143acd634a0f401b638072d8b57860e9d25c264e875adf51ec90557bb04.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:cd2b3d2e3953dff262e38887f9e53b615c4f9727d28f4d022af54494c72e8f0a.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:9e77b25363f7f2200291119884fb9550ce1d2ef38fc079202de8153f95947b39; certificate
sha256:4cac9b900a8cb2a269c58b46ca1e9d99261d6a133af96035c6cbc7cdc9aff1a2; external-validation
identity sha256:b81d7a1e9692778558b930b93baf6e9373047bbfb779cbb6520aebcf887dfa.

Post-seal empirical comparison. Directness:

`authoritative_observation_or_primary_measurement_boundary_correspondence`. Disposition:
`purpose_matched_authoritative_source_features_present`. Predicted consequence: `earth:finite-ti`

me-atmospheric-state-carrier__state-transition-and-observation-relation__initial-state-forcing-resolution-ensemble-and-outcomes-held__forecast-horizon-bounded. Source-derived consequence: earth:finite-time-atmospheric-state-carrier__state-transition-and-observation-relation__initial-state-forcing-resolution-ensemble-and-outcomes-held__forecast-horizon-bounded. Exact boundary correspondence: true. External evidence selected the survivor: false. Formal structure relabelled as direct measurement: false. Model, forecast, proxy or retrieval relabelled as observation: false.

Sources and complete feature accounting:

- NOAA-GHCND-001 - NOAA National Centers for Environmental Information, **Global Historical Climatology Network Daily**; class authoritative_station_climate_archive; transport captured; features 5/5 present; present: station identity and metadata, daily weather observations, quality flags, record length and variable coverage, archive version; absent and preserved: none.
- NASA-CERES-EBAF-001 - NASA Langley Research Center CERES, **CERES Energy Balanced and Filled data products**; class authoritative_radiative_budget_data_product; transport captured; features 5/5 present; present: top-of-atmosphere and surface boundary, shortwave and longwave flux distinction, cloud and clear-sky distinction, spatial and temporal resolution, data-quality documentation; absent and preserved: none.
- NOAA-GML-CO2-001 - NOAA Global Monitoring Laboratory, **Carbon-cycle greenhouse-gas measurements**; class authoritative_atmospheric_composition_archive; transport captured; features 3/5 present; present: gas and measurement programme identity, site and altitude, calibration and data-use policy; absent and preserved: time and frequency, source and sink interpretation separated from measurement.
- EPA-AIRDATA-001 - U.S. Environmental Protection Agency, **AirData pre-generated monitoring files**; class authoritative_air_quality_archive; transport captured; features 4/5 present; present: site and monitor identity, pollutant parameter and units, hourly daily and annual distinction, blank and missing records; absent and preserved: method and quality metadata.

Comparison record:

- sft-earth-weather-001-external-earth-evidence: predicted earth:finite-time-atmospheric-state-carrier__state-transition-and-observation-relation__initial-state-forcing-resolution-ensemble-and-outcomes-held__forecast-horizon-bounded; source-derived earth:finite-time-atmospheric-state-carrier__state-transition-and-observation-relation__initial-state-forcing-resolution-ensemble-and-outcomes-held__forecast-horizon-bounded; exact match True
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if forecast skill selects the governing law, if adverse forecasts are dropped, or if a model trajectory is relabelled as measured weather.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation

- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json;
 source manifest sha256:47d99709bf5955501eacd23b2971b65c08218f8edc8efa9a136eb2bd1de6ed3b;
 derivation seal sha256:c69a83e2b54ca4eaa93deefdb4b9d7b33acfa9a46a3510a26f398c791642f10e;
 engine receipt sha256:fddb62c115f6dd565cf4c008720a7f8619170ce4104c2a5017f15ceafe7502bb at
 receipts/engine/model_admitted/SFT-EARTH-WEATHER-001-fddb62c115f6dd56.json; empirical
 validation sha256:c413c9257d26026784bb5cb20516bf67febf3b5707e442cd26a5100edd52db0d;
 measurement receipt
 sha256:92f942d23847c9c1cdcccae9db646385172c12c9eabe5dfaf4a22a3469c54add; isolation
 sha256:897c86cee1ec752781bb115ee74b83731339282e9cf45cfb3eab9e745e534448; custody
 sha256:b6248f05c4726bfbc2ad983fafe5b1797854db5d72eecf6ad1e3d14e7dc5c257.

43. Earth-ionosphere resonant cavity

Claim identity: SFT-EARTH-EARTH-IONOSPHERE-RESONANCE-001

Question and exact theorem. Earth-ionosphere resonance composes the admitted bounded-wave relation with the measured Earth-atmosphere cavity geometry and retains the complete source spectrum and uncertainty.

earth-surface-atmosphere-ionosphere-boundary-carrier__earth-cavity-observation-bounded__bounded-propagation-mode-recurrence__geometry-medium-source-spectrum-time-and-uncertainty-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001

- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-CRYOSPHERE-001
- SFT-EARTH-WEATHER-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	earth-surface-atmosphere-ionosphere-boundary-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	earth-cavity-observation-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	bounded-propagation-mode-recurrence	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	geometry-medium-source-spectrum-time-and-uncertainty-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: earth-surface-atmosphere-ionosphere-boundary-carrier__earth-cavity-observation-bounded__bounded-propagation-mode-recurrence__geometry-medium-source-spectrum-time-and-uncertainty-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `carrier-present`: the claim retains a nonempty named carrier; passed `true`.
- `relation-distinct`: the carrier and relation are not silently conflated; passed `true`.
- `evidence-boundary`: the evidence boundary remains distinct from the result carrier; passed `true`.
- `exact-fold-witness`: the family's pre-source exact Fold construction reproduces; passed `true`.

Scientific meaning. A bounded cavity and positive finite recurrence force more than one distinguishable mode. The theorem does not read a frequency from theory or turn a measured hertz value into a proof parameter; primary magnetometer evidence tests the separate dimensional boundary.

Atmospheric composition, radiative transfer, moisture, circulation and weather are source- and timescale-bounded. A bounded Earth-ionosphere cavity forces distinct recurrent modes structurally; dimensional frequencies remain separately measured records rather than imported proof parameters.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:b86d5b228b335656a7e0dd44ddd9e090f63e63d46f79f4e1b51a564b74d86c57.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:b5400e46b6e3ed48a10eada81f02a57ebc9de483684a780b2f6b55c4164aa33d.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:c7f1b026a06459dd9ba16962d8c0aa85d3786fa3b50dad4e2e1797e459da1c4a.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:e97e979b2cafbfd9ddf773b3936db93cdee5fa8a68483cdaf819959526a37c80a.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:d7fd6d035d16d109f7f9ed44c0d164de657ab8f04362be21474383b43df796e3; certificate
sha256:c4233f0e44f003f0fd96063662b7880270ee661a12ad0ddb6227a7e1f9343f46; external-validation
identity sha256:98ef97b362e1dd72c395456cd9b2d89044efc65ca79ccc3065572462dcceca5c.

Post-seal empirical comparison. Directness:

`authoritative_observation_or_primary_measurement_boundary_correspondence`. Disposition:
`purpose_matched_authoritative_source_features_present`. Predicted consequence: `earth:earth-surface-atmosphere-ionosphere-boundary-carrier__bounded-propagation-mode-recurrence__geometry-medium-source-spectrum-time-and-uncertainty-held__earth-cavity-observation-bounded__bounded-discrete-modes-dimensional-frequency-measured`. Source-derived consequence: `earth:earth-surface-atmosphere-ionosphere-boundary-carrier__bounded-propagation-mode-recurrence__geometry-medium-source-spectrum-time-and-uncertainty-held__earth-cavity-observation-bounded__bounded-discrete-modes-dimensional-frequency-measured`. Exact boundary correspondence: `true`. External evidence selected the survivor: `false`. Formal structure relabelled as direct measurement: `false`. Model, forecast, proxy or retrieval relabelled as observation: `false`.

Sources and complete feature accounting:

- SCHUMANN-MAGNETOMETER-2018-001 - Virgo and KAGRA magnetometer measurement collaboration, **Measurement and subtraction of Schumann resonances at gravitational-wave interferometers**; class `primary_measurement_study`; transport captured; features 4/5 present; present: magnetometer spectrum, correlated Earth-scale resonance, frequency-resolved measured peaks, instrumental subtraction boundary; absent and preserved: independent site identity.
- EARTHSCOPE-SAGE-SERVICES-001 - NSF SAGE Facility operated by EarthScope Consortium, **Seismological and geophysical web services**; class `authoritative_geophysical_data_service`; transport captured; features 3/4 present; present: station and channel identity, waveform and event access, instrument response; absent and preserved: observed and derived product distinction.

Comparison record:

- `sft-earth-earth-ionosphere-resonance-001-external-earth-evidence`: predicted `earth:earth-surface-atmosphere-ionosphere-boundary-carrier__bounded-propagation-mode-recurrence__geometry-medium-source-spectrum-time-and-uncertainty-held__earth-cavity-observation-bounded__bounded-discrete-modes-dimensional-frequency-measured`; source-derived `earth:earth-surface-atmosphere-ionosphere-boundary-carrier__bounded-propagation-mode-recurrence__geometry-medium-source-spectrum-time-and-uncertainty-held__earth-cavity-observation-bounded__bounded-discrete-modes-dimensional-frequency-measured`; exact match `True`
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if the observed resonant frequency selects the cavity law, if a universal wave law is renowned, or if engineering transmission claims are inferred from resonance alone.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

`experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json`;
 source manifest sha256: 9ff88e2486e4f52030bb49618a5c24a9a50b249901816b299c7ab13d670e81c4;
 derivation seal sha256: ceb27881b156c712e194e3eafaae798106347c00bcc6963cce0d849c1dd895b1;
 engine receipt sha256: 089d63910614824c5d2a9424a2308b2ceb4cb6a0381ec7fd89a5d1d77fcfd8ae at receipts/engine/model_admitted/SFT-EARTH-EARTH-IONOSPHERE-RESONANCE-001-089d63910614824c.json; empirical validation
 sha256: 5522c5ef3b35137eacfc20f02307c2867b56297603a15ab60fbb420c9ba03d98; measurement receipt sha256: 0d5caac1f9e4005df6808186e5f9e6aebab08c0f24aae8ba298dbc61c7342c51; isolation sha256: 9b70a1f44f857ccdf611605e590baccd1c0aee0dc634bb2d5867dc4f48b38d5e; custody sha256: 438185990330ed64a35fe355f806b7ab4921e7f805f82f123bd0ea9492be0cd4.

Ledger family 8: Ocean and Coast

Ocean identity retains composition, depth, stratification, circulation and interfaces. Air-sea transfer is two-sided. Coastal and tidal observations retain geometry, forcing, local response, datum and time rather than treating a site-dependent magnitude as a universal constant.

44. Ocean system identity

Claim identity: SFT-EARTH-OCEAN-001

Question and exact theorem. The ocean is a basin- and boundary-resolved seawater organization retaining composition, state, depth, motion and exchanges with atmosphere, land, ice and seabed.

bounded-seawater-basin-interface-carrier__ocean-observation-bounded__storage-motion-composition-and-exchange-relation__basin-depth-composition-state-boundaries-and-time-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001

- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-EARTH-IONOSPHERE-RESONANCE-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	bounded-seawater-basin-interface-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	ocean-observation-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	storage-motion-composition-and-exchange-relation	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	basin-depth-composition-state-boundaries-and-time-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: bounded-seawater-basin-interface-carrier__ocean-observation-bounded__storage-motion-composition-and-exchange-relation__basin-depth-composition-state-boundaries-and-time-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- carrier-present: the claim retains a nonempty named carrier; passed true.
- relation-distinct: the carrier and relation are not silently conflated; passed true.
- evidence-boundary: the evidence boundary remains distinct from the result carrier; passed true.

- `exact-fold-witness`: the family's pre-source exact Fold construction reproduces; passed `true`.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar. Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

Ocean identity retains composition, depth, stratification, circulation and interfaces. Air-sea transfer is two-sided. Coastal and tidal observations retain geometry, forcing, local response, datum and time rather than treating a site-dependent magnitude as a universal constant.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:a8531fd94c42f15aec67605e9537d5b44216fded62e52d9f17814b4dcd89300c.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:f8667c5608ea6b56fa6f31b82936e0a0f212a45df59966f99558c25c751b58b1.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:d9b2d4a89bf1107bd322a102ceb926c6d784d85c100856d414cdc2f06cd00c09.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:54306a9088b73d00b398ba0e5cf8b52ba9fd8d6d76bded3903fb113cc8408554.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:e6cf2c67ac4dde85dd767675d77d88f15c8d6b477a9c6efdc45bb19ed0c833ba; certificate
sha256:e648779814a8f6fe7444502dd157ecf69bcb6c63e178dd64efce28b5ca3a8b8aa; external-validation
identity sha256:3fd11fb2a39927fa6555d0e1f4fcaea3a78175e356ef6c1348bb5dae8c8b09c0.

Post-seal empirical comparison. Directness:

authoritative_observation_or_primary_measurement_boundary_correspondence. Disposition:
purpose_matched_authoritative_source_features_present. Predicted consequence: earth:bounded-seawater-basin-interface-carrier__storage-motion-composition-and-exchange-relation__basin-depth-composition-state-boundaries-and-time-held__ocean-observation-bounded.
Source-derived consequence: earth:bounded-seawater-basin-interface-carrier__storage-motion-composition-and-exchange-relation__basin-depth-composition-state-boundaries-and-time-held__ocean-observation-bounded. Exact boundary correspondence: true. External evidence selected the survivor: false. Formal structure relabelled as direct measurement: false. Model, forecast, proxy or retrieval relabelled as observation: false.

Sources and complete feature accounting:

- NOAA-WOD-001 - NOAA National Centers for Environmental Information, [World Ocean Database](#); class
authoritative_ocean_profile_archive; transport captured; features 5/5 present; present: cast cruise and accession identity, depth location and time, temperature salinity nutrients and plankton, quality flags, original-source provenance; absent and preserved: none.
- ARGO-DATA-001 - Argo Data Management Team, [Argo profile and trajectory data access](#); class
authoritative_ocean_observing_program; transport captured; features 4/4 present; present: float and profile identity, depth temperature salinity and trajectory, real-time and delayed-mode quality distinction, quality-control flags; absent and preserved: none.
- NSIDC-SEA-ICE-INDEX-001 - National Snow and Ice Data Center, [Sea Ice Index data and image archive](#); class
authoritative_cryosphere_data_product; transport captured; features 4/4 present; present: hemisphere and date identity, extent and area distinction, daily and monthly data, missing-data and product-version documentation; absent and preserved: none.

Comparison record:

- sft-earth-ocean-001-external-earth-evidence: predicted earth:bounded-seawater-basin-interface-carrier__storage-motion-composition-and-exchange-relation__basin-depth-composition-state-boundaries-and-time-held__ocean-observation-bounded; source-derived earth:bounded-seawater-basin-interface-carrier__storage-motion-composition-and-exchange-relation__basin-depth-composition-state-boundaries-and-time-held__ocean-observation-bounded; exact match True
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if a surface observation is generalised through depth or if basin and global records are mixed without weighting and provenance.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json;
 source manifest sha256:ff41352e4528a1363942e0ce513ef1332b25b3768df463658e58acd2fda26a0d;
 derivation seal sha256:e576405b23ff14ff7a4de1429a837d35fe0a9f76942b22849d9204ade432caf2;
 engine receipt sha256:e84ef511820f3cd37b7f8df20793111ca7f971e47f3ed9b0835934d9ff09deed at
 receipts/engine/model_admitted/SFT-EARTH-OCEAN-001-e84ef511820f3cd3.json; empirical
 validation sha256:c075a01c197302996cec503234a9b43ce098d15ab79db814e866a1191b0a8b43;
 measurement receipt
 sha256:74df6b141b99e794e8a11dc9af673da33403512d598797a198a19384e8591250; isolation
 sha256:ab9fca7bf975831625cc573ca4bc8cbf3682a532ff530aa262ec3e1415587144; custody
 sha256:08617c7b9f5d719497192642e7c1703b5bb9174ad6c6368671686407493d2716.

45. Ocean composition and salinity

Claim identity: SFT-EARTH-OCEAN-COMPOSITION-001

Question and exact theorem. Ocean composition retains dissolved, particulate and gaseous carriers, amount basis, depth, location, time, method and transformations rather than salinity as an answer-only scalar.

water-dissolved-particulate-gas-carrier__sample-program-bounded__composition-fract
 ion-and-transformation-relation__species-basis-depth-location-time-and-method-held
 __evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-r
 etains-all-rows__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-EARTH-IONOSPHERE-RESONANCE-001
- SFT-EARTH-OCEAN-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	water-dissolved-particulate-gas-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	sample-program-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	composition-fraction-and-transformation-relation	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	species-basis-depth-location-time-and-method-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: water-dissolved-particulate-gas-carrier__sample-program-bounded__composition-fraction-and-transformation-relation__species-basis-depth-location-time-and-method-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- carrier-present: the claim retains a nonempty named carrier; passed true.
- relation-distinct: the carrier and relation are not silently conflated; passed true.
- evidence-boundary: the evidence boundary remains distinct from the result carrier; passed true.
- exact-fold-witness: the family's pre-source exact Fold construction reproduces; passed true.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar. Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

Ocean identity retains composition, depth, stratification, circulation and interfaces. Air-sea transfer is two-sided. Coastal and tidal observations retain geometry, forcing, local response, datum and time rather than treating a site-dependent magnitude as a universal constant.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:df7d860797cbf664c61c656fdd75e561bfe555417fd8caab6a1ee10d52cdbbd9.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt

sha256:8c4419b62aa906652370e27ed910b0f9fc99b3f8b5c2828017abb000598afcac.

- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:e48b3f520f3c66dd32dc742ac960b54c2f162ea6b45ada77590d49efa806ea52.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:096cfae02ba3aadaa3013ebb85750a8c5e7021c9a47401db9425d1a7db0f80c5.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:09a46dbe69924d6952159a8d2f7dbff3c31e294342426f52442d6f759f7ee168; certificate

sha256:d44ea9376dd727dc0a648472e35a4793d0f1d4a17ddeb7d403ccb2a6840a411e; external-validation identity sha256:79ealbccbece3af0ac501dbf63a6a248fe84cead2ea3e64c376f4504e15503af.

Post-seal empirical comparison. Directness:

authoritative_observation_or_primary_measurement_boundary_correspondence. Disposition: purpose_matched_authoritative_source_features_present. Predicted consequence: earth:water-dissolved-particulate-gas-carrier__composition-fraction-and-transformation-relation__species-basis-depth-location-time-and-method-held__sample-program-bounded. Source-derived consequence: earth:water-dissolved-particulate-gas-carrier__composition-fraction-and-transformation-relation__species-basis-depth-location-time-and-method-held__sample-program-bounded. Exact boundary correspondence: true. External evidence selected the survivor: false. Formal structure relabelled as direct measurement: false. Model, forecast, proxy or retrieval relabelled as observation: false.

Sources and complete feature accounting:

- NOAA-WOD-001 - NOAA National Centers for Environmental Information, [World Ocean Database](#); class authoritative_ocean_profile_archive; transport captured; features 5/5 present; present: cast cruise and accession identity, depth location and time, temperature salinity nutrients and plankton, quality flags, original-source provenance; absent and preserved: none.
- ARGO-DATA-001 - Argo Data Management Team, [Argo profile and trajectory data access](#); class authoritative_ocean_observing_program; transport captured; features 4/4 present; present: float and profile identity, depth temperature salinity and trajectory, real-time and delayed-mode quality distinction, quality-control flags; absent and preserved: none.
- NSIDC-SEA-ICE-INDEX-001 - National Snow and Ice Data Center, [Sea Ice Index data and image archive](#); class authoritative_cryosphere_data_product; transport captured; features 4/4 present; present: hemisphere and date identity, extent and area distinction, daily and monthly data, missing-data and product-version documentation; absent and preserved: none.

Comparison record:

- sft-earth-ocean-composition-001-external-earth-evidence: predicted earth:water-dissolved-particulate-gas-carrier__composition-fraction-and-transformation-relation__species-basis-depth-location-time-and-method-held__sample-program-bounded; source-derived earth:water-dissolved-particulate-gas-carrier__composition-fraction-and-transformation-relation__species-basis-depth-location-time-and-method-held__sample-program-bounded; exact match True
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if practical and absolute composition bases are silently mixed or if missing species are treated as absent.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity

- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json;
 source manifest sha256:621bdee8419e6b8e45c710c25affcf1e9ce3b267949545acc677e8d92969daea;
 derivation seal sha256:915f5fde4a941a14d629296bc24d3a3c997163e9d4130d2bf4023bf6d11f47f6;
 engine receipt sha256:84914bfb33d1be20ed6a57dbbbaeb1c7802636d6dcf8fd6dabb6c766aaff8f2b at
 receipts/engine/model_admitted/SFT-EARTH-OCEAN-COMPOSITION-001-84914bfb33d1be20.json;
 empirical validation sha256:89eb1d2bfc6ac9f600309fd7c9d1ba4c080365dc6e32afdd1f1e44416cffe6a;
 measurement receipt
 sha256:64cfaeb5cc923ce824bfde9e6286212e1b42031038bf16aec37f0b7b363c6831; isolation
 sha256:c10c5213cd1376235d6bce93d9088b665956e7278201e28c1ca61a80a06af564; custody
 sha256:38f8822bc8198ca04e26c3aea1ef39e130262030ddf5c71f581f8be6375b4cdc.

46. Ocean stratification

Claim identity: SFT-EARTH-OCEAN-STRATIFICATION-001

Question and exact theorem. Ocean stratification is a depth- and time-resolved organization of water masses defined by retained property contrasts, interfaces, mixing and observation uncertainty.

depth-indexed-water-mass-carrier__profile-network-bounded__property-contrast-and-l
 ayer-stability-relation__temperature-composition-pressure-mixing-and-time-held__ev
 idence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retai
 ns-all-rows__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001

- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-EARTH-IONOSPHERE-RESONANCE-001
- SFT-EARTH-OCEAN-COMPOSITION-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	depth-indexed-water-mass-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	profile-network-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	property-contrast-and-layer-stability-relation	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	temperature-composition-pressure-mixing-and-time-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: `depth-indexed-water-mass-carrier__profile-network-bounded__property-contrast-and-layer-stability-relation__temperature-composition-pressure-mixing-and-time-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule`.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `carrier-present`: the claim retains a nonempty named carrier; passed `true`.
- `relation-distinct`: the carrier and relation are not silently conflated; passed `true`.
- `evidence-boundary`: the evidence boundary remains distinct from the result carrier; passed `true`.
- `exact-fold-witness`: the family's pre-source exact Fold construction reproduces; passed `true`.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar.

Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

Ocean identity retains composition, depth, stratification, circulation and interfaces. Air-sea transfer is two-sided. Coastal and tidal observations retain geometry, forcing, local response, datum and time rather than treating a site-dependent magnitude as a universal constant.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:016bc5c02d2a9cd75a852a7c97a27ab2e7cb1699439b9aa312dcfc11bdc27e76.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:1a789e02523ff58a91bad0209d6f4ee32abbf8e86f2c57d4597f15d400dbc637.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:9bd008fda74e09b33158273efa847f1a2b557ba04f5188a20512bb29e49320a7.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:9a7372b2b3edd1828715c2d675cc542dfc6f6d2f8e728044f53b0e193abac697.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:f82d92ab11b61eadc0f0355c26bee43e6be6c6e1d1f35127b56f47b12ef8fc16; certificate
sha256:dbe95413e680a2412d26ac4e0991d18852f2479f6a1ab673afa7e746f6beb14e; external-validation
identity sha256:3e200d51f9dbd149ec9c8c3143e53e570b1508713c2dc819d7c11ca909146fc8.

Post-seal empirical comparison. Directness:

authoritative_observation_or_primary_measurement_boundary_correspondence. Disposition: purpose_matched_authoritative_source_features_present. Predicted consequence: earth:depth-indexed-water-mass-carrier__property-contrast-and-layer-stability-relation__temperature-composition-pressure-mixing-and-time-held__profile-network-bounded. Source-derived consequence: earth:depth-indexed-water-mass-carrier__property-contrast-and-layer-stability-relation__temperature-composition-pressure-mixing-and-time-held__profile-network-bounded. Exact boundary correspondence: true. External evidence selected the survivor: false. Formal structure relabelled as direct measurement: false. Model, forecast, proxy or retrieval relabelled as observation: false.

Sources and complete feature accounting:

- NOAA-WOD-001 - NOAA National Centers for Environmental Information, [World Ocean Database](#); class authoritative_ocean_profile_archive; transport captured; features 5/5 present; present: cast cruise and accession identity, depth location and time, temperature salinity nutrients and plankton, quality flags, original-source provenance; absent and preserved: none.
- ARGO-DATA-001 - Argo Data Management Team, [Argo profile and trajectory data access](#); class authoritative_ocean_observing_program; transport captured; features 4/4 present; present: float and profile identity, depth temperature salinity and trajectory, real-time and delayed-mode quality distinction, quality-control flags; absent and preserved: none.
- NSIDC-SEA-ICE-INDEX-001 - National Snow and Ice Data Center, [Sea Ice Index data and image archive](#); class authoritative_cryosphere_data_product; transport captured; features 4/4 present; present: hemisphere and date identity, extent and area distinction, daily and monthly data, missing-data and product-version documentation; absent and preserved: none.

Comparison record:

- sft-earth-ocean-stratification-001-external-earth-evidence: predicted earth:depth-indexed-water-mass-carrier__property-contrast-and-layer-stability-relation__temperature-composition-pressure-mixing-and-time-held__profile-network-bounded; source-derived earth:depth-indexed-water-mass-carrier__property-contrast-and-layer-stability-relation__temperature-composition-pressure-mixing-and-time-held__profile-network-bounded; exact match True
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if a single profile fixes a persistent layer globally or if density is imported without its admitted physical and compositional dependencies.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law

- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json;
 source manifest sha256:20f49fad10e6c46317a0d4104eef743703ac3bfc9617e5791fe759da27d6baba;
 derivation seal sha256:43d25e648c7d1110e1c241bb9677d8657d3cde0afbddd3e226699cafb27da2f7;
 engine receipt sha256:fa688fb11daffe7c0e6dce1047f26d6e26db9148c390cf939e78479b77261d41 at re
 ceipts/engine/model_admitted/SFT-EARTH-OCEAN-STRATIFICATION-001-fa688fb11daffe7c.json
 ; empirical validation
 sha256:a9721bf5a52f4a60569eb7f420d58a5bc459a5e22170ba8b3066c2bb45601664; measurement
 receipt sha256:6dde6518902d97aaf293790a68d9c24e73a3114ce4a6d1105845347819408d85; isolation
 sha256:0e3ec9de26d9874af007ae582449853017f390a1428c5f55a03dcfe5a01c694d; custody
 sha256:311b9803cabe8bde5820eb76ea8571589c50718dc084ac22c3a267cd4367acf5.

47. Ocean circulation

Claim identity: SFT-EARTH-OCEAN-CIRCULATION-001

Question and exact theorem. Ocean circulation is a carrier-resolved transport network retaining water-mass identity, direction, depth, mixing, exchange and observation interval.

water-mass-pathway-network__basin-time-window-bounded__transport-overturning-and-m
 ixing-relation__mass-properties-direction-depth-time-and-boundaries-held__evidence
 -class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all
 -rows__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001

- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-EARTH-IONOSPHERE-RESONANCE-001
- SFT-EARTH-OCEAN-STRATIFICATION-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	water-mass-pathway-network	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	basin-time-window-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	transport-overturning-and-mixing-relation	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	mass-properties-direction-depth-time-and-boundaries-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: water-mass-pathway-network__basin-time-window-bounded__transport-overturning-and-mixing-relation__mass-properties-direction-depth-time-and-boundaries-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `carrier-present`: the claim retains a nonempty named carrier; passed `true`.
- `relation-distinct`: the carrier and relation are not silently conflated; passed `true`.
- `evidence-boundary`: the evidence boundary remains distinct from the result carrier; passed `true`.
- `exact-fold-witness`: the family's pre-source exact Fold construction reproduces; passed `true`.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar.

Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

Ocean identity retains composition, depth, stratification, circulation and interfaces. Air-sea transfer is two-sided. Coastal and tidal observations retain geometry, forcing, local response, datum and time rather than treating a site-dependent magnitude as a universal constant.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:395a56ffbd2e1180b4eaa8e1b5ffb8338f9ce3f64f9096502b6d5b5df14fbf22.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:fb897baca97cc25f60dbb4a33e2222f706af97e4c10d87716934213f378c0ffe.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:ab0d35d527c86a6f7915c21f543e92b640a891b7b950c2cd2cd3607f9e5d61d1.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:e492831e98e73e848ace448a53930adbeece88267d765b9cea9085bf5dab5e45.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:b6dcc634adbefe3c45c204eccc3299bd2c4f41f4acb371e3c286aef1c4d78dc1; certificate

sha256:1f8d0beeac609f680621c27eb9f48a365507fc1b710326665f8c87fa2609e55f; external-validation
identity sha256:fe8d5058aeca6be599590e0c8397325b0c0cc051b7234a346055bde53bdfecbc.

Post-seal empirical comparison. Directness:

`authoritative_observation_or_primary_measurement_boundary_correspondence`. Disposition:
`purpose_matched_authoritative_source_features_present`. Predicted consequence: `earth:water-mass-pathway-network__transport-overturning-and-mixing-relation__mass-properties-direction-depth-time-and-boundaries-held__basin-time-window-bounded`. Source-derived consequence: `earth:water-mass-pathway-network__transport-overturning-and-mixing-relation__mass-properties-direction-depth-time-and-boundaries-held__basin-time-window-bounded`. Exact boundary correspondence: `true`. External evidence selected the survivor: `false`. Formal structure relabelled as direct measurement: `false`. Model, forecast, proxy or retrieval relabelled as observation: `false`.

Sources and complete feature accounting:

- NOAA-WOD-001 - NOAA National Centers for Environmental Information, [World Ocean Database](#); class `authoritative_ocean_profile_archive`; transport captured; features 5/5 present; present: cast cruise and accession identity, depth location and time, temperature salinity nutrients and plankton, quality flags, original-source provenance; absent and preserved: none.

- ARGO-DATA-001 - Argo Data Management Team, [Argo profile and trajectory data access](#); class `authoritative_ocean_observing_program`; transport captured; features 4/4 present; present: float and profile identity, depth temperature salinity and trajectory, real-time and delayed-mode quality distinction, quality-control flags; absent and preserved: none.
- NSIDC-SEA-ICE-INDEX-001 - National Snow and Ice Data Center, [Sea Ice Index data and image archive](#); class `authoritative_cryosphere_data_product`; transport captured; features 4/4 present; present: hemisphere and date identity, extent and area distinction, daily and monthly data, missing-data and product-version documentation; absent and preserved: none.

Comparison record:

- `sft-earth-ocean-circulation-001-external-earth-evidence`: predicted earth:water-mass-pathway-network__transport-overturning-and-mixing-relation__mass-properties-direction-depth-time-and-boundaries-held__basin-time-window-bounded; source-derived earth:water-mass-pathway-network__transport-overturning-and-mixing-relation__mass-properties-direction-depth-time-and-boundaries-held__basin-time-window-bounded; exact match True
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if a streamfunction or inferred path is treated as direct parcel history or if unresolved recirculation is erased.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

```
experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json;
source manifest sha256:2f69d3d7521d928756fc4fa4c10c3678aecca973f9bf593666bf80c75264ee9e;
derivation seal sha256:d346696cf5361970590d75f1f1fc7b49fb3b445b383a8d652b0b5db59b11fa53;
engine receipt sha256:63f2c8b81e8b1f448db38a1cce67d952dc03a5f6f0855d269cec74486b54a6fa at
receipts/engine/model_admitted/SFT-EARTH-OCEAN-CIRCULATION-001-63f2c8b81e8b1f44.json;
empirical validation sha256:fd6e03c2133f636e4ecf59a23ec755c755e13ddce2d265ca0d8c4c040f2f5031;
measurement receipt
sha256:c4c43d54b8a138444743a738989db1f9c6c1cc4a6335051907ffd5e7cf4ef436; isolation
sha256:d396ec49c5fffb6394c00c1bc3e1974e84dfa397cbc4b7758b8029e348fef55c2; custody
sha256:6c62d74acf9f9d54c2d6b62faf592ee48a8f549c109097126f2028163a9be971.
```

48. Air-sea exchange

Claim identity: SFT-EARTH-AIR-SEA-EXCHANGE-001

Question and exact theorem. Air-sea exchange is a two-way, interface-bounded transfer ledger retaining carrier identity, direction, area, state, time and method uncertainty.

```
atmosphere-ocean-interface-carrier__interface-method-bounded__two-way-material-ene
rgy-transfer-relation__species-energy-area-direction-state-and-time-held__evidence
-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all
-rows__no-extra-rule
```

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001

- SFT-EARTH-EARTH-IONOSPHERE-RESONANCE-001
- SFT-EARTH-OCEAN-CIRCULATION-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	atmosphere-ocean-interface-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	interface-method-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	two-way-material-energy-transfer-relation	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	species-energy-area-direction-state-and-time-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: atmosphere-ocean-interface-carrier__interface-method-bounded__two-way-material-energy-transfer-relation__species-energy-area-direction-state-and-time-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- carrier-present: the claim retains a nonempty named carrier; passed true.
- relation-distinct: the carrier and relation are not silently conflated; passed true.
- evidence-boundary: the evidence boundary remains distinct from the result carrier; passed true.
- exact-fold-witness: the family's pre-source exact Fold construction reproduces; passed true.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar.

Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

Ocean identity retains composition, depth, stratification, circulation and interfaces. Air-sea transfer is two-sided. Coastal and tidal observations retain geometry, forcing, local response, datum and time rather than treating a site-dependent magnitude as a universal constant.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:7032dd1476930aafde86d131871996c0e9cbfe5f7e9d40d23fbe714b690ac346.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:b96b104b07aad7091232f982d22b7018900c590304c7b18b7dde7e49b5313dad.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:6fba4c9467f9367b3280b39e60d13d6053c3d85809ffe06eec7cf75c4200d005.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:f60e9ac865ad97cc1846a5f1e4e67772904c02e495c7c5ed2d51a3f6483d8fd9.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:0600ff86779ac62fa95c0cb9eb539a4127f39a1ed035f38bb3883d4fdf7b720d; certificate
sha256:05d973266688dfeb48f8bb7f692e41bbf88414313d75f0838d12871e9650ec05; external-validation
identity sha256:9343df05fa035d237163227e6f8158600739d35c2b097e93ac6147d22d6df18e.

Post-seal empirical comparison. Directness:

authoritative_observation_or_primary_measurement_boundary_correspondence. Disposition:
purpose_matched_authoritative_source_features_present. Predicted consequence: earth:atmospher
e-ocean-interface-carrier__two-way-material-energy-transfer-relation__species-energy-
area-direction-state-and-time-held__interface-method-bounded. Source-derived consequence: eart
h:atmosphere-ocean-interface-carrier__two-way-material-energy-transfer-relation__spec
ies-energy-area-direction-state-and-time-held__interface-method-bounded. Exact boundary
correspondence: true. External evidence selected the survivor: false. Formal structure relabelled as direct measurement:
false. Model, forecast, proxy or retrieval relabelled as observation: false.

Sources and complete feature accounting:

- NOAA-WOD-001 - NOAA National Centers for Environmental Information, [World Ocean Database](#); class
authoritative_ocean_profile_archive; transport captured; features 5/5 present; present: cast cruise and
accession identity, depth location and time, temperature salinity nutrients and plankton, quality flags, original-source
provenance; absent and preserved: none.
- ARGO-DATA-001 - Argo Data Management Team, [Argo profile and trajectory data access](#); class
authoritative_ocean_observing_program; transport captured; features 4/4 present; present: float and
profile identity, depth temperature salinity and trajectory, real-time and delayed-mode quality distinction, quality-control
flags; absent and preserved: none.
- NSIDC-SEA-ICE-INDEX-001 - National Snow and Ice Data Center, [Sea Ice Index data and image archive](#); class
authoritative_cryosphere_data_product; transport captured; features 4/4 present; present: hemisphere
and date identity, extent and area distinction, daily and monthly data, missing-data and product-version documentation;
absent and preserved: none.

Comparison record:

- sft-earth-air-sea-exchange-001-external-earth-evidence: predicted earth:atmosphere-ocean-interface-carrier__two-way-mate
rial-energy-transfer-relation__species-energy-area-direction-state-and-time-held__interface-method-bounded;

source-derived earth:atmosphere-ocean-interface-carrier__two-way-material-energy-transfer-relation__species-energy-area-direction-state-and-time-held__interface-method-bounded; exact match True

- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if a transfer coefficient is fitted to close the budget and then presented as a forced Earth law.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json;
 source manifest sha256:dccd344db35ee6fd7ef6bf3102e7d26063cb22086762083893c53551479502c4;
 derivation seal sha256:b0b4d5facf1606fca98162ea02d14761da990401ca2f24e01ed038317ef3dfff;
 engine receipt sha256:47570f3a637629341d5ecc1de42037f109c5c6e3aef57870e9e0964d1a998a8 at
 receipts/engine/model_admitted/SFT-EARTH-AIR-SEA-EXCHANGE-001-47570f3a63762934.json;
 empirical validation sha256:46f9a9c6b893fd47ffa3112043f061750b8951de3b073ef8316eb36af3e92795;
 measurement receipt
 sha256:eb5139deb5e281d2cb61e221078e70f68de9d4f4f354c8e172804c845421d76b; isolation
 sha256:a84ad20b4f72a119a804923391c759257c076c7a49ef3a0a9d2be54d9ca30cc8; custody
 sha256:5b90d8dad9bbc128d26415c7a299c8adbe055419725957a995a328ee741a4bdb.

49. Coastal and tidal Earth-system response

Claim identity: SFT-EARTH-COAST-TIDE-001

Question and exact theorem. A coastal tidal response consumes Astronomy-owned orbital forcing and Physics-owned wave laws while Earth Science owns the measured water, current, sediment and boundary response.

ocean-land-orbital-forcing-interface-carrier__coastal-region-bounded__periodic-forcing-propagation-and-dissipation-relation__geometry-water-level-current-sediment-time-and-handoff-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001

- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-EARTH-IONOSPHERE-RESONANCE-001
- SFT-EARTH-AIR-SEA-EXCHANGE-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	ocean-land-orbital-forcing-interface-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	coastal-region-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	periodic-forcing-propagation-and-dissipation-relation	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	geometry-water-level-current-sediment-time-and-handoff-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: ocean-land-orbital-forcing-interface-carrier__coastal-region-bounded__periodic-forcing-propagation-and-dissipation-relation__geometry-water-level-current-sediment-time-and-handoff-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- carrier-present: the claim retains a nonempty named carrier; passed true.
- relation-distinct: the carrier and relation are not silently conflated; passed true.
- evidence-boundary: the evidence boundary remains distinct from the result carrier; passed true.
- exact-fold-witness: the family's pre-source exact Fold construction reproduces; passed true.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar.

Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

Ocean identity retains composition, depth, stratification, circulation and interfaces. Air-sea transfer is two-sided. Coastal and tidal observations retain geometry, forcing, local response, datum and time rather than treating a site-dependent magnitude as a universal constant.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:f4499bdca4d8ce85559feac716b8291efacb58205f3365dc727322e14cc47eeb.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:ecf5fec1eb9ff7891a57b33540758e52a1be38fef12689776a16f151188ab314.

- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:e862e5b782e02671bfd861dcbc75fbf1c2984b24d53ecbdd6dc65666dca8e4e7.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:27205075f951dafa2d9d07123bfa6ae86ff5e81554d752d311535f9edc714908.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:ffdb4f89cacdf4f01453da77aa635266512851371e13efd5e3fa06c35afb62bd; certificate
sha256:c54ab29f393d2bfe3403b5316be6337f48fec24f5a067f205eedd44668d2c0da; external-validation
identity sha256:7f4fd8feba1280cdbd45b9e2567895550c88b9e91df04e0f80f61ea6fee295da.

Post-seal empirical comparison. Directness:

`authoritative_observation_or_primary_measurement_boundary_correspondence`. Disposition:
`purpose_matched_authoritative_source_features_present`. Predicted consequence: earth:ocean-land-orbital-forcing-interface-carrier__periodic-forcing-propagation-and-dissipation-relation__geometry-water-level-current-sediment-time-and-handoff-held__coastal-region-bounded. Source-derived consequence: earth:ocean-land-orbital-forcing-interface-carrier__periodic-forcing-propagation-and-dissipation-relation__geometry-water-level-current-sediment-time-and-handoff-held__coastal-region-bounded. Exact boundary correspondence: true. External evidence selected the survivor: false. Formal structure relabelled as direct measurement: false. Model, forecast, proxy or retrieval relabelled as observation: false.

Sources and complete feature accounting:

- NOAA-WOD-001 - NOAA National Centers for Environmental Information, [World Ocean Database](#); class `authoritative_ocean_profile_archive`; transport captured; features 5/5 present; present: cast cruise and accession identity, depth location and time, temperature salinity nutrients and plankton, quality flags, original-source provenance; absent and preserved: none.
- ARGO-DATA-001 - Argo Data Management Team, [Argo profile and trajectory data access](#); class `authoritative_ocean_observing_program`; transport captured; features 4/4 present; present: float and profile identity, depth temperature salinity and trajectory, real-time and delayed-mode quality distinction, quality-control flags; absent and preserved: none.
- NSIDC-SEA-ICE-INDEX-001 - National Snow and Ice Data Center, [Sea Ice Index data and image archive](#); class `authoritative_cryosphere_data_product`; transport captured; features 4/4 present; present: hemisphere and date identity, extent and area distinction, daily and monthly data, missing-data and product-version documentation; absent and preserved: none.

Comparison record:

- sft-earth-coast-tide-001-external-earth-evidence: predicted earth:ocean-land-orbital-forcing-interface-carrier__periodic-forcing-propagation-and-dissipation-relation__geometry-water-level-current-sediment-time-and-handoff-held__coastal-region-bounded; source-derived earth:ocean-land-orbital-forcing-interface-carrier__periodic-forcing-propagation-and-dissipation-relation__geometry-water-level-current-sediment-time-and-handoff-held__coastal-region-bounded; exact match True
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if planetary tidal locking is renowned, if a local response is generalised globally, or if shoreline change is inferred from water level alone.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value

- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json;
 source manifest sha256:7ee733af2e59a71a857ddd024282808e3bf9a25296f4dbcef681418e6a48f0bc;
 derivation seal sha256:8b79b5b0413da2d16b4a04974e0087c16c49996d227a4637ad7f129637a7c119;
 engine receipt sha256:0ecc3db9c0519d9908e662e6dbf15c3b0fff8221b69b55fbc05f12a5b0a88f3b at
 receipts/engine/model_admitted/SFT-EARTH-COAST-TIDE-001-0ecc3db9c0519d99.json; empirical
 validation sha256:e2b68425c98ef38106363d03c881165b2b6595dac26ce35aec10fd094ceb51a2;
 measurement receipt
 sha256:5b08fd379d52bd14ec217a2e5b3e1ef3a5479c5e557c9b43c43e3d16f85542f5; isolation
 sha256:fb29d2eed5284dee7f35c630c7d484d6d2281872c2b636b6f725bcae8d5b6b6f; custody
 sha256:9ac19e3e2e6d0a2c3926899157f1a2267df5aabc9ebdac6b974dee5e8ae1a797.

Ledger family 9: Climate, Forcing, Feedback, Attribution and Tipping

Weather and climate differ by retained timescale and state history. Forcing, feedback, variability and attribution require complete counterfactual and evidence records. The exact Fold witness supplies multiple basins and a shared image; it does not fabricate a universal physical half-threshold.

50. Climate state

Claim identity: SFT-EARTH-CLIMATE-001

Question and exact theorem. Climate is a source- and interval-bounded distribution of Earth-system states, variations and extremes rather than one instantaneous weather state or mean.

distribution-of-earth-state-histories-carrier__declared-climate-window-bounded__time-window-recurrence-and-change-relation__variables-distributions-extremes-boundaries-and-methods-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001

- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-COAST-TIDE-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	distribution-of-earth-state-histories-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	declared-climate-window-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	time-window-recurrence-and-change-relation	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
record	favourable-output-with-incomplete-record	variables-distributions-extremes-boundaries-and-methods-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: `distribution-of-earth-state-histories-carrier__declared-climate-window-bounded__time-window-recurrence-and-change-relation__variables-distributions-extremes-boundaries-and-methods-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule`.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `carrier-present`: the claim retains a nonempty named carrier; passed `true`.
- `relation-distinct`: the carrier and relation are not silently conflated; passed `true`.
- `evidence-boundary`: the evidence boundary remains distinct from the result carrier; passed `true`.
- `exact-fold-witness`: the family's pre-source exact Fold construction reproduces; passed `true`.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar.

Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

Weather and climate differ by retained timescale and state history. Forcing, feedback, variability and attribution require complete counterfactual and evidence records. The exact Fold witness supplies multiple basins and a shared image; it does not fabricate a universal physical half-threshold.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:9bfbb67383b07471595d77d9f01c89d7b7924422171df39942bb8f34126aa7a3.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:2281939ca79d4bd1b87d72f7063fe4bf3bf43eac442bac938c84b22218fd125f.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:e9feb160f39bfe09bfdb03114854896e101e7fc0754c12303bc27556e9a06f4.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:0ae4df656a4793a4c867fe451da7fda892b3d36a0233e5fb2ad05d8bd995c3d0.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
 sha256:28957c36027492e83c083f2068da6f44826af469d07933d41ef4d87bd2175186; certificate
 sha256:cee38265cc29bef81325dd53e5effba32b1d8271479ca1e360c41980964c8c8c; external-validation
 identity sha256:c8ccc86342073c0bc99863e8d61f72b9d61fcdf036fb0469e6e2c1616526cb47.

Post-seal empirical comparison. Directness:

authoritative_observation_or_primary_measurement_boundary_correspondence. Disposition:
 purpose_matched_authoritative_source_features_present. Predicted consequence: earth:distribution-of-earth-state-histories-carrier__time-window-recurrence-and-change-relation__variables-distributions-extremes-boundaries-and-methods-held__declared-climate-window-bounded. Source-derived consequence: earth:distribution-of-earth-state-histories-carrier__time-window-recurrence-and-change-relation__variables-distributions-extremes-boundaries-and-methods-held__declared-climate-window-bounded. Exact boundary correspondence: true. External evidence selected the survivor: false. Formal structure relabelled as direct measurement: false. Model, forecast, proxy or retrieval relabelled as observation: false.

Sources and complete feature accounting:

- NOAA-GHCND-001 - NOAA National Centers for Environmental Information, **Global Historical Climatology Network Daily**; class authoritative_station_climate_archive; transport captured; features 5/5 present; present: station identity and metadata, daily weather observations, quality flags, record length and variable coverage, archive version; absent and preserved: none.
- NASA-CERES-EBAF-001 - NASA Langley Research Center CERES, **CERES Energy Balanced and Filled data products**; class authoritative_radiative_budget_data_product; transport captured; features 5/5 present; present: top-of-atmosphere and surface boundary, shortwave and longwave flux distinction, cloud and clear-sky distinction, spatial and temporal resolution, data-quality documentation; absent and preserved: none.
- NOAA-GML-CO2-001 - NOAA Global Monitoring Laboratory, **Carbon-cycle greenhouse-gas measurements**; class authoritative_atmospheric_composition_archive; transport captured; features 3/5 present; present: gas and measurement programme identity, site and altitude, calibration and data-use policy; absent and preserved: time and frequency, source and sink interpretation separated from measurement.
- NSIDC-SEA-ICE-INDEX-001 - National Snow and Ice Data Center, **Sea Ice Index data and image archive**; class authoritative_cryosphere_data_product; transport captured; features 4/4 present; present: hemisphere and date identity, extent and area distinction, daily and monthly data, missing-data and product-version documentation; absent and preserved: none.
- NOAA-PALEO-001 - NOAA National Centers for Environmental Information World Data Service for Paleoclimatology, **Paleoclimatology data archive**; class authoritative_proxy_archive; transport captured; features 3/4 present; present: archive and study identity, proxy and reconstructed-variable distinction, location and investigator provenance; absent and preserved: dating and temporal coverage.

Comparison record:

- sft-earth-climate-001-external-earth-evidence: predicted earth:distribution-of-earth-state-histories-carrier__time-window-recurrence-and-change-relation__variables-distributions-extremes-boundaries-and-methods-held__declared-climate-window-bounded; source-derived earth:distribution-of-earth-state-histories-carrier__time-window-recurrence-and-change-relation__variables-distributions-extremes-boundaries-and-methods-held__declared-climate-window-bounded; exact match True
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if the averaging interval, distribution, extremes, spatial support or observation changes are omitted.

Explicit exclusions.

- semantic numerical zero

- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

```
experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json;
source manifest sha256:e3618a45d67b97bd33c35cabbfd420e352fe4d6f974f87a61cc0e2c5047f8b47;
derivation seal sha256:1f6ac3eb716c192ffe20057d50b456bccfa4aee6dfe72d129cbfd01ca704b7da;
engine receipt sha256:d99699708929f2ac543e9559642735ff08de2dd5b5e4adb589a1880831cd945e at
receipts/engine/model_admitted/SFT-EARTH-CLIMATE-001-d99699708929f2ac.json; empirical
validation sha256:433aabc3b1e20ec76bb25bb664c0598d5a49d8aca578a77e754947fcada70479;
measurement receipt
sha256:528ef63142fb7f5a3ff2c04914809e55a6c1ba3b5edccd12cbf1bcde69cb1aea; isolation
sha256:ca08106107732272da7cae418b7eae7ba16cad5422afc55e8d07b3dfb5c96ad9; custody
sha256:2b92b639a7089c1fe7260b02672b2368cede7b57e9e72301e53e1e3222203f6e.
```

51. Weather-climate timescale distinction

Claim identity: SFT-EARTH-WEATHER-CLIMATE-TIMESCALE-001

Question and exact theorem. Weather and climate are distinct evidence classes linked by declared aggregation and memory relations; neither may be inferred from an unlabeled timescale.

```
nested-time-indexed-state-carrier__timescale-explicit__short-transition-to-long-di
stribution-relation__sampling-window-memory-forcing-and-variability-held__evidence
-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all
-rows__no-extra-rule
```

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001

- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-COAST-TIDE-001
- SFT-EARTH-CLIMATE-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	nested-time-indexed-state-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	timescale-explicit	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	short-transition-to-long-distribution-relation	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	sampling-window-memory-forcing-and-variability-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: nested-time-indexed-state-carrier__timescale-explicit__short-transition-to-long-distribution-relation__sampling-window-memory-forcing-and-variability-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- carrier-present: the claim retains a nonempty named carrier; passed true.
- relation-distinct: the carrier and relation are not silently conflated; passed true.
- evidence-boundary: the evidence boundary remains distinct from the result carrier; passed true.
- exact-fold-witness: the family's pre-source exact Fold construction reproduces; passed true.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar.

Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

Weather and climate differ by retained timescale and state history. Forcing, feedback, variability and attribution require complete counterfactual and evidence records. The exact Fold witness supplies multiple basins and a shared image; it does not fabricate a universal physical half-threshold.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:b52baaa40316cf7dc28a2bbd9722031def4679ffa5edb2d2ca2196f4af67df13.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:9b122f05c782ac1ca0b8d4c4d51cd21cd99328649555645ab1c6f72632f1d6df.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:318eb91dbe3b65ecc46fec76ac93b5c80db517edd13380f6747b6c51cacfbbbc8.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:03e950be9f7240a8a9a753a43fb403527e27c6ce46832e62b744021f32095a72.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:343931a2a01159311b475587cebfaf255a9eb5db291d42a5940fb3efb10e4ce7; certificate

sha256:4fb2221b7b8f8465315aa6009fc5cb08cd6eae2d966bab361a6685e1a8eae6; external-validation
identity sha256:7c0dd7a3d1d8c5b31894d35f9200085e219db9bccf4f327604673d341dd730f0.

Post-seal empirical comparison. Directness:

authoritative_observation_or_primary_measurement_boundary_correspondence. Disposition: purpose_matched_authoritative_source_features_present. Predicted consequence: earth:nested-time-indexed-state-carrier__short-transition-to-long-distribution-relation__sampling-window-memory-forcing-and-variability-held__timescale-explicit. Source-derived consequence: earth:nested-time-indexed-state-carrier__short-transition-to-long-distribution-relation__sampling-window-memory-forcing-and-variability-held__timescale-explicit. Exact boundary correspondence: true. External evidence selected the survivor: false. Formal structure relabelled as direct measurement: false. Model, forecast, proxy or retrieval relabelled as observation: false.

Sources and complete feature accounting:

- NOAA-GHCND-001 - NOAA National Centers for Environmental Information, **Global Historical Climatology Network Daily**; class authoritative_station_climate_archive; transport captured; features 5/5 present; present: station identity and metadata, daily weather observations, quality flags, record length and variable coverage, archive version; absent and preserved: none.
- NASA-CERES-EBAF-001 - NASA Langley Research Center CERES, **CERES Energy Balanced and Filled data products**; class authoritative_radiative_budget_data_product; transport captured; features 5/5 present; present: top-of-atmosphere and surface boundary, shortwave and longwave flux distinction, cloud and clear-sky distinction, spatial and temporal resolution, data-quality documentation; absent and preserved: none.
- NOAA-GML-CO2-001 - NOAA Global Monitoring Laboratory, **Carbon-cycle greenhouse-gas measurements**; class authoritative_atmospheric_composition_archive; transport captured; features 3/5 present; present: gas and measurement programme identity, site and altitude, calibration and data-use policy; absent and preserved: time and frequency, source and sink interpretation separated from measurement.
- NSIDC-SEA-ICE-INDEX-001 - National Snow and Ice Data Center, **Sea Ice Index data and image archive**; class authoritative_cryosphere_data_product; transport captured; features 4/4 present; present: hemisphere and date identity, extent and area distinction, daily and monthly data, missing-data and product-version documentation; absent and preserved: none.
- NOAA-PALEO-001 - NOAA National Centers for Environmental Information World Data Service for Paleoclimatology, **Paleoclimatology data archive**; class authoritative_proxy_archive; transport captured; features 3/4 present; present: archive and study identity, proxy and reconstructed-variable distinction, location and investigator provenance; absent and preserved: dating and temporal coverage.

Comparison record:

- sft-earth-weather-climate-timescale-001-external-earth-evidence: predicted earth:nested-time-indexed-state-carrier__short-transition-to-long-distribution-relation__sampling-window-memory-forcing-and-variability-held__timescale-explicit; source-derived earth:nested-time-indexed-state-carrier__short-transition-to-long-distribution-relation__sampling-window-memory-forcing-and-variability-held__timescale-explicit; exact match True
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if one event proves a climate law or if a long-term distribution is used as a deterministic event forecast.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter

- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json;
 source manifest sha256:bf7405b89cbcf841b25e0c4a80971fc6862ced7750f7cc0482fc611cbbd17d;
 derivation seal sha256:146f041e61ae97b34a911febc6853350cdfb27dcd64a90d8579a159b6c1e945;
 engine receipt sha256:fde55140e6f9ad1022bfad630103a55469f2bbd884740799fa9d09093dc8alc5 at re
 cepts/engine/model_admitted/SFT-EARTH-WEATHER-CLIMATE-TIMESCALE-001-fde55140e6f9ad10
 .json; empirical validation
 sha256:f0f5dbc388f3e339be39716ec485447185395418c2d87eaab5ad254cb94c0083; measurement
 receipt sha256:cfc0d23510ec6a92a280ce9ce6a02d7d201d662a2d27e20444e82a5fd592c25e; isolation
 sha256:3e0b9b1ef18a8a5328c3055581917188e5406c921c69a418627efbad10add9af; custody
 sha256:69e395225766fd63add6d3a74569c5f812238b0307e1edc8e08dbdfbd9ca03e8.

52. Climate forcing

Claim identity: SFT-EARTH-CLIMATE-FORCING-001

Question and exact theorem. A climate forcing is a registered driver that changes a declared Earth-system boundary ledger while retaining identity, timing, spectrum, location and uncertainty.

external-or-boundary-driver-carrier__system-boundary-bounded__driver-to-energy-or-matter-budget-change__identity-magnitude-timing-spectrum-location-and-uncertainty-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-success-or-retains-all-rows__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001

- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-COAST-TIDE-001
- SFT-EARTH-WEATHER-CLIMATE-TIMESCALE-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	external-or-boundary-driver-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	system-boundary-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	driver-to-energy-or-matter-budget-change	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	identity-magnitude-timing-spectrum-location-and-uncertainty-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: external-or-boundary-driver-carrier__system-boundary-bounded__driver-to-energy-or-matter-budget-change__identity-magnitude-timing-spectrum-location-and-uncertainty-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- carrier-present: the claim retains a nonempty named carrier; passed true.
- relation-distinct: the carrier and relation are not silently conflated; passed true.
- evidence-boundary: the evidence boundary remains distinct from the result carrier; passed true.
- exact-fold-witness: the family's pre-source exact Fold construction reproduces; passed true.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar. Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

Weather and climate differ by retained timescale and state history. Forcing, feedback, variability and attribution require complete counterfactual and evidence records. The exact Fold witness supplies multiple basins and a shared image; it does not fabricate a universal physical half-threshold.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:4e0e77ea706f1efda50b056d04c4f5ab71873ac68fdb7abb04c1c6b888536c2e.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:2d60062bb8688dbf2190136f7ccc9e39a4e97d4d5a6b16a3dda45b9abeea87bf.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:036db0b7de429ad3cd207076a7cfb8bfd1ad3214ab0bcd7bbbeff09672df9750.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:068b90341a2ec59b3f1ce6b99b4f9b5fe7da12046b686de68667383506c929ff.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:e77252ef3762bb57e818e846e35aaf402ef198bb97510997b497856da9b66f13; certificate
sha256:52688a53feea9e2c79d88eb2de34d3a12a3247cee248270f5f9088b21a5fec9d; external-validation
identity sha256:fc1b1fa299fa42a2bc70f7ca79945588dae0c2a12cefc1c2a29ec7b7ff767265.

Post-seal empirical comparison. Directness:

authoritative_observation_or_primary_measurement_boundary_correspondence. Disposition:
purpose_matched_authoritative_source_features_present. Predicted consequence: earth:external-or-boundary-driver-carrier__driver-to-energy-or-matter-budget-change__identity-magnitude-timing-spectrum-location-and-uncertainty-held__system-boundary-bounded. Source-derived

consequence: earth:external-or-boundary-driver-carrier__driver-to-energy-or-matter-budget-change__identity-magnitude-timing-spectrum-location-and-uncertainty-held__system-boundary-bounded. Exact boundary correspondence: true. External evidence selected the survivor: false. Formal structure relabelled as direct measurement: false. Model, forecast, proxy or retrieval relabelled as observation: false.

Sources and complete feature accounting:

- NOAA-GHCND-001 - NOAA National Centers for Environmental Information, **Global Historical Climatology Network Daily**; class authoritative_station_climate_archive; transport captured; features 5/5 present; present: station identity and metadata, daily weather observations, quality flags, record length and variable coverage, archive version; absent and preserved: none.
- NASA-CERES-EBAF-001 - NASA Langley Research Center CERES, **CERES Energy Balanced and Filled data products**; class authoritative_radiative_budget_data_product; transport captured; features 5/5 present; present: top-of-atmosphere and surface boundary, shortwave and longwave flux distinction, cloud and clear-sky distinction, spatial and temporal resolution, data-quality documentation; absent and preserved: none.
- NOAA-GML-CO2-001 - NOAA Global Monitoring Laboratory, **Carbon-cycle greenhouse-gas measurements**; class authoritative_atmospheric_composition_archive; transport captured; features 3/5 present; present: gas and measurement programme identity, site and altitude, calibration and data-use policy; absent and preserved: time and frequency, source and sink interpretation separated from measurement.
- NSIDC-SEA-ICE-INDEX-001 - National Snow and Ice Data Center, **Sea Ice Index data and image archive**; class authoritative_cryosphere_data_product; transport captured; features 4/4 present; present: hemisphere and date identity, extent and area distinction, daily and monthly data, missing-data and product-version documentation; absent and preserved: none.
- NOAA-PALEO-001 - NOAA National Centers for Environmental Information World Data Service for Paleoclimatology, **Paleoclimatology data archive**; class authoritative_proxy_archive; transport captured; features 3/4 present; present: archive and study identity, proxy and reconstructed-variable distinction, location and investigator provenance; absent and preserved: dating and temporal coverage.

Comparison record:

- sft-earth-climate-forcing-001-external-earth-evidence: predicted earth:external-or-boundary-driver-carrier__driver-to-energy-or-matter-budget-change__identity-magnitude-timing-spectrum-location-and-uncertainty-held__system-boundary-bounded; source-derived earth:external-or-boundary-driver-carrier__driver-to-energy-or-matter-budget-change__identity-magnitude-timing-spectrum-location-and-uncertainty-held__system-boundary-bounded; exact match True
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if an internal response is relabelled as forcing or if the forcing is inferred solely from the response it is meant to explain.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise

- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json;
 source manifest sha256:f110f2929663e8d937ba48543181cbd20e543e236789dfb72350e193520c7cf4;
 derivation seal sha256:57dbc1f87296c28d6669b132158117f89493ce7e4f1262d29d23d9cddd5dc95f;
 engine receipt sha256:8336679d3056e872210936ddf1d326bec025eed573cc4bc7c2cbd60bda5dc7de at
 receipts/engine/model_admitted/SFT-EARTH-CLIMATE-FORCING-001-8336679d3056e872.json;
 empirical validation sha256:6c684362991a5ee650ea5cb2c129b48e15444b225989b815a1ba5d42a2fe5226;
 measurement receipt
 sha256:8f596c22a4fd8924e3de8e72821e1e40d12d8ff864c8764b1291d025cc190dd5; isolation
 sha256:f66570eb2ddd74f68cbd4cbeafee36b030fd668794e9d1b3e80ea6fe6e79f9d; custody
 sha256:925d9955b2edf78e5a15caa26fc58503e9d0189aa5d0d74aceaa8a79d92c034d.

53. Climate feedback

Claim identity: SFT-EARTH-CLIMATE-FEEDBACK-001

Question and exact theorem. A climate feedback is a closed response path that modifies a registered driver or budget; its orientation, path, delay and boundary must remain structural labels rather than negative proof scalars.

state-response-return-path-carrier__process-timescale-bounded__response-modifies-d
 river-or-budget-relation__sign-as-orientation-gain-path-delay-and-boundaries-held_
 _evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-re
 tains-all-rows__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001

- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-COAST-TIDE-001
- SFT-EARTH-CLIMATE-FORCING-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	state-response-return-path-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	process-timescale-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	response-modifies-driver-or-budget-relation	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	sign-as-orientation-gain-path-delay-and-boundaries-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflicted	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: state-response-return-path-carrier__process-timescale-bounded__response-modifies-driver-or-budget-relation__sign-as-orientation-ga

`in-path-delay-and-boundaries-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule.`

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `carrier-present`: the claim retains a nonempty named carrier; passed `true`.
- `relation-distinct`: the carrier and relation are not silently conflated; passed `true`.
- `evidence-boundary`: the evidence boundary remains distinct from the result carrier; passed `true`.
- `exact-fold-witness`: the family's pre-source exact Fold construction reproduces; passed `true`.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar.

Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

Weather and climate differ by retained timescale and state history. Forcing, feedback, variability and attribution require complete counterfactual and evidence records. The exact Fold witness supplies multiple basins and a shared image; it does not fabricate a universal physical half-threshold.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:a3d1c719899b48945c58ddf85c911f1545a1844f13477c8ec7e19d87dfc7a142.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:a54c920be3f01dbf06cec65a68c27bd8a1768a871a2e1e38b8dde2983a919c37.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:4e68f46a6dcfcc229d2aec808a1a16420b12f5a9bd190df919b1519c081449cb.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:f6057a1aaa6a0fefb1f7219aceb75b2f3444e6b4ee3cc7b562f9a94920d221c2.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:470b7fbb5281765f7f3fbb951d37984c26e5ab552e5b89cba7f77723ba44eb18; certificate
sha256:dbaecf6bce4de9d37aab2e670a5b6a6accffbc6ff3b13b87598fb0a46e478b68; external-validation
identity sha256:0cb43c9c61d48c51ccb9d398cb59ba040c83bf001d1ff379561bfbe9c3121912.

Post-seal empirical comparison. Directness:

`authoritative_observation_or_primary_measurement_boundary_correspondence`. Disposition:
`purpose_matched_authoritative_source_features_present`. Predicted consequence: `earth:state-response-return-path-carrier__response-modifies-driver-or-budget-relation__sign-as-orientation-gain-path-delay-and-boundaries-held__process-timescale-bounded`. Source-derived consequence: `earth:state-response-return-path-carrier__response-modifies-driver-or-budget-relation__sign-as-orientation-gain-path-delay-and-boundaries-held__process-timescale-bounded`. Exact boundary correspondence: `true`. External evidence selected the survivor: `false`. Formal structure relabelled as direct measurement: `false`. Model, forecast, proxy or retrieval relabelled as observation: `false`.

Sources and complete feature accounting:

- NOAA-GHCND-001 - NOAA National Centers for Environmental Information, **Global Historical Climatology Network Daily**; class `authoritative_station_climate_archive`; transport captured; features 5/5 present; present: station identity and metadata, daily weather observations, quality flags, record length and variable coverage, archive version; absent and preserved: none.
- NASA-CERES-EBAF-001 - NASA Langley Research Center CERES, **CERES Energy Balanced and Filled data products**; class `authoritative_radiative_budget_data_product`; transport captured; features 5/5 present; present: top-of-atmosphere and surface boundary, shortwave and longwave flux distinction, cloud and clear-sky distinction, spatial and temporal resolution, data-quality documentation; absent and preserved: none.
- NOAA-GML-CO2-001 - NOAA Global Monitoring Laboratory, **Carbon-cycle greenhouse-gas measurements**; class `authoritative_atmospheric_composition_archive`; transport captured; features 3/5 present; present: gas and measurement programme identity, site and altitude, calibration and data-use policy; absent and preserved: time and frequency, source and sink interpretation separated from measurement.
- NSIDC-SEA-ICE-INDEX-001 - National Snow and Ice Data Center, **Sea Ice Index data and image archive**; class `authoritative_cryosphere_data_product`; transport captured; features 4/4 present; present: hemisphere and date identity, extent and area distinction, daily and monthly data, missing-data and product-version documentation; absent and preserved: none.
- NOAA-PALEO-001 - NOAA National Centers for Environmental Information World Data Service for Paleoclimatology, **Paleoclimatology data archive**; class `authoritative_proxy_archive`; transport captured; features 3/4 present; present: archive and study identity, proxy and reconstructed-variable distinction, location and investigator provenance; absent and preserved: dating and temporal coverage.

Comparison record:

- `sft-earth-climate-feedback-001-external-earth-evidence`: `predicted earth:state-response-return-path-carrier__response-modifies-driver-or-budget-relation__sign-as-orientation-gain-path-delay-and-boundaries-held__process-timescale-bounded`; source-derived `earth:state-response-return-path-carrier__response-modifies-driver-or-budget-relation__sign-as-orientation-gain-path-delay-and-boundaries-held__process-timescale-bounded`; exact match True
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if correlation substitutes for a closed path or if the feedback magnitude is fitted from the same target used to validate it.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json;
 source manifest sha256:0c72c23c6f61725b68130400599d0bdd3c08fbd99671ae3dd9881992a0d5eb62;
 derivation seal sha256:e825ce9fddf04fd2923d938512039d4b10d3f38b63ad78c4a7d4729d4433eef7;
 engine receipt sha256:d0293c869cef7f636ae317f35e7e03e169fcdbecce06bd3b524df427d1939a43 at
 receipts/engine/model_admitted/SFT-EARTH-CLIMATE-FEEDBACK-001-d0293c869cef7f63.json;
 empirical validation sha256:1069095a23e9c9cfd6c00e1e4527a6b49cbd612da7638398f06c6600dbd90579;
 measurement receipt
 sha256:114845022c43023fe7f72ec40dc0b74e7ce92554c9874aaf4335cb5155cf6d9e; isolation
 sha256:9e087a952ebca61e4b18e637fc4284413442c3ada94bd51faf214d8e1115e8c9; custody
 sha256:52e4cf80295ca5d4dd8ce012608eccdd645f4a5a2fd300bb00fdc556bb3d75c4.

54. Internal climate variability

Claim identity: SFT-EARTH-CLIMATE-VARIABILITY-001

Question and exact theorem. Internal climate variability is the source-bounded state-transition structure remaining after registered external drivers are retained, with recurrence and nonrecurrence both recorded.

coupled-state-history-carrier__record-and-scale-bounded__recurrent-and-nonrecurren
 t-transition-partition__modes-phases-forcing-record-length-and-missingness-held__e
 vidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-reta
 ins-all-rows__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001

- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-COAST-TIDE-001
- SFT-EARTH-CLIMATE-FEEDBACK-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	coupled-state-history-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	record-and-scale-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	recurrent-and-nonrecurrent-transition-partition	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	modes-phases-forcing-record-length-and-missingness-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: coupled-state-history-carrier__record-and-scale-bounded__recurrent-and-nonrecurrent-transition-partition__modes-phases-forcing-record-length-and-missingness-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `carrier-present`: the claim retains a nonempty named carrier; passed `true`.
- `relation-distinct`: the carrier and relation are not silently conflated; passed `true`.
- `evidence-boundary`: the evidence boundary remains distinct from the result carrier; passed `true`.
- `exact-fold-witness`: the family's pre-source exact Fold construction reproduces; passed `true`.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar. Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

Weather and climate differ by retained timescale and state history. Forcing, feedback, variability and attribution require complete counterfactual and evidence records. The exact Fold witness supplies multiple basins and a shared image; it does not fabricate a universal physical half-threshold.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:b5ad86101fc00d4c6f8a4f0f40d0767555e3fca169cf9e43ff9fbe914867250.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:e09cdb6701bcbb03b5a10746920dd2d5120c8eb4a679c6163b5dffefc1b333f.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:606dfc129650dde6fde81383c9884ac48427f94df0c447a2632e565adf2dabc1.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:5768156c5975cde4554c21a19ed072dbe91e0de77bcabb8a7252b30d14bfb42.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:a813a9ee9c97dc8c35c1acb3222bd6d44e62909f74371886acc5085fde4b66aa; certificate
sha256:30c4e847511afcbba4294b9b7f2258c330f7b86a7927823e536ff2d803c5fa35; external-validation
identity sha256:3c73827400692bc3a4c73a77ec1f31429f9c80fcae3c2aee4885441f4c94be1e.

Post-seal empirical comparison. Directness:

`authoritative_observation_or_primary_measurement_boundary_correspondence`. Disposition:
`purpose_matched_authoritative_source_features_present`. Predicted consequence: `earth:coupled-state-history-carrier__recurrent-and-nonrecurrent-transition-partition__modes-phases-forcing-record-length-and-missingness-held__record-and-scale-bounded`. Source-derived consequence: `earth:coupled-state-history-carrier__recurrent-and-nonrecurrent-transition-partition__modes-phases-forcing-record-length-and-missingness-held__record-and-scale-bounded`. Exact boundary correspondence: `true`. External evidence selected the survivor: `false`. Formal structure relabelled as direct measurement: `false`. Model, forecast, proxy or retrieval relabelled as observation: `false`.

Sources and complete feature accounting:

- NOAA-GHCND-001 - NOAA National Centers for Environmental Information, [Global Historical Climatology Network Daily](#); class `authoritative_station_climate_archive`; transport captured; features 5/5 present; present: station identity and metadata, daily weather observations, quality flags, record length and variable coverage, archive version; absent and preserved: none.
- NASA-CERES-EBAF-001 - NASA Langley Research Center CERES, [CERES Energy Balanced and Filled data products](#); class `authoritative_radiative_budget_data_product`; transport captured; features 5/5 present; present: top-of-atmosphere and surface boundary, shortwave and longwave flux distinction, cloud and clear-sky distinction, spatial and temporal resolution, data-quality documentation; absent and preserved: none.

- NOAA-GML-CO2-001 - NOAA Global Monitoring Laboratory, [Carbon-cycle greenhouse-gas measurements](#); class `authoritative_atmospheric_composition_archive`; transport captured; features 3/5 present; present: gas and measurement programme identity, site and altitude, calibration and data-use policy; absent and preserved: time and frequency, source and sink interpretation separated from measurement.
- NSIDC-SEA-ICE-INDEX-001 - National Snow and Ice Data Center, [Sea Ice Index data and image archive](#); class `authoritative_cryosphere_data_product`; transport captured; features 4/4 present; present: hemisphere and date identity, extent and area distinction, daily and monthly data, missing-data and product-version documentation; absent and preserved: none.
- NOAA-PALEO-001 - NOAA National Centers for Environmental Information World Data Service for Paleoclimatology, [Paleoclimatology data archive](#); class `authoritative_proxy_archive`; transport captured; features 3/4 present; present: archive and study identity, proxy and reconstructed-variable distinction, location and investigator provenance; absent and preserved: dating and temporal coverage.

Comparison record:

- sft-earth-climate-variability-001-external-earth-evidence: predicted earth:coupled-state-history-carrier__recurrent-and-nonrecurrent-transition-partition__modes-phases-forcing-record-length-and-missingness-held__record-and-scale-bounded; source-derived earth:coupled-state-history-carrier__recurrent-and-nonrecurrent-transition-partition__modes-phases-forcing-record-length-and-missingness-held__record-and-scale-bounded; exact match True
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if unexplained change is automatically labelled internal or if a short record establishes a depth-independent cycle.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json;
 source manifest sha256:5ecf12014ff914d781db9a993b915115bb804d6bed749d1591a39c5d99aac51b;
 derivation seal sha256:41e7951130335a2f385b71925724107b6b959e60524945354360934a56a21e5e;
 engine receipt sha256:5348ca53751d27ca4262738ebecfe7a166730424e91eb42bd46c9647dc950dc6 at receipts/engine/model_admitted/SFT-EARTH-CLIMATE-VARIABILITY-001-5348ca53751d27ca.json;
 empirical validation sha256:32595ccb07a0a73731f01b23efd40f67c8829effe5888674b875b4fedb558d10;
 measurement receipt
 sha256:546ae364d43fea2954afbbae8f0542d5c22ed929c06f01e3ed839ee18efc6871; isolation

sha256:dcdb469d78b8098528730d8afa5bfcd90f152beebf481253c2aa0a17a73b9d6e; custody
 sha256:2cbc80df462055a312f3fd58f1f6c2665c323b1450c119751e6ca812570e5a94.

55. Climate detection and attribution

Claim identity: SFT-EARTH-CLIMATE-ATTRIBUTION-001

Question and exact theorem. Climate attribution requires a preregistered observation, candidate driver set, complete alternatives, controls and source-bound uncertainty; detection of change alone is not attribution.

observation-driver-response-alternative-carrier__question-and-period-bounded__gene
 rated-alternative-elimination-relation__forcings-internal-variation-methods-contro
 ls-and-uncertainty-held__evidence-class-explicit__root-bound-forward-forcing__posi
 tive-finite-successor-retains-all-rows__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001

- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-COAST-TIDE-001
- SFT-EARTH-CLIMATE-VARIABILITY-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	observation-driver-response-alternative-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	question-and-period-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	generated-alternative-elimination-relation	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	forcings-internal-variation-methods-controls-and-uncertainty-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: observation-driver-response-alternative-carrier__question-and-period-bounded__generated-alternative-elimination-relation__forcings-internal-variation-methods-controls-and-uncertainty-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- carrier-present: the claim retains a nonempty named carrier; passed true.
- relation-distinct: the carrier and relation are not silently conflated; passed true.
- evidence-boundary: the evidence boundary remains distinct from the result carrier; passed true.
- exact-fold-witness: the family's pre-source exact Fold construction reproduces; passed true.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar. Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

Weather and climate differ by retained timescale and state history. Forcing, feedback, variability and attribution require complete counterfactual and evidence records. The exact Fold witness supplies multiple basins and a shared image; it does not fabricate a universal physical half-threshold.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:6d2e78b62d489a92bca6fdb48662c9515c53b714d73ee30c498e30b871ee67a3.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:11e009a98d6d6581e23b10924f2a09a1bf589b99e61511846d84963ebb3d23ce.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:09691283cd0f16c0153a06d9d343731e6b2ae07ca67dbb8bd2b6ef6835ce5dbb.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:343f26fcc603f7939adde5edb6fe63a0607d2f552fc49d06228b16ac1fb212a2.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:c9da3a11f73f00009fb0798d4459c331fbf632c78926f9d0cdb796cd624a821e; certificate
sha256:ffb6b295064452d58a917adf2d805d6951a984c82ea51b43f3f8c8c36fc24c96; external-validation
identity sha256:31aa965badf7b0c066a97d64fbd5acdc11a2c39c4febb24b5a0e0eaf59b51007.

Post-seal empirical comparison. Directness:

authoritative_observation_or_primary_measurement_boundary_correspondence. Disposition:
purpose_matched_authoritative_source_features_present. Predicted consequence: earth:observati
on-driver-response-alternative-carrier__generated-alternative-elimination-relation__f
orcings-internal-variation-methods-controls-and-uncertainty-held__question-and-period
-bounded. Source-derived consequence: earth:observation-driver-response-alternative-carrier__g
enerated-alternative-elimination-relation__forcings-internal-variation-methods-contro
ls-and-uncertainty-held__question-and-period-bounded. Exact boundary correspondence: true.
External evidence selected the survivor: false. Formal structure relabelled as direct measurement: false. Model, forecast,
proxy or retrieval relabelled as observation: false.

Sources and complete feature accounting:

- NOAA-GHCND-001 - NOAA National Centers for Environmental Information, **Global Historical Climatology Network Daily**; class authoritative_station_climate_archive; transport captured; features 5/5 present; present: station identity and metadata, daily weather observations, quality flags, record length and variable coverage, archive version; absent and preserved: none.
- NASA-CERES-EBAF-001 - NASA Langley Research Center CERES, **CERES Energy Balanced and Filled data products**; class authoritative_radiative_budget_data_product; transport captured; features 5/5 present; present: top-of-atmosphere and surface boundary, shortwave and longwave flux distinction, cloud and clear-sky distinction, spatial and temporal resolution, data-quality documentation; absent and preserved: none.
- NOAA-GML-CO2-001 - NOAA Global Monitoring Laboratory, **Carbon-cycle greenhouse-gas measurements**; class authoritative_atmospheric_composition_archive; transport captured; features 3/5 present; present: gas and measurement programme identity, site and altitude, calibration and data-use policy; absent and preserved: time and frequency, source and sink interpretation separated from measurement.
- NSIDC-SEA-ICE-INDEX-001 - National Snow and Ice Data Center, **Sea Ice Index data and image archive**; class authoritative_cryosphere_data_product; transport captured; features 4/4 present; present: hemisphere

and date identity, extent and area distinction, daily and monthly data, missing-data and product-version documentation; absent and preserved: none.

- NOAA-PALEO-001 - NOAA National Centers for Environmental Information World Data Service for Paleoclimatology, **Paleoclimatology data archive**; class authoritative_proxy_archive; transport captured; features 3/4 present; present: archive and study identity, proxy and reconstructed-variable distinction, location and investigator provenance; absent and preserved: dating and temporal coverage.

Comparison record:

- sft-earth-climate-attribution-001-external-earth-evidence: predicted earth:observation-driver-response-alternative-carrier__generated-alternative-elimination-relation__forcings-internal-variation-methods-controls-and-uncertainty-held__question-and-period-bounded; source-derived earth:observation-driver-response-alternative-carrier__generated-alternative-elimination-relation__forcings-internal-variation-methods-controls-and-uncertainty-held__question-and-period-bounded; exact match True
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if the tested driver set omits a viable alternative, if a model ensemble selects its own target, or if disagreement and failed transport rows are removed.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json;
 source manifest sha256:0d81d3fa80c379c534a55eb56707b739aac76e8d5608d250741a95ce3be6de4a;
 derivation seal sha256:59e1464cc8a2e380d129fd72a4acdd9557124565bba8502750586ca9000796fd;
 engine receipt sha256:5fe64d4e0896c6844feb6762a7a74cee285c7a2104488379a5cdf9c543ad4f4d at receipts/engine/model_admitted/SFT-EARTH-CLIMATE-ATTRIBUTION-001-5fe64d4e0896c684.json;
 empirical validation sha256:6fcb5d71219cec96d539c9532e6e5c8976be1c8518729061cbf8290d7915ce5d;
 measurement receipt
 sha256:494433200a04c38bdd7ef493bb24cc8c8d821f57225080ef1d1465ea86a0bb58; isolation
 sha256:f63c6c9c0e3e7e3eec3838c20707727b52d587890e5d493f7ec45af3ebb7ba7b; custody
 sha256:e56e59fb7c04bc8584063a46099e74053517acde9e8b56f344d679e17999dc1d.

56. Earth-system tipping boundary

Claim identity: SFT-EARTH-EARTH-SYSTEM-TIPPING-001

Question and exact theorem. An Earth-system tipping claim requires observed or independently reconstructed basins, forcing path, threshold crossing, hysteresis or return behavior and ordinary-variability controls; no universal physical half-One threshold is assumed.

```
multi-basin-state-forcing-carrier__system-proxy-and-timescale-bounded__threshold-crossing-hysteresis-and-return-relation__state-variable-driver-path-rate-controls-and-time-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule
```

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-COAST-TIDE-001

- SFT-EARTH-CLIMATE-ATTRIBUTION-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	multi-basin-state-forcing-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	system-proxy-and-timescale-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	threshold-crossing-hysteresis-and-return-relation	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	state-variable-driver-path-rate-controls-and-time-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: multi-basin-state-forcing-carrier__system-proxy-and-timescale-bounded__threshold-crossing-hysteresis-and-return-relation__state-variable-driver-path-rate-controls-and-time-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- carrier-present: the claim retains a nonempty named carrier; passed true.
- relation-distinct: the carrier and relation are not silently conflated; passed true.
- evidence-boundary: the evidence boundary remains distinct from the result carrier; passed true.
- exact-fold-witness: the family's pre-source exact Fold construction reproduces; passed true.

Scientific meaning. The exact two-basin witness forces distinguishable basins sharing an image and supports path dependence. It explicitly rejects conversion of the abstract half-One holding coordinate into a universal measured Earth-system threshold.

Weather and climate differ by retained timescale and state history. Forcing, feedback, variability and attribution require complete counterfactual and evidence records. The exact Fold witness supplies multiple basins and a shared image; it does not fabricate a universal physical half-threshold.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:3a85b24f39ac4b39dcb54a4f4958a6eaaa640c20f3bd6cc3037e741e7275a375.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:e293cde917080a7c9548aad24a8e173bc5bcc8793066a65bac44fb1704930a3a.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:alac6fa33eef93212f7654eb0b7eca9568f448c409424cc70997e0c2af44c6e3.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:32419c8d0036430929dac2d816138cd43eaa4095114bc8853af2b7cebbbfafa274.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:1b3b2379cabb30485306d74b4f31e65a9e3e2c360cbcf61fa9e7bf66598817c5; certificate
sha256:c81989b188aeb1ae9a2fa5f24543076fe08b9a2820afee99457d0b7dd721d3da; external-validation
identity sha256:5c99bf979c615eachbb8de914abeeb09a58524d91bc95cd8393566572b3350b71.

Post-seal empirical comparison. Directness:

authoritative_observation_or_primary_measurement_boundary_correspondence. Disposition:
purpose_matched_authoritative_source_features_present. Predicted consequence: earth:multi-bas
in-state-forcing-carrier__threshold-crossing-hysteresis-and-return-relation__state-va
riable-driver-path-rate-controls-and-time-held__system-proxy-and-timescale-bounded__t
wo-basin-structure-no-universal-physical-half-threshold. Source-derived consequence: earth:mul
ti-basin-state-forcing-carrier__threshold-crossing-hysteresis-and-return-relation__st
ate-variable-driver-path-rate-controls-and-time-held__system-proxy-and-timescale-boun
ded__two-basin-structure-no-universal-physical-half-threshold. Exact boundary correspondence:
true. External evidence selected the survivor: false. Formal structure relabelled as direct measurement: false. Model,
forecast, proxy or retrieval relabelled as observation: false.

Sources and complete feature accounting:

- NOAA-PALEO-001 - NOAA National Centers for Environmental Information World Data Service for Paleoclimatology, [Paleoclimatology data archive](#); class authoritative_proxy_archive; transport captured; features 3/4 present; present: archive and study identity, proxy and reconstructed-variable distinction, location and investigator provenance; absent and preserved: dating and temporal coverage.
- NOAA-GHCND-001 - NOAA National Centers for Environmental Information, [Global Historical Climatology Network Daily](#); class authoritative_station_climate_archive; transport captured; features 5/5 present; present: station identity and metadata, daily weather observations, quality flags, record length and variable coverage, archive version; absent and preserved: none.
- NSIDC-SEA-ICE-INDEX-001 - National Snow and Ice Data Center, [Sea Ice Index data and image archive](#); class authoritative_cryosphere_data_product; transport captured; features 4/4 present; present: hemisphere and date identity, extent and area distinction, daily and monthly data, missing-data and product-version documentation; absent and preserved: none.

Comparison record:

- sft-earth-earth-system-tipping-001-external-earth-evidence: predicted earth:multi-basin-state-forcing-carrier__threshold-crossing-hysteresis-and-return-relation__state-variable-driver-path-rate-controls-and-time-held__system-proxy-and-timescale-bounded__two-basin-structure-no-universal-physical-half-threshold; source-derived earth:multi-basin-state-forcing-carrier

`__threshold-crossing-hysteresis-and-return-relation__state-variable-driver-path-rate-controls-and-time-held__system-proxy-and-timescale-bounded__two-basin-structure-no-universal-physical-half-threshold; exact match True`

- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if abrupt change alone establishes tipping, if the threshold is read from the validation target, or if reversible and ordinary-variable alternatives survive.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json;
 source manifest sha256:7438b2dabf613e89eed1b2add8a15c4dd2abdd51aecc0558fccbffc82a527595;
 derivation seal sha256:186605d03001baac2b820b876856449173e7217e588570a76dee02561fe2b395;
 engine receipt sha256:af4ee0a52a6067727c14b16376c790f8338df98389581f09f063583e069fd1eb at receipts/engine/model_admitted/SFT-EARTH-EARTH-SYSTEM-TIPPING-001-af4ee0a52a606772.json;
 empirical validation
 sha256:2b31a0334c0928e32b9d37f4c782effb2e3b0856e0ac2471746f52a12ede7144; measurement
 receipt sha256:73b7062f54c6adc8ed521f12ac7811555087b1cf3e0c9978e787b9405044369b; isolation
 sha256:ceaaf922f0470c0e410e9988b384de3ce94a63914c7420bd4f4ee87b697ed3ca; custody
 sha256:a0429b9dbdea722fcb1a70a3b62da78305dc0a541e3294657975401e7d374143.

Ledger family 10: Biogeochemical and Ecological Coupling

Carbon, nitrogen and phosphorus cycles retain chemical species, reservoirs, organisms, transformations and flows. Biosphere exchange is explicitly two-way. Ecology remains at the Biology handoff: environmental carriers constrain biological response without re-owning universal biological law.

57. Earth carbon cycle

Claim identity: SFT-EARTH-CARBON-CYCLE-001

Question and exact theorem. The Earth carbon cycle is a species- and reservoir-resolved transformation and transfer network with isotopic, stock, flow, boundary and uncertainty records retained.

`carbon-species-reservoir-network__earth-carbon-domain-bounded__transformation-storage-and-transfer-relation__species-isotopes-stocks-flows-boundaries-and-time-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-re`

tains-all-rows__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-EARTH-SYSTEM-TIPPING-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	carbon-species-reservoir-network	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	earth-carbon-domain-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	transformation-storage-and-transfer-relation	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	species-isotopes-stocks-flows-boundaries-and-time-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: carbon-species-reservoir-network__earth-carbon-domain-bounded__transformation-storage-and-transfer-relation__species-isotopes-stocks-flows-boundaries-and-time-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- carrier-present: the claim retains a nonempty named carrier; passed true.
- relation-distinct: the carrier and relation are not silently conflated; passed true.
- evidence-boundary: the evidence boundary remains distinct from the result carrier; passed true.
- exact-fold-witness: the family's pre-source exact Fold construction reproduces; passed true.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar.

Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

Carbon, nitrogen and phosphorus cycles retain chemical species, reservoirs, organisms, transformations and flows. Biosphere exchange is explicitly two-way. Ecology remains at the Biology handoff: environmental carriers constrain biological response without re-owning universal biological law.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:540d2e2f10e4cc49a2fe38b833e625b0281fa8edea0c89bf474adfc3cc48cc64.

- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:8a6131b05edf4356db4c58f74846c663cc435ababdb987f5d9b220b7595f1967.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:7429110b1bd31620fe634375b0ff949d27a9ac07d6284e8273fbc94b1dc3791d.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:ed86f9a75ef1a604372bb4821f50f8fd7278173a611a77df41279d4c4669c4de.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:203005f5457faa497366e0290b3070f855548161e59e9574086a740cbbf06d8c; certificate
sha256:c77df0d9d2090be8c9d34f2cd572759051513ba5a252fb008ff218d689660122; external-validation
identity sha256:bb2cacef9bb8cd32877576de1796f2ddf4198050b26211c6367ecbce257f1e40.

Post-seal empirical comparison. Directness:

authoritative_observation_or_primary_measurement_boundary_correspondence. Disposition:
purpose_matched_authoritative_source_features_present. Predicted consequence: earth:carbon-sp
ecies-reservoir-network__transformation-storage-and-transfer-relation__species-isotop
es-stocks-flows-boundaries-and-time-held__earth-carbon-domain-bounded. Source-derived
consequence: earth:carbon-species-reservoir-network__transformation-storage-and-transfer-
relation__species-isotopes-stocks-flows-boundaries-and-time-held__earth-carbon-domain
-bounded. Exact boundary correspondence: true. External evidence selected the survivor: false. Formal structure
relabelled as direct measurement: false. Model, forecast, proxy or retrieval relabelled as observation: false.

Sources and complete feature accounting:

- NOAA-GML-CO2-001 - NOAA Global Monitoring Laboratory, [Carbon-cycle greenhouse-gas measurements](#); class
authoritative_atmospheric_composition_archive; transport captured; features 3/5 present; present:
gas and measurement programme identity, site and altitude, calibration and data-use policy; absent and preserved: time and
frequency, source and sink interpretation separated from measurement.
- NOAA-WOD-001 - NOAA National Centers for Environmental Information, [World Ocean Database](#); class
authoritative_ocean_profile_archive; transport captured; features 5/5 present; present: cast cruise and
accession identity, depth location and time, temperature salinity nutrients and plankton, quality flags, original-source
provenance; absent and preserved: none.
- WQP-WEB-SERVICES-001 - U.S. Geological Survey, U.S. Environmental Protection Agency and National Water
Quality Monitoring Council, [Water Quality Portal web services](#); class
authoritative_water_quality_federation; transport captured; features 4/4 present; present: station and
organization identity, sample and result identity, characteristic and unit, quality and metadata fields; absent and preserved:
none.
- NEON-OPEN-DATA-001 - U.S. National Science Foundation National Ecological Observatory Network, [NEON open
ecosystem data](#); class authoritative_ecosystem_observation_program; transport captured; features 4/4
present; present: site and data-product identity, organismal and abiotic observations, collection method, availability and
latency; absent and preserved: none.
- GBIF-OCCURRENCE-API-001 - Global Biodiversity Information Facility, [GBIF Occurrence API](#); class
authoritative_biodiversity_data_federation; transport captured; features 2/4 present; present:
occurrence and dataset identity, publishing organization; absent and preserved: taxon location date and basis of record,
paging limits and download distinction.

Comparison record:

- sft-earth-carbon-cycle-001-external-earth-evidence: predicted earth:carbon-species-reservoir-network__transformation-stor
age-and-transfer-relation__species-isotopes-stocks-flows-boundaries-and-time-held__earth-carbon-domain-bounded;
source-derived earth:carbon-species-reservoir-network__transformation-storage-and-transfer-relation__species-isotopes-st
ocks-flows-boundaries-and-time-held__earth-carbon-domain-bounded; exact match True

- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if chemically distinct carriers are merged, if an unmeasured sink closes the budget by assumption, or if biological uptake is rederived rather than cited.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json;
 source manifest sha256:2806ef6f701f258de55c3a1d8208e20d7f25dfadbbac779cb70c415c2241e814;
 derivation seal sha256:eac7dd0f86c2abb1a9efb0d44009efb4378097dce6be056a8a73909ce918cc66;
 engine receipt sha256:a0ccc2c74781615163b5773ed202120c3aac7d1941dad35734a11598251392f4 at
 receipts/engine/model_admitted/SFT-EARTH-CARBON-CYCLE-001-a0ccc2c747816151.json;
 empirical validation sha256:b2adf65f579e0ee230f5ab1cae2dc1f7bc0e8ee5adc532c26d8b3afa84d9f500;
 measurement receipt
 sha256:0d9e8648542afe5add8765f7bd2be0ad5477ea19a6450cfe4fc78af81a209b15; isolation
 sha256:866ff5cc49a05b6ae7adabdee33bf255efcafb0d4ce45660f0a546a6023ec16; custody
 sha256:1cf161cf2320fb3c777ac9271ca830951ff3cf82a60534548c41fdb3c9c7da21.

58. Earth nitrogen cycle

Claim identity: SFT-EARTH-NITROGEN-CYCLE-001

Question and exact theorem. The Earth nitrogen cycle retains each chemical species, abiotic and biological transformation, reservoir, transfer and observation boundary.

nitrogen-species-reservoir-network__earth-nitrogen-domain-bounded__transformation-
 storage-and-transfer-relation__species-isotopes-organisms-stocks-flows-and-time-he
 ld__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor
 -retains-all-rows__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001

- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-EARTH-SYSTEM-TIPPING-001
- SFT-EARTH-CARBON-CYCLE-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	nitrogen-species-reservoir-network	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	earth-nitrogen-domain-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	transformation-storage-and-transfer-relation	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	species-isotopes-organisms-stocks-flows-and-time-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: `nitrogen-species-reservoir-network__earth-nitrogen-domain-bounded__transformation-storage-and-transfer-relation__species-isotopes-organisms-stocks-flows-and-time-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule`.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `carrier-present`: the claim retains a nonempty named carrier; passed `true`.
- `relation-distinct`: the carrier and relation are not silently conflated; passed `true`.
- `evidence-boundary`: the evidence boundary remains distinct from the result carrier; passed `true`.
- `exact-fold-witness`: the family's pre-source exact Fold construction reproduces; passed `true`.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar.

Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

Carbon, nitrogen and phosphorus cycles retain chemical species, reservoirs, organisms, transformations and flows. Biosphere exchange is explicitly two-way. Ecology remains at the Biology handoff: environmental carriers constrain biological response without re-owning universal biological law.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
`sha256:cc02efeee9fd9b361dafdec1166ccd487bf4d33463855b1ed8d798c3e62e17e6`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
`sha256:86da56bb1c20d2bc9ae956a9eaa337d5648a1ef7d3ae6ee4194d860885ae3ba1`.

- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:7afd403d7f42da838cc44230d0019c4c9c890660c37b152c26a5f493c6eab7d2.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:96b0b5ffdc547bb6a5a23f36490dca1bcf616458c0cd9c30452a23cd851ea872.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:46bac82b0d3a85241e642766e56b67980343cce94d19afc0f32ba2266234dd6a; certificate
sha256:29c76bb742fffad5415168cd83f9f83996be7274df1ca02f9a7bea6c53ce7d92; external-validation
identity sha256:a804b2c6802e2ac650bcc143290a6046b1e3a359a35b6647645c6a08aa72c814.

Post-seal empirical comparison. Directness:

`authoritative_observation_or_primary_measurement_boundary_correspondence`. Disposition:
`purpose_matched_authoritative_source_features_present`. Predicted consequence: `earth:nitrogen-species-reservoir-network__transformation-storage-and-transfer-relation__species-isotopes-organisms-stocks-flows-and-time-held__earth-nitrogen-domain-bounded`. Source-derived consequence: `earth:nitrogen-species-reservoir-network__transformation-storage-and-transfer-relation__species-isotopes-organisms-stocks-flows-and-time-held__earth-nitrogen-domain-bounded`. Exact boundary correspondence: `true`. External evidence selected the survivor: `false`. Formal structure relabelled as direct measurement: `false`. Model, forecast, proxy or retrieval relabelled as observation: `false`.

Sources and complete feature accounting:

- NOAA-GML-CO2-001 - NOAA Global Monitoring Laboratory, [Carbon-cycle greenhouse-gas measurements](#); class `authoritative_atmospheric_composition_archive`; transport captured; features 3/5 present; present: gas and measurement programme identity, site and altitude, calibration and data-use policy; absent and preserved: time and frequency, source and sink interpretation separated from measurement.
- NOAA-WOD-001 - NOAA National Centers for Environmental Information, [World Ocean Database](#); class `authoritative_ocean_profile_archive`; transport captured; features 5/5 present; present: cast cruise and accession identity, depth location and time, temperature salinity nutrients and plankton, quality flags, original-source provenance; absent and preserved: none.
- WQP-WEB-SERVICES-001 - U.S. Geological Survey, U.S. Environmental Protection Agency and National Water Quality Monitoring Council, [Water Quality Portal web services](#); class `authoritative_water_quality_federation`; transport captured; features 4/4 present; present: station and organization identity, sample and result identity, characteristic and unit, quality and metadata fields; absent and preserved: none.
- NEON-OPEN-DATA-001 - U.S. National Science Foundation National Ecological Observatory Network, [NEON open ecosystem data](#); class `authoritative_ecosystem_observation_program`; transport captured; features 4/4 present; present: site and data-product identity, organismal and abiotic observations, collection method, availability and latency; absent and preserved: none.
- GBIF-OCCURRENCE-API-001 - Global Biodiversity Information Facility, [GBIF Occurrence API](#); class `authoritative_biodiversity_data_federation`; transport captured; features 2/4 present; present: occurrence and dataset identity, publishing organization; absent and preserved: taxon location date and basis of record, paging limits and download distinction.

Comparison record:

- `sft-earth-nitrogen-cycle-001-external-earth-evidence`: predicted `earth:nitrogen-species-reservoir-network__transformation-storage-and-transfer-relation__species-isotopes-organisms-stocks-flows-and-time-held__earth-nitrogen-domain-bounded`; source-derived `earth:nitrogen-species-reservoir-network__transformation-storage-and-transfer-relation__species-isotopes-organisms-stocks-flows-and-time-held__earth-nitrogen-domain-bounded`; exact match `True`
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved

- a deliberately changed external consequence was rejected

Falsification boundary: Reject if oxidation states, biological processes or atmospheric and aquatic boundaries are silently merged.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json;
 source manifest sha256:0642885947567f4c223d1af3ed5edacb869d2295553731a69db2b07f3622f900;
 derivation seal sha256:d1a1d3c85470ec02e5801c56651586b46c985582f822c4ce76bdc9b0809c76a3;
 engine receipt sha256:0db8e35b2dda3e781c0aa20c5fc1727f0861e19640e5f214d1ba4e6163707e45 at
 receipts/engine/model_admitted/SFT-EARTH-NITROGEN-CYCLE-001-0db8e35b2dda3e78.json;
 empirical validation sha256:31af4720001466b56dca0c16d4a4238943cb45c72e20d55cb81fe86fe6f11b44;
 measurement receipt
 sha256:79e36d22c2e3b1d9a78d1ace595ddebcec8207104c9759d891cca3821a23e885; isolation
 sha256:222951a52ef4bb11037fa80ed83e386fdc660bc453c935674702ec69340a4ec5; custody
 sha256:0fa9bdea6a18fec0cc330b3b0301f866cdab552bcacf0d0eb38e52358758fd12.

59. Earth phosphorus cycle

Claim identity: SFT-EARTH-PHOSPHORUS-CYCLE-001

Question and exact theorem. The Earth phosphorus cycle composes geological, chemical and biological carriers while retaining weathering, uptake, transfer, burial, return and boundary exports.

phosphorus-species-rock-soil-water-biota-network__earth-phosphorus-domain-bounded_
 _weathering-uptake-transfer-return-relation__species-stocks-flows-organisms-sedime
 nt-and-time-held__evidence-class-explicit__root-bound-forward-forcing__positive-fi
 nite-successor-retains-all-rows__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001

- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-EARTH-SYSTEM-TIPPING-001
- SFT-EARTH-NITROGEN-CYCLE-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	phosphorus-species-rock-soil-water-biota-network	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	earth-phosphorus-domain-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
relation	imported-fitted-or-missing-relation	weathering-uptake-transfer-return-relation	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	species-stocks-flows-organisms-sediment-and-time-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: phosphorus-species-rock-soil-water-biota-network__earth-phosphorus-domain-bounded__weathering-uptake-transfer-return-relation__species-stocks-flows-organisms-sediment-and-time-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- carrier-present: the claim retains a nonempty named carrier; passed true.
- relation-distinct: the carrier and relation are not silently conflated; passed true.
- evidence-boundary: the evidence boundary remains distinct from the result carrier; passed true.
- exact-fold-witness: the family's pre-source exact Fold construction reproduces; passed true.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar.

Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

Carbon, nitrogen and phosphorus cycles retain chemical species, reservoirs, organisms, transformations and flows. Biosphere exchange is explicitly two-way. Ecology remains at the Biology handoff: environmental carriers constrain biological response without re-owning universal biological law.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:c168e0a1d92761ce2181739868cbd382e7d8dee64d6d10116259bf3b2602beff.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:21f1672171b48584dd91e871ca2bfb93de039a428696ba75ba36877cfe220dc7.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:f9b2c45e1e2ff9e334d57b22d4fc7d7a9dbc15a47c8a55d0d9d3abade6e6efb3.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt

sha256:e789069fd1b5b67699d8c7b654b8d4d4b22362e1b55ae2bcc163f7014f15614b.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:cd4f79fe72c46e774b48f3bc2f326ff4a188068cfb5881fe78772c0ff2e797b2; certificate

sha256:57ec2dc589143f8c22e869fb3f51c10a7b1f14bf1f622502c263b528b2cf014c; external-validation identity sha256:666231ddb9f01bd5842f263ce2cde7b3f796e941c277cdc44110efb2d29ecf38.

Post-seal empirical comparison. Directness:

authoritative_observation_or_primary_measurement_boundary_correspondence. Disposition: purpose_matched_authoritative_source_features_present. Predicted consequence: earth:phosphorus-species-rock-soil-water-biota-network__weathering-uptake-transfer-return-relation__species-stocks-flows-organisms-sediment-and-time-held__earth-phosphorus-domain-bounded. Source-derived consequence: earth:phosphorus-species-rock-soil-water-biota-network__weathering-uptake-transfer-return-relation__species-stocks-flows-organisms-sediment-and-time-held__earth-phosphorus-domain-bounded. Exact boundary correspondence: true. External evidence selected the survivor: false. Formal structure relabelled as direct measurement: false. Model, forecast, proxy or retrieval relabelled as observation: false.

Sources and complete feature accounting:

- NOAA-GML-CO2-001 - NOAA Global Monitoring Laboratory, **Carbon-cycle greenhouse-gas measurements**; class authoritative_atmospheric_composition_archive; transport captured; features 3/5 present; present: gas and measurement programme identity, site and altitude, calibration and data-use policy; absent and preserved: time and frequency, source and sink interpretation separated from measurement.
- NOAA-WOD-001 - NOAA National Centers for Environmental Information, **World Ocean Database**; class authoritative_ocean_profile_archive; transport captured; features 5/5 present; present: cast cruise and accession identity, depth location and time, temperature salinity nutrients and plankton, quality flags, original-source provenance; absent and preserved: none.
- WQP-WEB-SERVICES-001 - U.S. Geological Survey, U.S. Environmental Protection Agency and National Water Quality Monitoring Council, **Water Quality Portal web services**; class authoritative_water_quality_federation; transport captured; features 4/4 present; present: station and organization identity, sample and result identity, characteristic and unit, quality and metadata fields; absent and preserved: none.
- NEON-OPEN-DATA-001 - U.S. National Science Foundation National Ecological Observatory Network, **NEON open ecosystem data**; class authoritative_ecosystem_observation_program; transport captured; features 4/4 present; present: site and data-product identity, organismal and abiotic observations, collection method, availability and latency; absent and preserved: none.
- GBIF-OCCURRENCE-API-001 - Global Biodiversity Information Facility, **GBIF Occurrence API**; class authoritative_biodiversity_data_federation; transport captured; features 2/4 present; present: occurrence and dataset identity, publishing organization; absent and preserved: taxon location date and basis of record, paging limits and download distinction.

Comparison record:

- sft-earth-phosphorus-cycle-001-external-earth-evidence: predicted earth:phosphorus-species-rock-soil-water-biota-network__weathering-uptake-transfer-return-relation__species-stocks-flows-organisms-sediment-and-time-held__earth-phosphorus-domain-bounded; source-derived earth:phosphorus-species-rock-soil-water-biota-network__weathering-uptake-transfer-return-relation__species-stocks-flows-organisms-sediment-and-time-held__earth-phosphorus-domain-bounded; exact match True
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if a named cycle asserts atmospheric equivalence or complete return without evidence.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json;
 source manifest sha256:d406c3bad960f14047d7b3ea6c8c6f0eca363fdb0122cc00c9c1cfb802a6b028;
 derivation seal sha256:7dc6ea939a805768c5b12870385722b5c0af5d4c49ad8c3b7201a040086b8c41;
 engine receipt sha256:49344c713351324ec81cbd41d0f6363b370d81d4ebc257c45b30664f743c599e at
 receipts/engine/model_admitted/SFT-EARTH-PHOSPHORUS-CYCLE-001-49344c713351324e.json;
 empirical validation sha256:16c52192c86770c969d6c759ecb4deece4c2f5656eedd1001313fd8edf19c346;
 measurement receipt
 sha256:0b466c223032939f4a9f6a39d04cbc0fe50bdb6358b858e75b762674c72c172; isolation
 sha256:203efdc1c405a70cfc424d1ab6ac5a48547804096117526685f9d054e9228fbe; custody
 sha256:011752385cccc97f0e8456a64da18dd03cca188fdd8e1f97acf983d492607a85.

60. Biosphere-Earth exchange

Claim identity: SFT-EARTH-BIOSPHERE-EXCHANGE-001

Question and exact theorem. Biosphere-Earth exchange cites Biology-owned organismal processes while Earth Science owns their measured coupling to atmosphere, water, soil and rock reservoirs.

organism-community-abiotic-interface-carrier__ecosystem-boundary-bounded__two-way-
 resource-waste-and-state-transfer__biological-owner-reservoir-carrier-rate-time-an
 d-controls-held__evidence-class-explicit__root-bound-forward-forcing__positive-fin
 ite-successor-retains-all-rows__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001

- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-EARTH-SYSTEM-TIPPING-001
- SFT-EARTH-PHOSPHORUS-CYCLE-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	organism-community-abiotic-interface-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	ecosystem-boundary-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	two-way-resource-waste-and-state-transfer	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
record	favourable-output-with-incomplete-record	biological-owner-reservoir-carrier-rate-time-and-control-s-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: organism-community-abiotic-interface-carrier__ecosystem-boundary-bounded__two-way-resource-waste-and-state-transfer__biological-owner-reservoir-carrier-rate-time-and-controls-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- carrier-present: the claim retains a nonempty named carrier; passed true.
- relation-distinct: the carrier and relation are not silently conflated; passed true.
- evidence-boundary: the evidence boundary remains distinct from the result carrier; passed true.
- exact-fold-witness: the family's pre-source exact Fold construction reproduces; passed true.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar.

Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

Carbon, nitrogen and phosphorus cycles retain chemical species, reservoirs, organisms, transformations and flows. Biosphere exchange is explicitly two-way. Ecology remains at the Biology handoff: environmental carriers constrain biological response without re-owning universal biological law.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:99117f88d8e7a400eb80c0be7ef828e1bce9ccc5d32b6c4d443c70b965dd4f86.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:e257ec8dbea69ef99fbeb039371acf96262b1905f79f776192212cc9664d3809.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:67b2130867bc886171c941a38189894c1918f6ff222d48aea3c705ac596efa39.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:cb036c3583f228a988d96e69bf93ee38793891b6484e14f72159f8006cb27ee0.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:ff92215355700a836670a7dc8b3d8dd507792303f2b7188586ef699d2b0f7842; certificate sha256:995a772d6f1d0135d9d9ce9235de0479ca39d92e00a213fdda4951c40a266acb; external-validation identity sha256:bdeee39a88068595b06265f581821f82f23dc2c6921162e39b45027da9b8a2cb.

Post-seal empirical comparison. Directness:

authoritative_observation_or_primary_measurement_boundary_correspondence. Disposition: purpose_matched_authoritative_source_features_present. Predicted consequence: earth:organism-community-abiotic-interface-carrier__two-way-resource-waste-and-state-transfer__biological-owner-reservoir-carrier-rate-time-and-controls-held__ecosystem-boundary-bounded. Source-derived consequence: earth:organism-community-abiotic-interface-carrier__two-way-resource-waste-and-state-transfer__biological-owner-reservoir-carrier-rate-time-and-controls-held__ecosystem-boundary-bounded. Exact boundary correspondence: true. External evidence selected the survivor: false. Formal structure relabelled as direct measurement: false. Model, forecast, proxy or retrieval relabelled as observation: false.

Sources and complete feature accounting:

- NOAA-GML-CO2-001 - NOAA Global Monitoring Laboratory, **Carbon-cycle greenhouse-gas measurements**; class authoritative_atmospheric_composition_archive; transport captured; features 3/5 present; present: gas and measurement programme identity, site and altitude, calibration and data-use policy; absent and preserved: time and frequency, source and sink interpretation separated from measurement.
- NOAA-WOD-001 - NOAA National Centers for Environmental Information, **World Ocean Database**; class authoritative_ocean_profile_archive; transport captured; features 5/5 present; present: cast cruise and accession identity, depth location and time, temperature salinity nutrients and plankton, quality flags, original-source provenance; absent and preserved: none.
- WQP-WEB-SERVICES-001 - U.S. Geological Survey, U.S. Environmental Protection Agency and National Water Quality Monitoring Council, **Water Quality Portal web services**; class authoritative_water_quality_federation; transport captured; features 4/4 present; present: station and organization identity, sample and result identity, characteristic and unit, quality and metadata fields; absent and preserved: none.
- NEON-OPEN-DATA-001 - U.S. National Science Foundation National Ecological Observatory Network, **NEON open ecosystem data**; class authoritative_ecosystem_observation_program; transport captured; features 4/4 present; present: site and data-product identity, organismal and abiotic observations, collection method, availability and latency; absent and preserved: none.
- GBIF-OCCURRENCE-API-001 - Global Biodiversity Information Facility, **GBIF Occurrence API**; class authoritative_biodiversity_data_federation; transport captured; features 2/4 present; present: occurrence and dataset identity, publishing organization; absent and preserved: taxon location date and basis of record, paging limits and download distinction.

Comparison record:

- sft-earth-biosphere-exchange-001-external-earth-evidence: predicted earth:organism-community-abiotic-interface-carrier__two-way-resource-waste-and-state-transfer__biological-owner-reservoir-carrier-rate-time-and-controls-held__ecosystem-boundary-bounded; source-derived earth:organism-community-abiotic-interface-carrier__two-way-resource-waste-and-state-transfer__biological-owner-reservoir-carrier-rate-time-and-controls-held__ecosystem-boundary-bounded; exact match True
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if Earth Science reowns biological mechanism or if an ecosystem observation is generalised without organism and environmental boundaries.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json;
 source manifest sha256:9907d81783024f2b6ea61a7e932ff3f9e3ac41f0673643be63187e0924a307f4;
 derivation seal sha256:3b39deed07fcf4c58cd2e45fa245349b1d8b2bb8a130623752199d5bf9feedad;
 engine receipt sha256:e074043cb8f77fff8031ebb3d85be04d9d1d4a2772c7449a39cb63f9729d5e19 at re
 cepts/engine/model_admitted/SFT-EARTH-BIOSPHERE-EXCHANGE-001-e074043cb8f77fff.json;
 empirical validation sha256:d413c58a3baeb310ce91cbbd451bde42588e5eed38141485a8439380c6741a73;
 measurement receipt
 sha256:a169a3fb86f74f8534ee1c87cfdc756f75f5c56dce45404ee68599451a1e51ad; isolation
 sha256:c5a1f7f888eca635eebb8b4535ff50f709f2f2ec32c3dc09a8f2d29fdcdcael1; custody
 sha256:c6cb4daeb8af451d31b38e82ef310e8a9318188fd05ae749e3414fedaec6d38b.

61. Coupled ecosystem-Earth organisation

Claim identity: SFT-EARTH-COUPLED-ECOSYSTEM-001

Question and exact theorem. A coupled ecosystem-Earth claim retains Biology-owned communities together with abiotic stocks, transfers, disturbances, observation effort and missing links.

community-abiotic-reservoir-joint-carrier__ecosystem-observation-bounded__biotic-a
 biotic-transfer-and-feedback-network__species-resources-disturbance-boundaries-eff
 ort-and-time-held__evidence-class-explicit__root-bound-forward-forcing__positive-f
 inite-successor-retains-all-rows__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001

- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-EARTH-SYSTEM-TIPPING-001
- SFT-EARTH-BIOSPHERE-EXCHANGE-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	community-abiotic-reservoir-joint-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	ecosystem-observation-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	biotic-abiotic-transfer-and-feedback-network	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	species-resources-disturbance-boundaries-effort-and-time-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: community-abiotic-reservoir-joint-carrier__ecosystem-observation-bounded__biotic-abiotic-transfer-and-feedback-network__species-resources-disturbance-boundaries-effort-and-time-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- carrier-present: the claim retains a nonempty named carrier; passed true.
- relation-distinct: the carrier and relation are not silently conflated; passed true.
- evidence-boundary: the evidence boundary remains distinct from the result carrier; passed true.
- exact-fold-witness: the family's pre-source exact Fold construction reproduces; passed true.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar.

Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

Carbon, nitrogen and phosphorus cycles retain chemical species, reservoirs, organisms, transformations and flows. Biosphere exchange is explicitly two-way. Ecology remains at the Biology handoff: environmental carriers constrain biological response without re-owning universal biological law.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:786c00624caebf133074070a69d64c8dae4de0c45c7d70f9aec0519ed162bd4e.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:19fc92e3658f7b108aa6f4bce66205f3c0c23283ff6504f4ca4617c1eb2e4894.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:d108bad8c9dc332357ce6eda849faf5f104fb543ffedd4ab57fec12cbaaa9a82.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:40558aa62028f77018985b219997cf405a12fc57082d062ed436d253ab2f5204.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:b3af1936ad38739d30761c4d41cca17add980d421d1a2a0a92e9d79ae3d7ef02; certificate

sha256:6359f3b4cbe9cecd596e7207658e26b2571602ce6612c092363e96b1851e0f7; external-validation identity sha256:e18dbd4d99584728e2738faae3a9989ea635b6c43ba6bee8b55714a7c997d7bb.

Post-seal empirical comparison. Directness:

authoritative_observation_or_primary_measurement_boundary_correspondence. Disposition: purpose_matched_authoritative_source_features_present. Predicted consequence: earth:community-abiotic-reservoir-joint-carrier__biotic-abiotic-transfer-and-feedback-network__species-resources-disturbance-boundaries-effort-and-time-held__ecosystem-observation-bounded. Source-derived consequence: earth:community-abiotic-reservoir-joint-carrier__biotic-abiotic-transfer-and-feedback-network__species-resources-disturbance-boundaries-effort-and-time-held__ecosystem-observation-bounded. Exact boundary correspondence: true. External evidence selected the survivor: false. Formal structure relabelled as direct measurement: false. Model, forecast, proxy or retrieval relabelled as observation: false.

Sources and complete feature accounting:

- NOAA-GML-CO2-001 - NOAA Global Monitoring Laboratory, **Carbon-cycle greenhouse-gas measurements**; class authoritative_atmospheric_composition_archive; transport captured; features 3/5 present; present: gas and measurement programme identity, site and altitude, calibration and data-use policy; absent and preserved: time and frequency, source and sink interpretation separated from measurement.
- NOAA-WOD-001 - NOAA National Centers for Environmental Information, **World Ocean Database**; class authoritative_ocean_profile_archive; transport captured; features 5/5 present; present: cast cruise and accession identity, depth location and time, temperature salinity nutrients and plankton, quality flags, original-source provenance; absent and preserved: none.
- WQP-WEB-SERVICES-001 - U.S. Geological Survey, U.S. Environmental Protection Agency and National Water Quality Monitoring Council, **Water Quality Portal web services**; class authoritative_water_quality_federation; transport captured; features 4/4 present; present: station and organization identity, sample and result identity, characteristic and unit, quality and metadata fields; absent and preserved: none.
- NEON-OPEN-DATA-001 - U.S. National Science Foundation National Ecological Observatory Network, **NEON open ecosystem data**; class authoritative_ecosystem_observation_program; transport captured; features 4/4 present; present: site and data-product identity, organismal and abiotic observations, collection method, availability and latency; absent and preserved: none.
- GBIF-OCCURRENCE-API-001 - Global Biodiversity Information Facility, **GBIF Occurrence API**; class authoritative_biodiversity_data_federation; transport captured; features 2/4 present; present: occurrence and dataset identity, publishing organization; absent and preserved: taxon location date and basis of record, paging limits and download distinction.

Comparison record:

- sft-earth-coupled-ecosystem-001-external-earth-evidence: predicted earth:community-abiotic-reservoir-joint-carrier__biotic-abiotic-transfer-and-feedback-network__species-resources-disturbance-boundaries-effort-and-time-held__ecosystem-observation-bounded; source-derived earth:community-abiotic-reservoir-joint-carrier__biotic-abiotic-transfer-and-feedback-network__species-resources-disturbance-boundaries-effort-and-time-held__ecosystem-observation-bounded; exact match True
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if ecological recurrence alone establishes planetary stability or if absent observations become absent interactions.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity

- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json;
 source manifest sha256:dacb473220e8ecdec552ebf8de52a5ea6b849b4868a423895825049392a4ab6b;
 derivation seal sha256:c54d7321d8a075f2a18633366e57822b170ec3c02eca382c3341d5e6f384b97d;
 engine receipt sha256:c0ca6cd5d9c6bf0d091bd1331e7c2da327e807b75fcc6386c8c11f636db6ab18 at
 receipts/engine/model_admitted/SFT-EARTH-COUPLED-ECOSYSTEM-001-c0ca6cd5d9c6bf0d.json;
 empirical validation sha256:5dee86f1a3c9a9c6cb4aa671c8fb842fa894078c7d7c887954e46059a6c69628;
 measurement receipt
 sha256:b7c9180e3e5f1eab9825f68e59d64794162c671f32ae750c759dbda7c2888102; isolation
 sha256:eba1308560e34e7c3c5c770e9def04187bfca24378a084e3be0de1dad1db68b1; custody
 sha256:08b5c976603384061bf21ff124f78f0f8189a980c3b7a02b3ee4de89071a4de0.

62. Biodiversity-environment boundary

Claim identity: SFT-EARTH-BIODIVERSITY-ENVIRONMENT-001

Question and exact theorem. A biodiversity-environment relation is a sampling- and design-bounded comparison retaining taxonomic identity, habitat, pressure, response, confounders and missingness.

taxon-trait-habitat-pressure-carrier__study-design-bounded__environmental-conditions-to-biological-response-comparison__sampling-taxonomy-space-time-confounders-and-missingness-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001

- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-EARTH-SYSTEM-TIPPING-001
- SFT-EARTH-COUPLED-ECOSYSTEM-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	taxon-trait-habitat-pressure-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	study-design-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	environmental-condition-to-biological-response-comparison	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	sampling-taxonomy-space-time-confounders-and-missingness-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: `taxon-trait-habitat-pressure-carrier__study-design-bounded__environmental-condition-to-biological-response-comparison__sampling-taxonomy-space-time-confounders-and-missingness-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule`.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `carrier-present`: the claim retains a nonempty named carrier; passed `true`.
- `relation-distinct`: the carrier and relation are not silently conflated; passed `true`.
- `evidence-boundary`: the evidence boundary remains distinct from the result carrier; passed `true`.
- `exact-fold-witness`: the family's pre-source exact Fold construction reproduces; passed `true`.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar.

Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

Carbon, nitrogen and phosphorus cycles retain chemical species, reservoirs, organisms, transformations and flows. Biosphere exchange is explicitly two-way. Ecology remains at the Biology handoff: environmental carriers constrain biological response without re-owning universal biological law.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:6a2f8b3c34c1d146616d93da474deec0b351677437c0f99bec1fb90441c5d1c0.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:fef1e6a12cae651de9deb8defb7cdc98c501f4cf25f37da6e0b9dfad4b2e3c84.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:4b87bcaaff6cdd6bdea2e129f40c42f229b72dc37c1aed209d0e4158206daade.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:5288dd652b3e86d0bc546990c3a84e75cc237b896bcd72c3c651da1663594157.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:0a9a98007653765610c8df2400ccf46bf47dd8f4371c6e7e7ad9af2dcb2d7bf4; certificate
sha256:262b4faf553978221ed907222ea7f847d710dc8818f3a752e8f27f9383b61b2f; external-validation
identity sha256:d42cac98f1b4954ea036dbbbb0b60f2795de254b58d15c65652740ebef61c10a.

Post-seal empirical comparison. Directness:

authoritative_observation_or_primary_measurement_boundary_correspondence. Disposition: purpose_matched_authoritative_source_features_present. Predicted consequence: earth:taxon-trait-habitat-pressure-carrier__environmental-condition-to-biological-response-comparison__sampling-taxonomy-space-time-confounders-and-missingness-held__study-design-bounded. Source-derived consequence: earth:taxon-trait-habitat-pressure-carrier__environmental-condition-to-biological-response-comparison__sampling-taxonomy-space-time-confounders-and-missingness-held__study-design-bounded. Exact boundary correspondence: true. External evidence selected the survivor: false. Formal structure relabelled as direct measurement: false. Model, forecast, proxy or retrieval relabelled as observation: false.

Sources and complete feature accounting:

- NOAA-GML-CO2-001 - NOAA Global Monitoring Laboratory, **Carbon-cycle greenhouse-gas measurements**; class authoritative_atmospheric_composition_archive; transport captured; features 3/5 present; present: gas and measurement programme identity, site and altitude, calibration and data-use policy; absent and preserved: time and frequency, source and sink interpretation separated from measurement.
- NOAA-WOD-001 - NOAA National Centers for Environmental Information, **World Ocean Database**; class authoritative_ocean_profile_archive; transport captured; features 5/5 present; present: cast cruise and accession identity, depth location and time, temperature salinity nutrients and plankton, quality flags, original-source provenance; absent and preserved: none.
- WQP-WEB-SERVICES-001 - U.S. Geological Survey, U.S. Environmental Protection Agency and National Water Quality Monitoring Council, **Water Quality Portal web services**; class authoritative_water_quality_federation; transport captured; features 4/4 present; present: station and organization identity, sample and result identity, characteristic and unit, quality and metadata fields; absent and preserved: none.
- NEON-OPEN-DATA-001 - U.S. National Science Foundation National Ecological Observatory Network, **NEON open ecosystem data**; class authoritative_ecosystem_observation_program; transport captured; features 4/4 present; present: site and data-product identity, organismal and abiotic observations, collection method, availability and latency; absent and preserved: none.
- GBIF-OCCURRENCE-API-001 - Global Biodiversity Information Facility, **GBIF Occurrence API**; class authoritative_biodiversity_data_federation; transport captured; features 2/4 present; present: occurrence and dataset identity, publishing organization; absent and preserved: taxon location date and basis of record, paging limits and download distinction.

Comparison record:

- sft-earth-biodiversity-environment-001-external-earth-evidence: predicted earth:taxon-trait-habitat-pressure-carrier__environmental-condition-to-biological-response-comparison__sampling-taxonomy-space-time-confounders-and-missingness-held__study-design-bounded; source-derived earth:taxon-trait-habitat-pressure-carrier__environmental-condition-to-biological-response-comparison__sampling-taxonomy-space-time-confounders-and-missingness-held__study-design-bounded; exact match True
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if correlation becomes causation or if unequal observation effort is treated as biological difference.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity

- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json;
 source manifest sha256:f7b376e695430c111c6794f3eedf74fff4c8de0e86067f02b3d4aff04a6988e;
 derivation seal sha256:53ffa71344c84ff186b482c85c2781d36b7387498cf734b69fd2c2ac86cb66f7;
 engine receipt sha256:9eca1b410e560534940ec34d28887c1c46ff8e9ebcde85549c8ac1f1ac0b0945 at re
 ceipts/engine/model_admitted/SFT-EARTH-BIODIVERSITY-ENVIRONMENT-001-9eca1b410e560534.
 json; empirical validation
 sha256:fd5151d1eba0b33e5013f4311f4b4482a28be3e1b6966b40b41d962b9ec2e13c; measurement
 receipt sha256:30cad6abd890bf122d16a665be470dd211b1f5acf2b0d17fc622ada499e7f901; isolation
 sha256:53ae06d392918ef4581b2ba6cd6d20b4d076dbed98b6556e5e185704a2e3ea4e; custody
 sha256:2117f18566545994378a292435f10b140b6c63fc37c9d99512155747ec6fdf9d.

Ledger family 11: Environmental Transport, Transformation and Quality

A pollutant requires substance, form, source, medium, exposure and endpoint. Source-path-receptor chains retain transport, transformation, accumulation and release. Environmental quality is purpose- and jurisdiction-bounded; a threshold cannot silently become a universal natural law.

63. Pollutant identity and boundary

Claim identity: SFT-EARTH-POLLUTANT-IDENTITY-001

Question and exact theorem. A pollutant is a source- and context-bound material form associated with a declared exposure and adverse endpoint; substance identity alone does not establish harm.

substance-form-source-context-carrier__environment-health-handoff-bounded__harm-re
 levant-exposure-classification__chemical-form-source-medium-dose-time-and-endpoint
 -held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-succes
 sor-retains-all-rows__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001

- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-BIODIVERSITY-ENVIRONMENT-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	substance-form-source-context-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	environment-health-handoff-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	harm-relevant-exposure-classification	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	chemical-form-source-medium-dose-time-and-endpoint-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: substance-form-source-context-carrier__environment-health-handoff-bounded__harm-relevant-exposure-classification__chemical-form-source-medium-dose-time-and-endpoint-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- carrier-present: the claim retains a nonempty named carrier; passed true.
- relation-distinct: the carrier and relation are not silently conflated; passed true.
- evidence-boundary: the evidence boundary remains distinct from the result carrier; passed true.
- exact-fold-witness: the family's pre-source exact Fold construction reproduces; passed true.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar. Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

A pollutant requires substance, form, source, medium, exposure and endpoint. Source-path-receptor chains retain transport, transformation, accumulation and release. Environmental quality is purpose- and jurisdiction-bounded; a threshold cannot silently become a universal natural law.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:db50f0823b0e40f63756cae1e4010d9f45f9258091405204ab62102ed5c2dfb2.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:a3378437bd2e3c342f1742295578bca7ef92712289fc1102b694b6f001b52b4b.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:b64a5a71b9e08b7db184825c01626fc810611e6f1af152655ceefb322537997f.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:6f108819d9566f6bceeacdde55f0190c2cc6a9ad3362cf8452b10a24c85002cd.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:c92c48305303d7a1a734cac95a7a18f4adb8b89dd9824283e41f9d2ed01eff4f; certificate

sha256:bcc761aeb26d45a02355988444ece03613c8c90828ff18817e68a36a385c98cd; external-validation identity sha256:44004389dfd3280d93a1a109ca463f11bc414adfaf600522ef926b8bc3221f7a.

Post-seal empirical comparison. Directness:

authoritative_observation_or_primary_measurement_boundary_correspondence. Disposition: purpose_matched_authoritative_source_features_present. Predicted consequence: earth:substance-form-source-context-carrier__harm-relevant-exposure-classification__chemical-form-source-medium-dose-time-and-endpoint-held__environment-health-handoff-bounded. Source-derived consequence: earth:substance-form-source-context-carrier__harm-relevant-exposure-classification__chemical-form-source-medium-dose-time-and-endpoint-held__environment-health-handoff-bounded. Exact boundary correspondence: true. External evidence selected the survivor: false. Formal structure relabelled as direct measurement: false. Model, forecast, proxy or retrieval relabelled as observation: false.

Sources and complete feature accounting:

- EPA-AIRDATA-001 - U.S. Environmental Protection Agency, [AirData pre-generated monitoring files](#); class authoritative_air_quality_archive; transport captured; features 4/5 present; present: site and monitor identity, pollutant parameter and units, hourly daily and annual distinction, blank and missing records; absent and preserved: method and quality metadata.
- WQP-WEB-SERVICES-001 - U.S. Geological Survey, U.S. Environmental Protection Agency and National Water Quality Monitoring Council, [Water Quality Portal web services](#); class authoritative_water_quality_federation; transport captured; features 4/4 present; present: station and organization identity, sample and result identity, characteristic and unit, quality and metadata fields; absent and preserved: none.
- USGS-LANDSAT-ACCESS-001 - U.S. Geological Survey Landsat Missions, [Landsat Data Access](#); class authoritative_remote_sensing_archive; transport captured; features 0/4 present; present: none; absent and preserved: sensor and mission identity, Earth surface observation, archive access, product and processing distinction.
- NEON-OPEN-DATA-001 - U.S. National Science Foundation National Ecological Observatory Network, [NEON open ecosystem data](#); class authoritative_ecosystem_observation_program; transport captured; features 4/4 present; present: site and data-product identity, organismal and abiotic observations, collection method, availability and latency; absent and preserved: none.

Comparison record:

- sft-earth-pollutant-identity-001-external-earth-evidence: predicted earth:substance-form-source-context-carrier__harm-relevant-exposure-classification__chemical-form-source-medium-dose-time-and-endpoint-held__environment-health-handoff-bounded; source-derived earth:substance-form-source-context-carrier__harm-relevant-exposure-classification__chemical-form-source-medium-dose-time-and-endpoint-held__environment-health-handoff-bounded; exact match True
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if chemical forms, exposure, dose, endpoint or Medicine-owned health evidence are omitted.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor

- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json;
 source manifest sha256:29a3261f9c7043990e97e831df6abf0d996f26bf35e1aa94c18c2e1b123fa097;
 derivation seal sha256:fd55fc88094860b22c45ef5b74dd8f180af5f90cc02d334fd0ad3d91641f28d2;
 engine receipt sha256:f70ce5119a12f8ad926fd279bbbe6fcb49b650d666e604dff2f6e586b7d1f993 at re
 ceipts/engine/model_admitted/SFT-EARTH-POLLUTANT-IDENTITY-001-f70ce5119a12f8ad.json;
 empirical validation sha256:ba4c39ef7f1e24f40bab7e9f6821455cdc165b4f107fabad2a3e74b1d4d07116;
 measurement receipt
 sha256:5e6cef45fac3e438aec699c791762a481c349fe4fe9d9c6a23f1044748a46c57; isolation
 sha256:a0488e3a99a4037a90e2dc6c67bdaaeab5458e6d3816b4aa989224661442de04; custody
 sha256:9bfca37fea35c5e9cdf2447d6ee6eef65ab35e6ef74bf8a62609ebdedb33a1d1.

64. Environmental source-path-receptor chain

Claim identity: SFT-EARTH-SOURCE-PATH-RECEPTOR-001

Question and exact theorem. An environmental exposure claim requires a reconstructible source-to-path-to-receptor chain retaining carrier identity, transformations, timing, alternatives and controls.

source-medium-path-receptor-carrier__system-and-study-bounded__causal-transfer-cha
 in__identity-route-timing-transformation-exposure-and-controls-held__evidence-clas
 s-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows
 __no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001

- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-BIODIVERSITY-ENVIRONMENT-001
- SFT-EARTH-POLLUTANT-IDENTITY-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	source-medium-path-receptor-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	system-and-study-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	causal-transfer-chain	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	identity-route-timing-transformation-exposure-and-controls-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: source-medium-path-receptor-carrier__system-and-study-bounded__causal-transfer-chain__identity-route-timing-transformation-exposure-and-controls-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- carrier-present: the claim retains a nonempty named carrier; passed true.
- relation-distinct: the carrier and relation are not silently conflated; passed true.
- evidence-boundary: the evidence boundary remains distinct from the result carrier; passed true.
- exact-fold-witness: the family's pre-source exact Fold construction reproduces; passed true.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar. Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

A pollutant requires substance, form, source, medium, exposure and endpoint. Source-path-receptor chains retain transport, transformation, accumulation and release. Environmental quality is purpose- and jurisdiction-bounded; a threshold cannot silently become a universal natural law.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:ada42adf2130038abcb4cb2f81aal36002674181af8fee0d9ed09f5e6ab60f1.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:95724c5032a48f18c902ee140d5b5f2dae1337ef6d2daf08be96e7ad64f8f964.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:3c30ef4f22a7a0f43675b147bb4ad75eb496b04bb02f2da644de3dde9917a797.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:b1118cc33a8c4c2bd7d801fbaeddcblf0619f9ab2410870741ec27ef0a096af4.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:3c0c6be0df0ddb0b42262e15d42623cffab0175abf3bbf9da2da174d475a7385; certificate
sha256:cf54ee8a59597dcb12203541c6f497824ca1343ee48179c7e26208a423b1ef47; external-validation
identity sha256:eb6ea0675c3be487d4ae613e406b05223845ac13accb012b0d0d1fc83b4190b4.

Post-seal empirical comparison. Directness:

authoritative_observation_or_primary_measurement_boundary_correspondence. Disposition: purpose_matched_authoritative_source_features_present. Predicted consequence: earth:source-medium-path-receptor-carrier__causal-transfer-chain__identity-route-timing-transformation-exposure-and-controls-held__system-and-study-bounded. Source-derived consequence: earth:source-medium-path-receptor-carrier__causal-transfer-chain__identity-route-timing-transformation-exposure-and-controls-held__system-and-study-bounded. Exact boundary correspondence: true. External evidence selected the survivor: false. Formal structure relabelled as direct measurement: false. Model, forecast, proxy or retrieval relabelled as observation: false.

Sources and complete feature accounting:

- EPA-AIRDATA-001 - U.S. Environmental Protection Agency, **AirData pre-generated monitoring files**; class `authoritative_air_quality_archive`; transport captured; features 4/5 present; present: site and monitor identity, pollutant parameter and units, hourly daily and annual distinction, blank and missing records; absent and preserved: method and quality metadata.
- WQP-WEB-SERVICES-001 - U.S. Geological Survey, U.S. Environmental Protection Agency and National Water Quality Monitoring Council, **Water Quality Portal web services**; class `authoritative_water_quality_federation`; transport captured; features 4/4 present; present: station and organization identity, sample and result identity, characteristic and unit, quality and metadata fields; absent and preserved: none.
- USGS-LANDSAT-ACCESS-001 - U.S. Geological Survey Landsat Missions, **Landsat Data Access**; class `authoritative_remote_sensing_archive`; transport captured; features 0/4 present; present: none; absent and preserved: sensor and mission identity, Earth surface observation, archive access, product and processing distinction.
- NEON-OPEN-DATA-001 - U.S. National Science Foundation National Ecological Observatory Network, **NEON open ecosystem data**; class `authoritative_ecosystem_observation_program`; transport captured; features 4/4 present; present: site and data-product identity, organismal and abiotic observations, collection method, availability and latency; absent and preserved: none.

Comparison record:

- `sft-earth-source-path-receptor-001-external-earth-evidence: predicted earth:source-medium-path-receptor-carrier__causal-transfer-chain__identity-route-timing-transformation-exposure-and-controls-held__system-and-study-bounded; source-derived earth:source-medium-path-receptor-carrier__causal-transfer-chain__identity-route-timing-transformation-exposure-and-controls-held__system-and-study-bounded`; exact match True
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if co-location or correlation substitutes for the transfer chain.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

`experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json`;
 source manifest sha256:0ac2fb837a10700e23a7ee44041f3b7fb7b7ecb3efbd53181988d7059c9a1a13;
 derivation seal sha256:a3f3c988a4c2d775f7e2e88679e1edae8076fe60618f93457521eb5e9e50c48c;

engine receipt sha256:29ff886be12a107ecbcbe035eb5573a52bc661557faa4c0bfafd9ff7f28a8727 at receipts/engine/model_admitted/SFT-EARTH-SOURCE-PATH-RECEPTOR-001-29ff886be12a107e.json ; empirical validation sha256:eca0981c67c514088b9db9ea6627cbb37c5af9d41c54ea8b5e1ea1257097e722; measurement receipt sha256:7f3c4752a09593c8816228109b358149f75c29999a91c8525512a46e87056499; isolation sha256:f4e41237617b421bedd92e4d5993b1f59a33eee337b539ed67af499f79babf46; custody sha256:b78ebe0ee0715abddb3b17dcd755de8aff0a5e2ef00646b3c399ee188c69ff.

65. Environmental contaminant transport

Claim identity: SFT-EARTH-ENVIRONMENTAL-TRANSPORT-001

Question and exact theorem. Environmental transport is a carrier-resolved pathway network that cites upstream physical and material laws and retains phase, direction, boundaries, transformations and uncertainty.

material-medium-pathway-network__site-and-medium-bounded__advection-dispersion-partition-and-transfer-relation__species-phase-direction-boundaries-time-and-uncertainty-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001

- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-BIODIVERSITY-ENVIRONMENT-001
- SFT-EARTH-SOURCE-PATH-RECEPTOR-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	material-medium-pathway-network	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	site-and-medium-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	advection-dispersion-partition-and-transfer-relation	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	species-phase-direction-boundaries-time-and-uncertainty-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: material-medium-pathway-network__site-and-medium-bounded__advection-dispersion-partition-and-transfer-relation__species-phase-direction-boundaries-time-and-uncertainty-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `carrier-present`: the claim retains a nonempty named carrier; passed `true`.
- `relation-distinct`: the carrier and relation are not silently conflated; passed `true`.
- `evidence-boundary`: the evidence boundary remains distinct from the result carrier; passed `true`.
- `exact-fold-witness`: the family's pre-source exact Fold construction reproduces; passed `true`.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar. Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

A pollutant requires substance, form, source, medium, exposure and endpoint. Source-path-receptor chains retain transport, transformation, accumulation and release. Environmental quality is purpose- and jurisdiction-bounded; a threshold cannot silently become a universal natural law.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:ea9f1689f2ed08369e70d3a0fe60593525c9f79a90a6634baf5c6be60fc6ebde.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:c3ac6ff9e593b40cf7ec4320a743a880b7e094526020e889215615903c49cd12.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:f20d6bac41d2056a2e6cf82b29b1d65bfba9dd757c25a718e115bba34333c7b9.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:69df3cb1bb83e7311c78c854d2a4bc91551a857b3e5b4c2119720adea99a82a0.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:1ba46c9a7d847cec1a07437fcab944358e19a1cbe319d9b1d4eec745fc2f7ebb; certificate
sha256:a53ebfc2c6872ce474e1b25c80cf9ed6148dce696a7ef1bf96db1fc49495e54f; external-validation
identity sha256:64b98c831112b388ec6726d804f10f8e8dca803dee9e9264b2db6f66697bf8a7.

Post-seal empirical comparison. Directness:

`authoritative_observation_or_primary_measurement_boundary_correspondence`. Disposition:
`purpose_matched_authoritative_source_features_present`. Predicted consequence: `earth:material-medium-pathway-network__advection-dispersion-partition-and-transfer-relation__species-phase-direction-boundaries-time-and-uncertainty-held__site-and-medium-bounded`.

Source-derived consequence: `earth:material-medium-pathway-network__advection-dispersion-partition-and-transfer-relation__species-phase-direction-boundaries-time-and-uncertainty-held__site-and-medium-bounded`. Exact boundary correspondence: `true`. External evidence selected the survivor: `false`. Formal structure relabelled as direct measurement: `false`. Model, forecast, proxy or retrieval relabelled as observation: `false`.

Sources and complete feature accounting:

- EPA-AIRDATA-001 - U.S. Environmental Protection Agency, [AirData pre-generated monitoring files](#); class `authoritative_air_quality_archive`; transport captured; features 4/5 present; present: site and monitor identity, pollutant parameter and units, hourly daily and annual distinction, blank and missing records; absent and preserved: method and quality metadata.
- WQP-WEB-SERVICES-001 - U.S. Geological Survey, U.S. Environmental Protection Agency and National Water Quality Monitoring Council, [Water Quality Portal web services](#); class `authoritative_water_quality_federation`; transport captured; features 4/4 present; present: station and organization identity, sample and result identity, characteristic and unit, quality and metadata fields; absent and preserved: none.

- USGS-LANDSAT-ACCESS-001 - U.S. Geological Survey Landsat Missions, **Landsat Data Access**; class authoritative_remote_sensing_archive; transport captured; features 0/4 present; present: none; absent and preserved: sensor and mission identity, Earth surface observation, archive access, product and processing distinction.
- NEON-OPEN-DATA-001 - U.S. National Science Foundation National Ecological Observatory Network, **NEON open ecosystem data**; class authoritative_ecosystem_observation_program; transport captured; features 4/4 present; present: site and data-product identity, organismal and abiotic observations, collection method, availability and latency; absent and preserved: none.

Comparison record:

- sft-earth-environmental-transport-001-external-earth-evidence: predicted earth:material-medium-pathway-network__advection-dispersion-partition-and-transfer-relation__species-phase-direction-boundaries-time-and-uncertainty-held__site-and-medium-bounded; source-derived earth:material-medium-pathway-network__advection-dispersion-partition-and-transfer-relation__species-phase-direction-boundaries-time-and-uncertainty-held__site-and-medium-bounded; exact match True
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if a fitted transport coefficient replaces missing path evidence or if unavailable downstream observations become absence.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json; source manifest sha256:a1c91eab8e9e04ebb6b806f420034c617008f1878e5761412b3db50a341c4fa5; derivation seal sha256:7f5cc29983f6832de26587cce7684f937ed4a5df7e943c45cb0f047cd458e0f5; engine receipt sha256:4406c9886170dbcd2c3b7a8348b855d249b42b605272afc2ba4a02ae4cde94c0 at receipts/engine/model_admitted/SFT-EARTH-ENVIRONMENTAL-TRANSPORT-001-4406c9886170dbcd.json; empirical validation sha256:18df7d9bd2a76854bcde16e07c1b316b8f6bec093bb922e1944922ba63613fff; measurement receipt sha256:e6804c53ae98d4b49dc569f8b98eafcdc78d38f799ae82ccf4ab5bdea5d3f513; isolation sha256:4e6e19dc3c53f8242613464b8987fe5a4118361088fd2c0659c1256415c845af; custody sha256:0bda8d688ad3a99aa0cee703bc1179880e928b8cf7d479a46dfde1e3dfb6242.

66. Environmental transformation and persistence

Claim identity: SFT-EARTH-ENVIRONMENTAL-TRANSFORMATION-001

Question and exact theorem. Environmental transformation is a condition-bounded chemical or biological transition network with precursors, products, alternatives, rates and persistence observations retained.

```
substance-condition-product-carrier__medium-condition-bounded__condition-dependent
-transition-network__precursors-products-pathways-rates-controls-and-time-held__ev
idence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retai
ns-all-rows__no-extra-rule
```

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-BIODIVERSITY-ENVIRONMENT-001
- SFT-EARTH-ENVIRONMENTAL-TRANSPORT-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	substance-condition-product-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	medium-condition-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	condition-dependent-transition-network	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	precursors-products-pathways-rates-controls-and-time-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: substance-condition-product-carrier__medium-condition-bounded__condition-dependent-transition-network__precursors-products-pathways-rates-controls-and-time-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- carrier-present: the claim retains a nonempty named carrier; passed true.
- relation-distinct: the carrier and relation are not silently conflated; passed true.
- evidence-boundary: the evidence boundary remains distinct from the result carrier; passed true.
- exact-fold-witness: the family's pre-source exact Fold construction reproduces; passed true.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar. Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

A pollutant requires substance, form, source, medium, exposure and endpoint. Source-path-receptor chains retain transport, transformation, accumulation and release. Environmental quality is purpose- and jurisdiction-bounded; a threshold cannot silently become a universal natural law.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:9be5214a9ca6a1e6178297cac2236e3e691881ac30d75e8a914589e9f6eca039.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:eca48e03b730f5816150afdd83af2d04dfd86907d38f2d835a3c62faee3a2d7f.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:58295557e800969369f4977893a421a84b0d7c0cc4e5e857ca4c83aa928de187.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:6618e5903b6b84b81188eef10ba13443bc06f117fbb4e5846f0ad6d82e8084be.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:0ca748fa0c991e8f58db897c9be35053c4c274bc41470a97a93722f29c34d26e; certificate
sha256:4e03d819400468a118491b4bb411ba6689c050ed85ac56d0cbd1b8bf3c843f3f; external-validation
identity sha256:3a4447296e25f62e59001cbd6940641eed29900206cfa93110ed9b7344406f86.

Post-seal empirical comparison. Directness:

authoritative_observation_or_primary_measurement_boundary_correspondence. Disposition:
purpose_matched_authoritative_source_features_present. Predicted consequence: earth:substance
-condition-product-carrier__condition-dependent-transition-network__precursors-produc
ts-pathways-rates-controls-and-time-held__medium-condition-bounded. Source-derived
consequence: earth:substance-condition-product-carrier__condition-dependent-transition-ne
twork__precursors-products-pathways-rates-controls-and-time-held__medium-condition-bo
unded. Exact boundary correspondence: true. External evidence selected the survivor: false. Formal structure relabelled as
direct measurement: false. Model, forecast, proxy or retrieval relabelled as observation: false.

Sources and complete feature accounting:

- EPA-AIRDATA-001 - U.S. Environmental Protection Agency, [AirData pre-generated monitoring files](#); class
authoritative_air_quality_archive; transport captured; features 4/5 present; present: site and monitor
identity, pollutant parameter and units, hourly daily and annual distinction, blank and missing records; absent and
preserved: method and quality metadata.
- WQP-WEB-SERVICES-001 - U.S. Geological Survey, U.S. Environmental Protection Agency and National Water
Quality Monitoring Council, [Water Quality Portal web services](#); class
authoritative_water_quality_federation; transport captured; features 4/4 present; present: station and
organization identity, sample and result identity, characteristic and unit, quality and metadata fields; absent and preserved:
none.
- USGS-LANDSAT-ACCESS-001 - U.S. Geological Survey Landsat Missions, [Landsat Data Access](#); class
authoritative_remote_sensing_archive; transport captured; features 0/4 present; present: none; absent
and preserved: sensor and mission identity, Earth surface observation, archive access, product and processing distinction.
- NEON-OPEN-DATA-001 - U.S. National Science Foundation National Ecological Observatory Network, [NEON open ecosystem data](#); class authoritative_ecosystem_observation_program; transport captured; features 4/4
present; present: site and data-product identity, organismal and abiotic observations, collection method, availability and
latency; absent and preserved: none.

Comparison record:

- sft-earth-environmental-transformation-001-external-earth-evidence: predicted earth:substance-condition-product-carrier__condition-dependent-transition-network__precursors-products-pathways-rates-controls-and-time-held__medium-condition-bounded; source-derived earth:substance-condition-product-carrier__condition-dependent-transition-network__precursors-products-pathways-rates-controls-and-time-held__medium-condition-bounded; exact match True
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if disappearance of the measured precursor is treated as destruction of matter or proof of harmless products.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json;
 source manifest sha256:aabce9961110f55b51b3d5a789cba0fd0ddfbaaa2226ea2aa914e6e61e5e2670;
 derivation seal sha256:4cc052cd7d4cb1104d250a12f827d67beffc756e51213c2b9fa02e79b482bbda;
 engine receipt sha256:90f90906ca26785c8256f3e093512560e27bfb09ca386250231eb0f66bc24fb8 at receipts/engine/model_admitted/SFT-EARTH-ENVIRONMENTAL-TRANSFORMATION-001-90f90906ca26785c.json; empirical validation
 sha256:8ee67eff5313f8d0b36b24eb177e8ab07598cc9114a67220fc03c3c4857a034d; measurement
 receipt sha256:39442ff9aff2cc6f40ff4a757b282d79b9776198f97f2c9dbfb56855ee49891e; isolation
 sha256:a89e82a0d7a2cabf0a294ee73e0e84e5fe5c2087c70126878fe3ac4a8174249d; custody
 sha256:27de46a3e17e1693d70ad7ea7317e911808d6e1ddfdb3135f5c98c649005a1c3.

67. Environmental accumulation

Claim identity: SFT-EARTH-ENVIRONMENTAL-ACCUMULATION-001

Question and exact theorem. Environmental accumulation is an input-storage-output ledger over a declared reservoir and interval, including transformation, remobilization and unresolved stock.

input-storage-output-history-carrier__reservoir-and-interval-bounded__retention-and-release-ledger__carrier-reservoir-input-output-transformation-and-time-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-BIODIVERSITY-ENVIRONMENT-001
- SFT-EARTH-ENVIRONMENTAL-TRANSFORMATION-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	input-storage-output-history-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	reservoir-and-interval-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	retention-and-release-ledger	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	carrier-reservoir-input-output-transformation-and-time-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: `input-storage-output-history-carrier__reservoir-and-interval-bounded__retention-and-release-ledger__carrier-reservoir-input-output-transformation-and-time-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule`.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `carrier-present`: the claim retains a nonempty named carrier; passed `true`.
- `relation-distinct`: the carrier and relation are not silently conflated; passed `true`.
- `evidence-boundary`: the evidence boundary remains distinct from the result carrier; passed `true`.
- `exact-fold-witness`: the family's pre-source exact Fold construction reproduces; passed `true`.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar. Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

A pollutant requires substance, form, source, medium, exposure and endpoint. Source-path-receptor chains retain transport, transformation, accumulation and release. Environmental quality is purpose- and jurisdiction-bounded; a threshold cannot silently become a universal natural law.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 41fd30d09698fb7783f3b0e3edd6e2e1f0e150cdc750fc5df6004ef673177387.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt

sha256:58224fa6c1365a78547f5b768b6952d6e48878d9938415802614a8ed59be9544.

- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:54f40c328b7f1a4358d2d81f054cfdb0fcc89639c922b1bc933fa3645c0699d7.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:885dddf33917784d09c2793696a97edf322bd30aa9a683040b6510b540583f9.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:812164a3bef7a83cead73c941276aadf522f0187bca9b9bf9460a02728776ff0; certificate

sha256:0f24bed7d787b744f4ce54a480eb9ef00427c2bd5fbf3bad056993966abab0305; external-validation identity sha256:834e6b1825e7c8c9a77ed42f32959fea364ad10935c31c2bb2957a07e67ca225.

Post-seal empirical comparison. Directness:

authoritative_observation_or_primary_measurement_boundary_correspondence. Disposition: purpose_matched_authoritative_source_features_present. Predicted consequence: earth:input-storage-output-history-carrier__retention-and-release-ledger__carrier-reservoir-input-output-transformation-and-time-held__reservoir-and-interval-bounded. Source-derived consequence: earth:input-storage-output-history-carrier__retention-and-release-ledger__carrier-reservoir-input-output-transformation-and-time-held__reservoir-and-interval-bounded. Exact boundary correspondence: true. External evidence selected the survivor: false. Formal structure relabelled as direct measurement: false. Model, forecast, proxy or retrieval relabelled as observation: false.

Sources and complete feature accounting:

- EPA-AIRDATA-001 - U.S. Environmental Protection Agency, [AirData pre-generated monitoring files](#); class authoritative_air_quality_archive; transport captured; features 4/5 present; present: site and monitor identity, pollutant parameter and units, hourly daily and annual distinction, blank and missing records; absent and preserved: method and quality metadata.
- WQP-WEB-SERVICES-001 - U.S. Geological Survey, U.S. Environmental Protection Agency and National Water Quality Monitoring Council, [Water Quality Portal web services](#); class authoritative_water_quality_federation; transport captured; features 4/4 present; present: station and organization identity, sample and result identity, characteristic and unit, quality and metadata fields; absent and preserved: none.
- USGS-LANDSAT-ACCESS-001 - U.S. Geological Survey Landsat Missions, [Landsat Data Access](#); class authoritative_remote_sensing_archive; transport captured; features 0/4 present; present: none; absent and preserved: sensor and mission identity, Earth surface observation, archive access, product and processing distinction.
- NEON-OPEN-DATA-001 - U.S. National Science Foundation National Ecological Observatory Network, [NEON open ecosystem data](#); class authoritative_ecosystem_observation_program; transport captured; features 4/4 present; present: site and data-product identity, organismal and abiotic observations, collection method, availability and latency; absent and preserved: none.

Comparison record:

- sft-earth-environmental-accumulation-001-external-earth-evidence: predicted earth:input-storage-output-history-carrier__retention-and-release-ledger__carrier-reservoir-input-output-transformation-and-time-held__reservoir-and-interval-bounded; source-derived earth:input-storage-output-history-carrier__retention-and-release-ledger__carrier-reservoir-input-output-transformation-and-time-held__reservoir-and-interval-bounded; exact match True
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if concentration alone establishes accumulation without volume, input, output and time records.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json;
 source manifest sha256:dc95e40e208f502c6d19f05a8d4cffa807a81f21a20929e8208bfc5bac2c6c6c;
 derivation seal sha256:45419ff6508e2e975481f270819a9b16a4cece82680a50d23b9348b37255eed2;
 engine receipt sha256:78aa279eb4f757e454c1d5148fc6722520c26e5ddcccb1a094b4272883fb3d48 at re
 cepts/engine/model_admitted/SFT-EARTH-ENVIRONMENTAL-ACCUMULATION-001-78aa279eb4f757e
 4.json; empirical validation
 sha256:60e719a3047adc6d9138fe63475ed360f205b036ebdb928ae448fc842335b512; measurement
 receipt sha256:6debb95b370681de580fd3fd85dc432fa88e0eb885f01d9339bc8d4630df0cef; isolation
 sha256:6df641714d4bb5a0bc23a231f0be2270ebc53ac7665fe3d49c36d6ecec38fb21; custody
 sha256:782d79dca28ec5e4d2d0bf222e3299fcebfe45f83f4b897c641d19962e63d49d.

68. Environmental quality state

Claim identity: SFT-EARTH-ENVIRONMENTAL-QUALITY-001

Question and exact theorem. Environmental quality is a purpose- and medium-specific evidence vector retaining measurement methods, thresholds, provenance, uncertainty and missing components; no single index may erase them.

multi-measurement-environmental-state-carrier__medium-purpose-jurisdiction-bounded
 __standard-purpose-and-ecological-comparison__variables-methods-threshold-provenan
 ce-missingness-and-time-held__evidence-class-explicit__root-bound-forward-forcing_
 _positive-finite-successor-retains-all-rows__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001

- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-BIODIVERSITY-ENVIRONMENT-001
- SFT-EARTH-ENVIRONMENTAL-ACCUMULATION-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	multi-measurement-environmental-state-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	medium-purpose-jurisdiction-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	standard-purpose-and-ecological-comparison	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	variables-methods-threshold-provenance-missingness-and-time-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: multi-measurement-environmental-state-carrier__medium-purpose-jurisdiction-bounded__standard-purpose-and-ecological-comparison__variables-methods-threshold-provenance-missingness-and-time-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- carrier-present: the claim retains a nonempty named carrier; passed true.
- relation-distinct: the carrier and relation are not silently conflated; passed true.
- evidence-boundary: the evidence boundary remains distinct from the result carrier; passed true.
- exact-fold-witness: the family's pre-source exact Fold construction reproduces; passed true.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar.

Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

A pollutant requires substance, form, source, medium, exposure and endpoint. Source-path-receptor chains retain transport, transformation, accumulation and release. Environmental quality is purpose- and jurisdiction-bounded; a threshold cannot silently become a universal natural law.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:27093fffbdec1d47b132fd346f64816e5df38aaa2b368d62bac7ec1d7b0ff721a.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:0e42f1142b1e41bceb022ef48c9afa85bdc279d230e71e67b3cd92008f470218.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:cf22a06a9d4b2055a294ffd30ceecabc0fc39ccde74d777d847522a6b2e1ad69.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:08c4b4ad6d1e5fc3bc51a3951565743097688a64a6a04dd025cba2d68a4d00b2.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:dd02126b62ca1720174b8fc28fdf7501878894363de1adc88ac46deb5906c689; certificate
 sha256:c31436ecff2d547908b1257f374e349435e38e50207dffe5e3629b4fb5d7404e; external-validation
 identity sha256:f3efb9446196b7d04862627a224dd766c21b03e73b177003762dda3c51787823.

Post-seal empirical comparison. Directness:

authoritative_observation_or_primary_measurement_boundary_correspondence. Disposition:
 purpose_matched_authoritative_source_features_present. Predicted consequence: earth:multi-measurement-environmental-state-carrier__standard-purpose-and-ecological-comparison__variables-methods-threshold-provenance-missingness-and-time-held__medium-purpose-jurisdiction-bounded. Source-derived consequence: earth:multi-measurement-environmental-state-carrier__standard-purpose-and-ecological-comparison__variables-methods-threshold-provenance-missingness-and-time-held__medium-purpose-jurisdiction-bounded. Exact boundary correspondence: true. External evidence selected the survivor: false. Formal structure relabelled as direct measurement: false. Model, forecast, proxy or retrieval relabelled as observation: false.

Sources and complete feature accounting:

- EPA-AIRDATA-001 - U.S. Environmental Protection Agency, [AirData pre-generated monitoring files](#); class authoritative_air_quality_archive; transport captured; features 4/5 present; present: site and monitor identity, pollutant parameter and units, hourly daily and annual distinction, blank and missing records; absent and preserved: method and quality metadata.
- WQP-WEB-SERVICES-001 - U.S. Geological Survey, U.S. Environmental Protection Agency and National Water Quality Monitoring Council, [Water Quality Portal web services](#); class authoritative_water_quality_federation; transport captured; features 4/4 present; present: station and organization identity, sample and result identity, characteristic and unit, quality and metadata fields; absent and preserved: none.
- USGS-LANDSAT-ACCESS-001 - U.S. Geological Survey Landsat Missions, [Landsat Data Access](#); class authoritative_remote_sensing_archive; transport captured; features 0/4 present; present: none; absent and preserved: sensor and mission identity, Earth surface observation, archive access, product and processing distinction.
- NEON-OPEN-DATA-001 - U.S. National Science Foundation National Ecological Observatory Network, [NEON open ecosystem data](#); class authoritative_ecosystem_observation_program; transport captured; features 4/4 present; present: site and data-product identity, organismal and abiotic observations, collection method, availability and latency; absent and preserved: none.

Comparison record:

- sft-earth-environmental-quality-001-external-earth-evidence: predicted earth:multi-measurement-environmental-state-carrier__standard-purpose-and-ecological-comparison__variables-methods-threshold-provenance-missingness-and-time-held__medium-purpose-jurisdiction-bounded; source-derived earth:multi-measurement-environmental-state-carrier__standard-purpose-and-ecological-comparison__variables-methods-threshold-provenance-missingness-and-time-held__medium-purpose-jurisdiction-bounded; exact match True
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if a weighted index imports fitted choices, hides an adverse component or silently becomes a universal natural law.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum

- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json;
 source manifest sha256:10ef399d1aac675f3cd5ba0dc9f38c04c85a1eb77cb635a967551d85537f006f;
 derivation seal sha256:311ea7b0f8e92085950aa74a0f6e0bbb50bf6a38c0985da02d969065e737f247;
 engine receipt sha256:1c71e9649e5baf46adbe3c1690b297a15cd85754ef8f0fd5cd59b37f3b3319e9 at re
 ceipts/engine/model_admitted/SFT-EARTH-ENVIRONMENTAL-QUALITY-001-1c71e9649e5baf46.jso
 n; empirical validation
 sha256:5d8ea35c6dc2236adbb54c2629119faf1da7c0dc0ecfc54ee36bc8f704e2f5bc; measurement
 receipt sha256:4db04649b126b13ade5adbd27006c953a1071b620aeb06a7edae66ad74c7c605; isolation
 sha256:c79b09f65ec6914b5363a44b113dab25b51f7f1cbe3b380cff8bdd4c2bfafeb8d; custody
 sha256:8912d8ff30d02bf3714d2cb18ef926168d7e47426d1cf237ed1d0237e9bbb633.

Ledger family 12: Field Evidence, Remote Sensing, Forecasts and Hazard Handoffs

Field observations, remote sensing, proxy reconstruction, model assimilation and forecasts are separate evidence classes. Hazard, exposure, vulnerability, capacity and consequence remain different carriers. Forecast success cannot admit an opaque relation, and a model output cannot be relabelled as observation.

69. Field observation and sampling

Claim identity: SFT-EARTH-FIELD-OBSERVATION-001

Question and exact theorem. A field observation retains site selection, sampling design, effort, calibration, processing, exclusions and representativeness limits.

site-specimen-instrument-result-carrier__representativeness-declared__sampling-to-
 population-relation__design-location-time-effort-calibration-and-exclusions-held__
 evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-ret
 ains-all-rows__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001

- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-ENVIRONMENTAL-QUALITY-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	site-specimen-instrument-result-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	representativeness-declared	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	sampling-to-population-relation	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	design-location-time-effort-calibration-and-exclusions-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: `site-specimen-instrument-result-carrier__representativeness-declared__sampling-to-population-relation__design-location-time-effort-calibration-and-exclusions-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule`.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `carrier-present`: the claim retains a nonempty named carrier; passed `true`.
- `relation-distinct`: the carrier and relation are not silently conflated; passed `true`.
- `evidence-boundary`: the evidence boundary remains distinct from the result carrier; passed `true`.
- `exact-fold-witness`: the family's pre-source exact Fold construction reproduces; passed `true`.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar.

Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

Field observations, remote sensing, proxy reconstruction, model assimilation and forecasts are separate evidence classes.

Hazard, exposure, vulnerability, capacity and consequence remain different carriers. Forecast success cannot admit an opaque relation, and a model output cannot be relabelled as observation.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:0df438472ce86cc135edbd150b7926270d7107616fcdf0adc4a09c9b3b874d41.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:b710a0da37e3301fd8ee4b35fa9f3a3b15f3b92b9f35aff9e81f825c67453ae5.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:b31fc3fd59340a57610408041401905112355d20acd3f87515567c6441ede381.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:18b1f4a23b5199286675da3ce35404b25615ba6eb210f56510afabd392126711.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:7bfcadfd38fda485d748c0c64728b04b2ce46b5a6a96ab117c9d08a13352fd8e; certificate
sha256:2941498c86690c9a4ba2e62d1f99861dcec22272488770e445f9684236917e08; external-validation
identity sha256:58de79f0fc91b2a85efde99cb7fe2e86601d3aebfbcff444fb784b224bed721c.

Post-seal empirical comparison. Directness:

authoritative_observation_or_primary_measurement_boundary_correspondence. Disposition: purpose_matched_authoritative_source_features_present. Predicted consequence: earth:site-specimen-instrument-result-carrier__sampling-to-population-relation__design-location-time-effort-calibration-and-exclusions-held__representativeness-declared. Source-derived consequence: earth:site-specimen-instrument-result-carrier__sampling-to-population-relation__design-location-time-effort-calibration-and-exclusions-held__representativeness-declared. Exact boundary correspondence: true. External evidence selected the survivor: false. Formal structure relabelled as direct measurement: false. Model, forecast, proxy or retrieval relabelled as observation: false.

Sources and complete feature accounting:

- NASA-EARTHDATA-SEARCH-001 - NASA Earth Science Data and Information System, [Earthdata Search data discovery and access](#); class authoritative_data_catalog; transport captured; features 4/4 present; present: dataset identity, spatial and temporal coverage, instrument or platform provenance, download or service access; absent and preserved: none.
- USGS-LANDSAT-ACCESS-001 - U.S. Geological Survey Landsat Missions, [Landsat Data Access](#); class authoritative_remote_sensing_archive; transport captured; features 0/4 present; present: none; absent and preserved: sensor and mission identity, Earth surface observation, archive access, product and processing distinction.
- NOAA-GHCND-001 - NOAA National Centers for Environmental Information, [Global Historical Climatology Network Daily](#); class authoritative_station_climate_archive; transport captured; features 5/5 present; present: station identity and metadata, daily weather observations, quality flags, record length and variable coverage, archive version; absent and preserved: none.
- NOAA-PALEO-001 - NOAA National Centers for Environmental Information World Data Service for Paleoclimatology, [Paleoclimatology data archive](#); class authoritative_proxy_archive; transport captured; features 3/4 present; present: archive and study identity, proxy and reconstructed-variable distinction, location and investigator provenance; absent and preserved: dating and temporal coverage.
- EARTHSCOPE-SAGE-SERVICES-001 - NSF SAGE Facility operated by EarthScope Consortium, [Seismological and geophysical web services](#); class authoritative_geophysical_data_service; transport captured; features 3/4 present; present: station and channel identity, waveform and event access, instrument response; absent and preserved: observed and derived product distinction.

Comparison record:

- sft-earth-field-observation-001-external-earth-evidence: predicted earth:site-specimen-instrument-result-carrier__sampling-to-population-relation__design-location-time-effort-calibration-and-exclusions-held__representativeness-declared; source-derived earth:site-specimen-instrument-result-carrier__sampling-to-population-relation__design-location-time-effort-calibration-and-exclusions-held__representativeness-declared; exact match True
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if convenience samples establish a global Earth relation or if failed and below-detection samples are removed.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter

- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json;
 source manifest sha256: fbae46e2029b360a23d57d9a976c1722e7f2a0e295cae41f6c23857eefdc05f8;
 derivation seal sha256: e262ee7c36da0e5a5c42eed2fb38412e5d7a6b1e3b0400f345d78d0fac0da26d;
 engine receipt sha256: 0a0767dc855b71b885537b3fb2f2fa2e4811e9f14219e6bc5ea4bc68cc251f28 at
 receipts/engine/model_admitted/SFT-EARTH-FIELD-OBSERVATION-001-0a0767dc855b71b8.json;
 empirical validation sha256: 946d94d4c6c3f819126447d3a3bc4f4acd539359b1eb50d67a75f2736435f7c4;
 measurement receipt
 sha256: 00f2530f8d0caab567e502d9c59dd6a7f50078d61c4ef9492c447e17906acc3d; isolation
 sha256: a4b308d09941d6603e7d0ba2b3586e71ca02ccf981e86d949b502b0c290f98de; custody
 sha256: 29f15bf517b34bff1f91d1bf6c060288a26a1b9dc0f64f6d53fe88907cf8b986.

70. Remote-sensing observation

Claim identity: SFT-EARTH-REMOTE-SENSING-001

Question and exact theorem. Remote sensing is a source-to-path-to-sensor-to-retrieval trace whose geophysical interpretation retains calibration, geometry, processing, validation and unavailable pixels.

source-path-sensor-retrieval-carrier__sensor-product-bounded__signal-to-geophysical-record-relation__geometry-spectrum-calibration-processing-validation-and-cloud-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001

- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-ENVIRONMENTAL-QUALITY-001
- SFT-EARTH-FIELD-OBSERVATION-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	source-path-sensor-retrieval-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	sensor-product-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	signal-to-geophysical-record-relation	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	geometry-spectrum-calibration-processing-validation-and-cloud-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: `source-path-sensor-retrieval-carrier__sensor-product-bounded__signal-to-geophysical-record-relation__geometry-spectrum-calibration-processing-validation-and-cloud-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule`.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `carrier-present`: the claim retains a nonempty named carrier; passed `true`.
- `relation-distinct`: the carrier and relation are not silently conflated; passed `true`.
- `evidence-boundary`: the evidence boundary remains distinct from the result carrier; passed `true`.
- `exact-fold-witness`: the family's pre-source exact Fold construction reproduces; passed `true`.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar. Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

Field observations, remote sensing, proxy reconstruction, model assimilation and forecasts are separate evidence classes.

Hazard, exposure, vulnerability, capacity and consequence remain different carriers. Forecast success cannot admit an opaque relation, and a model output cannot be relabelled as observation.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:97546d891605bd2b35669cdafb5c33ef1b90396e90eef77f1faf51c5f37976a7.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:80c7ce56a521a56b2238488d5458a80bc0f0e4b86f09c46c46671072f29c7efc.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:5c5a7371d58525be57bf808e891535c49cabfe2d114b028aaed91dcf7f7cd8c1.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:8068ed7f4db4a0e6ca5b3a9c0d567bcea9ea5d165550cf8bd21753995d6ca375.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:6eaded0b52f460ba0aaf6b20dd8a3ad869b2260dbc4ccd82dff0d338416baf99; certificate
sha256:9ea77af83b380fc540ee56f731b7c81d369ab35af71dc7ebf01544471c4a8c5b; external-validation
identity sha256:ce8e3b7fbfa2d2eafa6d66ddfcce178642d27f02337d9ab7ae4ed1ad565441a4.

Post-seal empirical comparison. Directness:

`authoritative_observation_or_primary_measurement_boundary_correspondence`. Disposition:
`purpose_matched_authoritative_source_features_present`. Predicted consequence: `earth:source-path-sensor-retrieval-carrier__signal-to-geophysical-record-relation__geometry-spectrum-calibration-processing-validation-and-cloud-held__sensor-product-bounded`. Source-derived

consequence: earth:source-path-sensor-retrieval-carrier__signal-to-geophysical-record-relation__geometry-spectrum-calibration-processing-validation-and-cloud-held__sensor-product-bounded. Exact boundary correspondence: true. External evidence selected the survivor: false. Formal structure relabelled as direct measurement: false. Model, forecast, proxy or retrieval relabelled as observation: false.

Sources and complete feature accounting:

- NASA-EARTHDATA-SEARCH-001 - NASA Earth Science Data and Information System, [Earthdata Search data discovery and access](#); class authoritative_data_catalog; transport captured; features 4/4 present; present: dataset identity, spatial and temporal coverage, instrument or platform provenance, download or service access; absent and preserved: none.
- USGS-LANDSAT-ACCESS-001 - U.S. Geological Survey Landsat Missions, [Landsat Data Access](#); class authoritative_remote_sensing_archive; transport captured; features 0/4 present; present: none; absent and preserved: sensor and mission identity, Earth surface observation, archive access, product and processing distinction.
- NOAA-GHCND-001 - NOAA National Centers for Environmental Information, [Global Historical Climatology Network Daily](#); class authoritative_station_climate_archive; transport captured; features 5/5 present; present: station identity and metadata, daily weather observations, quality flags, record length and variable coverage, archive version; absent and preserved: none.
- NOAA-PALEO-001 - NOAA National Centers for Environmental Information World Data Service for Paleoclimatology, [Paleoclimatology data archive](#); class authoritative_proxy_archive; transport captured; features 3/4 present; present: archive and study identity, proxy and reconstructed-variable distinction, location and investigator provenance; absent and preserved: dating and temporal coverage.
- EARTHSCOPE-SAGE-SERVICES-001 - NSF SAGE Facility operated by EarthScope Consortium, [Seismological and geophysical web services](#); class authoritative_geophysical_data_service; transport captured; features 3/4 present; present: station and channel identity, waveform and event access, instrument response; absent and preserved: observed and derived product distinction.

Comparison record:

- sft-earth-remote-sensing-001-external-earth-evidence: predicted earth:source-path-sensor-retrieval-carrier__signal-to-geophysical-record-relation__geometry-spectrum-calibration-processing-validation-and-cloud-held__sensor-product-bounded; source-derived earth:source-path-sensor-retrieval-carrier__signal-to-geophysical-record-relation__geometry-spectrum-calibration-processing-validation-and-cloud-held__sensor-product-bounded; exact match True
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if a retrieval product is called direct ground truth or if cloud, saturation and algorithm changes are erased.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation

- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json;
 source manifest sha256:1959b82d8b69db009a21901e62595b0461d13c95132cabe2606522f8ae2c2259;
 derivation seal sha256:e3a2d0965cb92a94de243984fc2801b96ea7f9af415815787d125528c9b38dd6;
 engine receipt sha256:826728a3fa2756f2cf9c9e97da5b14a7245d5e6c1cedaed03b98404a90013be2 at
 receipts/engine/model_admitted/SFT-EARTH-REMOTE-SENSING-001-826728a3fa2756f2.json;
 empirical validation sha256:aa5fdc29de1b4c573324e8a8d0756fc3bb9b8468c25571b787aad9ad2dc6c006;
 measurement receipt
 sha256:68922ccace6af40770ff426620ce63c11b7c6e9f30ce136518b1d4fb55fd59aa; isolation
 sha256:cf8c65fa5ed25c9d79d520726d16b8dd809264375458d16c25cd2abd212445f4; custody
 sha256:4fdcb3348c6d336948e523abb84d9901f44a642660a31b743a662841abbc907d.

71. Proxy and Earth-history reconstruction

Claim identity: SFT-EARTH-PROXY-RECONSTRUCTION-001

Question and exact theorem. A proxy reconstruction preserves archive formation, dating, calibration, resolution, uncertainty and competing environmental controls and remains distinct from direct observation.

archive-proxy-calibration-target-carrier__proxy-region-period-bounded__proxy-to-pa
 st-state-inference__archive-formation-dating-calibration-resolution-and-alternativ
 es-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-succ
 esssor-retains-all-rows__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001

- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-ENVIRONMENTAL-QUALITY-001
- SFT-EARTH-REMOTE-SENSING-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	archive-proxy-calibration-target-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	proxy-region-period-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	proxy-to-past-state-inference	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	archive-formation-dating-calibration-resolution-and-alternatives-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: archive-proxy-calibration-target-carrier__proxy-region-period-bounded__proxy-to-past-state-inference__archive-formation-dating-calibration-resolution-and-alternatives-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `carrier-present`: the claim retains a nonempty named carrier; passed `true`.
- `relation-distinct`: the carrier and relation are not silently conflated; passed `true`.
- `evidence-boundary`: the evidence boundary remains distinct from the result carrier; passed `true`.
- `exact-fold-witness`: the family's pre-source exact Fold construction reproduces; passed `true`.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar.

Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

Field observations, remote sensing, proxy reconstruction, model assimilation and forecasts are separate evidence classes.

Hazard, exposure, vulnerability, capacity and consequence remain different carriers. Forecast success cannot admit an opaque relation, and a model output cannot be relabelled as observation.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:15ee87336877391905973880317bae5a81db6941ab32180a8f33017462f70def.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:b401108df72e8eb2fdef1e21671101c4466078c94f25d979659ae1aa773996f5.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:f07be49ceb9df0f4dcebfe57417c38212a2d7872dc40e87873ccabe6c2c6ec8.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:aea2c16a2a5cc2e0a5f594dcdf74ff8789adc12c1015998c85bc8dde939a5a2c.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:1a469436c8872268d066a89c2c4733a4c0b3fdb87cd291bcbalab1009fe505a; certificate
sha256:ffa7a9666d7aba0ea79b92cc6970c0d2cb456570f84d5205981cd98c4823bc72; external-validation
identity sha256:58abcc5bfedfd5ca112da720fa12245e8c6c507c25bedc420b2343c67eb6142a.

Post-seal empirical comparison. Directness:

`authoritative_observation_or_primary_measurement_boundary_correspondence`. Disposition:
`purpose_matched_authoritative_source_features_present`. Predicted consequence: `earth:archive-proxy-calibration-target-carrier__proxy-to-past-state-inference__archive-formation-dating-calibration-resolution-and-alternatives-held__proxy-region-period-bounded`.
Source-derived consequence: `earth:archive-proxy-calibration-target-carrier__proxy-to-past-state-inference__archive-formation-dating-calibration-resolution-and-alternatives-held__proxy-region-period-bounded`. Exact boundary correspondence: `true`. External evidence selected the survivor:
`false`. Formal structure relabelled as direct measurement: `false`. Model, forecast, proxy or retrieval relabelled as observation: `false`.

Sources and complete feature accounting:

- NASA-EARTHDATA-SEARCH-001 - NASA Earth Science Data and Information System, [Earthdata Search data discovery and access](#); class `authoritative_data_catalog`; transport captured; features 4/4 present; present: dataset

identity, spatial and temporal coverage, instrument or platform provenance, download or service access; absent and preserved: none.

- USGS-LANDSAT-ACCESS-001 - U.S. Geological Survey Landsat Missions, **Landsat Data Access**; class `authoritative_remote_sensing_archive`; transport captured; features 0/4 present; present: none; absent and preserved: sensor and mission identity, Earth surface observation, archive access, product and processing distinction.
- NOAA-GHCND-001 - NOAA National Centers for Environmental Information, **Global Historical Climatology Network Daily**; class `authoritative_station_climate_archive`; transport captured; features 5/5 present; present: station identity and metadata, daily weather observations, quality flags, record length and variable coverage, archive version; absent and preserved: none.
- NOAA-PALEO-001 - NOAA National Centers for Environmental Information World Data Service for Paleoclimatology, **Paleoclimatology data archive**; class `authoritative_proxy_archive`; transport captured; features 3/4 present; present: archive and study identity, proxy and reconstructed-variable distinction, location and investigator provenance; absent and preserved: dating and temporal coverage.
- EARTHSOPE-SAGE-SERVICES-001 - NSF SAGE Facility operated by EarthScope Consortium, **Seismological and geophysical web services**; class `authoritative_geophysical_data_service`; transport captured; features 3/4 present; present: station and channel identity, waveform and event access, instrument response; absent and preserved: observed and derived product distinction.

Comparison record:

- `sft-earth-proxy-reconstruction-001-external-earth-evidence: predicted earth:archive-proxy-calibration-target-carrier__proxy-to-past-state-inference__archive-formation-dating-calibration-resolution-and-alternatives-held__proxy-region-period-bounded`; source-derived `earth:archive-proxy-calibration-target-carrier__proxy-to-past-state-inference__archive-formation-dating-calibration-resolution-and-alternatives-held__proxy-region-period-bounded`; exact match True
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if one modern calibration is assumed invariant through untested history or if proxy disagreement is discarded.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

`experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json`;
source manifest sha256: 71fc04d7b4d1165dfde943177401bf56957020412b899c2b6747c56443731294;

derivation seal sha256:7d71cdac39ff3605f0936d9fcc420900efd8a7cb69d99516d1bc2a0a939e95f1;
 engine receipt sha256:d02ad77618b5f20aca6067b7a28ef8002c34b40d80166a08f1ed588df9e68c90 at re
 ceipts/engine/model_admitted/SFT-EARTH-PROXY-RECONSTRUCTION-001-d02ad77618b5f20a.json
 ; empirical validation
 sha256:225fdc05201db10869eeec55b519ae2c20d82ddbb1d80e5bae0102d56d46c897; measurement
 receipt sha256:a2b018ec662ed49b64342a2b236c5ce7fc79efd77501325e9c8951eff5366d8a; isolation
 sha256:0c098cf3dd3bca3466323a32a4b7fb8a9df60b853b07f847f52405ba5e5a4be1; custody
 sha256:c9adf73c693fc8541f9f30296b1c86052c5e96bc9b13e5de0ff7ab0fe28f0c06.

72. Earth model and data-assimilation boundary

Claim identity: SFT-EARTH-MODEL-DATA-ASSIMILATION-001

Question and exact theorem. An Earth model is a declared conditional simulation and data assimilation is a traceable state update; neither can select an SFT law or become an observation.

law-state-observation-update-carrier__model-purpose-bounded__conditional-simulatio
 n-and-state-update-relation__equations-as-correspondence-inputs-parameters-observa
 tions-errors-and-outputs-held__evidence-class-explicit__root-bound-forward-forcing
 __positive-finite-successor-retains-all-rows__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001

- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-ENVIRONMENTAL-QUALITY-001
- SFT-EARTH-PROXY-RECONSTRUCTION-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	law-state-observation-update-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	model-purpose-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	conditional-simulation-and-state-update-relation	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	equations-as-correspondence-inputs-parameters-observations-errors-and-outputs-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: law-state-observation-update-carrier__model-purpose-bounded__conditional-simulation-and-state-update-relation__equations-as-correspondence-inputs-parameters-observations-errors-and-outputs-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- carrier-present: the claim retains a nonempty named carrier; passed true.
- relation-distinct: the carrier and relation are not silently conflated; passed true.
- evidence-boundary: the evidence boundary remains distinct from the result carrier; passed true.
- exact-fold-witness: the family's pre-source exact Fold construction reproduces; passed true.

Scientific meaning. A model remains a conditional correspondence with inputs, parameters, observations, errors and outputs retained. Assimilation does not promote the model into a direct observation or allow forecast performance to select a hidden law.

Field observations, remote sensing, proxy reconstruction, model assimilation and forecasts are separate evidence classes. Hazard, exposure, vulnerability, capacity and consequence remain different carriers. Forecast success cannot admit an opaque relation, and a model output cannot be relabelled as observation.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:ba57517a2f276f72c835cfaca970b4bd85e2e3fca30c750cefd51454f35e3035.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:d5da08603fecb0d89edb215840f0dddaa99aa04fe9de9cc31ddd2260c8080303.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:d33a3edbf05b4dd8183899700b404bac651e3ebcf7ddc879bb9fd97dc3e56a4.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:73e3af134d12068baef5ce204504d4e3d9484c9531fbe6b6ff72e93973630b30.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:a65379ef9d1b438e3715cf4471a2301b52ce790e917c1aba59cd135639a615e8; certificate
sha256:b76c6d281dbf1f5617929dcb6eea9f91d77bdf2a24e45e8c169bala74b275f23; external-validation
identity sha256:9e571cfffda77cd8e210b5c88ce97f6904091bb8fa93f5a009ee5aff6d80b70b9.

Post-seal empirical comparison. Directness:

authoritative_observation_or_primary_measurement_boundary_correspondence. Disposition:
purpose_matched_authoritative_source_features_present. Predicted consequence: earth:law-state-
-observation-update-carrier__conditional-simulation-and-state-update-relation__equati
ons-as-correspondence-inputs-parameters-observations-errors-and-outputs-held__model-p
urpose-bounded. Source-derived consequence: earth:law-state-observation-update-carrier__condit
ional-simulation-and-state-update-relation__equations-as-correspondence-inputs-parame
ters-observations-errors-and-outputs-held__model-purpose-bounded. Exact boundary
correspondence: true. External evidence selected the survivor: false. Formal structure relabelled as direct measurement:
false. Model, forecast, proxy or retrieval relabelled as observation: false.

Sources and complete feature accounting:

- NASA-EARTHDATA-SEARCH-001 - NASA Earth Science Data and Information System, [Earthdata Search data discovery and access](#); class authoritative_data_catalog; transport captured; features 4/4 present; present: dataset identity, spatial and temporal coverage, instrument or platform provenance, download or service access; absent and preserved: none.
- USGS-LANDSAT-ACCESS-001 - U.S. Geological Survey Landsat Missions, [Landsat Data Access](#); class authoritative_remote_sensing_archive; transport captured; features 0/4 present; present: none; absent and preserved: sensor and mission identity, Earth surface observation, archive access, product and processing distinction.

- NOAA-GHCND-001 - NOAA National Centers for Environmental Information, **Global Historical Climatology Network Daily**; class authoritative_station_climate_archive; transport captured; features 5/5 present; present: station identity and metadata, daily weather observations, quality flags, record length and variable coverage, archive version; absent and preserved: none.
- NOAA-PALEO-001 - NOAA National Centers for Environmental Information World Data Service for Paleoclimatology, **Paleoclimatology data archive**; class authoritative_proxy_archive; transport captured; features 3/4 present; present: archive and study identity, proxy and reconstructed-variable distinction, location and investigator provenance; absent and preserved: dating and temporal coverage.
- EARTHSOPE-SAGE-SERVICES-001 - NSF SAGE Facility operated by EarthScope Consortium, **Seismological and geophysical web services**; class authoritative_geophysical_data_service; transport captured; features 3/4 present; present: station and channel identity, waveform and event access, instrument response; absent and preserved: observed and derived product distinction.

Comparison record:

- sft-earth-model-data-assimilation-001-external-earth-evidence: predicted earth:law-state-observation-update-carrier__conditional-simulation-and-state-update-relation__equations-as-correspondence-inputs-parameters-observations-errors-and-outputs-held__model-purpose-bounded; source-derived earth:law-state-observation-update-carrier__conditional-simulation-and-state-update-relation__equations-as-correspondence-inputs-parameters-observations-errors-and-outputs-held__model-purpose-bounded; exact match True
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if tuned performance is offered as structural proof, if observations leak into a sealed prediction, or if model and observation errors are conflated.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

```
experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json;
source manifest sha256:a0a8e317f41e8f1bd8f8419491e82656a25664703301d88ddd71e9dce6b5c202;
derivation seal sha256:c63e25c79220cd06034b64a67dc592a1a1b5b7a7475a95f1c355f7515a5c84ed;
engine receipt sha256:0589bd2c187b02828686eaf6e38bf8a1fb29aa9bd0f194a739594d06b824f at receipts/engine/model_admitted/SFT-EARTH-MODEL-DATA-ASSIMILATION-001-0589bd2c187b0282.json; empirical validation
sha256:fa025eb867ad6165c62743b03ea210ee20db11226b70572375f25050f48091f0; measurement
```

receipt sha256:6f5c89629b617ee08f0ba3a0be1fa71f163c08635401f7ae67045307a870b8d5; isolation sha256:fdgcd0d976f1d3ba07c398279859678e5ca27e1a9ee1aa58c0761f1491eb81d2; custody sha256:e86716e130a8ab3f6026db66bf7ae8012fc9d887190c2361d9cdaecba34b8039.

73. Earth forecast validation

Claim identity: SFT-EARTH-FORECAST-VALIDATION-001

Question and exact theorem. Forecast validation compares a sealed forecast with later source-bound outcomes using preregistered targets, metrics, baselines and complete adverse and missing rows.

```
sealed-forecast-outcome-carrier__forecast-question-bounded__predeclared-comparison
-relation__targets-metrics-baselines-adverse-cases-missingness-and-time-held__evid
ence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains
-all-rows__no-extra-rule
```

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001

- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-ENVIRONMENTAL-QUALITY-001
- SFT-EARTH-MODEL-DATA-ASSIMILATION-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	sealed-forecast-outcome-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	forecast-question-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	predeclared-comparison-relation	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	targets-metrics-baselines-adverse-cases-missingness-and-time-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: sealed-forecast-outcome-carrier__forecast-question-bounded__predeclared-comparison-relation__targets-metrics-baselines-adverse-cases-missingness-and-time-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule.

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- carrier-present: the claim retains a nonempty named carrier; passed true.
- relation-distinct: the carrier and relation are not silently conflated; passed true.
- evidence-boundary: the evidence boundary remains distinct from the result carrier; passed true.

- `exact-fold-witness`: the family's pre-source exact Fold construction reproduces; passed `true`.

Scientific meaning. This closes the exact foundational relation stated above within its declared positive-finite grammar. Earth-specific dimensional magnitudes remain source-, place-, time-, instrument- and protocol-bounded observation records rather than fitted proof parameters.

Field observations, remote sensing, proxy reconstruction, model assimilation and forecasts are separate evidence classes. Hazard, exposure, vulnerability, capacity and consequence remain different carriers. Forecast success cannot admit an opaque relation, and a model output cannot be relabelled as observation.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:d0807f0e701f12f88a6766b8c1cb948eac541bea39d5ac7980e81ed4c5b3f63e.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:e3b4eaed7ec2037ec85992ccf1fb24c48d598ec7ce7b2b267fc12c84ecd65d.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:e47537579c0c0579fa87050658679f64bbb2adead2ba7487a27bf39354eb6917.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:d8dc7f62dee07a59fa28f56049047aef65b4e418ef18f1467cc6b3fde7d986e8.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:4b4d5c96a009fd7ed3839e25619764059d20620c6ac57d461a306fdaf5d87a06; certificate
sha256:52a4d2b86820fd4439ad3870c300859111d0f7086ad32a96f6426e833261621c; external-validation
identity sha256:4d727f91d0b521cb1cec76d59d7a328863f3717ae6730f3027a6ee3aa1160a47.

Post-seal empirical comparison. Directness:

authoritative_observation_or_primary_measurement_boundary_correspondence. Disposition:
purpose_matched_authoritative_source_features_present. Predicted consequence: earth:sealed-fo
recast-outcome-carrier__predeclared-comparison-relation__targets-metrics-baselines-ad
verse-cases-missingness-and-time-held__forecast-question-bounded. Source-derived consequence:
earth:sealed-forecast-outcome-carrier__predeclared-comparison-relation__targets-metri
cs-baselines-adverse-cases-missingness-and-time-held__forecast-question-bounded. Exact
boundary correspondence: true. External evidence selected the survivor: false. Formal structure relabelled as direct
measurement: false. Model, forecast, proxy or retrieval relabelled as observation: false.

Sources and complete feature accounting:

- NASA-EARTHDATA-SEARCH-001 - NASA Earth Science Data and Information System, [Earthdata Search data discovery and access](#); class authoritative_data_catalog; transport captured; features 4/4 present; present: dataset identity, spatial and temporal coverage, instrument or platform provenance, download or service access; absent and preserved: none.
- USGS-LANDSAT-ACCESS-001 - U.S. Geological Survey Landsat Missions, [Landsat Data Access](#); class authoritative_remote_sensing_archive; transport captured; features 0/4 present; present: none; absent and preserved: sensor and mission identity, Earth surface observation, archive access, product and processing distinction.
- NOAA-GHCND-001 - NOAA National Centers for Environmental Information, [Global Historical Climatology Network Daily](#); class authoritative_station_climate_archive; transport captured; features 5/5 present; present: station identity and metadata, daily weather observations, quality flags, record length and variable coverage, archive version; absent and preserved: none.
- NOAA-PALEO-001 - NOAA National Centers for Environmental Information World Data Service for Paleoclimatology, [Paleoclimatology data archive](#); class authoritative_proxy_archive; transport captured; features 3/4 present; present: archive and study identity, proxy and reconstructed-variable distinction, location and investigator provenance;

absent and preserved: dating and temporal coverage.

- EARTHSCOPE-SAGE-SERVICES-001 - NSF SAGE Facility operated by EarthScope Consortium, **Seismological and geophysical web services**; class authoritative_geophysical_data_service; transport captured; features 3/4 present; present: station and channel identity, waveform and event access, instrument response; absent and preserved: observed and derived product distinction.

Comparison record:

- sft-earth-forecast-validation-001-external-earth-evidence: predicted earth:sealed-forecast-outcome-carrier__predeclared-comparison-relation__targets-metrics-baselines-adverse-cases-missingness-and-time-held__forecast-question-bounded; source-derived earth:sealed-forecast-outcome-carrier__predeclared-comparison-relation__targets-metrics-baselines-adverse-cases-missingness-and-time-held__forecast-question-bounded; exact match True
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if the target or score enters before sealing, if unfavourable forecasts are omitted, or if accuracy is treated as proof of an opaque law.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json;
 source manifest sha256:34b420fdc55923f3eb18fdcbd9bc4aba0fa04515e7dc2a1624e13c49780bc246;
 derivation seal sha256:6e28fefaf70600ba9ca42f6fd021f7ef8879397c34ff3a9bc8a8dca911f4968ef;
 engine receipt sha256:6a4d0b638c170dbb8dd13c7f253637e3b1c4127806a2e1f68fab74a05e49563d at receipts/engine/model_admitted/SFT-EARTH-FORECAST-VALIDATION-001-6a4d0b638c170dbb.json;
 empirical validation sha256:47f81f4cb6fa7b1d02060c79235b16d1f6f9d96258b0a27f632f3ead920776cc;
 measurement receipt
 sha256:42f5d612289238a9a992c8aafc98d18ea6096b1a5439019379b3bf4da191f7a; isolation
 sha256:3cd979050b3a8c28fbd04a7b6e8f221d28b66639adb69d4849dcbbf428b44452; custody
 sha256:c3beaaf362c493400a51aa85841d8fcb1f083625b3a346ef92494ac5f03d8d20.

74. Hazard, risk, resilience and disciplinary handoff

Claim identity: SFT-EARTH-HAZARD-RISK-HANDOFF-001

Question and exact theorem. Earth Science owns the physical or environmental hazard record; risk composes exposure and vulnerability at explicit Social, Medicine and Engineering handoffs, while resilience retains recovery capacities and failures.

```
process-exposure-vulnerability-consequence-carrier__decision-context-bounded__haza
rd-to-impact-and-recovery-chain__event-probability-exposure-vulnerability-capacity
-time-and-ownership-held__evidence-class-explicit__root-bound-forward-forcing__pos
itive-finite-successor-retains-all-rows__no-extra-rule
```

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free or fitted parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-PROBABILITY-STATISTICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-ENTROPY-UNCERTAINTY-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MECH-CONSERVATION-001
- SFT-PHYS-THERMO-FIRST-LAW-001
- SFT-PHYS-THERMO-SECOND-LAW-001
- SFT-PHYS-FLUID-CONSERVATION-001
- SFT-PHYS-WAVE-PROPAGATION-001
- SFT-CHEM-MEAS-SUBSTANCE-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-MAT-MEAS-MATERIAL-001
- SFT-MAT-MEAS-PROPERTY-001
- SFT-MAT-THERM-CONDUCTION-001
- SFT-BIO-ECOSYSTEM-001
- SFT-BIO-BIO-HANDOFF-001
- SFT-EARTH-ENVIRONMENTAL-QUALITY-001
- SFT-EARTH-FORECAST-VALIDATION-001

Every dependency precedes this claim in the admitted census and recursively reaches the premise-free root. A branch label, publication or consensus statement cannot substitute for that receipt graph.

Generated grammar. Generate the literal Cartesian product of eight registered binary carrier, boundary, relation, record, evidence, provenance, generality and extension dimensions before external Earth-source identities or outcomes are opened. Filter only by the preregistered property required by each dimension.

Boundary: Exactly eight binary preservation dimensions, giving 256 forms. Form closure is depth-independent under the stated positive-finite successor; empirical correspondence remains source, place, time, method and protocol bounded.

The literal eight-axis product contains 256 candidates, 256 stored candidate identities and 256 one-for-one decisions. Exactly one survives and 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	answer-only-or-unbounded-carrier	process-exposure-vulnerability-consequence-carrier	An answer-only value or unbounded object erases the Earth carrier. The complete claim-specific carrier is retained.
boundary	boundary-or-interface-erased	decision-context-bounded	Erasing the spatial, temporal or categorical boundary makes stocks, flows and observations incomparable. The exact observation and system boundary remains explicit.
relation	imported-fitted-or-missing-relation	hazard-to-impact-and-recovery-chain	An imported, fitted or absent relation can manufacture the intended Earth result. Only generated Fold relations and admitted upstream dependencies act.
record	favourable-output-with-incomplete-record	event-probability-exposure-vulnerability-capacity-time-and-ownership-held	A favourable output without stocks, transfers, time, missing and adverse records is not reconstructible. The complete required record is retained.
evidence	model-proxy-forecast-or-observation-conflated	evidence-class-explicit	Earth evidence classes cannot silently substitute for one another. Direct observation, retrieval, proxy, reconstruction, model and forecast remain distinguishable.
provenance	prior-target-consensus-or-application-selected	root-bound-forward-forcing	Earlier SFT answers, target values and consensus models cannot select a V3 law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-case-with-erased-alternatives	positive-finite-successor-retains-all-rows	One favourable site, event or interval cannot close a generated class. Every supplied finite extension retains favourable, adverse, absent and unresolved rows.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	Fitting, exceptions and opaque predictors can force a preferred answer. No rule beyond the admitted dependencies and registered preservation conditions is added.

Unique survivor and depth independence. Sole survivor: process-exposure-vulnerability-consequence-carrier__decision-context-bounded__hazard-to-impact-and-recovery-chain__event-probability-exposure-vulnerability-capacity-time-and-ownership-held__evidence-class-explicit__root-bound-forward-forcing__positive-finite-successor-retains-all-rows__no-extra-rule .

Base: The least complete Earth carrier retains its boundary, relation, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite reservoir, interface, observation, event, time, species or location preserves all earlier identities, transfers, adverse rows and boundaries and appends the new trace.

Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- carrier-present: the claim retains a nonempty named carrier; passed true.
- relation-distinct: the carrier and relation are not silently conflated; passed true.
- evidence-boundary: the evidence boundary remains distinct from the result carrier; passed true.
- exact-fold-witness: the family's pre-source exact Fold construction reproduces; passed true.

Scientific meaning. Hazard is not risk. Exposure, vulnerability, capacity and consequence are additional carriers whose ownership passes to social, medical and engineering branches at explicit boundaries.

Field observations, remote sensing, proxy reconstruction, model assimilation and forecasts are separate evidence classes.

Hazard, exposure, vulnerability, capacity and consequence remain different carriers. Forecast success cannot admit an opaque relation, and a model output cannot be relabelled as observation.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:bf4905122fe65dc3d4183eb3ab880cd9e87acd676f167e425bf43f2d51e49693.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:f0ebf9be8f49b146940b8e85bac133bf746800355701289a95da19b9474aa43a.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:d918e3cacc991641dbcad4c296887632280b0fa7895571bcb1309df828bc9079.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:246f88c7e3545d03957d87ad567b97378c9b2f0dcb21a09e6ed26976848b8e7b.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:d6ed46880b80ac03b9b0f12ce0c5915a3caba2101ae6dfc694386ddd954cddb5; certificate
sha256:06254fdbb04b826d658f25d1cf39c7ee1b5654f97fb53dab693471d41052a56; external-validation
identity sha256:a3178278d22b8c0ab5bfd1c1a664a872dcde0c1cdd61c96c1ed7e89713652d25.

Post-seal empirical comparison. Directness:

authoritative_observation_or_primary_measurement_boundary_correspondence. Disposition:
purpose_matched_authoritative_source_features_present. Predicted consequence: earth:process-e
xposure-vulnerability-consequence-carrier_hazard-to-impact-and-recovery-chain_event
-probability-exposure-vulnerability-capacity-time-and-ownership-held__decision-contex
t-bounded. Source-derived consequence: earth:process-exposure-vulnerability-consequence-carrie
r_hazard-to-impact-and-recovery-chain_event-probability-exposure-vulnerability-cap
acity-time-and-ownership-held__decision-context-bounded. Exact boundary correspondence: true.
External evidence selected the survivor: false. Formal structure relabelled as direct measurement: false. Model, forecast,
proxy or retrieval relabelled as observation: false.

Sources and complete feature accounting:

- NASA-EARTHDATA-SEARCH-001 - NASA Earth Science Data and Information System, [Earthdata Search data discovery and access](#); class authoritative_data_catalog; transport captured; features 4/4 present; present: dataset identity, spatial and temporal coverage, instrument or platform provenance, download or service access; absent and preserved: none.
- USGS-LANDSAT-ACCESS-001 - U.S. Geological Survey Landsat Missions, [Landsat Data Access](#); class authoritative_remote_sensing_archive; transport captured; features 0/4 present; present: none; absent and preserved: sensor and mission identity, Earth surface observation, archive access, product and processing distinction.
- NOAA-GHCND-001 - NOAA National Centers for Environmental Information, [Global Historical Climatology Network Daily](#); class authoritative_station_climate_archive; transport captured; features 5/5 present; present: station identity and metadata, daily weather observations, quality flags, record length and variable coverage, archive version; absent and preserved: none.
- NOAA-PALEO-001 - NOAA National Centers for Environmental Information World Data Service for Paleoclimatology, [Paleoclimatology data archive](#); class authoritative_proxy_archive; transport captured; features 3/4 present; present: archive and study identity, proxy and reconstructed-variable distinction, location and investigator provenance; absent and preserved: dating and temporal coverage.
- EARTHSCOPE-SAGE-SERVICES-001 - NSF SAGE Facility operated by EarthScope Consortium, [Seismological and geophysical web services](#); class authoritative_geophysical_data_service; transport captured; features 3/4 present; present: station and channel identity, waveform and event access, instrument response; absent and preserved: observed and derived product distinction.

Comparison record:

- sft-earth-hazard-risk-handoff-001-external-earth-evidence: predicted earth:process-exposure-vulnerability-consequence-carrier__hazard-to-impact-and-recovery-chain__event-probability-exposure-vulnerability-capacity-time-and-ownership-held__decision-context-bounded; source-derived earth:process-exposure-vulnerability-consequence-carrier__hazard-to-impact-and-recovery-chain__event-probability-exposure-vulnerability-capacity-time-and-ownership-held__decision-context-bounded; exact match True
- source features: 67 present; 24 absent preserved; 91 registered
- the original failed transport and every adverse, absent and unresolved row remain preserved
- a deliberately changed external consequence was rejected

Falsification boundary: Reject if hazard is equated with impact, if a probability is asserted without a reference class, or if Earth Science reowns social, clinical or engineered consequences.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- conventional Earth model as premise
- proxy or retrieval relabelled as direct observation
- model or forecast relabelled as observation
- opaque predictor as law
- erased adverse, absent, censored, missing or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/earth_environment_foundation_complete_pre_source.json;
 source manifest sha256:25856fb71ff5f20cb4e42ca47e3ca435df898004dce816e60ea7209377c5b10f;
 derivation seal sha256:d2ddd25590c3a8263bf31a43df6ad588be556d7f45abff1189ac55be340ec38d;
 engine receipt sha256:27d3e0cf47bb4199bbd260c9643df2afd0ea24975e8799ab5ada6ca9a9985eed at receipts/engine/model_admitted/SFT-EARTH-HAZARD-RISK-HANDOFF-001-27d3e0cf47bb4199.json;
 empirical validation sha256:20d709d2724757a2de2f5e50a70c005db2bf5d125a20ad1b9290aae90febd137;
 measurement receipt
 sha256:5d384f47cf1ec12e78d38e7508a6cc689d16231144d0a88391234b64b19e5432; isolation
 sha256:5c3ebbcc6c642c34652fd134c375eeb7879b9b32f073001a2036d2aa1683b873; custody
 sha256:e3cb606cac9db47799cab145e26766cb380dba35eb0ac289c588f41be50d9ca5.

Complete result and verification summary

The foundation contains exactly 74 admitted laws and 18,944 generated candidates. Exactly 74 forms survive. All 296 structural controls pass. Seventy-four independent implementations reproduce candidate order, decisions and sole survivor. Seventy-four post-seal comparisons pass at their declared boundary, while the mixed-catalogue earthquake test remains explicitly adverse inside the one quantitative claim. All three inherited atomic questions reconcile. Every dependency path reaches There Is No Nothing. The engine seal is

sha256:4f4cdd7986808e6a6102d650c85e6093d6425e49f14a5f05d70fa05e6031d46a; the authority seal is sha256:bf810a190b504f0f874a778a52e23251904b17b40a7364135e74b34e8ba0c3b8. Neither protected surface was modified.

The branch is current-evidence closed and extension-open, not permanently locked. New evidence or a new Earth relation may extend, correct or invalidate a claim only through the same public process: exact statement, frozen grammar, complete enumeration, unique survivor or preserved halt, independent reconstruction, adverse controls, purpose-matched evidence and unchanged-engine receipt.

Full-field continuation roadmap

This foundational edition is the dependency-safe base. Later versioned editions will extend, in order, through:

- complete solid-Earth, geodesy and geomagnetism census
- complete mineralogy, petrology, sedimentology, stratigraphy and palaeontology handoff
- complete geomorphology, soils and landscape dynamics
- complete hydrology, cryosphere, oceanography, atmosphere and meteorology
- complete climate, palaeoclimate, forcing, feedback and attribution
- complete ecology, biodiversity and coupled Earth systems
- complete pollution, toxic transport and environmental quality
- complete hazards, risk, resilience, resources and sustainability accounting
- complete Earth observation, data assimilation and forecast-validation census

Other planets and cosmic populations remain Astronomy and Cosmology-owned. Human-environment systems pass to Social and Collective Systems. Intervention design passes to Engineering Translation. Applications may test the laws but never select them.

Limitations

- This is the complete frozen Layer One foundation, not the complete Layer Two reconstruction of every Earth and environmental discipline.
- Seventy-three claims make structural and evidence-boundary comparisons; they do not pretend that site-, time-, instrument- or method-dependent magnitudes are universal constants.
- The earthquake exponent is the only exact quantitative natural-data prediction in this edition. Its first test is adverse and remains adverse; the independent homogeneous holdout is compatible and does not erase it.
- Bounded Earth-ionosphere modes are derived without a dimensional frequency prediction in this edition.
- The two-basin tipping witness is structural and explicitly makes no universal physical threshold claim.
- Twenty-four absent source features and one failed original transport remain visible. They are not support.
- External catalogues and documentation can change; this release binds captured bytes, source identities, dates and transport history.
- Current-evidence closure is revocable by lawful new evidence and does not constitute permanent ownership of knowledge.

Conclusion

Foundational Earth and Environmental Sciences is current-evidence closed and extension-open: 74 laws, 18,944 exact candidates, 74 sole survivors, 296 passing controls, 74 independent reconstructions, 74 post-seal comparisons and 3 of 3 inherited atomic obligations reconciled.

The scientific result is not a new opaque Earth model. It is the opposite: an exact constitution for what an Earth object, boundary, ledger, history, observation, prediction and adverse result must retain before it can become scientific evidence. It derives a quantitative earthquake exponent before measurement and preserves both its adverse first test and compatible holdout. It derives bounded recurrence without fitting a frequency and tipping structure without pretending that a mathematical half is a measured planetary threshold.

This is how Earth science remains empirical without becoming answer-selected. A sensor is not a law. A model is not an observation. A missing row is not zero. A forecast score is not derivation. An institutional label is not proof. Open criticism remains unrestricted; scientific admission is the reproducible act of satisfying the same public standard.

Repository and publication status

- Canonical repository: <https://github.com/MettaMazza/ernos-labs-sft-platform>
- Zenodo DOI: <https://doi.org/10.5281/zenodo.21640810>
- Author: Maria Smith, Ernos Labs
- Contact: Maria.Smith.Sftoe@gmail.com
- Submissions: <https://discord.gg/ucwGryVxGr>
- Current state: published open access

References

Primary and authoritative Earth evidence:

- NASA Earth Science Data and Information System. [Earthdata Search data discovery and access](#). Registered class: authoritative_data_catalog.
- U.S. Geological Survey Landsat Missions. [Landsat Data Access](#). Registered class: authoritative_remote_sensing_archive.
- U.S. Geological Survey National Cooperative Geologic Mapping Program. [National Geologic Map Database](#). Registered class: authoritative_geoscience_archive.
- U.S. Geological Survey. [Soil geochemistry and mineralogy data portal](#). Registered class: authoritative_soil_observation_archive.
- NSF SAGE Facility operated by EarthScope Consortium. [Seismological and geophysical web services](#). Registered class: authoritative_geophysical_data_service.
- U.S. Geological Survey Geomagnetism Program. [USGS geomagnetic observatory operations](#). Registered class: authoritative_geomagnetic_observation_network.
- U.S. Geological Survey Earthquake Hazards Program. [FDSN Earthquake Catalog web service](#). Registered class: authoritative_event_catalog_api.
- Smithsonian Institution Global Volcanism Program. [Volcanoes of the World web services](#). Registered class: authoritative_volcano_eruption_database.
- U.S. Geological Survey Water Resources. [National Water Information System](#). Registered class: authoritative_hydrological_observation_network.
- U.S. Geological Survey, U.S. Environmental Protection Agency and National Water Quality Monitoring Council. [Water Quality Portal web services](#). Registered class: authoritative_water_quality_federation.
- National Snow and Ice Data Center. [Sea Ice Index data and image archive](#). Registered class: authoritative_cryosphere_data_product.
- NOAA National Centers for Environmental Information. [Global Historical Climatology Network Daily](#). Registered class: authoritative_station_climate_archive.
- NASA Langley Research Center CERES. [CERES Energy Balanced and Filled data products](#). Registered class: authoritative_radiative_budget_data_product.
- NOAA Global Monitoring Laboratory. [Carbon-cycle greenhouse-gas measurements](#). Registered class: authoritative_atmospheric_composition_archive.
- NOAA National Centers for Environmental Information. [World Ocean Database](#). Registered class: authoritative_ocean_profile_archive.

- Argo Data Management Team. [Argo profile and trajectory data access](#). Registered class: authoritative_ocean_observing_program.
- NOAA National Centers for Environmental Information World Data Service for Paleoclimatology. [Paleoclimatology data archive](#). Registered class: authoritative_proxy_archive.
- U.S. Environmental Protection Agency. [AirData pre-generated monitoring files](#). Registered class: authoritative_air_quality_archive.
- U.S. National Science Foundation National Ecological Observatory Network. [NEON open ecosystem data](#). Registered class: authoritative_ecosystem_observation_program.
- Global Biodiversity Information Facility. [GBIF Occurrence API](#). Registered class: authoritative_biodiversity_data_federation.
- Virgo and KAGRA magnetometer measurement collaboration. [Measurement and subtraction of Schumann resonances at gravitational-wave interferometers](#). Registered class: primary_measurement_study.

Smithian Fold Theory branch dependencies:

- Smith, Maria. *From Nothing to Fold*. doi:10.5281/zenodo.21515629.
- Smith, Maria. *From Fold to Mathematics*. doi:10.5281/zenodo.21516146.
- Smith, Maria. *From Distinction to Information*. doi:10.5281/zenodo.21516916.
- Smith, Maria. *From Fold to Physics*. doi:10.5281/zenodo.21520881.
- Smith, Maria. *From Fold to Chemistry*. Canonical branch paper in the repository and Zenodo series.
- Smith, Maria. *From Fold to Materials*. Canonical branch paper in the repository and Zenodo series.
- Smith, Maria. *From Fold to Life*. doi:10.5281/zenodo.21630203.

Open-science evidence supporting the institutional argument:

- Lundh A et al. *Conflicts of interest at the European Medicines Agency: a policy analysis*. 2018. <https://doi.org/10.1007/s00134-018-5293-7>.
- Fabbri A et al. *The influence of industry sponsorship on the research agenda*. 2018. <https://pubmed.ncbi.nlm.nih.gov/30252531/>.
- Gallo SA et al. *Reliability and fairness in peer review of research funding*. 2023. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10553257/>.
- Demicheli V and Di Pietrantonj C. *Peer review for improving the quality of grant applications*. Cochrane review and archive. <https://pmc.ncbi.nlm.nih.gov/articles/PMC8973940/>.
- UNESCO. *Recommendation on Open Science*. 2021. <https://www.unesco.org/en/legal-affairs/recommendation-open-science>.
- NIST. *Artificial Intelligence Risk Management Framework 1.0*. <https://airc.nist.gov/airmf-resources/airmf/3-sec-characteristics/>.

Complete current-census successor supplement

The authoritative current scope of this paper is 75 live model-admitted claims (earth_environment 75). Counts retained inside the preceding-version narrative describe the dated freeze at which those passages were written; this successor table and the machine census control the current total.

The preceding manuscript did not contain 1 later-admitted claim section(s). They are included below in dependency order so the successor covers the complete current branch surface.

Normalized Earth tipping Fold lock

Claim ID: SFT-EARTH-TIPPING-FOLD-LOCK-002

Exact statement: The two exact normalized basin carriers are the quarter-One and three-quarter-One preimages of half-One; dimensional thresholds remain system-specific measurements.

Dependency route: SFT-EARTH-EARTH-SYSTEM-TIPPING-001 -> SFT-FOUNDATION-HALF-ONE-001 -> SFT-MATH-DYNAMICAL-SYSTEMS-001

Candidate generator: Generate the complete eight-axis product for SFT-EARTH-TIPPING-FOLD-LOCK-002 and reconstruct its exact witness independently.

Grammar boundary: Every antipodal two-basin normalized Fold carrier with dimensional-threshold controls.

Complete census: 256 candidates; 1 unique survivor; 4 controls

Survivor: exact-positive-fold-parts-and-counts__quarter-three-quarter-common-half-one-lock__normalized-law-measured-correspondence-separate__complete-source-and-adverse-record__complete-declared-product__derivation-seal-before-target__favorable-adverse-absent-heterogeneous-unresolved__no-extra-rule

Exact result: Quarter-One and three-quarter-One are distinct basins, both Fold to half-One and together complete One. Crossing into the shared normalized lock takes one Fold. This does not assert one universal dimensional Earth threshold.

Closure: depth_independent; minimality True; named-shape uniqueness True

External status: independently_replicated

Engine receipt: sha256:3a8ebb9a7cc49c18c5615b8744e02128bcd6414c2300362b1f8458e2b78eec9a at receipts/engine/model_admitted/SFT-EARTH-TIPPING-FOLD-LOCK-002-3a8ebb9a7cc49c18.json

Control	Passed	Expected	Observed
false_promise	True	reject the form missing the first structural preservation	reject the form missing the first structural preservation
tampered_source	True	reject a changed source identity	reject a changed source identity
tampered_artifact	True	reject a missing, duplicate or additional survivor	reject a missing, duplicate or additional survivor
boundary	True	reject forbidden values, target feedback and extra laws	reject forbidden values, target feedback and extra laws

Preceding-version abstract preserved for chronology

The following abstract is reproduced from version 1.0.0. Its counts and status language describe that dated version and do not override the current abstract or census above.

This paper reports the current-evidence-closed, extension-open foundation of Earth and Environmental Sciences in the third clean-room Smithian Fold Theory reconstruction. Seventy-four obligations in twelve ordered families generate 18,944 exact candidate forms and decisions, seventy-four unique survivors, seventy-four depth-independent certificates, 296 passing adverse controls, seventy-four implementation-distinct reconstructions and seventy-four post-seal empirical-boundary comparisons. Every dependency chain reaches the single premise-free theorem, There Is No Nothing. The branch uses zero axioms, zero free or fitted parameters, no negative proof quantities, no irrational or imaginary proof values, no target-selected law and no opaque predictor.

The foundation reconstructs Earth-system identity, reservoirs, boundaries, stocks, flows, energy and matter ledgers, geological order and history, interior and geodynamic carriers, rupture and seismic records, volcanism, water and ice, atmosphere, weather, ocean, climate, biogeochemical cycling, environmental transport and quality, observational evidence classes and hazard handoffs. It explicitly distinguishes observation from retrieval, proxy, reconstruction, model and forecast; observed absence from non-detection, censoring and missingness; stock from flow; hazard from risk; and universal upstream law from contingent Earth history.

The quantitative headline is a pre-source exponent prediction for earthquake magnitude-frequency counts. Exact candidate enumeration uniquely selected exponent 1. The first global 2010-2020 mixed-magnitude catalogue was adverse: $N \geq 5/N \geq 6 = 18469/1494$ and its registered exceedance interval $0.076998990840681139-0.084918164588559525$ excludes $1/10$. That result remains adverse and unclassified. A separately preregistered homogeneous 2020-2026 mww holdout returned $N \geq 6/N \geq 7 = 253/27$, or $27/253$ exceedance; its interval $0.085657503235220589-0.13089188889528297$ contains $1/10$. The branch also

derives bounded distinct Earth-ionosphere recurrence modes without importing a dimensional frequency and derives a two-basin tipping structure while rejecting a universal physical half-threshold.

All 356 V1 rows and 407 V2 steps were atomically reviewed. The three Earth-owned prior questions are all reconciled to current receipts. Twenty-one post-seal external source identities contribute ninety-one preregistered features: sixty-seven are present and twenty-four are absent and preserved. One original failed transport remains recorded; an independently preregistered alternate transport is an addendum, not a silent replacement.

Scope and ownership boundary

This paper owns the `earth_environment` surface comprising 75 current claims. Ownership identifies responsibility for derivation and falsification; it does not prevent criticism or lawful cross-branch use. Application performance, consensus, credentials and Lean acceptance cannot transfer ownership or select a law.

Mathematical, computational and empirical constitutions

The mathematical constitution retains exact generated carriers, relations, candidate grammars, decisions, survivors and closure boundaries. The computational constitution retains executable identities, complete traces, controls, independent reconstructions and receipts. The empirical constitution retains source registration, transport, pre-seal and post-seal chronology, custody, measurement or observation, uncertainty, favourable and adverse rows, missingness, unresolved records and falsification conditions. None of these evidence classes substitutes for another.

Registered source surface

Human-readable scholarly references remain in the inherited paper. Machine source IDs, custodians, releases, locators, source captures, access outcomes, hashes and transport status remain in the current claim packages and external-source registries. Source prestige is not evidential authority; each source is used only in its registered role. The Lean source-binding gate checks custody and identity, not empirical truth by reputation.

Data and code availability

The successor Markdown is

`publications/successors/earth_environment/FROM_ONE_WORLD_TO_EARTH_PAPER_001_V1_1.md`
and its scientific predecessor is

`publications/current/earth_environment/FROM_ONE_WORLD_TO_EARTH.md`. The Lean sources and report are under `generated/lean4_validation/`; the report is `generated/lean4_validation/reports/whole_model_validation.json`. Current claims are indexed by `census/claims.json`; complete candidates, decisions, controls, certificates, observations and receipts remain under `claims/<claim-id>/` and `receipts/`. The paper's evidence map and draft deposit metadata sit beside the successor source. No remote action was performed.

Successor machine-identity appendix

- Preceding source: `publications/current/earth_environment/FROM_ONE_WORLD_TO_EARTH.md`
- Preceding source SHA-256:
sha256:044d381d988635cb69a368ff9dbfe2fad8d87b31cec7f4edd76faee959c60354
- Lean report: `generated/lean4_validation/reports/whole_model_validation.json`
- Lean report SHA-256:
sha256:6b6a5a9a9a0b74687a6f97fb3b4fcc7455968e35c82395290837cda60d7501cf
- Ordered census SHA-256:
sha256:f3f540f77a9b677bf7ddb9f5c6ea1ea41f08e57bb07163e25f385fd4d1dcde71
- Execution manifest SHA-256:
sha256:517fd288f4484f43fdc0343b17fcabd06ea9e12c977a4512d4a891dd038fce41

- Protected engine seal:
sha256:4f4cdd7986808e6a6102d650c85e6093d6425e49f14a5f05d70fa05e6031d46a
- Verification-authority seal:
sha256:bf810a190b504f0f874a778a52e23251904b17b40a7364135e74b34e8ba0c3b8
- Publication authorised: false
- Remote actions performed: false

Lean 4 independent verification update

Verified result

The pinned Lean toolchain `leanprover/lean4:v4.32.0` compiled the verification project and the executable verifier returned `PASS`. Its immutable report identity for this successor is

sha256:6b6a5a9a9a0b74687a6f97fb3b4fcc7455968e35c82395290837cda60d7501cf.

Verified surface	Result
Ordered census claims	2,777
Proof-bearing accepted claim gates	2,777
Source-bound claims	2,777
Candidate records	898,902
Decision records	898,902
Passed controls	11,108
Registered branches	17
Issues	0

This paper's registered branch surface contributes **75** of the accepted claims.

Native theorem proof and artifact verification are different layers

`SFTValidation/Root.lean` formalises the exact registered two-member operational grammar:

`unpresentedAbsence` and `presentedOccurrence`. The decision function rejects the former and accepts the latter.

Lean proves existence by the accepted constructor and uniqueness by exhaustive case analysis over the two constructors.

`#print axioms` reports no imported or user-declared axioms for the exported root theorems. This is a kernel-checked proof of the unique survivor inside the exact registered operational grammar.

`SFTValidation/Gates.lean` defines twelve Boolean acceptance obligations and can construct an inhabitant of `ClaimGate.Accepted` gate only when all twelve equal `true`. The runtime verifier then parses the complete ordered census and execution manifest, recomputes identities, cardinalities, one-to-one decision coverage, survivor count, controls, closure fields, empirical boundaries, certificate bindings and authoritative receipts, and calls that proof-bearing constructor for every current claim.

The exact scientific statements of the other claims are not all re-expressed as 2,776 separate Lean propositions in this version. They are validated as the repository's complete registered machine artifacts through Lean-executed checks. The source-manifest gate uses a read-only Python bridge solely to instantiate the already registered source factories; Lean consumes the result. This layer does not replace the protected admission engine, does not write receipts and does not substitute for source experiments or empirical replication.

Fail-closed custody event

The first whole-model run halted on six source-capture byte hashes. The discrepancy was traced to line-ending normalisation in the working tree, not to a changed scientific target or a failed theorem. The registered source bytes were restored, exact-byte Git attributes were added for those six captures, all 385 registered external source bindings were re-audited with no mismatch, and the unchanged Lean verification was rerun to `PASS`. The preserved halt demonstrates that source drift is detected rather than silently forgiven.

What the result supports

The result materially strengthens the model's case for formal coherence, exhaustive current-census coverage, unique-survivor enforcement, dependency and provenance integrity, cross-branch consistency and reproducibility by an implementation outside the admission engine. It does not by itself establish that every empirical claim is true in nature, that the registered grammar is the only conceivable grammar outside its stated boundary, or that SFT is superior to every rival theory. Those questions remain governed by the papers' declared experiments, falsification conditions and comparative tests.

Successor conclusion

Version 1.1.0 preserves the preceding paper's derivations, evidence classes, corrections, adverse and unresolved records, ownership limits and open frontier, while integrating the independent Lean 4 result and every later current claim in its declared scope. This paper contributes 75 current claim(s) within `earth_environment` to the total dependency spine.

The new formal result establishes that the registered two-class operational root has one Lean-proved survivor and that all 2,777 current claims satisfy the proof-bearing artifact gates. The major conflation resolved by this update is between native theorem proof and whole-repository artifact verification: both are valuable, but they are not the same evidential act. No empirical status is strengthened merely because the artifact graph passes, and no historical halt or correction is erased.

The current evidence therefore establishes formal coherence, current-census completeness, unique-survivor enforcement and intact source, dependency, certificate and receipt custody for this version. Field-specific empirical conclusions remain exactly those authorised by their current records. Lawful discovery, falsification, stronger evidence and versioned extension remain open, and publication itself remains pending Maria Smith's explicit confirmation.